Paper 4 – Recent Advances in Urology: All Answers by Section

Compiled from: 2015–2025 actual Paper 4 (12 papers) and Practice Sets 1–6 Paper 4, including both the current and the earlier versions of Practice Sets 3–6 Paper 4 — 165 answers.

Repeats merged: questions asked more than once are printed once, with every wording and source listed under "Asked as"; their answers are combined so that no fact from any version is lost. 35 merged topics replace 91 original answers; 109 topics in all.

Layout: the 17 revision-book sections; in each, long answer questions (LAQ) first, then short answer questions (SAQ), then a Rapid Revision Sheet of that section's key points.

Contents

Section 1 · Basic Principles — Long answer questions (LAQ)

1. Robotics in urology (including robotic vs standard laparoscopy) *(merged: 4 questions)*
2. LASERs in urology: applications and newer generation laser technologies (new LASER for stone treatment) *(merged: 5 questions)*
3. Flexible endourological instruments, digital and current ureteroscopes *(merged: 3 questions)*
4. Natural orifice endoscopic surgery.
5. Tissue engineering in urology, including recent advances *(merged: 2 questions)*
6. Laparoendoscopic single-site (single-port) surgery: LESS *(merged: 2 questions)*
7. Basic energy modalities in urological surgery, including bipolar energy *(merged: 2 questions)*

Section 1 · Basic Principles — Short answer questions (SAQ)

1. Urethral stents and metallic stents in urology *(merged: 3 questions)*
2. Describe evidence based medicine.
3. Operating room safety : the role of surgeon.
4. Urological applications in nanotechnology.
5. Different types of trocars in laparoscopy.
6. General complications related to laparoscopy.
7. Thalium laser in urology
8. Morcellation
9. KTP*** laser in urology
10. Stem cell therapy
11. Holmium laser
12. Hemostasis in lap surgery
13. Advances in lap surgeries
14. Artificial intelligence in urology
15. 3D laparoscopy
16. Telemedicine in urology
17. Simulators in urology
18. Virtual and augmented reality in urology
19. 3D printing in urology

Section 2 · Oncology, DSD, Gender & Ageing — Long answer questions (LAQ)

1. Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T.
2. Discuss in brief, present concepts in various GU cancers

Section 2 · Oncology, DSD, Gender & Ageing — Short answer questions (SAQ)

1. Prepucioplasty.
2. Enlist the steps in female to male gender reassignment surgery.
3. Penile Inversion Vaginoplasty Technique
4. Antioxidants in urological malignancy
5. Gene therapy in urology
6. Molecular targeted therapy in urological cancers

Section 3 · Infections & Inflammation — Long answer questions (LAQ)

1. Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management *(merged: 2 questions)*

Section 3 · Infections & Inflammation — Short answer questions (SAQ)

1. Chronic non bacterial prostatitis.
2. Nitrofurantoin.

Section 4 · Paediatric Urology 1 — Long answer questions (LAQ)

1. Hypospadias: present concepts in management, and hypospadias fistula *(merged: 4 questions)*
2. VUR: etiopathogenesis, recent trends in management and STING technique *(merged: 3 questions)*

Section 5 · Paediatric Urology 2 — Short answer questions (SAQ)

1. Vesicostomy.
2. Management of Defecation Disorders resulting into LUTS

Section 6 · Female Urology — Short answer questions (SAQ)

1. Acute urinary retention in females.

Section 7 · Voiding Dysfunction — Long answer questions (LAQ)

1. Neuromodulation for voiding dysfunction, including sacral neuromodulation *(merged: 4 questions)*

Section 7 · Voiding Dysfunction — Short answer questions (SAQ)

1. Video Urodynamics
2. Botulinum toxin in urology
3. Detrussor underactivity.

Section 9 · Andrology & Sexual Medicine — Long answer questions (LAQ)

1. Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival.

Section 9 · Andrology & Sexual Medicine — Short answer questions (SAQ)

1. Male menopause (andropause, late-onset hypogonadism) *(merged: 3 questions)*
2. Klinefelter's syndrome.
3. Semen analysis nomenclature.
4. Penile implants.

Section 10 · Testis, Penis & Urethra — Short answer questions (SAQ)

1. Testis sparing surgeries for CA testis
2. Videoendoscopic inguinal lymphadenectomy
3. Surgical management of primary scrotal cancer.
4. Muscle assisted urethroplasty *(merged: 2 questions)*
5. Robotic RPLND
6. Gracilis myocutaneous flap in Uro.

Section 11 · Stones, Obstruction & Trauma — Long answer questions (LAQ)

1. Urolithiasis: etiopathogenesis, medical management of (recurrent) stone disease and recent advances in RIRS instrumentation *(merged: 2 questions)*
2. Energy sources for breaking stones in urology *(merged: 2 questions)*

Section 11 · Stones, Obstruction & Trauma — Short answer questions (SAQ)

1. Tubeless PCNL *(merged: 2 questions)*
2. Miniperc: miniaturised PCNL, its nomenclature and newer PCNL techniques *(merged: 3 questions)*
3. Struvite stones.
4. Role of open and laparoscopic stone surgeries in modern era of endourology.
5. Diet in stone disease *(merged: 2 questions)*
6. Newer methods of stone analysis and their significance in management of calculus disease *(merged: 2 questions)*
7. ECIRS (endoscopic combined intrarenal surgery)

Section 12 · Bladder Cancer & Diversion — Long answer questions (LAQ)

1. Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion *(merged: 6 questions)*
2. Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy *(merged: 3 questions)*
3. Discuss bladder preservation surgery for TCC of urinary bladder.
4. Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer?

Section 12 · Bladder Cancer & Diversion — Short answer questions (SAQ)

1. Optical coherence tomography in carcinoma bladder *(merged: 2 questions)*
2. Re-look (second-look) TUR bladder tumour *(merged: 2 questions)*
3. Secondary tumours after urinary diversion *(merged: 2 questions)*
4. VI-RADS
5. Virtual cystoscopy
6. Bladder transplantation
7. Artificial urinary bladder

Section 13 · Kidney & UTUC — Long answer questions (LAQ)

1. Radiology for solid renal masses.

Section 13 · Kidney & UTUC — Short answer questions (SAQ)

1. Nephron Saving Surgery (NSS)
2. Immuno modulation in Renal, cell carcinoma
3. Zero-ischaemia partial nephrectomy (laparoscopic and robotic)

Section 14 · Adrenal & Retroperitoneum — Long answer questions (LAQ)

1. Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management *(merged: 2 questions)*

Section 15 · Renal, Nephrology & Transplant — Long answer questions (LAQ)

1. Allograft rejection, immunobiology of renal transplant, marginal kidney donor and newer immunosuppressant drugs *(merged: 4 questions)*

Section 15 · Renal, Nephrology & Transplant — Short answer questions (SAQ)

1. Brain-stem death: guidelines for declaration and the Board of Medical Experts (brain dead committee) *(merged: 2 questions)*
2. Vascular access for dialysis.
3. Female kidney donors: special precautions *(merged: 2 questions)*

4. Hospital authorization committee for live related kidney donation.
5. Marginal donor and Remuzzi score for pre-implantation biopsy *(merged: 2 questions)*
6. Laparoscopic donor nephrectomy (including pros and cons) *(merged: 2 questions)*
7. Steroid-free immunosuppression in renal transplantation
8. Artificial kidney
9. Robotic kidney transplantation

Section 16 · BPH — Long answer questions (LAQ)

1. HoLEP (holmium laser enucleation of the prostate) *(merged: 2 questions)*
2. Prostate anatomy, newer surgical concepts in CA prostate and neurovascular bundle preservation in radical prostatectomy *(merged: 2 questions)*

Section 16 · BPH — Short answer questions (SAQ)

1. Silodosin *(merged: 2 questions)*
2. UroLift (prostatic urethral lift) *(merged: 2 questions)*
3. Green light lasers

Section 17 · Prostate Cancer — Long answer questions (LAQ)

1. Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate?
2. Discuss advances in management of CA prostate

Section 17 · Prostate Cancer — Short answer questions (SAQ)

1. Cryotherapy for carcinoma prostate
2. Saturation biopsies of prostate
3. Prostate cancer prevention trials
4. GnRH (LHRH) analogues: agonists and antagonists *(merged: 2 questions)*
5. Combined Androgen Blockade.
6. PSMA PET Ct scan
7. Methods to predict prognosis of patient of Cancer prostate
8. PARP inhibitors in Prostate cancer.
9. PI-RADS and MRI–ultrasound fusion biopsy
10. Indications of genomic testing in carcinoma prostate

Section 1 · Basic Principles — Long answer questions (LAQ)

Robotics in urology (including robotic vs standard laparoscopy)

Asked as:

- Role of robotics in urology — *Summer 2015 Paper IV Q1 (LAQ)*
- Discuss Robot and its role in urology. — *Winter 2017 Paper IV Q1 (LAQ)*
- Robots in urology — *Practice Set 4 Paper IV Q2 (LAQ)*
- Robotic vs std lap surgeries — *Practice Set 3 Paper IV Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2017-Winter Paper 4, Q1; 2015-Summer Paper 4, Q1) (robotics also answered in 2017-Winter Paper 4, Q1 and 2015-Summer Paper 4, Q1)

Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- Robotic surgery is **robot-assisted laparoscopy** with a **master–slave telemanipulator** (surgeon console, patient-side cart, vision cart). Access and physiology are the same; the interface differs.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled. RARP now dominates in the USA.

History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)); "Mona" first human trial (1997).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP (**Binder and Kramer**) in **2000**; robotic adrenalectomy 1999; first paediatric robotic procedure **2002**; robotic RPLND (retroperitoneal lymph node dissection) 2006.

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) view magnified about 10 times, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; most have a 10-use limit; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

Generation	Added feature
Standard (2000)	Three arms
S (2006) / Si (2009)	High definition, fourth arm / dual console for training
Xi (2014)	8-mm camera in any port, finer arms, side docking (docks from either side), no stirrups
SP (2018)	All instruments through one 2.5-cm cannula
da Vinci 5 (2024)	Force feedback

- Patents expired in 2019 (after patent expiry), so other platforms appeared (e.g. **Hugo, Versius** and the low-cost Indian **SSI Mantra**).

- **Also know:** ports 8–10 cm apart to avoid arm collisions; Xi docks from any side; retroperitoneal cases dock over the head.

Robotic vs standard laparoscopy: comparison

Feature	Standard laparoscopy	Robotic
Vision	Mostly 2D; assistant holds camera	**3D**, 10× magnified, surgeon controls camera
Instruments	Rigid, **4 DOF (degrees of freedom)**; fulcrum effect reverses movement	**EndoWrist, 7 DOF**, 90° articulation; intuitive movement
Precision	Tremor transmitted	**Tremor filtration, motion scaling**
Ergonomics	Standing, awkward	Seated, less fatigue
Suturing	Hard; steep learning curve	Easy; shorter learning curve
Haptic feedback	Present (limited)	Absent (force feedback only in newest system)
Port placement	Flexible	Ports 8–10 cm apart; docking time
Bedside	Surgeon at patient	Surgeon remote; emergency undocking delays
Cost	Low, reusable	High capital; instruments limited to about 10 uses
Failure	—	Robot failure about 0.4%

Advantages and disadvantages

Advantages	Disadvantages
3D vision, wristed instruments, easy suturing	High capital and instrument cost; cost-effective only at high volume
Tremor filtration, seated ergonomics	Limited haptic feedback
Less blood loss and shorter stay than open; fewer conversions than laparoscopy	Docking time, bulk, surgeon away from patient

Applications (with evidence)

1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, with less blood loss and shorter stay; **Coughlin 2018**: equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable. Bladder neck stricture under 2% vs up to 16% open.
- **Retzius-sparing** and **"hood"** techniques preserve continence structures and give earlier continence; ERAS (enhanced recovery after surgery) allows 23-hour or same-day discharge. Robotic surgery also widens candidacy (prior mesh, prior pelvic surgery) and allows salvage surgery.
- Versus laparoscopy: laparoscopic prostatectomy stalled on the vesicourethral anastomosis; the robot made it routine and dominant. A small RCT (randomised controlled trial; Porpiglia, 2013) favoured robotic continence and potency.

2. RAPN (robot-assisted partial nephrectomy)

- Versus laparoscopy (robotic vs laparoscopic gives): **fewer conversions**, **shorter warm ischaemia**, less (smaller) eGFR (estimated glomerular filtration rate) loss/fall and shorter

stay; complications and margins similar. So more complex and hilar tumours are done minimally invasively.
- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.
- ICG (indocyanine green) guides selective clamping.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), with less blood loss and fewer transfusions (less transfusion).
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications. Continent diversion fell (11% in iROC).

4. Other procedures

Procedure	Role of the robot / why the robot helps
Pyeloplasty, ureteric reimplantation	Easy intracorporeal suturing; also in children and infants (small working space: about 1 L vs 6 L in adults); similar success to laparoscopy; easier suturing
BMG (buccal mucosa graft) ureteroplasty	Long proximal or mid-ureteric strictures
RPLND (retroperitoneal lymph node dissection)	Template, nerve-sparing; short stay
Adrenalectomy, nephroureterectomy	Equivalent to laparoscopy; adrenalectomy suits the retroperitoneal approach
Simple prostatectomy	Large glands; less bleeding than open
Sacrocolpopexy, VVF (vesicovaginal fistula) / fistula repair	Deep pelvic suturing
RAKT (robot-assisted kidney transplantation)	Fewer wound complications in obese recipients

- **Radical nephrectomy, adrenalectomy:** no outcome advantage over laparoscopy; robot only adds cost.
- Outcomes depend more on the **surgeon** than on the platform.

Choosing robot vs laparoscopy

- Robot: complex excision plus reconstruction. Laparoscopy: extirpation, low-resource settings.

Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

Physiology and positioning

- **CO_2 pneumoperitoneum** (12–15 mm Hg, lower in children; another source: work at 10–12 mm Hg) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery increases venous return and cardiac output, and raises intracranial and intraocular pressure.
- The only strict contraindication to minimally invasive surgery is inability to tolerate pneumoperitoneum.

Complications specific to robotics

Problem	Details
Steep Trendelenburg (~22–25°)	Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion
Nerve injury (up to ~6%)	Brachial plexus (avoid shoulder braces); femoral, peroneal
Compartment syndrome, rhabdomyolysis	Long lithotomy or flank position; fasciotomy if >30 mmHg with symptoms
Thermal bowel injury / bowel burns	Insulation failure, capacitive coupling; presents late
Robot malfunction	About 0.4% (another source: <1%); half found before the patient enters; rarely forces conversion
Port-site hernia	Close larger port sites; transverse incisions cut it from 5.4% to 0.4% after RARP

- **Prevention**: secure inaccessible arms and lines before tilting, pad pressure points, limit tilt and lithotomy time, protect the eyes.

Training and credentialing

- Box trainer, **VR (virtual reality)** simulator (simulators), animal and cadaver labs, then proctored cases (proctoring). **FRS (Fundamentals of Robotic Surgery)** curriculum improved skills in an RCT. Five-day mini-residencies also help. Dual console and simulators for training. Proficiency comes after about 40 cases, but outcomes keep improving (outcomes improve) with volume.

Recent advances

- Single-port transvesical and extraperitoneal surgery; da Vinci SP (single port); force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency; latency limits it).
- New platforms (Hugo RAS, Versius, Indian SSI Mantra) may cut costs.
- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

Diagram to draw: console, vision cart and patient cart; RARP port placement; EndoWrist (7 DOF) beside a rigid instrument (4 DOF).

Key points to remember:

- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
- Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

LASERs in urology: applications and newer generation laser technologies (new LASER for stone treatment)

Asked as:

- Describe applications of various LASERs in urology. — *Summer 2017 Paper IV Q2 (LAQ)*
- Newer generation laser technologies. — *Winter 2024 Paper IV Q3c (SAQ)*
- New LASER for stone Treatment — *Winter 2025 Paper IV Q3b (SAQ)* (also answered in 2024-Winter Paper 4, Q3c)
- Newer generation laser technologies. — *Practice Set 3 Paper IV Q3e (earlier version) (SAQ)* (also answered in 2024-Winter Paper 4, Q3c; stone lasers in 2025-Winter Paper 4, Q3b)
- New LASER for stone Treatment — *Practice Set 3 Paper IV Q3f (earlier version) (SAQ)* (See Practice Set 3 Paper 4, Q3e (full answer).)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Principle

- LASER (light amplification by stimulated emission of radiation): a "pumped" gain medium releases identical photons. A mirror resonator amplifies them, and a partial mirror lets the beam out.
- Light is **monochromatic, coherent and collimated**, so it passes down thin fibres.
- **Chromophore** decides the effect. Water absorbs holmium and thulium light. Haemoglobin absorbs KTP (potassium titanyl phosphate) light.
- Absorption sets **penetration**. Shallow penetration vaporises and cuts. Deep penetration coagulates and later sloughs.
- Coagulation occurs at about 60 °C; vaporisation at ≥100 °C.
- **Pulse duration** decides the stone mechanism. Short pulses (<10 µs) act photomechanically (shock waves). Long pulses (>250 µs: Ho:YAG [holmium:yttrium-aluminium-garnet], TFL [thulium fibre laser]) act **photothermally**.

Lasers used

Laser	Wavelength (nm)	Chromophore	Penetration	Main uses
Ho:YAG	2120 (Ch 19 quotes 2100; often quoted as 2100 nm)	Water	0.3–0.4 mm (other sources: 0.4 mm; 0.31 mm)	Stones, HoLEP, incisions, tumours
TFL	1940	Water (peak)	<0.1 mm (other sources: about 0.2 mm; 0.08 mm)	Dusting, enucleation
Tm:YAG	~2010 (other sources: 2013)	Water	0.2 mm	Vaporesection, enucleation
KTP/LBO	532	Haemoglobin	0.8 mm	PVP
Diode	940–1470	Water + haemoglobin	0.5–5 mm	Prostate vaporisation
Nd:YAG	1064	Neither strongly	Up to 10 mm	Coagulation of tumours
CO_2	10,600	Water	~0.1 mm	Skin and penile lesions

1. Stones

- **Ho:YAG** is the gold standard (standard for stones and enucleation). It breaks all stone types. Used in URS (ureteroscopy), RIRS (retrograde intrarenal surgery), PCNL

(percutaneous nephrolithotomy) and for bladder stones. It is water-absorbed, pulsed and **photothermal**: long pulses heat the stone until it decomposes.

- **Moses effect:** the start of each pulse makes a vapour channel; the rest crosses it to the stone. So keep the fibre close and perpendicular.
- Fragmenting: 0.6–1.0 J × 6–10 Hz. Dusting: 0.2–0.3 J × 40–80 Hz.
- **Retropulsion** rises with pulse energy (not frequency) and falls with longer pulses or pulse modulation.
- **TFL:** higher water absorption and low peak power give finer dust and less retropulsion. It beat low-power holmium in RCTs (randomised controlled trials) and equalled 120 W pulse-modulated holmium.
- **Heat** depends on power, not laser type. Keep ≤10 W in the ureter; use good flow and an access sheath.

2. BPH (benign prostatic hyperplasia)

- **HoLEP (holmium laser enucleation of prostate):**
  - Endoscopic equivalent of open prostatectomy, **size-independent**. Enucleate at about 2 J × 40–50 Hz, then morcellate.
  - Only endoscopic procedure superior to TURP (transurethral resection of prostate). Less bleeding, shorter catheter time.
- **PVP (photoselective vaporisation, KTP/LBO 120–180 W):**
  - Haemoglobin absorption gives bloodless vaporisation, so anticoagulants can continue.
  - **GOLIATH RCT:** non-inferior to TURP. Avoid >80 cc (retreatment).
- **Thulium:** vaporisation, vaporesection, ThuLEP (equivalent to HoLEP). Diode vaporisation also used.
- Nd:YAG coagulation was abandoned (1 cm penetration, sloughing).
- AUA (American Urological Association): HoLEP, ThuLEP or PVP for **high bleeding risk**.

3. Incisions

- Laser urethrotomy has no advantage over cold knife.
- Holmium incision of bladder neck contracture, ureteric stricture, endopyelotomy, ureterocele and calyceal diverticulum neck.

4. Urothelial tumours

- **NMIBC (non-muscle-invasive bladder cancer):** laser **en bloc resection** (Ho, Tm, KTP) gives detrusor in the specimen and no obturator jerk. Office ablation of small low-grade recurrences.
- **UTUC (upper tract urothelial carcinoma):** biopsy, then ablate by ureteroscopy:
  - Ho:YAG: <0.5 mm, haemostatic.
  - Nd:YAG: 5–6 mm coagulation, so stricture risk.
  - Thulium: <0.2 mm.
- **PDT (photodynamic therapy):** a photosensitiser plus laser light makes singlet oxygen.

5. Penile and genital lesions

- PeIN (penile intraepithelial neoplasia) and small Ta/T1: **Nd:YAG preferred**, with base biopsies.
- CO_2 is too superficial for T1 (recurrence up to 50%). CO_2 or Nd:YAG for genital warts.

Safety

- Wavelength-specific eyewear for all, including the patient. Door signage, covered windows, trained staff and a laser safety officer.

- Check fibres; never fire inside the scope. Guard against fire and thermal injury.
- **Heat follows power (total power), not laser type (applies to all):** ≤10 W in the ureter, ≤25 W in the kidney; use low power, pauses, high flow, cool irrigation and an **access sheath**; damage (urothelial injury) begins above about 43 °C.

Recent advances: newer generation laser technologies

- Pulse modulation (Moses) and high-frequency holmium. TFL and thin fibres. Suction sheaths. AI (artificial intelligence) settings.

Background

- Newer lasers aim at **less retropulsion, finer dust, faster work** (Aims: less retropulsion, finer dust, faster work).

1. High-power holmium (60–120 W; other sources: 100–120 W machines) with pulse modulation

- High frequency (100–120 W machines allow it) allows **dusting** (low energy, high frequency: 0.2–0.3 J × 40–80 Hz) and "popcorning"; long-pulse modes reduce retropulsion.

2. Pulse modulation (Moses technology and similar)

- **Moses effect:** the start (first part) of a pulse boils a **vapour channel**; vapour absorbs less light, so the rest reaches the stone (target).
- **Moses technology** splits each pulse in two: the first opens the channel, the second hits the target (strikes the stone).
- Less retropulsion; may shorten **HoLEP (holmium laser enucleation of the prostate)**.

3. TFL (thulium fibre laser), 1940 nm

- **Diode-pumped silica fibre** laser, at (near) the **water absorption peak** (higher absorption than holmium: about **4× holmium absorption**); penetration about 0.2 mm (other sources: **0.08 mm**).
- **Low peak power, long pulses, very high frequency (above 2000 Hz)** → finer (fine) dust, **minimal retropulsion**.
- **Thin fibres** (down to about 150 µm): more (better) irrigation and scope deflection.
- **Stones:** finer dust, minimal retropulsion. **RCTs (randomised controlled trials):** TFL beat **low-power (30 W) holmium** (stone-free rate, time); it **equalled 120 W pulse-modulated holmium** in stone-free rate (**equal** to 120 W pulse-modulated holmium).
- **Prostate:** ThuFLEP (thulium fibre laser enucleation).
- Compact and quiet.

4. Thulium:YAG (2013 nm)

- Originally continuous wave (0.2 mm penetration, carbonisation a drawback): ThuVARP (vaporesection), ThuVEP (vapoenucleation), **ThuLEP (laser enucleation**, size-independent like HoLEP).
- **Pulsed Tm:YAG (thulium:YAG):** newer pulsed solid-state version of the continuous-wave thulium laser for photothermal lithotripsy and enucleation; it also does lithotripsy.

Comparison

	Ho:YAG	TFL
Wavelength	2120 nm	1940 nm
Water absorption	High	About 4× higher
Penetration	0.31 mm	0.08 mm
Peak power	High	Low
Retropulsion	More	Minimal (Less)

	Ho:YAG	TFL
Fragments	Small	Smaller
Fibre	≥200 µm	≥150 µm

Principles of settings

- **Retropulsion rises with pulse energy**, not frequency; it falls with longer pulses.
- Keep the fibre close and perpendicular to the stone.
- TFL settings ≤40 Hz often enough.

Others

- **Also know:** KTP/LBO (potassium titanyl phosphate/lithium triborate) 532 nm vaporisation (180 W GreenLight (532 nm) vaporisation); diode lasers; suction sheaths; burst wave lithotripsy (investigational).

Diagram to draw: water and haemoglobin absorption curves marking 532, 1064, 1940 and 2120 nm (for newer lasers: water absorption curve marking 1940 and 2120 nm).

Key points to remember:

- Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
- Holmium lithotripsy is photothermal (Moses); TFL gives less retropulsion.
- HoLEP size-independent, superior to TURP; PVP for anticoagulated men (GOLIATH).
- Nd:YAG for PeIN; eyewear for everyone.
- TFL 1940 nm: water peak, low peak power, minimal retropulsion.
- Moses: two-part pulse through a vapour channel.
- TFL beats low-power, equals high-power pulse-modulated holmium.
- Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney.

Key points to remember (stone angle):

- TFL (1940 nm): fine dust, minimal retropulsion, thin fibres; settings ≤40 Hz often enough.
- Pulse-modulated (Moses) high-power holmium: vapour channel, less retropulsion.
- Safety: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

Flexible endourological instruments, digital and current ureteroscopes

Asked as:

- Briefly outline flexible endourological instrument and its role in management of urological disease — *Practice Set 1 Paper IV Q1 (LAQ)*
- Digital flexible ureteroscopes — *Summer 2019 Paper IV Q3c (SAQ)*
- Current ureter scopes. — *Summer 2022 Paper IV Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(flexible ureteroscopy in detail: see Practice Set 1 Paper 2, Q3b; digital scopes also in 2019-Summer Paper 4, Q3c)

Summer 2022 Q3d taken as "current ureteroscopes"; see also 2019-Summer Paper 4, Q3c.

Introduction

- A flexible endoscope has a **deflectable tip** and carries its image by a fibre bundle or a tip-mounted chip. It follows the natural curves of the urethra, ureter and collecting system.
- Physical basis: light travels along a curved glass fibre by **total internal reflection** (Tyndall).
- Marshall reported the first flexible URS (ureteroscopy) in 1964. It became therapeutic only when a **working channel** and **active tip deflection** were added in the early 1980s.
- **Fibreoptic flexible URS (ureteroscopy):** coherent glass-fibre bundle carries the image; fibres break → black spots; honeycomb (moiré) pattern.
- **Digital scopes** (first 2006): a sensor at the tip (**"chip-on-tip"**) converts light to an electrical signal, so the image does not degrade.

Classification of current ureteroscopes

Type	Key features
Rigid rod-lens	Larger calibre; largely historical
Semirigid	**Fibreoptic** image bundle; tip **≤7 Fr**; channel >3 Fr; often **two channels**; in-line or offset eyepiece
Flexible fibreoptic	Tip 4.5–7.5 Fr (smallest); image degrades as fibres break (black dots)
Flexible digital	"Chip-on-tip" CMOS (complementary metal-oxide semiconductor) or CCD (charge-coupled device) sensor plus LED (light-emitting diode); no camera head
Single-use flexible	Digital; LithoVue (2016) first; now several makers
Ureteral resectoscope	11.5 Fr; largely replaced by laser

Semirigid ureteroscope

- Uses **coherent fibre bundles** for the image (random fibres carry only light).
- **Two channels** are preferred: irrigation through one while an instrument is in the other, or a laser through one to break a stone held in a basket.
- **Offset eyepiece** gives a straight channel for rigid probes (pneumatic, biopsy).
- **Rule:** semirigid below the iliac vessels, flexible above.

Components

Component	Key features
Optics	**Fibreoptic:** a *coherent* image bundle carries the picture; a *random* light bundle carries illumination; cladding improves reflection. **Digital:** CCD (charge-coupled device) or CMOS (complementary metal oxide semiconductor) chip at the tip with LED (light-emitting diode) light; no camera head or light cable
Deflection	Lever → control wires through movable metal rings → wire fixed at the tip (deflection mechanism: lever pulls wires). "Intuitive" = tip moves with the lever. The reticle (notch) marks the plane of deflection
Working channel	Single channel shared by irrigation and instruments, so instruments reduce flow and deflection
Shaft	Tapered tip; passive (secondary) deflection segment behind the tip helps lower-pole access

- Flexible ureteroscope parts: optics, deflection mechanism (lever pulls wires), working channel.

Sensor technology (digital)

	CCD (charge-coupled device)	CMOS (complementary metal-oxide semiconductor)
Heat and power	More	Less (runs cooler)
Cost, speed	Higher, slower	Cheaper, faster
Status	Phased out	Current standard

- **LED (light-emitting diode)** at the tip; no camera head or light cable → better ergonomics.

Types

Instrument	Specifications	Main use
Flexible cystoscope	14–17 Fr; 37–40 cm; deflection 120–220°; channel 6.4–7.5 Fr; field about 120°	Office diagnosis and minor therapy
Flexible ureteroscope	Tip 4.5–7.5 Fr; deflection ≥270° both ways; channel **3.6 Fr**; dual-channel scope (2 × 3.3 Fr) exists	Kidney and upper ureter
Flexible nephroscope (or flexible cystoscope used through the tract)	Larger calibre; too wide for the UPJ (ureteropelvic junction)	Clearing calyces during PCNL (percutaneous nephrolithotomy)

Deflection and channel (flexible ureteroscope)

- Active deflection **≥270° both directions** (another source: **>270° both ways** in one plane); enhanced primary deflection eases lower-pole access / reaches the lower pole (ureteroinfundibular angle up to about 175°).
- Working channel **3.6 Fr**; dual-channel scopes exist.

Fibreoptic vs digital (digital versus fibreoptic)

Feature	Fibreoptic	Digital
Image	Honeycomb pattern; black dots as fibres break; degrades	Sharper, larger (magnified), stable image; no focusing or white balance

Feature	Fibreoptic	Digital
Size / Tip	Smallest tips, tapered	Larger (about 8.5 Fr), flatter; more access-sheath use
Outcome / Operative time	Reference; longer	Shorter operating time, equal (similar) stone-free rate (SFR)
Durability	Fragile; over-rotation damages fibres; repair about every 15 cases	**Not better** in later comparisons; not clearly better in trials
Processor	Platform-agnostic	Proprietary

- Surprise finding: in vitro, illumination was better with the fibreoptic cystoscope.
- Digital: sharper image, **shorter operating time, same stone-free rate**. But larger tip, so **more access sheaths and sheath complications**.
- Larger tip → more access-sheath use. Keep a fibreoptic scope for narrow UPJ (pelviureteric junction) or infundibulum.

Single-use scopes (single-use vs reusable)

- **LithoVue (2016):** first single-use digital ureteroscope; CMOS, LED, tip 7.7 Fr, shaft 9.5 Fr, 3.6 Fr channel, 270° each way. Single-use cystoscopes followed (2020).
- Rationale (reason): reusable ureteroscopes may need repair after only **3–5 cases**; reprocessing and repair add cost.
- Single-use advantages: maximal deflection each case, sterile, no reprocessing or repair; cost **lower below about 99 cases/year**.
- Best value in low-volume centres and for **large lower-pole stones**.
- Reusable: cheaper in high-volume centres; needs reprocessing (peracetic acid, glutaraldehyde, ethylene oxide).
- Cost advantage depends on case volume; environmental impact is similar to reusables. Most centres use a **hybrid fleet**.

Accessories

- **Guidewires:** 0.035/0.038 inch PTFE (polytetrafluoroethylene)-coated safety wire; hydrophilic wires to pass impacted stones (exchange, as too slippery for safety); nitinol-core and hybrid wires.
- **Access sheath** (11–16 Fr): repeated passes, lower intrarenal pressure; small risk of ureteric injury.
- **Laser fibres** 200–365 µm; **nitinol tipless baskets** ≤3 Fr; three-prong graspers; 3 Fr cup biopsy forceps; 3 Fr balloon dilating to 12 Fr; fulguration electrodes; pressure irrigation pumps.
- Flexible EHL (electrohydraulic lithotripsy) probes work; flexible pneumatic probes restrict deflection.

Role in urological disease

Area	Role
Lower tract	Haematuria work-up; bladder cancer surveillance, office biopsy and fulguration; NBI (narrow-band imaging) and blue light; stent removal or exchange; bladder stones, foreign bodies, strictures; botulinum toxin injection; diversions through a catheterisable channel; difficult ureteric access via a suprapubic tract
Stones	RIRS (retrograde intrarenal surgery): renal stones <20 mm, lower pole <10 mm; >2 cm when PCNL is unsafe

Area	Role
UTUC (upper tract urothelial carcinoma)	Diagnostic URS and biopsy (predicts grade in about 9 of 10); laser ablation of low-grade tumours; prefer to percutaneous route (seeding)
Obstruction	Endopyelotomy without crossing vessels; endoureterotomy for strictures <1.5 cm
Diagnosis	Positive cytology, filling defects, lateralising haematuria (haemangioma fulgurated), recurrent one-sided infection; foreign body retrieval
PCNL	Antegrade or retrograde flexible endoscopy raises stone-free rate; ECIRS (endoscopic combined intrarenal surgery)

Care and complications

- **Scope damage:** deflection wear; laser fibre scraping and perforating the channel. Pass fibres with the scope straight; **fire laser only when the fibre tip is seen touching the stone** (fire only with the fibre tip seen on the stone); leave the bladder empty to avoid buckling.
- **Sterilisation:** sterilise with ethylene oxide, glutaraldehyde or peracetic acid.
- **Patient:** ureteric injury, stricture, sepsis from high pressure. Use saline irrigation and antibiotic prophylaxis for all upper tract procedures.

Recent advances

- **Intrarenal pressure sensor** in the scope (LithoVue Elite): high pressure causes fever, SIRS (systemic inflammatory response syndrome), sepsis. Temperature sensing next.
- **AI (artificial intelligence):** stone recognition, laser setting adjustment, navigation.
- **Robotic flexible URS (ureteroscopy):** ergonomics, less radiation to surgeon.
- **Also know:** suction (access) sheaths, further miniaturisation.

Diagram to draw: cross-section of a flexible ureteroscope: chip, LED, working channel and deflection wires (scope tip cross-section showing CMOS chip, LEDs, 3.6 Fr channel and 270° deflection).

Key points to remember:

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
- Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
- Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
- Semirigid: two channels; below the iliac vessels.
- Pressure sensors and AI are the newest developments.
- Golden rule: never fire unless the fibre tip is seen on the stone.

Natural orifice endoscopic surgery.

Type: LAQ (25 marks)

Source: 2022-Summer Paper 4, Q1

Taken as NOTES (natural orifice transluminal endoscopic surgery) and its relatives (hybrid NOTES, LESS [laparoendoscopic single-site surgery]).

Definition and concept

- **NOTES:** entry into the peritoneal or retroperitoneal cavity through a deliberate incision in a hollow viscus reached through a natural orifice (mouth, vagina, bladder, rectum). The aim is **"scarless" surgery**: no abdominal wall wound, so no wound pain, infection or hernia.
- It differs from **endoluminal** surgery (e.g. ureteroscopy), which stays inside the lumen.
- **Pure NOTES:** all instruments through the orifice. **Hybrid NOTES:** natural orifice plus one or more small abdominal ports. Hybrid is what is actually used in humans.
- Chapter 142 places NOTES at the end of a spectrum: transperitoneal → retroperitoneal → hand-assisted → LESS → SP (single-port) robotic → NOTES. **Each step to fewer incisions buys cosmesis and less pain at the cost of working space and triangulation.**

History (milestones)

Year	Event
2002	Transvaginal laparoscopic nephrectomy in pigs (Gettman, Cadeddu)
2004	Transgastric peritoneoscopy in pigs (Kalloo)
2006	NOSCAR (Natural Orifice Surgery Consortium for Assessment and Research) white paper listed barriers
2008–2010	Hybrid, then pure, transvaginal nephrectomy in humans

Routes of access

Route	Features	Urological use
Transvaginal (posterior colpotomy)	Commonest; easy closure under vision; large specimens pass; women only	Nephrectomy, donor nephrectomy, specimen extraction
Transgastric	Upper abdomen reachable; gastrotomy closure and leak are the main risk	Experimental (peritoneoscopy, renal surgery in animals)
Transvesical	Sterile urine; bladder dome port; small calibre; closure difficult	Peritoneoscopy (animal, Lima), transvesical access to pelvis
Transurethral	Urologist's familiar route	Experimental transurethral prostatectomy in cadavers
Transcolonic/transanal	Faecal contamination risk	Rarely urological

Hybrid NOTES and natural orifice extraction

- Standard laparoscopy does the dissection; **the vagina is the access for a camera or instrument and the extraction route** (Chapter 142). This is the only NOTES form with real clinical urological use.
- **NOSE (natural orifice specimen extraction):** kidney removed through the vagina after laparoscopic nephrectomy, avoiding the extraction incision, which is the largest and most painful wound.

- **NOTES-assisted donor nephrectomy:** laparoscopic or robotic donor nephrectomy with **transvaginal kidney extraction**. Reported benefits: less pain, faster recovery, no extraction scar; warm ischaemia acceptable.

Steps of transvaginal (hybrid) nephrectomy

1. Lithotomy then modified flank; bowel and vaginal preparation; antibiotics.
2. Umbilical port (camera or instrument) plus one or two small ports.
3. **Posterior colpotomy** under laparoscopic vision; long port or flexible scope through the vagina.
4. Colon reflection, hilar dissection, ureter and vessels stapled.
5. Kidney in a bag, delivered through the colpotomy.
6. Colpotomy closed with absorbable sutures under vision.

LESS: the close relative

- All ports through **one skin incision** (umbilicus or below the waistline) using several low-profile trocars or a multichannel port (TriPort; da Vinci SP 25–27-mm entry guide).
- Problems: **no triangulation**, instrument clashing, in-line view. Solutions: articulating or bent instruments, flexible-tip scopes, instruments of different lengths.
- Evidence (Chapter 142): randomised trial versus laparoscopy showed the same time, blood loss and complications, with less analgesia. Donor meta-analysis: lower pain scores. Oncological outcome unchanged.
- Drawback (Chapter 20): **more incisional and umbilical hernias** because of the larger single incision.
- **SP robot (FDA [Food and Drug Administration] approval 2018)** has revived single-site surgery.

vNOTES (vaginal NOTES)

- A gynaecological development: a single-port device in the vagina gives access to the peritoneum.
- **HALON RCT (randomised controlled trial):** vNOTES hysterectomy was non-inferior to laparoscopic hysterectomy, with more same-day discharge and less pain.
- Urogynaecological uses: vNOTES apical prolapse repair (uterosacral suspension), adnexal surgery.

Challenges (NOSCAR list)

- **Safe access** and **secure closure** of the viscerotomy (leak, fistula); peritoneal **infection**.
- **Spatial orientation**: retroflexed, off-axis view.
- **Platform**: flexible scopes lack stiffness, traction and triangulation.
- Control of **haemorrhage** and other complications.
- Training and cost; dyspareunia and fertility concerns after colpotomy; transvaginal route unavailable in men.
- Chapter 142: adoption is limited by **no clear therapeutic benefit**, safety concerns and very selective patients; **pure NOTES has very few urological cases.**

Recent advances

- **Flexible robotic platforms** (Chapter 39): MASTER (NOTES, haptics), ViaCath.
- **Transvesical single-port** (LESS or da Vinci SP) surgery for diverticulectomy and foreign bodies (Chapter 124).
- Magnetic anchoring and guidance systems to restore triangulation.

Also know: LESSCAR consortium standardised the term LESS; "embryonic NOTES" (umbilicus as the natural scar); hidden-incision endoscopic surgery (HIdES) in children.

Diagram to draw: sagittal female pelvis showing posterior colpotomy with a transvaginal port and kidney extraction, plus umbilical port.

Key points to remember:

- NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
- Transvaginal extraction removes the most painful wound.
- LESS: one incision, lost triangulation, more hernias.
- Barriers: closure, infection, orientation, platform; no proven benefit yet.

Tissue engineering in urology, including recent advances

Asked as:

- Tissue engineering in urology. — *Summer 2022 Paper IV Q2 (LAQ)*
- Discuss recent advances in tissue engineering with respect to urology — *Practice Set 6 Paper IV Q2 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2023-Winter Paper 1, Q1 and 2022-Summer Paper 4, Q2; this answer [Practice Set 6] stresses recent advances)

Definition and rationale

- **Tissue engineering:** applying (applies) engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Why in urology:** bowel used for bladder and diversion (bowel in the urinary tract) causes mucus, stones, metabolic acidosis, infection and malignancy (cancer); grafts for urethra (for long urethral defects) are limited in supply (scarce). The aim is **to replace "wrong tissue" with "right tissue"**.
- **Regenerative medicine** is the wider term: cells, scaffolds, growth factors and gene therapy.

The tissue-engineering triad (basic principle)

```
Cells  +  Scaffold  +  Signals (growth factors, mechanical cycling in a bioreactor)
          -> construct -> implanted -> vascularised and remodelled by the host
```

```
Biopsy → cells expanded in culture → seeded on scaffold (+ signals)
       → conditioned in bioreactor → implanted → host vascularises and remodels
```

Cell sources

Source	Features / advance and limitation
Autologous differentiated cells (urothelial and smooth muscle cells from a bladder biopsy)	Standard; no rejection; expanded in culture; unusable if the bladder is malignant or diseased
Adult stem cells: bone marrow or adipose mesenchymal, muscle-derived, **urine-derived**	Easy, non-invasive harvest; paracrine effect (act mainly by paracrine growth factors)
Embryonic stem cells	Pluripotent; ethics, teratoma risk
iPSCs (induced pluripotent stem cells)	Adult cells reprogrammed; patient-specific, no ethical issue; tumour risk

- Gene-edited and gene-modified cells are being explored. Cell sheet engineering builds layers without a scaffold.

Scaffolds (biomaterials)

- **Requirements:** biocompatible; biodegradable at the rate of tissue ingrowth (degrade as tissue forms); porous for cell attachment and vessel ingrowth; mechanically suitable/suited (compliance for bladder; compliant bladder, non-collapsing tube).

Type	Examples	Comment
Natural	Collagen, alginate, silk fibroin	Good cell recognition, weaker

Type	Examples	Comment
Acellular tissue matrices	Bladder acellular matrix, SIS (small intestinal submucosa), decellularised dermis	Retain (keep) native architecture and growth factors
Synthetic polymers	PGA (polyglycolic acid), PLA (polylactic acid), PLGA (their copolymer)	Controlled degradation and shape; no bioactivity
Newer	Electrospun nanofibres, hydrogels (mimicking extracellular matrix); **"smart" scaffolds** releasing growth factors such as VEGF (vascular endothelial growth factor); silk fibroin	Mimic extracellular matrix

- **Unseeded (acellular)** grafts (matrices) depend on host cell migration (host cells migrating in) and work (suit) only for short or onlay defects (about 0.5 cm for tubular urethra [tubes] in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

Bioreactors

- Perfuse nutrients through thick constructs and apply **cyclic stretch and filling**, so engineered bladder muscle is conditioned before implantation.
- **Also know:** bioreactors apply cyclic stretch to train smooth muscle.

Solving vascularity (the main barrier)

- Oxygen diffuses only about 100–200 µm, so thick constructs die centrally.
- Strategies: omental wrap (Atala), angiogenic factors, co-seeding endothelial cells, printed vascular channels.

Clinical applications (where we stand)

1. Bladder

- **Atala et al. (2006):** first seeded biodegradable bladder construct (autologous urothelial and muscle cells on collagen–PGA, wrapped in omentum) for augmentation in **7 myelomeningocele patients**; compliance improved.
- **Industry (Tengion) multi-institution phase II trial** in spina bifida: **no clinically or statistically significant improvement in compliance or capacity** (Chapter 48).
- Chapter 48 conclusion: a bladder with specialised urothelium and contractile muscle has **not** been achieved; it remains experimental.
- Chapter 111: an engineered bladder gives **storage only**; innervation for voluntary emptying cannot (yet) be recreated.

2. Urinary diversion: Neo-Urinary Conduit (Tengion, NCT01087687) (Chapter 135)

- Seeded biodegradable conduit to avoid bowel. Small enrolment; conduits survived about 250 days on average; **half developed stenosis or stricture.**
- Lessons: stiff, non-collapsible material; no cells formed around the connective-tissue sleeve; **no blood supply**.

3. Urethra

- Acellular bladder matrix onlay for hypospadias and stricture: success in short onlay repairs (short strictures).
- **Raya-Rivera et al. (2011, Lancet):** tubular PGA–PLGA scaffolds seeded with autologous cells in **5 boys** with posterior urethral defects; functional urethras at up to 6 years.
- Tissue-engineered oral mucosa (MukoCell) for urethroplasty, avoiding donor-site morbidity.

4. Other organs

- **Sphincter, detrusor:** autologous muscle-derived cells for SUI (stress urinary incontinence); also for underactive detrusor (Chapter 111: open-label improvement; open-label studies improved voiding efficiency). Biodegradable scaffold slings (Chapter 48).
- **Vagina:** engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59).
- **Penis:** engineered corporal tissue, animal stage. ED (erectile dysfunction) stem cell therapy is investigational (Chapter 72).
- **Kidney:** decellularised–recellularised scaffolds (animal); renal assist devices; kidney organoids.
- **Testis:** spermatogonial stem cell banking for prepubertal cancer patients (future, Chapter 4).

Organ	Status
Bladder	**Atala (2006):** seeded collagen–PGA construct with omental wrap in **7 myelomeningocele patients**, compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Still experimental (Chapter 48)
Conduit	Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis or stricture**; stiff material, no blood supply (Chapter 135)
Urethra	Acellular matrix onlay for short strictures; **Raya-Rivera (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa avoids donor-site morbidity
Sphincter, detrusor	Autologous muscle-derived cells for SUI (stress urinary incontinence); for underactive detrusor, open-label studies improved voiding efficiency (Chapter 111)
Vagina	Engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59)
Penis, testis	Corporal tissue (animal); stem cells for ED (erectile dysfunction) investigational; spermatogonial stem cell banking

3D bioprinting

- Layer-by-layer deposition (laid layer by layer) of **"bio-ink"** (cells in hydrogel) with or without polymer, driven by patient CT (computed tomography) or MRI (magnetic resonance imaging) data.
- Advantages: exact patient-specific shape, multiple (several) cell types placed precisely (placing several cell types precisely), built-in channels for vessels.
- Current urological use: 3D-printed (non-living) kidney and prostate models for planning and teaching (already aid planning and teaching); bioprinted urethra and bladder constructs are still at the animal stage.

Organoids

- 3D self-organising mini-organs grown from stem cells or tumour tissue.
- **Kidney organoids** from iPSCs: models of development and disease (model disease), drug toxicity testing.
- **Tumour organoids** (bladder, prostate): personalised drug testing (allow personalised drug testing) and biology (WNT/β-catenin work, Chapter 129).

Whole-organ decellularisation

- Kidneys are stripped of cells by detergent perfusion, leaving the matrix and vascular tree, then recellularised with the patient's cells; still at the animal stage. Renal assist devices are a parallel route.

Challenges

- **Vascularisation** of thick constructs (diffusion limit), **innervation** and contractility.
- Fibrosis and contracture; scale-up, cost, culture time; regulatory approval (regulation).
- Safety of stem cells (tumour formation; stem cell tumour risk). Disappointing controlled trials after promising pilots (promising pilots followed by disappointing controlled trials).

Recent advances

- Robotic bladder auto-transplantation and vascularised composite bladder allograft (Chapter 135; robotic retrieval of a **vascularised composite bladder allograft**): a transplant, not engineered, route to "right tissue", needing immunosuppression.
- Smart scaffolds releasing growth factors; urine-derived stem cells; gene-edited cells.

Future

- Vascularised bioprinted tissue, off-the-shelf cell-free smart scaffolds, combined gene and cell therapy.

Diagram to draw: biopsy → cell culture → seeding on scaffold → bioreactor → implant with omental wrap.

> **Key points to remember:**
>
> - Triad: cells, scaffold, signals; vascularity (100–200 µm diffusion) is the main barrier.
> - Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
> - Neo-urinary conduit (conduit trial): half stenosed; no blood supply.
> - Raya-Rivera 2011: seeded tubular urethras in 5 boys.
> - New: iPSCs, urine-derived cells, bioprinting, organoids.

Laparoendoscopic single-site (single-port) surgery: LESS

Asked as:

- Laproendoscopic single site surgery: LESS — *Winter 2023 Paper IV Q1 (LAQ)*
- Single port lap. Surgeries in urology — *Practice Set 2 Paper IV Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered as a long answer in 2023-Winter Paper 4, Q1)

Read as laparoendoscopic single-site surgery (LESS), i.e. single-port (single-site) laparoscopic and robotic surgery in urology. NOTES (natural orifice transluminal endoscopic surgery) is covered in 2022-Summer Paper 4, Q1.

Definition and concept

- **LESS:** laparoscopic or robotic surgery in which all instruments and the camera enter (pass) through **one skin incision**, usually hidden in the umbilicus (umbilical) or below the waistline.
- Aim: further reduce access trauma, giving **better cosmesis, less pain and shorter stay**. Chapter 142 places it on a spectrum: multiport → hand-assisted → LESS → SP (single-port) robot → NOTES. Each step to fewer incisions buys cosmesis at the price of working space and triangulation.
- First urological single-port nephrectomies were reported in 2007; the first single-port robotic urological surgery followed in 2009.

Nomenclature (LESSCAR)

- Early names were many: SILS (single-incision laparoscopic surgery), SPA (single-port access), OPUS (one-port umbilical surgery), TUES (transumbilical endoscopic surgery), e-NOTES (embryonic NOTES).
- **LESSCAR (Laparoendoscopic Single-Site Surgery Consortium for Assessment and Research, 2008)** agreed the single umbrella term **LESS** and set standards for reporting, training and research.
- The term was agreed by LESSCAR (LESS Consortium for Assessment and Research, 2008).
- Sub-types (variants): transumbilical or extraumbilical; **pure LESS**; **"LESS + 1"** or needlescopic-assisted (an extra 2–5-mm port); **R-LESS** (robotic LESS); purpose-built **SP (single-port) robot**.

Access: ports and devices

Option	Description
Several low-profile trocars	Placed close together through one fascial incision (the future extraction incision)
Multichannel ports	TriPort 15 (two 5-mm + one 15-mm); TriPort+ (three 5-mm + one 10-mm); SILS port; GelPOINT (gel cap, trocars placed anywhere on it)
da Vinci SP entry guide	One 25–27-mm cannula carrying three articulating (wristed) 6-mm instruments + one flexible articulating camera

- Position: modified flank (transperitoneal) or full flank (retroperitoneal SP, 2.5–3-cm incision near the 12th rib tip, balloon dilatation).

The triangulation problem

- In standard laparoscopy ports are spread so instruments meet the target at an angle (the "diamond"). In LESS all shafts are **parallel and in-line** with the camera (instruments lie parallel to the camera).

- Results: **instrument clashing**, **"crossing swords"**, poor depth perception, poor retraction and hard suturing.
- Solutions:
  - **Articulating instruments** (tip bends inside the abdomen) and **pre-bent or curved instruments**: they cross at the fascia, so the right hand works on the left side (**chiral manipulation**).
  - **Flexible-tip or 5-mm low-profile scopes**, offset optics, extra-long instruments and instruments of **different lengths** to avoid handle clash.
  - A 2-mm needlescopic instrument or extra 5-mm port (an extra needlescopic port) without hesitation.
  - **Robotic platforms** restore wristed movement.

Robotic single-site surgery

Platform	Features
da Vinci Single-Site (Si/Xi)	Curved crossing cannulas; semi-rigid **non-wristed** instruments; software swaps hand control
da Vinci SP (FDA [Food and Drug Administration] approval 2018)	Purpose-built; one arm, no external collisions; instruments with elbow and wrist; 3D articulating camera; on-screen navigator shows instrument positions

- SP needs **10–25 cm between (from) cannula and (to) target** to deploy the joints; a GelPort lets instruments "float" outside the skin. Assistance through an **"SP + 1" port** or ROSI (remote-operated suction-irrigation). Limited by cost and availability.

Indications

- **Kidney:** simple and **donor nephrectomy**, radical nephrectomy, partial nephrectomy, **pyeloplasty**, nephroureterectomy, cyst decortication, cryoablation.
- **Adrenalectomy**; varicocelectomy; orchidopexy and pyeloplasty in children (paediatric).
- **Pelvis:** SP radical prostatectomy (transperitoneal, extraperitoneal); **SP transvesical simple prostatectomy** and bladder diverticulectomy.
- **Best case:** thin patient (patients), small or benign lesion (lesions), experienced surgeon.
- **Relative contraindications (avoid in):** obesity, large or hilar tumours, complex vascular anatomy, dense adhesions, unfit for pneumoperitoneum.

Outcomes versus conventional laparoscopy (Chapter 142)

- **Randomised trial (RCT, randomised controlled trial):** same operative time, blood loss, complications and stay; **less analgesia**.
- **Donor nephrectomy meta-analysis:** same stay, recovery, transfusion and conversion (otherwise equal); **lower pain scores**.
- **Renal mass:** same perioperative outcome and survival.
- **SP robotic partial nephrectomy vs multiport:** shorter stay, less opioid; same margins and complications; **longer warm ischaemia** in one review.
- An **extra port is often needed**; conversion is mostly to laparoscopy, rarely open.
- **Net effect:** gains are confined (limited) to **pain, cosmesis and stay**; oncological and functional outcomes are unchanged, but the operation is technically harder and longer.

Complications

- All laparoscopic complications (usual laparoscopic injuries: access injury, vascular and bowel injury, CO_2 effects).
- **Incisional and umbilical hernia** is higher (larger fascial defect): about 5% with da Vinci Xi single-site, lower with SP.

- Wound infection; conversion; longer operation; surgeon ergonomic strain. Prevent by careful fascial closure and case selection.

Recent advances

- **da Vinci SP** with same-day discharge pathways; **SP retroperitoneal** renal surgery.
- **Transvesical SP** prostatectomy; hidden incision endoscopic surgery (HIdES) in children.
- New single-port platforms; magnetic anchoring and guidance systems / instruments (restore triangulation).

Diagram to draw: umbilical multichannel port with articulating (crossed) instruments crossing at the fascia (chiral) versus a triangulated multiport diamond (layout).

Key points to remember:

- LESS = one incision; LESSCAR (2008) standardised the term.
- Core problem: lost triangulation; solved by articulating or curved instruments and robots.
- Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
- da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

Basic energy modalities in urological surgery, including bipolar energy

Asked as:

- Describe Basic Energy Modalities in Urological surgery? — *Winter 2025 Paper IV Q2 (LAQ)*
- Bipolar energy — *Practice Set 1 Paper IV Q3b (SAQ)* (also covered in 2025-Winter Paper 4, Q2)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction

- Every modality delivers energy that tissue absorbs and releases as **heat, mechanical stress or cavitation**. Frequency sets penetration; absorption sets the effect.
- Tissue effects of heat: about 60 °C coagulates protein; 100 °C vaporises cell water; higher temperatures carbonise.

Classification

Energy	Examples
Electrical	Monopolar, bipolar, vessel sealers, RFA (radiofrequency ablation), IRE (irreversible electroporation), EMDA (electromotive drug administration)
Electromagnetic	Lasers, MWA (microwave ablation)
Acoustic	Ultrasonic dissector, SWL (shock wave lithotripsy), HIFU (high-intensity focused ultrasound), histotripsy
Direct contact	Cryoablation, water vapour, waterjet (Aquablation)

1. Electrosurgery

- **Physics:** alternating current at about **500 kHz** (radiofrequency). Mains current (50 Hz) depolarises nerve and muscle; at 500 kHz ions only oscillate and produce heat.
- **Ohm's law** ($V = I \times R$): tissue is the resistance. **Current density** decides the effect: a small tip concentrates current and heats; a large pad disperses it.
- **Monopolar:** generator → active electrode → patient → dispersive pad → generator. Partial pad contact causes burns.
- **Bipolar:** current passes only between two jaws (**only between two electrodes on the same instrument**); less power, less spread, no pad (no dispersive pad; the patient is not in the circuit).

Bipolar energy in detail

- Radiofrequency current (about 500 kHz). Less power and less lateral thermal spread (1–6 mm vs 5–7 mm monopolar). In endoscopy, **saline** irrigation can be used.

	Monopolar	Bipolar
Circuit	Loop → patient → pad	Active and return poles on the instrument
Irrigant	Glycine 1.5%, sorbitol–mannitol	Normal saline
TUR (transurethral resection) syndrome	Yes	Theoretically eliminated
Capacitive coupling, pad burns	Yes	No

- **Endoscopic uses:**

    - **Bipolar TURP (transurethral resection of prostate):**
    - *Plasma system:* energy vaporises saline around the loop into plasma (orange glow of sodium); cuts at lower temperature while sealing vessels; platinum–iridium loop.
    - *TURis (transurethral resection in saline):* the sheath acts as return electrode; chapter notes report strictures and fistulae.
    - *True bipolar:* dual loops.
    - Outcomes: equal IPSS (International Prostate Symptom Score) and Qmax; less transfusion and clot retention; shorter catheter time and stay. NNT (number needed to treat) about 50 to prevent one TUR syndrome.
    - **Bipolar TUVP (transurethral vaporisation):** PlasmaButton; AUA (American Urological Association) option for <30 and 30–80 cc.
    - **Bipolar enucleation:** plasmakinetic energy can perform AEEP (anatomic endoscopic enucleation of prostate) satisfactorily; comparative data limited.
    - **TURBT (transurethral resection of bladder tumour):** fewer obturator jerks and less bleeding in reviews; one RCT (randomised controlled trial) showed only less thermal artefact.
- **Laparoscopic and open uses:** bipolar forceps for pinpoint haemostasis; vessel sealers (below); uses: prostatic pedicles, cystectomy.
- **Bipolar safety:** safer near pacemakers; touching tines complete the circuit without tissue. Obturator reflex reduced, not abolished.
- **Electrocautery** (heated wire) differs: no current enters the patient.

Waveform	Character	Effect
Cut	Continuous, low voltage; 50–80 W	Rapid boiling, clean incision, little char
Coagulation	Interrupted (about 6% duty cycle), high voltage; 30–50 W	Desiccation; spread up to 3 mm
Fulguration (spray)	High voltage arcs across air	Non-contact surface haemostasis
Blend	Cut with interruptions	Cutting plus haemostasis

- **Vessel sealers (advanced vessel sealers: LigaSure, EnSeal, Caiman):** bipolar current plus jaw compression fuses collagen and elastin; a feedback generator stops when impedance rises (impedance feedback stops output). Seal vessels up to 7 mm (**≤7 mm**) in one application (one application only), with little spread (spread 1–3 mm).
- **Endoscopic:** monopolar TURP (transurethral resection of prostate) needs non-conducting **1.5% glycine**, risking hyponatraemia (TUR syndrome); **bipolar** uses saline, with comparable efficacy and better safety.
- **Obturator reflex:** lateral-wall TURBT (transurethral resection of bladder tumour) stimulates the nerve → adduction → perforation. Prevent with paralysis, obturator block, partial filling, short bursts, bipolar.
- **Safety:** pad burns; insulation failure; **direct coupling** (touching metal); **capacitive coupling** (current induced through intact insulation); fire with oxygen; inadvertent activation. Use AEM (active electrode monitoring), lowest power, bipolar near pacemakers.

2. Ultrasonic energy

- **Ultrasonic dissector:** blade vibrates at about 55.5 kHz; friction and cavitation denature protein. No current through the patient; little lateral spread, but the blade stays hot.
- Intracorporeal ultrasonic lithotripter: vibrating probe, suction of fragments.

3. Lasers

- **Principle:** pumped gain medium in a mirrored cavity → stimulated emission → **monochromatic, coherent, collimated** beam.
- **Effect** depends on wavelength (chromophore: water, haemoglobin) and pulse duration.

Laser	nm	Penetration (mm)	Use
KTP/LBO (potassium-titanyl-phosphate/lithium triborate)	532	0.8	Prostate photovaporisation (haemoglobin)
Nd:YAG (neodymium:yttrium-aluminium-garnet)	1064	10	Deep coagulation
TFL (thulium fibre laser)	1940	0.08	Stones, enucleation
Tm:YAG (thulium:YAG)	2000	0.2	Vaporesection, enucleation
Ho:YAG (holmium:YAG)	2120	0.31	Stones, HoLEP (holmium laser enucleation of prostate), strictures

- **Mechanisms:** photothermal (long pulses), photomechanical (short pulses, shock waves), photochemical (PDT, photodynamic therapy).
- **Penetration predicts behaviour:** deep (Nd:YAG) coagulates widely with delayed sloughing; shallow (thulium) vaporises precisely with char.
- Safety: eye protection for all; irrigant heating; fibre damage.

4. Ablative energies

Modality	Mechanism	Key points
RFA	375–900 kHz current; ionic agitation	Target 60 °C; char at 150 °C; heat sink
MWA	915 MHz or 2.45 GHz; dielectric hysteresis	Hotter, larger zones; less heat sink
Cryoablation	Argon expansion (Joule–Thomson)	Below −40 °C; **two freeze–thaw cycles**; ice ball visible
HIFU	Focused ultrasound absorbed → necrosis	1500–10,000 W/cm²; rice-grain lesions; transrectal prostate
IRE	High-voltage microsecond pulses → permanent membrane pores	Non-thermal; spares vessels and ducts; focal prostate

- Cryo mechanism: extracellular ice → cell dehydration → intracellular ice → membrane rupture on thawing; ischaemia and apoptosis follow.
- IRE mechanism: electric pulses open pores in cell membranes that do not reseal, so cells die by apoptosis without heat; collagen scaffolds survive. It needs general anaesthesia with muscle paralysis and ECG (electrocardiogram)-synchronised pulses.
- **Also know:** histotripsy (cavitation liquefaction, investigational); water vapour (Rezum, 103 °C, convective, no char); Aquablation (non-thermal saline jet); chemohyperthermia (42 °C) and EMDA for mitomycin.

5. Lithotripsy energies

- **SWL:** electrohydraulic, electromagnetic or piezoelectric sources. Stones break by **spallation, shear and cavitation**. Slower rates (about 1 Hz) work better and injure less.
- **Intracorporeal:** pneumatic (ballistic), ultrasonic, combined, electrohydraulic, laser. **Also know:** burst wave lithotripsy, ultrasonic propulsion.

Diagram to draw: monopolar vs bipolar circuit, with cut and coagulation waveforms.

Diagram to draw: monopolar loop-to-pad circuit beside bipolar loop with local return.

Key points to remember:

- Electrosurgery at about 500 kHz; current density decides heating.
- Cut continuous low voltage; coagulation interrupted high voltage.
- Capacitive coupling and insulation failure cause unseen burns.
- Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink.
- Bipolar: no pad, saline irrigation, no TUR syndrome.
- Bipolar TURP equals monopolar efficacy with fewer complications.
- Vessel sealers: vessels ≤7 mm, spread 1–3 mm.

Section 1 · Basic Principles — Short answer questions (SAQ)

Urethral stents and metallic stents in urology

Asked as:

- Urethral stents — *Summer 2015 Paper IV Q3d (SAQ)*
- Metallic stents in urology — *Practice Set 3 Paper IV Q3c (SAQ)*
- Urethral stents. — *Practice Set 4 Paper IV Q3f (earlier version) (SAQ)*

Type: SAQ (10 marks)

(urethral stents also answered in 2015-Summer Paper 4, Q3d; ureteric stents in 2015-Summer Paper 3, Q1)

Definition and introduction

- A urethral stent is an endoscopically placed tube that keeps the **prostatic or bulbar urethra** open, mainly in men unfit for surgery.
- Polymer stents (polyurethane, silicone) fail early under **extrinsic compression** and need frequent changes. Metal stents **resist compression** and stay longer, but suffer **encrustation, tissue ingrowth and migration**.

Classification (urethral stents)

Type	Examples	Features
Temporary	**Memokath** (nitinol), **Allium**	Not epithelialised (designed to avoid epithelial incorporation); removable; left 6–12 months
Permanent, epithelialising	**UroLume**	Urothelium grows over it; **withdrawn**
Biodegradable	Experimental	Dissolve

Classification of metallic stents (by site)

Site	Type	Example	Features
Ureter	Full-length coiled metal, double pigtail	**Resonance** (2006)	Tightly coiled alloy; changed every 12 months
Ureter	Segmental self-expanding mesh, bare or covered	Wallstent-type; covered nitinol (Allium)	Placed across malignant stricture; bare mesh allows ingrowth
Ureter	Thermo-expandable nitinol coil	Memokath 051	Removable
Prostate	Temporary thermo-expandable nitinol coil	**Memokath**	Not epithelialised; removable
Prostate, bulbar urethra	Covered self-expanding, removable	Allium	Left 6–12 months
Bulbar urethra	Permanent epithelialising mesh	**UroLume**	Withdrawn

Ureteric metal stents

- **Indications:** malignant extrinsic obstruction; retroperitoneal fibrosis (first Resonance use); benign stricture in a patient unfit for repair; ureteroenteric stricture with limited life

expectancy; failed tandem polymer stents.
- **Outcomes:** full-length metal stent: up to 3 years of drainage, pooled success about 72.5%, migration about 1%, but about one-third failed in one malignant series. Ureteroenteric strictures: patency >80% at 6–22 months.
- **Problems:** encrustation, difficult exchange, ingrowth, migration.

Urethral and prostatic stents

Indications

1. BPH (benign prostatic hyperplasia) retention in men unfit for TURP (transurethral resection of prostate) / unfit for surgery.
2. Recurrent short bulbar stricture in men unfit for urethroplasty.
3. **DSD (detrusor–sphincter dyssynergia)**: formerly (once) an alternative to sphincterotomy; now **abandoned** for complications.

Technique

- Local or regional anaesthesia with endoscopic or fluoroscopic guidance. Memokath expands when warmed (with warm water) and softens when cooled (with cold water) for removal. A bulbar stent must **not cross the external sphincter**, or incontinence follows.

Complications

- **Migration** (temporary stents), **encrustation**, **hyperplastic ingrowth** with restenosis (UroLume), incontinence, pain and painful erections.

Results (outcomes)

- Prostatic stents free most unfit men from catheters short term, but many are later removed. Memokath variable. Removable bulbar stents succeed only in small series. UroLume worked early (good early results, then ingrowth) but failed long term from **hyperplastic ingrowth** and restenosis, so withdrawn, and many men still carry one.
- No stent removes **spongiofibrosis**, so none cures a stricture.

Guidelines

- **AUA (American Urological Association)**: **no (urethral) stent if urethroplasty is considered** (planned); stents make reconstruction harder.

Recent advances

- **Optilume** paclitaxel balloon (FDA (Food and Drug Administration) approval 2023 for anterior stricture): **ROBUST III**, after failed endoscopy, about 75% vs 27% stricture-free. Optilume BPH (EVEREST-1).
- **iTind** (temporary nitinol prostatic device): nitinol device left 5–7 days; AUA 2023 option for 25–75 mL glands without median lobe.
- **Also know:** biodegradable and drug-eluting stents (experimental).

Also know: a soft silicone "urethral stent" (splint) after hypospadias repair or urethroplasty is a different device; in complicated hypospadias repairs it reduces meatal stenosis and fistula.

Diagram to draw: sagittal urethra with prostatic stent and bulbar stent distal to sphincter; prostatic Memokath ending above the external sphincter.

Key points to remember:

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

Describe evidence based medicine.

Type: SAQ (10 marks)

Source: 2020-Summer Paper 4, Q3f

Definition

- **EBM (evidence-based medicine)**: "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients" (**Sackett, 1996**).
- It integrates best research evidence, **clinical expertise** and **patient values**.

Steps ("5 As")

1. **Ask** a focused question in **PICO** form (Population, Intervention, Comparison, Outcome).
2. **Acquire**: PubMed, Cochrane Library, guidelines.
3. **Appraise**: validity (randomisation, blinding), effect size (ARR (absolute risk reduction), **NNT (number needed to treat) = 1/ARR**), precision (95% CI (confidence interval)), applicability.
4. **Apply** with expertise and patient preference.
5. **Assess** (audit) the outcome.

Hierarchy: levels of evidence (Oxford CEBM (Centre for Evidence-Based Medicine), as adapted by EAU (European Association of Urology))

Level	Type of evidence
1a	Systematic review/meta-analysis of RCTs (randomised controlled trials)
1b	At least one RCT
2a	Well-designed controlled study without randomisation
2b	Quasi-experimental (cohort) study
3	Non-experimental: case-control, correlation, case reports
4	Expert committee reports or opinion

- **Grades**: A (consistent level 1), B (level 2–3), C (level 4 or consensus); D (some systems) = no recommendation possible.
- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; RCTs start high, observational studies low; **downgrade** for bias, inconsistency, indirectness, imprecision, publication bias; **upgrade** for large effect or dose–response. Recommendations **strong or weak** (now used by EAU). AUA (American Urological Association): evidence strength A/B/C; strong, moderate or conditional recommendations.

Pitfalls in appraisal

- **Bias**: selection, confounding (comorbidity, frailty: needs risk adjustment), **lead-time**, **length-time**, stage migration (Will Rogers), attribution.
- **Surrogate end points** must meet the **Prentice criteria** (prostate cancer: only metastasis-free survival does).
- **Also know:** Clavien–Dindo; validated PROs (patient-reported outcomes); CONSORT, PRISMA, STROBE checklists; Donabedian model.

Limitations

- Few surgical RCTs; trial patients differ from real ones; publication bias; "cookbook medicine" if patient values are ignored.

Diagram to draw: evidence pyramid and the three overlapping circles of EBM.

Key points to remember:

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.

Operating room safety : the role of surgeon.

Type: SAQ (10 marks)

Source: 2021-Summer Paper 4, Q3a

Introduction

- About half of hospital adverse events involve surgery; half of these are preventable. The surgeon is **team leader** and ultimately responsible.
- **Swiss cheese model (Reason):** harm occurs when holes in several barriers line up, so safety needs layered defences.

WHO (World Health Organization) Surgical Safety Checklist (2008, 19 items)

Phase	Key checks
Sign in (before induction)	Identity, site, procedure, consent; site marked; allergy; airway; blood loss >500 mL
Time out (before incision)	Team introductions; confirm patient, site, procedure; surgeon states critical steps, duration, blood loss; antibiotic within 60 min; imaging displayed
Sign out (before leaving theatre)	Procedure recorded; counts correct; specimen labelled; equipment problems; recovery concerns

- Pilot in 8 hospitals: deaths fell 47%, major complications 36%.

Surgeon's responsibilities

1. **Universal Protocol** (Joint Commission, 2004): verification, **site marking** by the operating surgeon, patient awake, correct side only, time-out. Wrong-site surgery is a never event.
2. **Lead the time-out:** whole team engaged; repeat when procedure or surgeon changes.
3. **Retained surgical items** (sponges commonest): a count discrepancy raises risk about 100-fold; do not close until resolved; X-ray or RF (radiofrequency) tags.
4. **Fire safety:** the surgeon owns the **ignition source** (diathermy, light cables, laser). Let alcohol prep dry; FiO_2 (inspired oxygen fraction) ≤30% near the airway.
5. **Sharps:** double gloves, blunt needles, neutral zone (no hand-to-hand passing).
6. **Distractions:** "sterile cockpit" during critical steps (e.g. hilar clamping).
7. **Communication** (failure underlies up to 70% of events): Use **TeamSTEPPS** (Team Strategies and Tools to Enhance Performance and Patient Safety) tools: brief, debrief, **SBAR** (situation, background, assessment, recommendation), closed-loop check-back, **CUS** ("concerned, uncomfortable, safety issue"), two-challenge rule. Flatten hierarchy so juniors speak up.
8. **Self-monitoring: I'M SAFE** (illness, medication, stress, alcohol, fatigue, eating).

- **Also know:** positioning and padding; VTE (venous thromboembolism) prophylaxis; simulation.

Diagram to draw: Swiss cheese model with aligned holes.

Key points to remember:

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

Urological applications in nanotechnology.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 4, Q3a

Taken as "applications of nanotechnology in urology".

Definition

- **Nanotechnology:** design and use of materials of about **1–100 nm**. At this scale, surface area and optical, magnetic and thermal properties change.
- **Why useful:** particles pass through leaky tumour vessels (**enhanced permeability and retention**), carry drugs or tracers, and can be targeted with ligands such as PSMA (prostate-specific membrane antigen) antibodies.

Applications

Area	Application
Imaging and staging	**USPIO (ultrasmall superparamagnetic iron oxide) lymphotropic MRI (magnetic resonance imaging):** normal nodes take up particles and lose T2 signal; metastatic nodes do not (Chapter 6; investigational in penile cancer, Chapter 84). Gold-nanoparticle CT (computed tomography) contrast, quantum dots
Sentinel node	Technetium-99m **nanocolloid** for dynamic sentinel node biopsy in penile cancer (Chapter 84)
Diagnosis	Nanowire and nanoparticle biosensors detecting PSA (prostate-specific antigen) and urinary markers at very low levels
Drug delivery	Nanoparticle (albumin-bound) paclitaxel in urothelial and prostate cancer (Chapter 164: tolerated, no pathological complete response); **NanoPac** intratumoral paclitaxel (Chapter 163); liposomal drugs; mucoadhesive intravesical carriers; siRNA (small interfering RNA) delivery
Focal ablation	**Gold nanoshells (AuroLase):** collect in tumour vessels and are heated by near-infrared laser; about three-quarters success on 12-month biopsy (Chapter 163). Magnetic nanoparticle hyperthermia
Irreversible electroporation	Electric pulses create **nanoscale membrane pores** ("NanoKnife"); spares matrix, vessels and ducts; investigational (Chapter 143)
Devices	Silver and other nanocoatings on catheters and stents against biofilm and encrustation
Tissue engineering	Electrospun nanofibre scaffolds that mimic extracellular matrix
Stone disease	"Calcifying nanoparticles" (nanobacteria) linked to Randall plaque, not proven causal (Chapter 95); nano-CT of stones and tissue

Limitations

- Toxicity and long-term tissue accumulation; uptake by liver and spleen; manufacturing cost; regulatory hurdles. Most uses remain experimental.

Diagram to draw: nanoparticle with drug core, polymer shell and surface PSMA ligand entering a leaky tumour vessel.

Key points to remember:

- 1–100 nm; enhanced permeability and retention allows tumour targeting.
- USPIO MRI: benign nodes lose signal; metastatic nodes do not.
- AuroLase gold nanoshells: photothermal focal therapy.

Different types of trocars in laparoscopy.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 4, Q3c

Definition

- A **trocar** is a **cannula (port) plus an obturator**; 3–20 mm diameter, 5–20 cm long (Chapter 18).
- Parts: obturator tip, cannula, **valve** (keeps gas in), insufflation side-arm, retention mechanism.

Classification

Basis	Types
Obturator tip	Bladed (cutting) with spring safety shield; **bladeless (blunt dilating)**; **optical (visual)**; blunt (Hasson)
Material	Reusable metal; disposable plastic
Valve	Flap or trumpet valve (needs reducer); **multiseal** (5–12 mm, no reducer); **valveless (AirSeal)**
Retention	Smooth (suture-anchored); threaded sleeve; expandable arms; **balloon (Blunt Tip)**
Special	Hasson open cannula; radially expanding; hand ports; single-site multichannel ports; robotic 8-mm metal ports

Main types explained

- **Bladed trocars:** cut through tissue; more epigastric injury and port-site hernia. Now of historical interest.
- **Bladeless (dilating) trocars:** spread muscle fibres. About **fivefold less inferior epigastric injury**; trocar-site bleeding 3% vs 9%. Switching cut port hernia from 1.83% to 0.19%. **Fascial closure only for midline ports >10 mm** or stretched sites.
- **Optical trocars:** clear tip with a 0° scope inside; each layer is seen during entry.
- **Hasson cannula:** blunt obturator and cone, fixed with fascial stay sutures after open entry; safer with adhesions; leaks more gas.
- **Radially expanding trocar:** a sleeve over the Veress needle is dilated to 12 mm; less port-site bleeding and pain (Chapter 39).
- **Balloon (Blunt Tip) trocar:** internal balloon and external foam cuff seal the wall; prevents leak and **subcutaneous emphysema** in retroperitoneoscopy.
- **AirSeal valveless port:** stable pneumoperitoneum at 8–12 mmHg despite suction.
- **Single-site ports:** TriPort; da Vinci SP (single-port) 25–27-mm guide. **Hand ports:** GelPort.

Safety points

- **Never use a metal trocar with a plastic retaining ring**: capacitive coupling burns viscera out of view.
- Insert under vision; ports 8–10 cm apart.

Also know: Veress needle (14-gauge, spring-loaded blunt stylet) is an entry needle, not a trocar.

Diagram to draw: trocar with obturator tip types (bladed, bladeless, optical) and valve.

Key points to remember:

- Trocar = cannula + obturator.
- Bladeless: fewer epigastric injuries and hernias.
- Balloon trocar prevents retroperitoneal emphysema.

General complications related to laparoscopy.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 4, Q3e (adapted from 2019-Summer Paper 1, Q2)

Incidence

- Historically about 4%, mortality 0.03–0.08%. Contemporary series 13–22% because procedures are complex. **Vascular, then adjacent-organ injuries are commonest.**
- Risk factors: complex procedures, operating time >4 h, creatinine >1.5 mg/dL, learning curve.

Complications by phase

Phase	Complication	Key points
Access (Veress, first trocar)	Preperitoneal insufflation; vessel or bowel puncture; **major vascular injury (0.11–2%)**	Confirm Veress tests; open access with adhesions. Major vessel: leave trocar in, laparotomy
Insufflation	**Gas embolism**; subcutaneous emphysema; pneumothorax (1.6–4%); pneumomediastinum, pneumopericardium; hypercarbia	Embolism: end-tidal CO_2 (carbon dioxide) rises then falls, mill-wheel murmur → stop gas, left lateral head-down, 100% oxygen, aspirate central line
Secondary trocars	Epigastric bleeding; crossing swords, rollover	Transilluminate; bladeless trocars; insert under vision
Dissection	Thermal or mechanical bowel injury; vascular injury; bladder or ureter injury	Four electrosurgical mechanisms below
Positioning	Nerve injury (about 2.8%: brachial plexus, femoral, sciatic); **rhabdomyolysis** (about 1% retroperitoneal)	Padding, kidney rest <1 h, cases <5 h
Exit	Bowel entrapment, late port bleeding	Inspect each site at 5 mmHg
Postoperative	Port-site hernia (0.3–5.4%), wound infection, shoulder-tip pain, DVT (deep vein thrombosis)	Close fascia of large ports

Electrosurgical bowel injury: four mechanisms

1. **Inappropriate activation** out of view.
2. **Direct coupling** with another metal instrument.
3. **Capacitive coupling** (metal cannula in a plastic retainer).
4. **Insulation failure.**

- Monopolar injury presents **late (up to 18 days)**; resect with margin.

Missed bowel injury (Bishoff)

- Not classic peritonitis: **pain at one trocar site, leucopenia, low fever, distension, diarrhoea** at 24–48 h.
- **CT (computed tomography) with oral contrast**; free gas is normal for up to 7–9 days.

Physiological complications

- Hypercarbia and acidosis; oliguria from renal vein compression; reduced venous return; raised intracranial pressure in head-down tilt.

Diagram to draw: abdominal wall showing capacitive coupling through a metal cannula in a plastic anchor.

Key points to remember:

- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down.
- Missed bowel injury: trocar-site pain, leucopenia.

Thalium laser in urology

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 4, Q3c (also covered in 2024-Winter Paper 4, Q3c)

Assumed meaning: thulium laser ("thalium" is a spelling error).

Types and physics

	Tm:YAG (thulium:yttrium-aluminium-garnet)	TFL (thulium fibre laser)
Wavelength	2013 nm	**1940 nm** (water absorption peak)
Source	Diode-pumped solid crystal	Diode-pumped thulium-doped silica fibre
Output	Continuous wave (now also pulsed)	Pulsed or continuous
Water absorption	High	About 4× holmium

- Optical penetration is shallowest of urological lasers: about 0.2 mm (chapter values for TFL: 0.08–0.2 mm), against 0.31–0.4 mm for Ho:YAG (holmium:YAG, 2120 nm).
- Shallow absorption gives precise vaporisation and clean incision, but **carbonisation**.

Stone lithotripsy (TFL)

- Frequency up to about 2000 Hz (holmium about 120 Hz); pulse energy as low as 0.025 J.
- **Low peak power, long pulse** → photothermal "dusting", minimal retropulsion.
- Fibres 50–150 µm → better irrigation and scope deflection. Small, quiet unit; no dedicated high-voltage outlet.
- Settings: start 0.6 J × 6 Hz; dusting 0.2 J × 40 Hz; avoid >40 Hz (pulses miss the stone). "Paint, do not drill."
- RCTs (randomised controlled trials): better than 30 W holmium; equal stone-free rate to 120 W pulse-modulated holmium with less retropulsion.

Prostate

- **Techniques:** ThuVARP (vaporesection), ThuVAP (vaporisation), ThuVEP (vapoenucleation), **ThuLEP** (enucleation, 2010; mostly blunt dissection), ThuFLEP (fibre laser enucleation).
- Settings: 70–90 W enucleation, 30 W coagulation; morcellation as for HoLEP (holmium laser enucleation).
- Outcomes: ThuLEP equals HoLEP; size-independent; safe on anticoagulants (AUA [American Urological Association]). Retrograde ejaculation about four-fifths.

Other uses

- **Also know:** en bloc bladder tumour resection; UTUC (upper tract urothelial carcinoma) ablation; incision of strictures.

Safety

- Heat follows power: ≤10 W in the ureter, high flow, access sheath. Eyewear for all.
- Concerns: efficacy on all stone types; thermal injury at high power.

Diagram to draw: water absorption curve marking 1940, 2013 and 2120 nm.

Key points to remember:

- TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
- TFL: low peak power, fine dust, thin fibres.
- ThuLEP is size-independent and equal to HoLEP.

Morcellation

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 4, Q3d

Definition

- Cutting a large specimen into small pieces so it leaves through a port, small incision or endoscope.
- **Manual:** ring forceps or Kelly clamp inside a bag. **Power (electromechanical):** rotating cylindrical blade (laparoscopy) or endoscopic tissue morcellator (prostate).

1. Endoscopic morcellation after enucleation

- Made **HoLEP (holmium laser enucleation of prostate)** practical; also ThuLEP (thulium laser enucleation).
- **Steps:**
  1. Achieve good haemostasis first.
  2. Pass a long nephroscope with a 5 mm channel.
  3. Run two inflow lines fully open to keep the bladder distended.
  4. Engage lobes with suction; cut only in mid-bladder, away from the wall.
  5. Collect fragments in a trap for histology.
- Devices: Piranha (rotating blades), VersaCut (oscillating).
- **Complications:** bladder mucosal injury (stop, release suction, disengage; mostly catheter only); intraperitoneal perforation needs laparotomy; **retained adenoma** (from malfunction or poor vision) needs repeat morcellation.

2. Laparoscopic morcellation (kidney, spleen)

- **LapSac** (nylon with polyurethane lining) is impermeable to tumour cells and bacteria.
- **Rules:** purpose-built bag; drapes; change gloves and instruments after; no deep forceps passes; keep pneumoperitoneum and watch the bag.

	Intact extraction	Morcellation
Incision	Larger	Small, better cosmesis
Pain, time, stay	Similar	Similar
Pathology	Full staging	Type and grade kept; size, capsule, vein invasion, and depth in UTUC (upper tract urothelial carcinoma) lost
Seeding	—	Rare port-site recurrence

- Splenectomy morcellation risks **splenosis** if the bag ruptures.

3. FDA 2014 warning (gynaecology)

- In 2014 the FDA (Food and Drug Administration) warned against laparoscopic power morcellation in hysterectomy or myomectomy for fibroids, because unsuspected uterine sarcoma (about 1 in 350) was spread and upstaged.
- A boxed warning followed; contained (in-bag) morcellation is advised.
- Lesson for urology: avoid uncontained power morcellation of possible malignancy.

Complications summary

- Bag rupture, seeding, lost staging, visceral or vascular injury, retained tissue.

Diagram to draw: enucleated lobes in a distended bladder with morcellator suction.

Key points to remember:

- Morcellator made HoLEP practical; distend bladder, haemostasis first.
- Morcellation loses stage information; histology still obtained.
- FDA 2014: uncontained power morcellation spreads occult sarcoma.

KTP*** laser in urology

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 4, Q3e (lasers in BPH also in 2017-Winter Paper 2, Q3a)

Assumed meaning: KTP (potassium titanyl phosphate) laser, the "GreenLight" laser; the asterisks are ignored.

Physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam passes through a **KTP or LBO (lithium triborate) crystal**. This doubles frequency and halves wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water. Optical penetration 0.8 mm; tissue effect 1–3 mm.
- Effect: **photoselective vaporisation** with a thin coagulation rim, so vaporisation is bloodless.

Platforms

Generation	Power
KTP	80 W
LBO HPS (High Performance System)	120 W
LBO XPS	**180 W**; 750 µm MoXy fibre (metal tip, fluid cooling)

- Side-firing fibre emits at 70°; 23–24 Fr cystoscope; saline irrigation.

PVP (photoselective vaporisation of prostate) technique

- Identify ureteric orifices; open the bladder neck first.
- Sweep 15–30° in 1–2 seconds at ≤2 mm working distance. Large bubbles mean efficient vaporisation.
- Pulsatile bleeding: coagulation mode (about 30 W). Energy >5 kJ/cc predicts complete vaporisation.

Indications

- AUA (American Urological Association): prostates <30 cc and 30–80 cc; men at **higher bleeding risk** (anticoagulation continued). Not >80 cc (high retreatment).

GOLIATH trial

- RCT (randomised controlled trial), 29 European centres, about 270 men, prostate ≤100 cc: **180 W PVP vs TURP** (transurethral resection of prostate).
- PVP **non-inferior** for IPSS (International Prostate Symptom Score), Qmax and complications at 6 months, 1 and 2 years.
- PVP: shorter catheter time and stay, fewer early reinterventions. TURP: faster. Retrograde ejaculation (about two-thirds) and 2-year reoperation equal.

Complications

- Dysuria and storage symptoms (highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum; capsular perforation 0.2–1%; retrograde ejaculation.
- No TUR (transurethral resection) syndrome; transfusion very rare.
- **No tissue** for histology.

Other uses

- **Also know:** laparoscopic non-contact cutting (largely supplanted); KTP enucleation possible.

Diagram to draw: side-firing fibre at 70° vaporising lateral lobe.

> **Key points to remember:**
> - 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
> - GOLIATH: 180 W PVP non-inferior to TURP.
> - Best for anticoagulated men; not >80 cc.

Stem cell therapy

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 4, Q3f (tissue engineering also in 2023-Winter Paper 1, Q1)

Definition

- Stem cells **self-renew** and **differentiate** into specialised cells. Therapy uses them to repair or replace damaged tissue.

Types

Type	Source	Comment
Embryonic	Blastocyst	Pluripotent; ethics, teratoma
Adult mesenchymal	Bone marrow, adipose (ADSC, adipose-derived stem cells), umbilical cord	Easy harvest; mainly paracrine
Muscle-derived (MDSC)	Skeletal muscle biopsy	Used for sphincter and detrusor
iPSCs (induced pluripotent stem cells)	Reprogrammed adult cells	Patient-specific; tumour risk
Urine-derived, spermatogonial	Urine; testis	Research

Mechanisms

- Differentiation into muscle, nerve or endothelium.
- **Paracrine action:** growth factors, anti-inflammatory, anti-fibrotic and neurotrophic signals.
- Homing to injured tissue; seeding of scaffolds.

Applications

- **SUI (stress urinary incontinence):**
  - Autologous muscle-derived cells: thigh muscle biopsy → culture → transurethral or periurethral injection into the sphincter.
  - Aim: regenerate the striated sphincter, not just bulk it.
  - Early studies improved; placebo-controlled trials have not shown clear superiority, so it remains investigational.
  - After childbirth injury, stem cell homing chemokines (CXCL12, CCL7) rise.
- **ED (erectile dysfunction):** intracavernosal bone-marrow, adipose or cord cells; animal and small uncontrolled human series only. SMSNA (Sexual Medicine Society of North America): experimental, IRB (institutional review board) protocols only.
- **Underactive detrusor:** muscle-derived cells injected into the bladder wall; two open-label studies improved voiding; research.
- **Tissue engineering:** cell-seeded scaffolds for bladder (Atala, 2006) and urethra (Raya-Rivera, 2011); adipose stem cell-seeded grafts for Peyronie disease (experimental).
- **Fertility:** spermatogonial stem cell banking in prepubertal cancer patients.
- **Kidney:** iPSC kidney organoids (research).

Challenges

- Poor cell survival and retention; tumour formation (embryonic, iPSC); standardisation, cost and regulation.

- Unregulated "stem cell clinics" offer unproven treatments outside trials, a growing ethical concern.
- **Also know:** cancer stem cells (a different concept) may drive tumour recurrence and resistance.

Diagram to draw: muscle biopsy → culture → periurethral injection into sphincter.

> **Key points to remember:**
> - Adult stem cells act mainly by paracrine signals.
> - SUI muscle-derived cells and ED stem cells are investigational.
> - Embryonic and iPSCs carry tumour risk.

Holmium laser

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 4, Q3e (holmium enucleation in detail in 2023-Winter Paper 4, Q2; stone energy sources in Practice Set 1 Paper 4, Q2)

Physics

- Ho:YAG (holmium:yttrium-aluminium-garnet), a solid-state, flashlamp-pumped, **pulsed** laser at **2120 nm** (often quoted as 2100 nm).
- Chromophore **water**; penetration about **0.4 mm**. It cuts and vaporises; heat spreads 2–3 mm and seals small vessels.
- Flexible quartz fibres; works in saline.

Mechanism on stones

- Long pulses (>250 µs) act **photothermally**, not by shock waves: the stone melts or decomposes.
- **Moses effect:** the start of each pulse makes a vapour channel; the rest of the pulse crosses it to the stone. So keep the fibre close and perpendicular.
- Breaks **all stone types**, including cystine and calcium oxalate monohydrate.

Settings

Mode	Settings	Effect
Fragmenting	0.6–1.0 J × 6–10 Hz	Pieces for basketing; more retropulsion
Dusting	0.2–0.3 J × 40–80 Hz	Fine dust; "popcorning" in a calyx
HoLEP enucleation	2.0 J × 40–50 Hz	Cutting along the capsule

- Retropulsion rises with pulse energy, falls with longer pulses or pulse modulation.
- Heat tracks power: ≤10 W in the ureter; good outflow.

Uses

- **Stones:** URS (ureteroscopy), RIRS (retrograde intrarenal surgery), PCNL (percutaneous nephrolithotomy), bladder stones.
- **BPH (benign prostatic hyperplasia):** HoLEP (holmium laser enucleation of the prostate): size-independent, the only endoscopic procedure superior to TURP (transurethral resection of the prostate), safe on anticoagulants.
- **Incisions:** strictures, bladder neck contracture, endopyelotomy, ureterocele.
- **Tumours:** UTUC (upper tract urothelial carcinoma) ablation; en bloc bladder tumour resection.

Safety

- Eyewear for all; thermal urothelial injury; steep HoLEP learning curve.

Recent advances

- High-power (120 W) pulse-modulated machines. TFL (thulium fibre laser, 1940 nm) beat low-power holmium but equalled high-power holmium in RCTs (randomised controlled trials).

Diagram to draw: vapour bubble (Moses channel) between fibre tip and stone.

Key points to remember:

- 2120 nm; water chromophore; pulsed; 0.4 mm penetration.
- Photothermal; Moses effect: fibre close and perpendicular.
- Fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz; ≤10 W in ureter.
- HoLEP: size-independent, superior to TURP.

Hemostasis in lap surgery

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 4, Q3f (haemostasis in laparoscopic partial nephrectomy in 2019-Summer Paper 3, Q1; energy sources in 2015-Summer Paper 2, Q3d)

Principles

- Devices seal by **denaturing protein**. Plan from imaging; keep a haemorrhage tray ready.

Effect of pneumoperitoneum

- At 12–15 mmHg CO_2 **tamponades venous bleeding**, hiding it.
- Raise to 20–25 mmHg briefly for venous bleeding. Before exit, **inspect at low pressure (5–10 mmHg)** to unmask hidden bleeders and port-site bleeding.

Energy devices

Device	Lateral spread	Note
Monopolar	5–7 mm	Capacitive coupling, insulation failure
Bipolar	1–6 mm	Current only between jaws
Vessel sealer (LigaSure)	1–3 mm	Seals vessels **≤7 mm**; one application
Ultrasonic (Harmonic)	≈1 mm	Blade >200 °C; cools slowly; no current
Argon beam	Superficial	Raw surfaces; gas embolism

Clips

- Titanium clips: no diathermy near them (slip).
- **Hem-o-lok** (polymer, locking): circumferential dissection, see the tip, feel the snap, at least two clips on the patient side with a 2-mm cuff.
- **FDA (Food and Drug Administration) warning:** after donor renal artery stump haemorrhage, the manufacturer contraindicated Hem-o-lok on the **living-donor renal artery**. A later series found it safe with correct technique, but use a stapler there.
- Lapra-Ty absorbable clips act as knots.

Staplers

- Endo GIA through a 12-mm port; **grey 2.0 mm or white 2.5 mm** for renal vessels; blue or green for thicker tissue.
- **Never fire across clips** (misfire). Reinforce a bleeding staple line with clips.

Sutures and haemostatic agents

- Intracorporeal or barbed sutures. Adjuncts: fibrin sealant, gelatin–thrombin matrix, oxidised cellulose.

Managing a venous injury

- Raise pressure → suction → pledget tamponade → sealant → extra port for suturing or Satinsky → hand-assist → convert. Keep the patient well filled (gas embolism).
- Arterial: minor, tamponade; renal artery, staple; aortic, convert.

Diagram to draw: Hem-o-lok placement with two patient-side clips and a 2-mm cuff.

Key points to remember:

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

Advances in lap surgeries

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 4, Q3a (related: LESS in 2023-Winter Paper 4, Q1; energy sources in 2015-Summer Paper 2, Q3d)

Introduction

- Since the first laparoscopic nephrectomy (Clayman, 1991), advances aim at **less access trauma, better vision, safer haemostasis, gentler physiology and faster training**.

1. Access and insufflation

- **Non-bladed dilating and optical trocars**: port-site hernia fell from 1.83% to 0.19%, wall bleeding from 0.83% to 0.16%.
- **Balloon dilators** for retroperitoneal and extraperitoneal space.
- **AirSeal** valveless system: stable pneumoperitoneum at **8–12 mmHg** despite suction, smoke evacuation, less CO_2 absorption.
- Lower pressure (12 rather than 15 mmHg): less pain and oliguria.

2. Fewer and smaller incisions

Technique	Gain	Limitation
LESS (laparoendoscopic single-site surgery)	Cosmesis, less analgesia	No triangulation, clashing
NOTES (natural orifice transluminal endoscopic surgery)	No abdominal scar (hybrid transvaginal extraction)	No proven benefit
Mini-laparoscopy (2–3 mm instruments)	Less pain, cosmesis	Instrument flexibility
Hand-assisted	Tactile feedback, fast bleeding control	Larger wound

3. Vision

- Chip-on-tip HD (high-definition) and **3D** laparoscopes; deflectable tips.
- **ICG (indocyanine green) near-infrared fluorescence**: vessel mapping for selective clamping in partial nephrectomy, ureteric identification, sentinel nodes.
- Laparoscopic ultrasound; 3D-printed planning models.

4. Haemostasis and suturing

- **Advanced bipolar sealers** (LigaSure: vessels ≤7 mm, 1–3 mm spread) and **ultrasonic shears** (about 1 mm spread, but very hot jaws).
- **Hem-o-lok** clips, **barbed sutures** (shorter anastomosis), Lapra-Ty clips, powered articulating staplers.

5. Perioperative care

- **ERAS (enhanced recovery after surgery)**: stay about 30% shorter, complications halved; no routine bowel preparation.

6. Training

- Box trainers, VR (virtual reality) simulators, EAU **E-BLUS** (European Basic Laparoscopic Urological Skills) certification, 5-day mini-residencies.

7. Robot-assisted laparoscopy

- Wristed 3D platforms (da Vinci Xi, SP 2018; Hugo, Versius, SSI Mantra) solved laparoscopic suturing (**see Q3b**).
- **Also know:** augmented reality, artificial intelligence, telesurgery.

Diagram to draw: LESS multichannel port in the umbilicus with articulating instruments.

> **Key points to remember:**
>
> - Blunt trocars cut port-site hernia to 0.19%.
> - AirSeal allows 8–12 mmHg working pressure.
> - ICG fluorescence guides selective clamping.
> - LESS: cosmesis at the cost of triangulation.

Artificial intelligence in urology

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 4, Q3d

Definitions

- **AI (artificial intelligence):** computer systems doing tasks that need human intelligence: recognising patterns, predicting, deciding.
- **ML (machine learning):** algorithms learn from data rather than fixed rules. **Supervised** (labelled data, e.g. tumour or not), **unsupervised** (finds clusters), reinforcement.
- **Deep learning:** multilayer **ANNs (artificial neural networks)**; **CNNs (convolutional neural networks)** are the commonest for images.
- **NLP (natural language processing)** and **LLMs (large language models)** handle text; **radiomics** extracts quantitative image features.

Development steps

```
Data collection and labelling → training → internal validation
 → external validation → regulatory approval → use and monitoring
```

Applications

Area	Examples
Imaging	CNNs detect stones on CT (computed tomography) with very high sensitivity; ANNs detect renal tumours; prostate MRI (magnetic resonance imaging) lesion detection; deep-learning reconstruction allows ultra-low-dose CT (about 1 mSv)
Pathology	Automated cancer detection and Gleason grading on digital slides; Paige Prostate was the first FDA (Food and Drug Administration)-authorised AI pathology tool (2021)
Endoscopy	Real-time bladder-tumour detection at cystoscopy; future ureteroscopes setting laser energy, irrigation, pressure and temperature
Stones	Predict composition and stone-free outcome
Prognosis	Radiomics, pathomics and genomics combined for RCC (renal cell carcinoma) prognosis; AI on biopsy slides predicting benefit of ADT (androgen deprivation therapy) with radiotherapy
Robotic surgery	Video and kinematic data for skill assessment, outcome prediction, real-time guidance; future semi-autonomy
Paediatrics	Deep learning on ultrasound predicts CKD (chronic kidney disease) progression in valves; VUR (vesicoureteral reflux) infection-risk models
Others	Urodynamics interpretation; LLM chatbots for patient information

Advantages

- Fast big-data processing; consistency; decision support.

Limitations and ethics

- Needs large, high-quality labelled data; **bias** and poor generalisability (overfitting); "black box" decisions; few prospective or external validations.
- Privacy and data sharing; medico-legal liability; LLM "hallucinations"; regulation.
- AI assists, not replaces, the urologist or radiologist.

Diagram to draw: neural network: input layer (image pixels), hidden layers, output (stone or no stone).

Key points to remember:

- Deep learning is part of ML, which is part of AI; CNNs for images.
- CNNs: very high sensitivity for CT stones.
- Paige Prostate: first FDA-authorised AI pathology tool (2021).
- Limits: data, bias, black box, validation, liability.

3D laparoscopy

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 4, Q3b

Introduction

- Standard laparoscopy gives a **two-dimensional (2D)** image. Depth is judged from monocular cues (shadow, size, motion parallax, touch), which slows suturing.
- **Three-dimensional (3D) laparoscopy** restores **stereoscopic depth perception**, as in robotic surgery (Chapter 18: two-lens binocular systems give depth perception).

Principle

- Two lenses or two chip-on-tip sensors, a few millimetres apart, capture two images from slightly different angles, like the two eyes.
- The monitor shows both images; **polarised glasses** send one to each eye. The brain fuses them, and the **binocular disparity** is perceived as depth.
- **Also know:** active-shutter glasses; head-mounted displays; glasses-free (autostereoscopic) screens; 3D scopes at 0° and 30°, and with a deflectable tip; 3D 4K systems.

Advantages

- Better depth judgement and hand–eye coordination.
- **Shorter time for complex tasks**, especially intracorporeal suturing and knotting, with fewer errors.
- **Shorter learning curve**; novices gain more than experts.
- Gives robotic-type vision at much lower cost.

Disadvantages

- Visual strain, headache, dizziness or nausea in some users; glasses dim the image.
- Every team member needs glasses and a correct viewing distance and angle.
- Higher cost than 2D; 30° rotation needs care.
- No wristed instruments, tremor filtering or motion scaling: these come only with the robot.

Urological uses

- Reconstructive steps: **pyeloplasty**, vesicourethral anastomosis in laparoscopic radical prostatectomy, **partial nephrectomy** renorrhaphy, ureteric reimplantation, donor nephrectomy.

Evidence

- Randomised trials and meta-analyses in laboratory tasks and laparoscopic surgery show shorter operating and suturing times with 3D. Differences in complications and oncological outcomes are not established.

Feature	2D laparoscopy	3D laparoscopy	Robot
Depth perception	No	Yes	Yes (magnified)
Wristed instruments	No	No	Yes
Cost	Low	Moderate	High

Diagram to draw: two lenses at the scope tip, two images, polarised glasses, fused 3D image in the brain.

Key points to remember:

- Two lenses, two images, polarised glasses: binocular disparity gives depth.
- Main gain: faster, more accurate suturing; shorter learning curve.
- Drawbacks: eye strain, headache, cost, glasses for all.
- Robotic vision at low cost, but without wristed instruments.

Telemedicine in urology

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 4, Q3c

Definition

- **Telemedicine** is delivery of health care at a distance using information and communication technology (video, phone, messaging, data transfer).
- **COVID-19 (coronavirus disease 2019)** distancing accelerated its adoption. Chapter 2: urologists' telehealth use rose sharply in 2019–2020.

Types

Type	Method	Urological example
Synchronous	Real-time video or phone consultation	Stone, PSA (prostate-specific antigen) and post-operative follow-up
Asynchronous (store-and-forward)	Reports, images, bladder diaries sent for later review	Scan reports, uroflow traces
Remote monitoring	Apps and home devices	Bladder diary apps, home uroflowmetry
Provider-to-provider	E-consults, telementoring, tele-tumour boards	Haematuria triage (Chapter 8)
Telesurgery	Remote robotic surgery	Experimental; limited by latency

Uses in urology

- **History-driven care:** LUTS (lower urinary tract symptoms) review, results, drug titration.
- ED (erectile dysfunction) evaluation by selected telemedicine (Chapter 72); men's health in large, sparse regions (Chapter 68).
- After mid-urethral sling, telemedicine follow-up is safe, and app-based follow-up gave higher satisfaction than office visits (Chapter 116).
- Internet-based pelvic floor training equalled a postal programme in an RCT (randomised controlled trial) (Chapter 114).

Benefits (Chapter 2)

- Access for rural and underserved patients; less travel; shorter waits; relatives join remotely; sustained satisfaction.

Limitations (Chapter 2)

- **No examination or in-office tests**: vital signs, cystoscopy, PVR (post-void residual).
- Needs connectivity and a secure, simple platform; hard for elderly or non-technical patients.
- No assistant to pre-gather results; more no-shows.
- **Medico-legal:** US licensure is state-based, and pandemic payment waivers have ended.

Indian framework

- **Telemedicine Practice Guidelines (2020)**: only a registered medical practitioner may consult; identity of patient and doctor confirmed; consent and records kept; narcotic and Schedule X drugs may not be prescribed.
- Government platform **eSanjeevani**; patient data covered by the Digital Personal Data Protection Act 2023.

Diagram to draw: patient (home or district hospital) linked by video and data to the urologist, with remote monitoring feeding in.

Key points to remember:

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

Simulators in urology

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 4, Q3d

Rationale

- Fewer training cases and complex technology mean skills must be learnt **outside the patient**.
- Chapter 1: simulation is used to **acquire and maintain skills and to warm up** before operating.

Types

Type	Examples	Strengths and limits
Bench / box trainer	Peg transfer, cutting, suturing; chicken oesophagus anastomosis; 3D (three-dimensional)-printed perfused hydrogel kidney	Cheap, proven skill gains; limited realism
VR (virtual reality)	LapSim, LapMentor, RobotiX Mentor, SimNow (robotic prostatectomy); endourology trainers for cystoscopy, ureteroscopy, PCNL (percutaneous nephrolithotomy), TURP (transurethral resection of the prostate)	Records metrics (precision, economy of motion), builds curricula, sets expert benchmarks; costly; poor tissue feel
Live animal	Pig	Real bleeding; ethics, cost
Cadaver	Fresh-frozen	Real human anatomy; scarce
Team (non-technical)	High-fidelity theatre crises	Communication and teamwork

Validity (needed before a simulator is trusted)

- **Face:** looks real. **Content:** teaches the right skills. **Construct:** experts score better than novices. **Concurrent:** agrees with an established test. **Predictive:** simulator score forecasts operating performance.

Evidence (Chapter 18, Chapter 12)

- VR training **beats no training or box training** for novices and improves live performance and operating time.
- Robotic simulator scores correlate with live RARP (robot-assisted radical prostatectomy) performance.
- **FRS (Fundamentals of Robotic Surgery):** a multi-institutional RCT (randomised controlled trial) showed better speed and accuracy.
- Ureteroscopy team simulation taught non-technical skills better than lectures (randomised trial).

Structured curricula and assessment

- **EAU (European Association of Urology) E-BLUS** (European Basic Laparoscopic Urological Skills): online theory, then dry-model tasks, then examination.
- **Proficiency-based progression:** move on only after reaching expert benchmarks.
- Scoring: **GEARS (Global Evaluative Assessment of Robotic Skills)** and **OSATS (Objective Structured Assessment of Technical Skills)**.

Limitations

- Cost, poor tissue feel, limited proof of better patient outcomes.

Diagram to draw: pyramid: box trainer at the base, then VR, then animal or cadaver, then supervised surgery at the top.

Key points to remember:

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
- VR beats box or no training for novices.
- E-BLUS and FRS; proficiency-based progression; GEARS.

Virtual and augmented reality in urology

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 4, Q3e

Definitions

- **Virtual reality (VR):** the user is immersed in a fully computer-generated environment, usually through a head-mounted display.
- **Augmented reality (AR):** digital images are laid over the real view, for example a 3D (three-dimensional) tumour model on the robotic camera image.
- **Mixed reality:** virtual holograms are anchored in the room and can be handled.

Workflow

```
CT (computed tomography) / MRI (magnetic resonance imaging)
   -> segmentation of organ, tumour, vessels, collecting system
   -> 3D virtual model -> viewed in VR, 3D-printed, or overlaid (AR)
   -> registration to the live surgical view
```

Applications

Use	Examples
Pre-operative planning	Partial nephrectomy: tumour depth, arterial branches, selective clamping. **PCNL (percutaneous nephrolithotomy)**: calyx and tract choice (Chapter 98: immersive VR and 3D printing show early promise). Prostatectomy: MRI lesion model to plan nerve-sparing
Intra-operative navigation	AR overlay in the robot console to locate an endophytic tumour, hilar vessels, or a prostate lesion near the capsule
Education	VR anatomy, 360° operation videos; simulation, see Q3d
Patient counselling	Patients understand their anatomy and planned operation; better-informed consent
Paediatrics	VR distraction reduces anxiety before surgery and procedures such as catheterisation and VCUG (voiding cystourethrogram) (Chapters 35 and 38)

Limitations

- **Registration error:** organs move and deform with breathing, pneumoperitoneum and dissection, so a rigid overlay drifts; manual alignment is still common.
- Segmentation time and cost.
- Chapter 161: promising but **resource-intensive and hard to validate**.
- Chapter 98: 3D reformatting added nothing over orthogonal CT measurements for predicting stone-free rate.
- Little outcome evidence; cybersickness with headsets.

Future

- **AI (artificial intelligence)** for automatic segmentation and real-time tracking.
- **Digital surgery:** reconstruction of the patient's own anatomy allows rehearsal before the operation (Chapter 147).

Diagram to draw: CT kidney, segmented 3D model, and overlay of the model on the robotic view of a renal tumour.

Key points to remember:

- VR immerses; AR overlays on reality.
- Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
- Biggest problem: registration on moving, deforming organs.
- Evidence for outcomes still limited.

3D printing in urology

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 4, Q3a

Definition

- **3D (three-dimensional) printing** is additive manufacturing: a solid object is built layer by layer from a digital model.
- In urology it turns a patient's CT (computed tomography) or MRI (magnetic resonance imaging) into a physical replica of kidney, tumour, vessels or prostate.

Workflow

```
Thin-slice CT/MRI (DICOM files) → segmentation of kidney, tumour, vessels, calyces
   → STL (standard tessellation language) mesh → printer → post-processing
```

- Takes hours to days; segmentation needs expertise.

Printing technologies

Method	Principle	Use
FDM (fused deposition modelling)	Melted plastic filament extruded	Cheap, rigid models
SLA (stereolithography)	Laser cures liquid resin	Fine detail, transparent models
Material jetting (PolyJet)	Jetted photopolymers, multi-colour	Tumour, vessels and calyces in different colours
SLS (selective laser sintering)	Laser fuses powder	Durable parts, devices
Bioprinting	Living cells in hydrogel "bio-ink"	Tissue engineering (see Practice Set 6 Paper 4, Q2)

Applications

1. **Partial nephrectomy planning:** complex hilar or endophytic tumours. The model shows the tumour against vessels and collecting system, so the surgeon plans the approach and rehearses steps. Promising.
2. **PCNL (percutaneous nephrolithotomy) planning:** staghorn stones, anomalous kidneys; choice of calyx. Early promise; 3D reformatting did not predict stone-free rate better.
3. **Prostate:** models with MRI lesions for margin and nerve-sparing planning and focal therapy.
4. **Patient education and consent:** patients understand their anatomy and the operation better.
5. **Training and simulation:** silicone or hydrogel phantoms for PCNL puncture, partial nephrectomy, pyeloplasty, anastomosis.
6. **Patient-specific devices:** prototype stents, catheters and surgical guides.

Limitations

- Cost, printing time and segmentation errors.
- Materials do not reproduce tissue feel or bleeding.
- Evidence mostly small series; no proven outcome benefit.

Future

- Virtual and augmented reality models on the robotic console, point-of-care printing, drug-eluting printed stents.

Diagram to draw: CT → segmentation → STL → printer → colour kidney model with tumour, artery and calyces.

> **Key points to remember:**
>
> - Additive manufacturing from CT/MRI via segmentation and an STL file.
> - Main uses: partial nephrectomy and PCNL planning, consent, simulation.
> - Limits: cost, time, unrealistic tissue, weak outcome evidence.

Rapid Revision Sheet – Section 1 · Basic Principles

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Robotics in urology (including robotic vs standard laparoscopy) (LAQ)

- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
- Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

LASERs in urology: applications and newer generation laser technologies (new LASER for stone treatment) (LAQ)

- Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
- Holmium lithotripsy is photothermal (Moses); TFL gives less retropulsion.
- HoLEP size-independent, superior to TURP; PVP for anticoagulated men (GOLIATH).
- Nd:YAG for PeIN; eyewear for everyone.
- TFL 1940 nm: water peak, low peak power, minimal retropulsion.
- Moses: two-part pulse through a vapour channel.
- TFL beats low-power, equals high-power pulse-modulated holmium.
- Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney.
- TFL (1940 nm): fine dust, minimal retropulsion, thin fibres; settings ≤40 Hz often enough.
- Pulse-modulated (Moses) high-power holmium: vapour channel, less retropulsion.
- Safety: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

Flexible endourological instruments, digital and current ureteroscopes (LAQ)

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
- Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
- Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
- Semirigid: two channels; below the iliac vessels.
- Pressure sensors and AI are the newest developments.
- Golden rule: never fire unless the fibre tip is seen on the stone.

Natural orifice endoscopic surgery. (LAQ)

- NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
- Transvaginal extraction removes the most painful wound.
- LESS: one incision, lost triangulation, more hernias.
- Barriers: closure, infection, orientation, platform; no proven benefit yet.

Tissue engineering in urology, including recent advances (LAQ)

- Triad: cells, scaffold, signals; vascularity (100–200 µm diffusion) is the main barrier.
- Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
- Neo-urinary conduit (conduit trial): half stenosed; no blood supply.
- Raya-Rivera 2011: seeded tubular urethras in 5 boys.

- New: iPSCs, urine-derived cells, bioprinting, organoids.

Laparoendoscopic single-site (single-port) surgery: LESS (LAQ)

- LESS = one incision; LESSCAR (2008) standardised the term.
- Core problem: lost triangulation; solved by articulating or curved instruments and robots.
- Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
- da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

Basic energy modalities in urological surgery, including bipolar energy (LAQ)

- Electrosurgery at about 500 kHz; current density decides heating.
- Cut continuous low voltage; coagulation interrupted high voltage.
- Capacitive coupling and insulation failure cause unseen burns.
- Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink.
- Bipolar: no pad, saline irrigation, no TUR syndrome.
- Bipolar TURP equals monopolar efficacy with fewer complications.
- Vessel sealers: vessels ≤7 mm, spread 1–3 mm.

Urethral stents and metallic stents in urology (SAQ)

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

Describe evidence based medicine. (SAQ)

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.

Operating room safety : the role of surgeon. (SAQ)

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

Urological applications in nanotechnology. (SAQ)

- 1–100 nm; enhanced permeability and retention allows tumour targeting.
- USPIO MRI: benign nodes lose signal; metastatic nodes do not.
- AuroLase gold nanoshells: photothermal focal therapy.

Different types of trocars in laparoscopy. (SAQ)

- Trocar = cannula + obturator.
- Bladeless: fewer epigastric injuries and hernias.
- Balloon trocar prevents retroperitoneal emphysema.

General complications related to laparoscopy. (SAQ)

- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down.
- Missed bowel injury: trocar-site pain, leucopenia.

Thalium laser in urology (SAQ)

- TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.

- TFL: low peak power, fine dust, thin fibres.
- ThuLEP is size-independent and equal to HoLEP.

Morcellation (SAQ)

- Morcellator made HoLEP practical; distend bladder, haemostasis first.
- Morcellation loses stage information; histology still obtained.
- FDA 2014: uncontained power morcellation spreads occult sarcoma.

KTP* laser in urology (SAQ)**

- 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
- GOLIATH: 180 W PVP non-inferior to TURP.
- Best for anticoagulated men; not >80 cc.

Stem cell therapy (SAQ)

- Adult stem cells act mainly by paracrine signals.
- SUI muscle-derived cells and ED stem cells are investigational.
- Embryonic and iPSCs carry tumour risk.

Holmium laser (SAQ)

- 2120 nm; water chromophore; pulsed; 0.4 mm penetration.
- Photothermal; Moses effect: fibre close and perpendicular.
- Fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz; ≤10 W in ureter.
- HoLEP: size-independent, superior to TURP.

Hemostasis in lap surgery (SAQ)

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

Advances in lap surgeries (SAQ)

- Blunt trocars cut port-site hernia to 0.19%.
- AirSeal allows 8–12 mmHg working pressure.
- ICG fluorescence guides selective clamping.
- LESS: cosmesis at the cost of triangulation.

Artificial intelligence in urology (SAQ)

- Deep learning is part of ML, which is part of AI; CNNs for images.
- CNNs: very high sensitivity for CT stones.
- Paige Prostate: first FDA-authorised AI pathology tool (2021).
- Limits: data, bias, black box, validation, liability.

3D laparoscopy (SAQ)

- Two lenses, two images, polarised glasses: binocular disparity gives depth.
- Main gain: faster, more accurate suturing; shorter learning curve.
- Drawbacks: eye strain, headache, cost, glasses for all.
- Robotic vision at low cost, but without wristed instruments.

Telemedicine in urology (SAQ)

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

Simulators in urology (SAQ)

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
- VR beats box or no training for novices.
- E-BLUS and FRS; proficiency-based progression; GEARS.

Virtual and augmented reality in urology (SAQ)

- VR immerses; AR overlays on reality.
- Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
- Biggest problem: registration on moving, deforming organs.
- Evidence for outcomes still limited.

3D printing in urology (SAQ)

- Additive manufacturing from CT/MRI via segmentation and an STL file.
- Main uses: partial nephrectomy and PCNL planning, consent, simulation.
- Limits: cost, time, unrealistic tissue, weak outcome evidence.

Section 2 · Oncology, DSD, Gender & Ageing — Long answer questions (LAQ)

Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T.

Type: LAQ (25 marks)

Source: 2024-Winter Paper 4, Q2

Introduction

- A **cancer vaccine** is a therapeutic agent that trains the patient's own T cells to recognise **tumour antigens**. It treats existing cancer; it does not prevent it.
- **BCG (bacille Calmette–Guérin)**, a tuberculosis vaccine, was among the first immune treatments for any cancer. It is non-specific. Cancer vaccines aim for a **specific** anticancer response.

Rationale in urology

- **Immune editing (three Es):** Elimination → Equilibrium → **Escape**. A clinical cancer has escaped by losing antigens, losing MHC (major histocompatibility complex), or recruiting Tregs (regulatory T cells). Vaccines try to restart the response.
- **Kidney:** rare spontaneous regression of metastases after nephrectomy; responses to IL-2 (interleukin-2) and interferon.
- **Prostate:** slow growth gives time to build a response; **organ-specific antigens** (PSA [prostate-specific antigen], PAP [prostatic acid phosphatase]).
- **Bladder:** BCG success; high mutation load gives many neoantigens.
- Low toxicity; possible lasting memory.

Mechanism of action

- Two components: **antigen** (overexpressed in tumour) + **adjuvant** (a danger signal, e.g. GM-CSF [granulocyte-macrophage colony-stimulating factor], which activates DCs [dendritic cells]).

```
Intradermal antigen + adjuvant
  → immature DC takes up antigen; adjuvant matures it
  → DC migrates to lymph node
  → peptide on MHC I/II = signal 1; B7 costimulation = signal 2
  → CD4 Th1 help (IL-2, interferon-γ) + CD8 cytotoxic T cells
  → T cells traffic to tumour → apoptosis of antigen-bearing cells
  → memory T cells
```

- Without signal 2 the T cell becomes tolerant, hence the adjuvant.
- **Prime–boost:** first dose primes, later doses boost.
- **Delayed effect:** survival may improve without tumour shrinkage or PSA fall.
- **Resistance:** PD-L1 (programmed death ligand-1) on tumour, Tregs, MDSCs (myeloid-derived suppressor cells), MHC loss. This is the basis for combining vaccines with checkpoint inhibitors.

Types (platforms)

Platform	Example
Whole-cell (autologous or allogeneic)	Reniale (RCC [renal cell carcinoma]); GVAX (prostate)
Peptide/protein	IMA901
Dendritic-cell (ex vivo loaded)	Sipuleucel-T; AGS-003 (rocapuldencel-T)
Viral vector (poxvirus)	PROSTVAC-VF; TroVax (MVA-5T4)
Heat-shock protein–peptide	Vitespen (HSPPC-96)
DNA/mRNA, personalised neoantigen	Phase I–II trials

Vaccines by organ

Renal cell carcinoma

- **IMA901:** 10 tumour-associated peptides + GM-CSF, low-dose cyclophosphamide to deplete Tregs; HLA-A*02+ (human leucocyte antigen) metastatic RCC with sunitinib. **No OS (overall survival) benefit.**
- **AGS-003 (rocapuldencel-T):** autologous DCs loaded with tumour RNA, with sunitinib. **Terminated for lack of efficacy.**
- **TroVax** with standard first-line therapy: no OS benefit. **Vitespen** (adjuvant): no overall benefit.
- **Reniale** (adjuvant autologous tumour-cell vaccine): claimed benefit, not adopted.
- Checkpoint inhibitors, not vaccines, are today's RCC immunotherapy.

Prostate cancer

- **Sipuleucel-T:** the **only approved** vaccine (below).
- **PROSTVAC-VF:** vaccinia prime, fowlpox boost, encoding PSA + TRICOM (three costimulatory molecules). Phase III **closed for futility**.
- **GVAX:** allogeneic whole-cell lines secreting GM-CSF. Phase III trials **failed**.
- Also: DCVAC/PCa (phase III negative); DNA vaccines against PAP (trials).

Bladder cancer

- **BCG:** intravesical live attenuated mycobacteria; standard for high-risk NMIBC (non-muscle-invasive bladder cancer).
- **Gene and viral therapy: nadofaragene firadenovec** (adenovirus carrying IFN-α2b gene; FDA [Food and Drug Administration] approved December 2022 for BCG-unresponsive CIS [carcinoma in situ]); **cretostimogene** (oncolytic adenovirus with GM-CSF gene).
- **N-803:** IL-15 superagonist given with BCG.
- Peptide and neoantigen vaccines: experimental only.

Why most vaccines failed

- Tolerance to self-antigens; immunosuppressive tumour environment; dysfunctional patient DCs; bulky late disease; progression endpoints miss delayed effects.
- **Pattern: only sipuleucel-T succeeded in phase III.**

Note on Sipuleucel-T

- **Nature:** autologous cellular immunotherapy: the patient's **CD54+ APCs (antigen-presenting cells)** activated with **PA2024**, a fusion protein of **PAP + GM-CSF**.
- **Manufacture and dosing:**
  1. Leukapheresis collects PBMCs (peripheral blood mononuclear cells).
  2. Cells incubated about 36 h with PA2024 at a central facility.
  3. Infused intravenously about 3 days after collection.

4. **Three doses at 2-week intervals** (one month course).

- **Mechanism:** primes PAP-specific CD4 and CD8 T cells; later infusions boost.
- **IMPACT trial (Kantoff, NEJM 2010):** 512 men, 2:1 versus placebo, asymptomatic or minimally symptomatic mCRPC (metastatic castration-resistant prostate cancer).
  - **OS 25.8 vs 21.7 months (HR [hazard ratio] 0.78)**, gain 4.1 months, despite placebo crossover.
  - **No change in time to progression or PSA.**
- **FDA approval April 2010:** first therapeutic vaccine for a solid tumour.
- **Indication:** asymptomatic or minimally symptomatic mCRPC, good performance status, **no visceral metastases, no opioid need**; usually before chemotherapy.
- **Toxicity:** mild infusion reactions (fever, chills, fatigue); premedicate with paracetamol and antihistamine.
- **Limitations:** cost, logistics, no response marker; not available in India.

Diagram to draw: sipuleucel-T flow: leukapheresis → PA2024 loading → reinfusion → T-cell priming.

Key points to remember:

- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months; no PSA or progression effect.
- BCG is the original urological immunotherapy.

Discuss in brief, present concepts in various GU cancers

Type: LAQ (25 marks)

Source: Practice Set 4 Paper 4, Q1

Read as: the current key concepts in diagnosis and management of each GU (genitourinary) cancer: kidney, upper tract, bladder, prostate, testis and penis.

Common themes

- **De-escalation:** surveillance and organ sparing for low-risk disease.
- **Precision:** molecular classification, germline testing and better imaging (mpMRI (multiparametric MRI, magnetic resonance imaging), PSMA (prostate-specific membrane antigen) PET (positron emission tomography)).
- **Perioperative systemic therapy:** immune checkpoint inhibitors (ICIs), ADCs (antibody–drug conjugates) and targeted drugs.
- **Robotic surgery** as the default approach.

1. Kidney: RCC (renal cell carcinoma)

- **WHO (World Health Organization) 2022:** new molecularly defined types (FH (fumarate hydratase)-deficient, SDH (succinate dehydrogenase)-deficient, TFE3-rearranged RCC).
- **Mechanism:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (vascular endothelial growth factor). This explains why VEGF TKIs (tyrosine kinase inhibitors) and the HIF-2α inhibitor **belzutifan** work.
- **Small renal mass:** active surveillance in the elderly or unfit; renal mass biopsy; **partial nephrectomy** for cT1 to avoid CKD (chronic kidney disease); thermal ablation for small tumours.
- **Adjuvant pembrolizumab** after nephrectomy for high-risk RCC (**KEYNOTE-564**: better disease-free and overall survival).
- **Metastatic:** IMDC (International Metastatic RCC Database Consortium) risk groups. First line is an **ICI combination**: nivolumab plus ipilimumab, or an ICI plus TKI (pembrolizumab–axitinib, nivolumab–cabozantinib, pembrolizumab–lenvatinib). **CARMENA (2018):** sunitinib alone was non-inferior to cytoreductive nephrectomy plus sunitinib in intermediate/poor risk, so cytoreduction is now selective. SBRT (stereotactic body radiotherapy) for oligometastases.

2. Upper tract urothelial carcinoma (UTUC)

- **Risk stratification (EAU (European Association of Urology)):** low risk (unifocal, <2 cm, low-grade, non-invasive on CT (computed tomography) urography) goes to **kidney-sparing** ureteroscopic ablation or segmental ureterectomy; UGN-101 (mitomycin gel) is an option. High risk goes to **RNU (radical nephroureterectomy) with bladder cuff** and template lymph node dissection.
- **Single postoperative intravesical mitomycin** after RNU cuts bladder recurrence (**ODMIT-C**: 28% to 16% at 1 year).
- **POUT:** adjuvant platinum chemotherapy improved disease-free survival in pT2–T4 or N+ disease.
- **Also know:** screen for **Lynch syndrome**; aristolochic acid (Balkan) nephropathy.

3. Bladder

- **NMIBC (non-muscle-invasive bladder cancer):** EAU risk groups; en bloc TURBT (transurethral resection of bladder tumour); blue-light or narrow-band cystoscopy; re-TURBT for T1; BCG (bacillus Calmette–Guérin) induction plus maintenance. **BCG-unresponsive**

disease: radical cystectomy is standard; bladder-sparing options are pembrolizumab, nadofaragene firadenovec (gene therapy) and N-803 with BCG.

- **MIBC (muscle-invasive):** VI-RADS (Vesical Imaging Reporting and Data System) on MRI; **cisplatin-based neoadjuvant chemotherapy** then cystectomy; **trimodality** bladder preservation in selected patients; adjuvant **nivolumab** (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months); perioperative durvalumab with chemotherapy (**NIAGARA, 2024**) improved event-free and overall survival.
- **Metastatic: enfortumab vedotin plus pembrolizumab** (**EV-302**) replaced platinum first line; avelumab maintenance; erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

4. Prostate

- **Diagnosis: MRI before biopsy** (**PRECISION**: more significant and fewer insignificant cancers); **transperineal biopsy** (less sepsis); **PSMA PET** staging (**proPSMA**: more accurate than CT plus bone scan); ISUP (International Society of Urological Pathology) grade groups; germline testing (BRCA2).
- **Localised: active surveillance** for low risk and selected favourable intermediate risk (**ProtecT, 15 years**: cancer mortality low and similar with surveillance, surgery or radiotherapy); robotic prostatectomy; **hypofractionated** radiotherapy; focal therapy in selected men.
- **Advanced:** mHSPC (metastatic hormone-sensitive) gets ADT (androgen deprivation therapy) plus an **ARPI** (androgen receptor pathway inhibitor); **triplet** with docetaxel (**PEACE-1, ARASENS**); prostate radiotherapy for low volume. EMBARK: enzalutamide for high-risk biochemical recurrence. mCRPC (metastatic castration-resistant): **PARP (poly-ADP ribose polymerase) inhibitors** for HRR (homologous recombination repair) mutations; **^{177}Lu-PSMA-617** (**VISION**: overall survival HR (hazard ratio) 0.62).

5. Testis

- **Stage I: surveillance** preferred for most seminomas and NSGCTs (non-seminomatous germ cell tumours); alternatives are one cycle of carboplatin (seminoma) or one BEP (bleomycin, etoposide, cisplatin) for high-risk NSGCT.
- **miR-371a-3p:** new serum marker for seminoma and NSGCT.
- **Low-volume stage II seminoma: primary RPLND (retroperitoneal lymph node dissection)** (SEMS, PRIMETEST) spares chemotherapy toxicity; robotic, nerve-sparing templates preserve ejaculation.
- **Also know:** IGCCCG (International Germ Cell Cancer Collaborative Group) 2021 update; survivorship care.

6. Penis

- **WHO 2022: HPV (human papillomavirus)-associated** vs **HPV-independent** squamous cell carcinoma; p16 staining; HPV vaccination for prevention.
- **Organ sparing:** topical treatment or laser for PeIN (penile intraepithelial neoplasia), glans resurfacing, wide excision, glansectomy; a negative margin of a few millimetres is enough.
- **Nodes:** for cN0 pT1b or higher, **DSNB (dynamic sentinel node biopsy)** or modified inguinal dissection. Bulky nodes: neoadjuvant TIP (paclitaxel, ifosfamide, cisplatin), then consolidative dissection.

Diagram to draw: six-row table: cancer, key diagnostic advance, localised treatment, advanced treatment.

Key points to remember:

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
- Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

Section 2 · Oncology, DSD, Gender & Ageing — Short answer questions (SAQ)

Prepucioplasty.

Type: SAQ (10 marks)

Source: 2017-Summer Paper 4, Q3b

Read as: preputioplasty, a foreskin-preserving operation for phimosis.

Definition

- An operation that widens the tight preputial ring but keeps the foreskin. It is an alternative to circumcision.

Background

- **Physiological phimosis** at birth is due to glans–preputial adhesions. Smegma and erections separate them. About 90% retract by 3 years; fewer than 1% are phimotic at 17 years.
- **Secondary phimosis** follows scarring (forced retraction, balanitis) or BXO (balanitis xerotica obliterans).

Indications

- Persistent symptomatic phimosis: recurrent balanoposthitis, UTI (urinary tract infection), paraphimosis, painful erection.
- Failed **topical steroid** (betamethasone 0.05% twice daily for about 4 weeks). About 1 in 6 fail.
- Wish to keep the foreskin (cultural reasons, cosmesis).

Contraindications

- **BXO** with scarring: circumcision is preferred. Triradiate incision with triamcinolone has been reported in selected boys.
- Physiological non-retractility needs only reassurance. Treat active infection first.

Techniques

Technique	Method
Dorsal slit, transverse closure	Longitudinal incision through the ring, closed transversely (Heineke–Mikulicz)
Triple incision	Three longitudinal cuts over the ring, each closed transversely
Y–V plasty	V-flap of outer skin advanced into a Y-cut

- General anaesthesia or penile block; confirm full retraction; absorbable sutures.

Postoperative care

- **Daily retraction from the first week**, continued for weeks, prevents re-stenosis. A topical steroid may help.

Outcomes and complications

- Most achieve a retractile foreskin. It is quicker and less painful than circumcision.
- Complications: re-phimosis (needs circumcision), bleeding, infection, poor cosmesis, paraphimosis.

- **Also know:** preputial reconstruction in hypospadias repair is also called preputioplasty; dehiscence is its risk.

Diagram to draw: dorsal incision through the ring, closed transversely.

Key points to remember:

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.
- BXO: circumcision preferred.

Enlist the steps in female to male gender reassignment surgery.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 4, Q3b

Preconditions

- Gender dysphoria diagnosed per **WPATH (World Professional Association for Transgender Health)** by an MDT (multidisciplinary team).
- **Testosterone:** chapter cites **at least 12 months** before genital surgery (clitoral growth plateaus at about 2 years). WPATH SOC8 (Standards of Care 8, 2022) relaxed this to about 6 months.
- **Urological screening:** uroflowmetry, PVR (post-void residual), renal ultrasound; treat LUTD (lower urinary tract dysfunction) first.

Stage 1: removal of female structures

1. **Subcutaneous mastectomy** (usually first).
2. **Hysterectomy + bilateral salpingo-oophorectomy**, preferably laparoscopic; remove whole cervix ± upper third of vagina.

Stage 2: genitoperineal transformation

3. **Vaginectomy** (needed if the urethra is lengthened):
    - Hydrodissection; circumferential introital incision.
    - Rectovaginal plane bluntly; urethrovaginal plane sharply, **no diathermy near the sphincter**.
    - Paravaginal haemostasis; levators closed in midline.
4. **Urethral lengthening, fixed (perineal) part:** vestibular mucosa and labia minora tubularised over 18 Fr; covered with bulbospongiosus.
5. **Phallus construction**, either:
    - **Metoidioplasty:** release clitoral chordee and urethral plate; lengthen urethra to the glans (preputial or labial flap ± buccal graft). Gives a 3–8 cm phallus with erection and sensation.
    - **Phalloplasty: RFFF (radial forearm free flap)** or ALT (anterolateral thigh) flap with **tube-in-tube urethra**; microvascular anastomosis; **clitoral nerve coapted** to a flap nerve.
6. **Urethral anastomosis** of phallic to fixed part; neomeatus at the tip.
7. **Scrotoplasty:** labia majora flaps rotated 180° in front (Ghent type). Monsplasty and perineal closure.
8. **SPC (suprapubic catheter)** + urethral catheter; urethral catheter out at 14–21 days.

Stage 3: prostheses

9. **Testicular prostheses** (17 mL) at 6 months or more.
10. **Erectile prosthesis** after 1 year, once protective sensation returns; single cylinder fixed to pubic bone (dedicated devices, e.g. ZSI 475 FTM [female-to-male]).

Complications

- **Fistula** (phallic base), **stricture** (fixed–phallic anastomosis), hair growth, dribbling, prosthesis infection or erosion, vaginectomy bleeding or rectal/bladder injury.

Diagram to draw: three parts of the neourethra (native, fixed, phallic).

Key points to remember:

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

Penile Inversion Vaginoplasty Technique

Type: SAQ (10 marks)

Source: 2025-Winter Paper 4, Q3d (also answered in 2022-Summer Paper 2, Q3b)

Introduction

- Commonest gender-affirming genital operation for transgender women: a sensate, functional vulva and vagina.
- **Principle:** homologous penile and scrotal tissues on their own blood supply.

Preparation

- **WPATH (World Professional Association for Transgender Health) SOC8 (Standards of Care 8):** one assessment, **6 months of hormones**.
- **Permanent hair removal** of scrotal graft skin; BMI (body mass index) usually ≤35 kg/m²; stop smoking; fertility preservation.
- **Choose depth:** full-depth vs **vulvoplasty (zero-depth)**: vulva without a canal, chosen for high surgical risk, prior prostatectomy, or no wish for penetrative intercourse.

Tissue transfer

Male structure	Becomes
Glans on dorsal neurovascular bundle	**Clitoris**
Preputial collar	Clitoral hood, labia minora
Spatulated urethra	Vestibule
Inverted penile skin	**Neovaginal lining**
Scrotal full-thickness graft	Proximal lining
Lateral scrotal skin	Labia majora

Steps

```
Lithotomy; tranexamic acid 1 g; catheter
→ harvest, defat, tubularise scrotal graft
→ bilateral orchidectomy
→ deglove penis, keep dartos with skin (blood supply)
→ dorsal neurovascular bundle + glans raised;
   excise ALL erectile tissue
→ clitoroplasty; shorten, spatulate urethra
→ canal dissection between prostate and rectum;
   keep Denonvilliers fascia intact; rectal checks
→ invert skin tube + graft; labiaplasty
→ pack canal
```

- **Why excise all erectile tissue:** residual tissue engorges and narrows the canal.
- **Peritoneal flaps (robotic)** add depth; **intestinal vaginoplasty** when skin is inadequate or for revision.

Postoperative care

- Pack and catheter out about day 5; **lifelong dilation** (several times daily at first, tapering); pelvic floor physiotherapy; continue oestrogen; most achieve orgasm.

Complications

- **Wound dehiscence** (commonest, 12–33%).

- **Rectovaginal fistula** (0.8% primary, 6.3% revision); urethrovaginal fistula (0.8–3.9%).
- Urethral stricture, meatal malposition (spraying); **vaginal stenosis** (poor dilation); prolapse 4–10%.

Diagram to draw: sagittal view of neovagina between prostate and rectum.

> **Key points to remember:**
>
> - Glans → clitoris; penile skin → canal; scrotal skin → graft.
> - Excise all erectile tissue; preserve Denonvilliers fascia.
> - Lifelong dilation; rectovaginal fistula is worst.

Antioxidants in urological malignancy

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 4, Q3a (prevention trials also in 2017-Summer Paper 4, Q3a)

Rationale

- ROS (reactive oxygen species) cause **oxidative DNA damage**; polyunsaturated fat and excess iron increase it. Antioxidants scavenge ROS, so diet studies suggested protection.
- Randomised trials of single high-dose supplements have been **null or harmful**.

Prostate cancer

Agent	Evidence	Result
SELECT (Selenium and Vitamin E Cancer Prevention Trial)	35,533 men; selenium 200 µg, vitamin E 400 IU, both, or placebo	Stopped early (median 5.46 years) for futility. No benefit. Vitamin E alone trended to more cancer (P = 0.06); extended follow-up showed a significant **increase (about 17%)**
ATBC (Alpha-Tocopherol, Beta-Carotene)	Low-dose vitamin E in heavy smokers	Secondary analysis: prostate cancer ↓32%, deaths ↓41%; beta-carotene ↑ lung cancer
Selenised yeast	Skin-cancer trial	Secondary fall in prostate cancer prompted SELECT
Lycopene (cooked tomato)	Meta-analysis: modestly lower risk, mainly advanced cancer	WCRF (World Cancer Research Fund) downgraded "probable" (2007) to "no conclusion" (2018); supplement trials mostly null
Green tea catechins	HGPIN (high-grade prostatic intraepithelial neoplasia)/ASAP trials	Large fall in progression to cancer (meta-analysis); dose impractical
Vitamin C, high-dose antioxidants	High doses act as pro-oxidants	Supplements above median → more advanced cancer

- **Also know:** vitamin D (VITAL, no effect); coffee (modestly lower advanced risk); pomegranate, soy and curcumin unproven.

Bladder cancer

- Supplemental beta-carotene, vitamins C and E and folate do not lower risk; SELECT showed no effect.
- Fruit and vegetable benefit disappears after adjusting for smoking. Smoking cessation is the real prevention.

Kidney cancer

- No antioxidant trial has shown benefit.

Why the trials failed

- Single nutrient at high dose; wrong chemical form; non-smoking populations; PSA-era indolent cancers; U-shaped dose effects.

Current position

- Antioxidant supplements are **not recommended** for cancer prevention. Advise a balanced diet, no smoking and normal weight.

Diagram to draw: ROS → DNA damage → cancer, with antioxidants blocking the first arrow.

Key points to remember:

- SELECT: no benefit; vitamin E alone increased prostate cancer.
- ATBC benefit limited to smokers on low dose.
- Lycopene downgraded to "no conclusion" (2018).

Gene therapy in urology

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 4, Q3f (intravesical gene therapy also answered in 2022-Summer Paper 2, Q3a)

Definition

- Delivery of nucleic acid (DNA or RNA) into cells to treat disease: add, replace or silence a gene.

Vectors

Vector	Features
Adenovirus	Efficient, non-integrating, transient expression; replication-incompetent or oncolytic
Retrovirus, lentivirus	Integrate: long expression, risk of insertional mutagenesis
AAV (adeno-associated virus)	Low immunogenicity, long expression
Non-viral	Naked plasmid, liposomes, nanoparticles: safer, less efficient
Gene silencing	siRNA (small interfering RNA), antisense; CRISPR (clustered regularly interspaced short palindromic repeats) editing experimental

Strategies in urological cancer

1. **Immunogene (cytokine) therapy: nadofaragene firadenovec**: replication-incompetent adenovirus type 5 carrying the **interferon-α2b gene** with **Syn3** surfactant, which crosses the bladder's protective barrier. The urothelium keeps secreting interferon, overcoming the 1–2-hour dwell of protein interferon. 75 mL every 3 months; CIS (carcinoma in situ) complete response **53% at 3 months, 24% at 12 months**. **FDA (Food and Drug Administration) approved December 2022** for BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer).
2. **Oncolytic virus: cretostimogene (CG0070)**: replication-competent adenovirus; E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity.
3. **Suicide gene:** HSV-tk (herpes simplex virus thymidine kinase) turns ganciclovir or valacyclovir into a toxin; prostate cancer with radiotherapy (trials).
4. **Toxin gene under a tumour promoter:** BC-819 (diphtheria toxin A, H19 promoter).
5. **Gene-modified cells:** CAR-T (chimeric antigen receptor T cells) against PSMA (prostate-specific membrane antigen): early trials, toxic. Viral-vector vaccine PROSTVAC failed.

Benign conditions

- **OAB (overactive bladder) and ED (erectile dysfunction):** URO-902 (hMaxi-K) plasmid for the BK potassium channel relaxes smooth muscle; phase 1 only.
- **Primary hyperoxaluria type 1:** lumasiran (RNA interference against glycolate oxidase) lowers oxalate.
- Underactive detrusor: experimental.

Challenges

- Delivery barriers, vector immunity, short expression, cost, biosafety.

Diagram to draw: adenovirus → urothelial cell → interferon secretion → tumour apoptosis.

Key points to remember:

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

Molecular targeted therapy in urological cancers

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 4, Q3c

Concept

- **Targeted therapy** blocks a molecule the cancer depends on (receptor, kinase, transcription factor, DNA-repair enzyme, surface antigen), not all dividing cells.
- Classes: small-molecule TKIs (tyrosine kinase inhibitors, "-nib"), monoclonal antibodies ("-mab"), mTOR (mammalian target of rapamycin) inhibitors, HIF-2α (hypoxia-inducible factor-2α) inhibitor, PARP (poly-ADP-ribose polymerase) inhibitors, ADCs (antibody–drug conjugates), radioligands.

RCC (renal cell carcinoma)

- **Basis:** VHL (von Hippel–Lindau) loss → HIF accumulates even in normoxia → constant VEGF (vascular endothelial growth factor), PDGF (platelet-derived growth factor) and other targets.

Target	Drugs	Key evidence
VEGF receptors	Sunitinib, pazopanib, axitinib, cabozantinib (also MET, AXL), lenvatinib	COMPARZ: pazopanib noninferior to sunitinib; METEOR: cabozantinib beat everolimus
VEGF ligand	Bevacizumab	Modest PFS (progression-free survival) gain
mTOR	Temsirolimus (IV, poor risk), everolimus (oral)	Temsirolimus improved OS (overall survival) over interferon
HIF-2α	**Belzutifan**	VHL disease tumours; LITESPARK-005 beat everolimus after immunotherapy and TKI

- First line now: TKI **plus** checkpoint inhibitor. Everolimus also treats tuberous sclerosis angiomyolipoma.

Urothelial carcinoma

- **FGFR3 (fibroblast growth factor receptor 3)** alterations, typical of luminal tumours: **erdafitinib** (oral) after prior therapy.
- **ADCs: enfortumab vedotin** (Nectin-4) improved OS after platinum and immunotherapy; sacituzumab govitecan (Trop-2); HER2 (human epidermal growth factor receptor 2)-directed ADCs emerging.
- **EV-302 (2024):** enfortumab vedotin plus pembrolizumab, first line, roughly doubled OS versus platinum chemotherapy and is now standard (chapter notes predate it).

Prostate cancer (mCRPC, metastatic castration-resistant)

- **PARP inhibitors** exploit **synthetic lethality** in HRR (homologous recombination repair)-deficient tumours, especially BRCA2. **PROfound:** olaparib beat a second hormonal agent (imaging PFS 7.4 vs 3.6 months in BRCA/ATM).
- **PSMA (prostate-specific membrane antigen) radioligand: 177Lu-PSMA-617; VISION**: OS 15.3 vs 11.3 months.

Toxicity

- VEGF TKIs: hypertension, hand–foot syndrome, hypothyroidism. mTOR: mucositis, hyperglycaemia. Erdafitinib: hyperphosphataemia. Enfortumab: neuropathy, rash. PARP inhibitors: anaemia.

Limitations

- Resistance develops; cures rare; biomarker testing needed.

Diagram to draw: VHL–HIF–VEGF pathway with drug sites (TKI at receptor, bevacizumab at VEGF, mTOR and HIF-2α inhibitors).

> **Key points to remember:**
> - RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
> - Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
> - Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION).

Rapid Revision Sheet – Section 2 · Oncology, DSD, Gender & Ageing

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T. (LAQ)

- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months; no PSA or progression effect.
- BCG is the original urological immunotherapy.

Discuss in brief, present concepts in various GU cancers (LAQ)

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
- Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

Prepucioplasty. (SAQ)

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.
- BXO: circumcision preferred.

Enlist the steps in female to male gender reassignment surgery. (SAQ)

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

Penile Inversion Vaginoplasty Technique (SAQ)

- Glans → clitoris; penile skin → canal; scrotal skin → graft.
- Excise all erectile tissue; preserve Denonvilliers fascia.
- Lifelong dilation; rectovaginal fistula is worst.

Antioxidants in urological malignancy (SAQ)

- SELECT: no benefit; vitamin E alone increased prostate cancer.
- ATBC benefit limited to smokers on low dose.
- Lycopene downgraded to "no conclusion" (2018).

Gene therapy in urology (SAQ)

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

Molecular targeted therapy in urological cancers (SAQ)

- RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
- Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
- Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION).

Section 3 · Infections & Inflammation — Long answer questions (LAQ)

Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management

Asked as:

- Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis. — *Winter 2024 Paper IV Q1 (LAQ)*
- Painful bladder syndrome management. — *Summer 2021 Paper IV Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2017-Winter Paper 2, Q3b and 2021-Summer Paper 4, Q3d)

Read as: recent trends in the evaluation, diagnosis and management of IC/BPS (interstitial cystitis/bladder pain syndrome).

Definition: how it has changed

- **NIDDK (National Institute of Diabetes and Digestive and Kidney Diseases, 1987):** research criteria; missed about 60% of patients, so the definition was broadened.
- **AUA (American Urological Association), current:** an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the bladder, with LUTS (lower urinary tract symptoms) for **more than 6 weeks** (**>6 weeks**), without infection or other identifiable cause.
- **ESSIC (European Society for the Study of Interstitial Cystitis, 2008; uses >6 months):** chronic (>6 months) bladder pain with at least one other urinary symptom, after confusable diseases are excluded.
- **ICD-11 (International Classification of Diseases, 11th revision):** "chronic primary bladder pain syndrome".
- **Trend: splitting the syndrome.** "IC" is reserved for **Hunner lesion (HL) disease**, a true inflammatory disease (true inflammation); BPS without HL is a symptom complex. Both sit within UCPPS (urological chronic pelvic pain syndrome).
- **Phenotypes:** Hunner lesion (HL) disease versus non-HL; also pelvic-floor-dominant and widespread (centralised) pain.

Pathophysiology (current view)

- Multifactorial: leaky GAG (glycosaminoglycan) layer; mast-cell activation; neurogenic inflammation; **central sensitisation**; stress-axis dysfunction; small-fibre neuropathy. No infection proven.
- HL disease shows dense lymphoplasmacytic infiltrate; non-HL bladders are often histologically normal.
- **Associated:** IBS (irritable bowel syndrome, >30%), fibromyalgia, vulvodynia, endometriosis; tender pelvic floor in about 80%.

Recent trends in evaluation (AUA 2022) — before treatment

- **Diagnosis of exclusion plus localisation:** tests aim to find the pain generator.
- **History:** pain worse on filling, relieved by voiding; "painful urgency" without urge incontinence, unlike OAB (overactive bladder). **Ask about ketamine.**
- **Examination: six-site pelvic-floor palpation** (anterior and posterior levator ani, obturator internus, each side); vulval swab test; DRE (digital rectal examination) in men.

- **Basic tests:** PVR (post-void residual), urinalysis, culture; cytology if risk factors; frequency–volume chart.
- **Scores:** O'Leary–Sant ICSI/ICPI (Interstitial Cystitis Symptom/Problem Index), GUPI (Genitourinary Pain Index), pain VAS (visual analogue scale); **score pain and urinary symptoms separately.**
- **Body map:** widespread pain = 3 or more of 12 regions besides the pelvis.
- **Office cystoscopy if over 50 or HL suspected** (AUA 2022 change): HL is a stellate red area with radiating vessels, found only by looking.
- **Local anaesthetic challenge** (lidocaine ± bupivacaine) localises pain to the bladder and selects intravesical therapy.
- **Dropped:** potassium sensitivity test (neither sensitive nor specific); routine hydrodistension (glomerulations non-specific); routine urodynamics.
- **Biopsy** only to exclude CIS (carcinoma in situ).

ESSIC classification (cystoscopy with hydrodistension + biopsy, e.g. "3C"):

Cystoscopy	Biopsy
X not done; 1 normal; 2 glomerulations; 3 Hunner lesion	X not done; A normal; B inconclusive; C positive (inflammation, mastocytosis, fibrosis)

Exclude: UTI (urinary tract infection) including tuberculosis, CIS (carcinoma in situ), stones, radiation or cyclophosphamide cystitis, **ketamine cystitis**, OAB (overactive bladder), diverticulum, endometriosis.

Phenotype-based diagnosis (key trend)

Phenotype	Identified by	Treatment
Hunner lesion	Cystoscopy	Fulguration and/or triamcinolone; ciclosporin A if refractory
Anaesthetic-responsive bladder pain	Local anaesthetic challenge	Intravesical therapy
Pelvic-floor tenderness	Pelvic-floor examination	Myofascial physiotherapy
Widespread (centralised) pain	Body map, associated pain conditions	Systemic: amitriptyline, gabapentinoids, CBT (cognitive behavioural therapy), pain team

- **MAPP (Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain) reanalysis, 2023:** hydroxyzine/PPS (pentosan polysulfate) helped localised pain; amitriptyline helped widespread pain.

Recent trends in management (AUA 2022): principles

- **Tiers abandoned: Old AUA (2011/2014): six tiers**, from self-care to surgery. **AUA 2022 dropped the tiers:** the six-tier ladder is replaced by grouped options, phenotype-guided, shared decisions; multimodal treatment (care) allowed early; major surgery stays last. Stop treatments that fail.

Treatment options

Group	Options
Behavioural	Education; avoid trigger foods (coffee, citrus, spicy food, alcohol); stress management; flare plan / flare plans
Physical	**Myofascial pelvic-floor therapy / physiotherapy** if tender (Standard); **avoid Kegel exercises**

Group	Options
Oral	**Amitriptyline** 10 mg at night, up to about (~) 50 mg; hydroxyzine; cimetidine; **PPS (pentosan polysulfate) 100 mg three times daily** (**pigmentary maculopathy**: baseline and periodic eye checks; retinal checks)
Intravesical	**DMSO (dimethyl sulfoxide) 50%, 50 mL, 15–20 min**; heparin; alkalinised lidocaine (flares); also hyaluronic acid, chondroitin
Procedures	**HL: fulguration and/or triamcinolone injection** (75–90% improve); short (1–3 min) low-pressure (60–80 cmH_2O) hydrodistension; **onabotulinumtoxinA 100 U into the trigone** (must accept self-catheterisation; risk of retention); SNM (sacral neuromodulation)
Refractory HL	**Oral ciclosporin A** (watch kidney function and blood pressure)
Last resort (<10%)	Cystoplasty (augmentation) or diversion ± cystectomy; pain may persist

- **Not offered / do not offer:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term oral steroids.
- Pain management: gabapentinoids; opioids only via pain specialists.
- Monitor with symptom scores; stop ineffective treatments. Most hydrodistension responses last under 2 months.
- Treat associated pain conditions (IBS (irritable bowel syndrome), fibromyalgia) with other specialists.

Recent advances

- Urinary biomarkers and functional MRI (magnetic resonance imaging) for phenotyping (research); phenotype-stratified trials.
- **Also know:** tanezumab (anti-NGF [nerve growth factor]) not adopted; liposomal intravesical botulinum toxin no better than placebo; pudendal neuromodulation.

Diagram to draw: cystoscopic Hunner lesion with radiating vessels, and the four-phenotype treatment chart.

> **Key points to remember:**
> - AUA definition: bladder pain with LUTS >6 weeks; ESSIC >6 months.
> - AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test.
> - HL: fulguration or triamcinolone (fulguration/triamcinolone); ciclosporin if refractory.
> - PPS causes maculopathy; antibiotics and BCG not offered.

Section 3 · Infections & Inflammation — Short answer questions (SAQ)

Chronic non bacterial prostatitis.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 4, Q3d

Definition and classification

- Now called **CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)**, NIH (National Institutes of Health) category III: pelvic pain for **≥3 of the last 6 months** without uropathogenic infection. 90–95% of prostatitis.

NIH category	Description
I	Acute bacterial prostatitis
II	Chronic bacterial prostatitis (recurrent same-organism UTI)
IIIA	CP/CPPS, **inflammatory** (leucocytes in EPS (expressed prostatic secretion), VB3 (post-massage urine) or semen)
IIIB	CP/CPPS, **non-inflammatory**
IV	Asymptomatic inflammatory

- IIIA vs IIIB has no treatment value.

Aetiology

- Interplay of past infection, neurogenic inflammation, **pelvic floor muscle dysfunction**, autoimmunity, psychological stress and **central sensitisation**. The prostate may not be the pain source.

Clinical features

- Perineal (commonest), testicular, suprapubic or penile pain; ejaculatory pain; LUTS (lower urinary tract symptoms); sexual dysfunction.

Evaluation (diagnosis of exclusion)

- **NIH-CPSI (Chronic Prostatitis Symptom Index):** pain, urinary and quality-of-life domains (0–43).
- DRE (digital rectal examination): prostate tender in fewer than half; palpate the pelvic floor for **trigger points** (tender in ~half).
- Urinalysis and culture; **two-glass test** (pre- and post-massage). Cytology if irritative symptoms or haematuria. PVR (post-void residual), flow.
- Selected: cystoscopy, urodynamics.
- **UPOINT phenotyping: U**rinary, **P**sychosocial, **O**rgan-specific, **I**nfection, **N**eurological/systemic, **T**enderness of muscles.

Management: multimodal, phenotype-directed (monotherapy trials fail)

Domain	Therapy
Urinary	**α-blockers** in recent-onset, α-blocker-naive men (<1 year); antimuscarinics
Psychosocial	CBT (cognitive behavioural therapy), stress reduction

Domain	Therapy
Organ (prostate)	**Quercetin**, **pollen extract** (small RCTs, randomised controlled trials), 5-ARI (5α-reductase inhibitor) if BPH (benign prostatic hyperplasia)
Infection	Quinolone or tetracycline for ≥6 weeks if treatment-naive and <1 year (EAU, European Association of Urology), or positive VB3; no repeated courses
Neurological	Pregabalin, amitriptyline, duloxetine
Tenderness	**Pelvic floor physiotherapy**, myofascial trigger-point release, biofeedback

- **Others: acupuncture** (beat sham in a large RCT), PTNS (percutaneous tibial nerve stimulation), shockwave therapy, aerobic exercise; NSAIDs (non-steroidal anti-inflammatory drugs) only in combination.
- **Not recommended:** TURP (transurethral resection of the prostate), radical prostatectomy.

Diagram to draw: UPOINT snowflake with a therapy for each domain.

> **Key points to remember:**
> - NIH III: pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
> - NIH-CPSI to score; UPOINT to phenotype.
> - Multimodal therapy; α-blocker plus antibiotic only early.

Nitrofurantoin.

Type: SAQ (10 marks)

Source: 2020-Summer Paper 4, Q3e

Introduction

- A synthetic **nitrofuran**; **first-line for acute uncomplicated cystitis** (EAU (European Association of Urology)) because it concentrates in urine and spares bowel and vaginal flora.

Mechanism

- Reduced by bacterial flavoproteins (nitroreductases) to **reactive intermediates** that damage ribosomal proteins, DNA and several enzyme systems.
- **Bacteriostatic**, bactericidal at high urinary concentration.
- Multiple targets keep **acquired resistance low**.

Pharmacokinetics

- Well absorbed (better with food); **rapidly excreted in urine**. Serum and tissue levels are sub-therapeutic.
- Hence **useless for pyelonephritis, prostatitis, febrile or bloodstream infection**.
- Macrocrystals: slower absorption, less nausea.

Spectrum

Susceptible	Resistant
E. coli (most), ***Enterococcus faecalis***, *Staphylococcus saprophyticus*, *S. aureus* (not MRSA (methicillin-resistant *S. aureus*)), many *Klebsiella*, *Citrobacter*	***Proteus*** (intrinsic), *Pseudomonas*, *Serratia*, *Providencia*, *Morganella*, *Acinetobacter*

- Active against many ESBL (extended-spectrum β-lactamase)-producing *E. coli*.

Uses and doses

Indication	Regimen
Uncomplicated cystitis (women)	**100 mg twice daily × 5 days** (modified release)
Recurrent UTI (urinary tract infection) prophylaxis, adults	50–100 mg at night, or post-coital
Children: treatment / prophylaxis	3–5 mg/kg/day in 2 doses / 1–2 mg/kg at night (VUR (vesicoureteral reflux), after 3 months of age)
Pregnancy	Second and third trimesters; first trimester only if no alternative; **avoid near term**

Adverse effects

- **GI (gastrointestinal) upset** (commonest).
- **Pulmonary**: acute hypersensitivity (fever, cough, dyspnoea); **chronic interstitial fibrosis** with long-term use.
- **Peripheral neuropathy** (worse with renal impairment, diabetes, anaemia, B-vitamin deficiency).
- **Haemolysis in G6PD (glucose-6-phosphate dehydrogenase) deficiency**, including the neonate (hence avoid at term and <3 months of age).
- Hepatotoxicity (cholestatic or chronic hepatitis), rare.

Precautions

- **Renal impairment**: inadequate urine levels and more toxicity. Historically avoided if CrCl (creatinine clearance) <60; Beers Criteria (2015) allow short courses down to **CrCl 30 mL/min**; avoid long-term use in the elderly.
- Interactions: probenecid (blocks excretion), magnesium antacids, quinolones (antagonism).

Diagram to draw: spectrum table (susceptible versus resistant).

Key points to remember:

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015).

Rapid Revision Sheet – Section 3 · Infections & Inflammation

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management (LAQ)

- AUA definition: bladder pain with LUTS >6 weeks; ESSIC >6 months.
- AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test.
- HL: fulguration or triamcinolone (fulguration/triamcinolone); ciclosporin if refractory.
- PPS causes maculopathy; antibiotics and BCG not offered.

Chronic non bacterial prostatitis. (SAQ)

- NIH III: pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
- NIH-CPSI to score; UPOINT to phenotype.
- Multimodal therapy; α-blocker plus antibiotic only early.

Nitrofurantoin. (SAQ)

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015).

Section 4 · Paediatric Urology 1 — Long answer questions (LAQ)

Hypospadias: present concepts in management, and hypospadias fistula

Asked as:

- Present concept in management of hypospadias — *Practice Set 6 Paper IV Q1 (LAQ)*
- Present concept in management of hypospadias — *Practice Set 6 Paper IV Q1 (earlier version) (LAQ)*
- Etiology and management of hypospadias fistula — *Summer 2015 Paper IV Q3a (SAQ)*
- Etiology and management of hypospadias fistula. — *Practice Set 6 Paper IV Q3d (earlier version) (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(embryology and proximal repair also in 2017-Winter Paper 3, Q1; Snodgrass repair in 2023-Winter Paper 3, Q3c)

Read as "present (current) concepts in the management of hypospadias".

Introduction

- Hypospadias is an arrest of the androgen-dependent "closing zipper" of urethral fold fusion (weeks 8–16). Triad: ventral ectopic meatus, ventral curvature, dorsal hooded prepuce.
- 1 in 150–300 male births; distal 70–85%, proximal 10–25%.
- **Goals:** a straight erection, a urethra to the tip (standing voiding, semen delivery) and acceptable appearance, with the fewest operations and a result lasting into adulthood.

Concept 1: severity is judged in theatre, not by the meatus

- A "distal" meatus may hide absent spongiosum and severe curvature, so classify **after degloving**.

GMS (glans–meatus–shaft) score	Glans	Meatus	Shaft curvature
1	Good size, deep groove	Glanular	None
2	Adequate, grooved	Coronal	<30°
3	Small; narrow or flat plate	Mid or distal shaft	30–60°
4	Very small; flat plate	Proximal shaft, penoscrotal	>60°

- Total 3–12; a higher score predicts more complications.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft over plain TIP (tubularised incised plate).

Concept 2: evaluation

- **DSD (disorder of sex development) work-up** if hypospadias plus undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).

- No routine VCUG (voiding cystourethrogram) or renal ultrasound. Associations: hernia or hydrocele 9–16%, cryptorchidism about 7% (10% in proximal), enlarged utricle.
- **Aetiology:** causes converge on deficient androgen action (placental insufficiency, assisted reproduction, genes, endocrine disruptors).

Concept 3: timing and preparation

- Repair at **6–18 months** (AAP (American Academy of Pediatrics): 6–12 months), before genital awareness at about 18 months. One short anaesthetic appears low-risk.
- A straight penis that voids standing may gain little from surgery: counsel honestly.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Evidence mixed; two RCTs (randomised controlled trials) showed fewer complications.

Concept 4: curvature decides the operation

```
Deglove + excise dysplastic dartos + artificial erection
  ├─ straight       → urethroplasty
  ├─ residual <30°  → dorsal midline plication (12 o'clock nerve-free zone)
  └─ >30°           → divide plate → still >30° → ventral corporotomy ± graft → staged repair
```

- Recurrence 28% after plication versus 9% after grafting. Under-correction causes repeated surgery, so correct fully first time.

Concept 5: choice of technique

Situation	Current choice
Glanular meatus, no curvature	Advancement: MAGPI (meatal advancement and glanuloplasty), MIV (M inverted V) glansplasty; selectively
Distal or midshaft, good plate	**TIP (Snodgrass)**: >90% of distal, about 80% of midshaft repairs; Thiersch–Duplay; GAP (glans approximation procedure) for a deep groove
Narrow or shallow plate	TIP with **dorsal inlay graft**; onlay island flap; Mathieu (SLAM modification)
Proximal, curvature >30°	**Two-stage**: STAG (staged tubularised autograft) or Bracka graft; Byars flaps; distal-first (Ulaanbaatar)
Proximal, curvature corrected, good tissue	One-stage onlay or TPIF (transverse preputial island flap); proximal TIP abandoned

- **Why stage:** a urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed. Stage 2 after ≥6 months.

Concept 6: protect blood supply

- Magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), rapidly absorbed fine sutures, no hair-bearing skin, haemostasis.
- Postoperative: stent 3–21 days by complexity, compressive dressing, oxybutynin; antibiotic prophylaxis debated.

Concept 7: standardised complications and outcomes

- Definitions: glans dehiscence (fusion <2 mm); meatal stenosis (symptoms plus <8 Fr before puberty, <12 Fr after); recurrent curvature >30°.
- TIP meta-analysis (Pfistermuller 2015): complications 10.6%, fistula 5.7%. Proximal complications 23–68%; STAG about 23% (lowest); Byars 43–81%.
- A long, rigid, spongiosum-free neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum. Fistula: see 2015-Summer Paper 4, Q3a (another source: see Practice Set 6 Paper 4, Q3d).

- **Also know:** BXO (balanitis xerotica obliterans), excise and stage with buccal graft; redo repairs use buccal (lip inside glans) mucosa, Cecil scrotal parasitisation, tissue expansion.

Concept 8: follow-up and patient-reported outcomes

- About three-quarters of complications appear by 3 years; stricture and curvature appear late. Follow proximal cases **through puberty**.
- Tools: HOSE (Hypospadias Objective Scoring Evaluation, surgeon), PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation, photo panels). Decision aids reduce parental regret.

Recent advances

- Registries, AI (artificial intelligence) video review, genomic risk profiling, a quality-of-life measure, engineered tissue.

Diagram to draw: curvature algorithm and two-stage graft repair.

Hypospadias fistula: aetiology and management

(also answered in 2015-Summer Paper 4, Q3a)

Definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak anywhere below the meatus: the **commonest complication** of hypospadias repair, just under 10% at short-term follow-up.
- After TIP (tubularised incised plate): 5.7% (**Pfistermuller 2015**); after proximal repair 3–45%. Usually at the original meatus or coronal margin.

Aetiology

- **Distal obstruction** (meatal stenosis, stricture): raises voiding pressure and turbulence; the **key correctable cause**.
- **Ischaemia**, oedema, infection, haematoma, tension.
- **Technical:** overlapping suture lines, poorly inverted epithelium, slowly absorbed sutures, **no barrier layer**.
- **Patient:** proximal meatus, glans <14 mm, redo surgery. A long, rigid neourethra raises resistance (Poiseuille's law).

Prevention

- **Vascularised barrier** (dartos; **tunica vaginalis** in proximal or redo), offset suture lines, stent in complex repairs.

Evaluation

- Split stream / multiple streams, usually within 2 years.
- Under anaesthesia: **bougie calibration** for distal narrowing, **urethroscopy** if irregular, and a **stress test** (compress proximally and distally, inject saline or methylene blue) for hidden tracts.

Management

- Rarely closes spontaneously. **Wait 6–12 months**; **correct distal obstruction at the same sitting**.

Fistula	Repair
Small, shaft	Excise the epithelialised tract, inverting subepithelial closure with fine absorbable suture, multiple offset layers, **dartos barrier**
Large or thin tissue	Trap-door or island skin flap plus second layer
Coronal margin (glans dehiscence)	Implies distal narrowing or poor tissue: **redo urethroplasty and glansplasty** (redo urethroplasty and glans)

Fistula	Repair
With diverticulum	Excise, or reuse the redundant wall as a flap

- Recurrence about **20%**; reduce with interposed vascularised layers (dartos, tunica vaginalis, Cecil modification when tissue is scarce).
- Catheter: none for simple repairs, 7–14 days for complex.
- **Also know:** preputial fistula does not involve the urethra; congenital urethral fistula is closed the same way; follow through puberty.

```
UCF → wait 6–12 months → calibrate, urethroscopy, stress test
 ├─ distal obstruction → correct at same sitting
 ├─ small → excise + layered closure + barrier flap
 └─ large/coronal → flap repair or redo urethroplasty
```

Diagram to draw (fistula): neourethra, dartos barrier and skin with offset suture lines.

Key points to remember:

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

VUR: etiopathogenesis, recent trends in management and STING technique

Asked as:

- Describe etiopathogenesis of VUR and recent trends in management — *Practice Set 2 Paper IV Q2 (LAQ)*
- STING technique for VUR. — *Summer 2020 Paper IV Q3c (SAQ)*
- STING technique for VUR. — *Practice Set 5 Paper IV Q3e (earlier version) (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(VUR also answered in Practice Set 1 Paper 1, Q3f and 2020-Summer Paper 1, Q2; detailed management of bilateral VUR with surgical techniques: see Practice Set 2 Paper 3, Q2) (STING also answered in 2020-Summer Paper 4, Q3c)

Definition and epidemiology

- VUR (vesicoureteral reflux) is retrograde flow of urine from the bladder into the upper tract.
- Found in about **30% of children with UTI (urinary tract infection)** and about 17% imaged for hydronephrosis. The younger the child, the likelier VUR.
- Infant boys predominate (higher voiding pressures, more UTI if uncircumcised); overall girls twice as often. Less common in Black children.

Normal antireflux mechanism

- Balance of three factors: **ureteric peristalsis**, **UVJ (ureterovesical junction) anatomy** and **bladder dynamics**.
- **Passive flap valve:** the intramural ureter is compressed against detrusor backing as the bladder fills. This needs adequate tunnel length and fixation.
- **Paquin:** tunnel length to diameter about **5:1** normally versus 1.4:1 in refluxing units. Orifice shape also matters.
- Waldeyer's sheath anchors the intramural ureter to the deep trigone.

Etiopathogenesis

1. Embryology (Mackie and Stephens)

	Early budding	Late budding
Orifice	High and lateral	Distal and medial
Result	Short tunnel, **VUR**	Ectopic, often obstructed ureter
Kidney	Abnormal bud–metanephros contact: **dysplasia**	—

- This single fault explains why high-grade VUR and dysplasia occur together.

2. Primary versus secondary

Primary	Secondary
Congenitally deficient UVJ: short tunnel, weak detrusor backing, lateral orifice	Normal UVJ overwhelmed by pressure: PUV (posterior urethral valves), neurogenic bladder (VUR in up to 30% of spina bifida), dysfunctional voiding; or UVJ anomaly (ureterocele, duplication, paraureteral diverticulum)

- They overlap: "primary" VUR often has a treatable secondary component.

3. Bladder and bowel dysfunction (BBD): high voiding pressures and incomplete emptying distort the UVJ. Detrusor overactivity is the commonest urodynamic finding. Infant boys have high voiding pressures that settle with maturation.

4. Genetics: sibling prevalence about 30%, highest under 3 years; identical twins almost fully concordant; polygenic.

5. Infection: cystitis lowers the reflux threshold; *E. coli* endotoxin causes ureteric atony.

Pathogenesis of renal damage

	Reflux-associated dysplasia	Acquired scar
Cause	Abnormal bud (congenital)	Infected urine plus VUR
Typical	Boy, grade IV–V, antenatal hydronephrosis	Girl, recurrent febrile UTI, BBD
Preventable	No	Yes

- **Intrarenal reflux** through flat compound papillae at the poles needs only 2 mmHg in neonates, so scars are polar.
- **Sterile reflux rarely scars** unless pressures are high (Ransley and Risdon). Most scars follow early infections ("big bang", now debated).
- Sequelae: hypertension in **34–38%** of scarred children; CKD (chronic kidney disease); poor growth; pre-eclampsia.

International grading (1981)

Grade	Appearance on VCUG (voiding cystourethrogram)
I	Ureter only
II	Up to calyces, no dilatation
III	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation and tortuosity; papillary impressions lost

- Natural history: grade II resolves in 63–85%, grade III about 50%; grade IV–V and bilateral III rarely.

Recent trends in management

1. **Treat infection risk, not reflux:** risk stratification by sex, age, grade, laterality, BBD, scars and presentation (Hidas model). Breakthrough UTI on prophylaxis at 2 years: 8.6% low, 26% intermediate, 62.5% high risk.
2. **Fewer VCUGs:** ultrasound first (AAP [American Academy of Pediatrics] 2011, NICE [National Institute for Health and Care Excellence]); or "top-down" DMSA (dimercaptosuccinic acid) first. Contrast-enhanced voiding urosonography avoids radiation.
3. **Objective measures:** UDR (ureteral diameter ratio) predicts resolution and breakthrough UTI better than grade; machine-learning models.
4. **Selective CAP (continuous antibiotic prophylaxis): RIVUR** halved recurrent UTI (23.6% to 12.9%) but did not reduce scarring. NNT (number needed to treat) is 5 in high-risk and 18 in low-risk children. The Swedish Reflux Trial showed benefit in girls with dilating VUR.
5. **BBD first:** timed voiding, constipation treatment, anticholinergics. Topical steroid for phimosis in boys.
6. **Endoscopic injection:** Dx/HA (dextranomer/hyaluronic acid); HIT and double HIT (hydrodistension implantation technique) give about 90% resolution. Vantris works better but can obstruct late. (Details: STING technique below.)

7. **RALUR-EV (robot-assisted extravesical reimplantation):** less pain and shorter stay; results approaching the open benchmark of **95.9%**.
8. **Less follow-up imaging:** VCUG may be omitted after surgery for low-grade VUR if ultrasound is normal.

STING technique for VUR

Definition and principle

- **STING (subureteric Teflon injection)**: endoscopic injection of a **bulking agent** beneath the ureteric orifice in VUR (vesicoureteral reflux). The mound supports the intramural ureter, restoring the **flap-valve**.
- Matouschek, 1981; popularised by **O'Donnell and Puri, 1986**.

Bulking agents

Agent	Particle	Problem
Teflon (PTFE (polytetrafluoroethylene) paste)	90% <40 µm, permanent	**Migration**; never FDA (Food and Drug Administration) approved
Dx/HA (dextranomer/hyaluronic acid, Deflux)	**80–250 µm**; gel resorbed	Volume loss about 23%; FDA approved 2001; standard agent
Polyacrylate (Vantris)	300 µm, permanent	Late ureteric obstruction

Technique

1. Antibiotic at induction; offset-lens cystoscope; rigid 3.7–5 Fr needle; bladder partly filled.
2. **STING**: needle enters **2–3 mm below the orifice at 6 o'clock**, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.
4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding. Day case.

Technique	Needle entry	Result
STING	Below orifice	Baseline
HIT (hydrodistension implantation technique)	Inside hydrodistended orifice	Better than STING, all grades
Double HIT	Two intraureteric sites	About 90% resolution

Indications

- Persistent primary VUR (grades II–IV) with breakthrough febrile UTI (urinary tract infection), new scars, poor adherence or parental preference; residual VUR after reimplantation.
- **Relative contraindications**: grade V, **UVJ (ureterovesical junction) obstruction**, untreated BBD (bladder and bowel dysfunction).

Results

- Low-grade VUR about 90% after one injection. **Elder 2006 meta-analysis**: grade I–II about 79%, III 72%, IV 63%, V 51%.
- **Swedish Reflux Trial**: 20% recurrence at 2 years.

Complications

- Failure or recurrence (repeat 1–2 injections, then reimplant); new contralateral VUR; **early or late obstruction**; calcified mound mimics a stone. **Also know:** Macroplastique.

Follow-up

- Antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound ± VCUG (voiding cystourethrogram) at 3 months.

Diagram to draw: normal long oblique tunnel versus short lateral refluxing tunnel, with an early-budding Y-diagram; needle sites: STING below orifice, HIT and double HIT inside the tunnel, with volcano mound.

Key points to remember:

- Paquin 5:1 normal versus 1.4:1 refluxing; early budding causes VUR and dysplasia.
- Sterile reflux rarely scars; 34–38% of scarred children become hypertensive.
- RIVUR: CAP halves UTI, not scars.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
- Watch late obstruction; calcified mound mimics stone.

Rapid Revision Sheet – Section 4 · Paediatric Urology 1

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Hypospadias: present concepts in management, and hypospadias fistula (LAQ)

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

VUR: etiopathogenesis, recent trends in management and STING technique (LAQ)

- Paquin 5:1 normal versus 1.4:1 refluxing; early budding causes VUR and dysplasia.
- Sterile reflux rarely scars; 34–38% of scarred children become hypertensive.
- RIVUR: CAP halves UTI, not scars.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
- Watch late obstruction; calcified mound mimics stone.

Section 5 · Paediatric Urology 2 — Short answer questions (SAQ)

Vesicostomy.

Type: SAQ (10 marks)

Source: 2016-Summer Paper 4, Q3f

Definition

- **Cutaneous vesicostomy**: the bladder dome is brought to the lower abdominal skin as an incontinent stoma draining into the nappy. It "vents" the bladder, protecting the upper tracts; usually a **temporary** bridge to reconstruction.

Indications

- **PUV (posterior urethral valves)**: infant too small for a resectoscope; upper tract deterioration despite drainage or ablation.
- **Prune-belly syndrome**: acute renal failure, urosepsis, urethral atresia. A larger stoma is needed because stenosis is common.
- **Neurogenic bladder**: high pressure despite CIC (clean intermittent catheterisation) and drugs, or CIC impossible; worsening hydronephrosis; febrile UTI (urinary tract infection) with high-grade reflux.
- **Also know:** urethral atresia; MMIHS (megacystis-microcolon-intestinal hypoperistalsis syndrome); urogenital sinus anomalies.

Blocksom technique (Duckett modification)

1. Transverse 2 cm incision midway between umbilicus and pubis. (In prune-belly, excise any urachal diverticulum.)
2. Incise the rectus sheath; separate the recti.
3. Fill the bladder; mobilise the **dome** up to the urachus, sweeping peritoneum off.
4. Bring the **dome** (not the anterior wall) to the wound. This reduces prolapse of the posterior wall.
5. Anchor detrusor to rectus fascia; stoma should take about 24 Fr.
6. Open the dome; suture mucosa to skin. No catheter needed.

Advantages

- Simple; no catheter or appliance.
- The bladder still fills and empties through the stoma at low leak pressure, so compliance is **not** lost; it remains usable for reconstruction or transplantation.

Complications

- **Stomal stenosis**, mucosal **prolapse**, peristomal dermatitis, stones, UTI; leakage harder to manage as the child grows.

Closure

- When the cause is treated (valve ablation, urethral growth) or at definitive reconstruction; check bladder function first (cystogram or urodynamics).
- Excise the stoma, free the bladder from fascia, close in two layers, catheter drainage for a few days.

Alternatives

Option	Use
Cutaneous ureterostomy or pyelostomy	Rare; persistent upper tract obstruction
Ileovesicostomy	Unable to catheterise; 10–15 cm ileum
Appendicovesicostomy (Mitrofanoff, 1980)	Continent catheterisable channel

Diagram to draw: sagittal view of bladder dome sutured to fascia and skin.

Key points to remember:

- Blocksom: dome-based stoma midway between umbilicus and pubis.
- Bladder keeps cycling; compliance preserved.
- Stenosis and prolapse are the main complications.

Management of Defecation Disorders resulting into LUTS

Type: SAQ (10 marks)

Source: 2025-Winter Paper 4, Q3f

Assumed meaning: bladder and bowel dysfunction, where constipation or defecation disorders cause LUTS (lower urinary tract symptoms).

Why the bowel causes LUTS

- **Shared space:** a loaded rectum compresses the bladder, reducing capacity and obstructing outflow.
- **Shared nerves:** bladder and bowel afferents converge on the same spinal segments; cross-organ sensitisation causes urgency and detrusor overactivity.
- **Shared muscle:** a pelvic floor that contracts during defecation (**dyssynergic defecation**) also contracts during voiding (**dysfunctional voiding**) → high PVR (post-void residual), UTI (urinary tract infection), reflux.
- Result: urgency, daytime incontinence, enuresis, retention, recurrent UTI.

Evaluation

- Stool frequency, pain, **Bristol chart**; 3-day bladder and bowel diary.
- **Rome IV (child):** at least 2, weekly for 1 month: ≤2 defecations/week; faecal incontinence weekly; retentive posturing; painful or hard stools; large rectal mass; large-diameter stools.
- **Rome IV (adult), for 3 months:** ≥2 of straining, hard stools, incomplete evacuation, blockage, manual manoeuvres (each in >25% of defecations), <3 stools/week.
- Examine abdomen, spine and perineum; exclude organic causes (Hirschsprung disease, hypothyroidism, neurogenic).
- **Also know:** rectal diameter ≥30 mm on ultrasound; no routine X-ray; manometry if refractory.

Management

Rule: treat the bowel first (EAU, European Association of Urology/ESPU, European Society for Paediatric Urology).

1. **Disimpaction:** oral PEG (polyethylene glycol) 3350, 1–1.5 g/kg/day for 3–6 days; enemas if needed.
2. **Maintenance:** PEG 0.4 g/kg/day (range 0.2–0.8), at least 2 months; lactulose alternative; stimulants (senna) for expulsion.
3. **Urotherapy:** toileting after meals (gastrocolic reflex); footstool posture; normal fluid and fibre.
4. **Biofeedback, pelvic floor physiotherapy** for dyssynergia.
5. **Refractory: transanal irrigation** (water via rectal cone; a quarter to a third stop); **ACE (antegrade continence enema, Malone)**; botulinum toxin to the sphincter; sacral or tibial neuromodulation.

- **Adults:** laxatives, prucalopride or linaclotide, biofeedback; rectocele repair; irrigation in neurogenic bowel.
- **Caution:** anticholinergics worsen constipation; LUTS return.

Diagram to draw: pelvic cross-section showing loaded rectum compressing bladder.

Key points to remember:

- Treat the bowel first.
- Rome IV: ≥2 criteria for 1 month (child).
- PEG disimpaction 1–1.5 g/kg/day for 3–6 days.

Rapid Revision Sheet – Section 5 · Paediatric Urology 2

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Vesicostomy. (SAQ)

- Blocksom: dome-based stoma midway between umbilicus and pubis.
- Bladder keeps cycling; compliance preserved.
- Stenosis and prolapse are the main complications.

Management of Defecation Disorders resulting into LUTS (SAQ)

- Treat the bowel first.
- Rome IV: ≥2 criteria for 1 month (child).
- PEG disimpaction 1–1.5 g/kg/day for 3–6 days.

Section 6 · Female Urology — Short answer questions (SAQ)

Acute urinary retention in females.

Type: SAQ (10 marks)

Source: 2017-Summer Paper 4, Q3f

Introduction

- AUR (acute urinary retention) is sudden inability to void. It is uncommon in women and is often non-obstructive, so search for the cause.

Causes

Group	Examples
Obstructive	Large cystocele or prolapse; tight sling after incontinence surgery; urethral diverticulum, stricture; pelvic mass, **retroverted gravid uterus**, haematocolpos; cancer
Functional	**Fowler syndrome**; dysfunctional voiding; primary bladder neck obstruction (rare)
Neurogenic	MS (multiple sclerosis; may present first with retention), cauda equina, transverse myelitis, diabetes
Inflammatory	Cystitis, vulvovaginitis, **genital herpes**, Bartholin abscess
Drugs	Anticholinergics, opioids, tricyclics, calcium-channel blockers
Postoperative/postpartum; psychogenic	Epidural, pain; diagnosis of exclusion

Fowler syndrome

- Young women (20–35) with painless retention and bladder capacity over 1 litre.
- **Urethral sphincter EMG (electromyography):** complex repetitive discharges, so the sphincter fails to relax. Urethral closure pressure >100 cm H_2O.

Evaluation

- **History:** drugs, surgery, delivery, neurological and bowel symptoms, genital lesions.
- **Examination:** pelvic (prolapse, mass); neurological (perineal sensation, anal tone).
- **Tests:** drained volume; urine culture; creatinine; pregnancy test; ultrasound.
- MRI (magnetic resonance imaging) spine if neurological signs; cystoscopy.
- **Later:** video-urodynamics (female obstruction, e.g. Qmax ≤12 mL/s with Pdet ≥20 cm H_2O); sphincter EMG.

Management

- **Immediate:** urethral catheter; suprapubic if urethral access fails.
- **Treat the cause:** stop drugs; treat infection or herpes; pessary or prolapse repair; reposition the uterus; sling division; diverticulectomy.
- **TWOC (trial without catheter)** after correction.
- **Persistent retention:**
  - **CISC (clean intermittent self-catheterisation)** rather than an indwelling catheter.
  - **SNM (sacral neuromodulation):** about 70% success in Fowler syndrome.

- Bethanechol not recommended. Bladder neck incision only with caution (incontinence).

Diagram to draw: flowchart: AUR → catheterise → find cause → treat → TWOC → CISC or SNM.

Key points to remember:

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention.

Rapid Revision Sheet – Section 6 · Female Urology

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Acute urinary retention in females. (SAQ)

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention.

Section 7 · Voiding Dysfunction — Long answer questions (LAQ)

Neuromodulation for voiding dysfunction, including sacral neuromodulation

Asked as:

- Discuss neuromodulation for voiding dysfunction. — *Summer 2019 Paper IV Q1 (LAQ)*
- Sacral neuromodulation — *Summer 2015 Paper IV Q3c (SAQ)*
- Sacral neuromodulation. — *Summer 2016 Paper IV Q3c (SAQ)*
- Sacral Neuromodulation — *Winter 2017 Paper IV Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

See 2015-Summer Paper 3, Q3a (identical question) (for Summer 2015 Paper IV Q3c; recent advances to add are listed under Recent advances below). **See 2017-Winter Paper 3, Q3a (full answer)** (for Winter 2017 Paper IV Q3b).

Definition and history

- **Neuromodulation:** electrical stimulation of a peripheral nerve that alters the activity of central pathways controlling the bladder. It acts on the nervous system, not directly on muscle.
- Direct muscle stimulation failed (painful, local). Stimulating **afferents** needs less current and affects the whole organ.
- **Tanagho and Schmidt** (about 1985): SNM (sacral neuromodulation). PTNS (percutaneous tibial nerve stimulation): FDA (Food and Drug Administration) approval 2000.
- **SNM (sacral neuromodulation)** (Tanagho and Schmidt): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator).
- **Rule:** exclude anatomical causes first (obstruction, stones, cancer, fistula). Stimulation cannot correct anatomy or restore muscle.

Mechanism

- Sacral roots are mixed nerves. Current excites **afferents → spinal cord and forebrain** (areas of awareness and attention) → altered PMC (pontine micturition centre) output. It acts on **afferent** nerves, not motor nerves.
- **OAB (overactive bladder):** heightened central control damps PMC firing → fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling (filling sensation) and pelvic floor tone → PMC switches on → **EUS (external urethral sphincter) relaxes**. Hence SNM treats opposite disorders (treats **both**).
- **Tibial:** nerve carries **S2–S3** fibres; inhibition of OAB-type (C-fibre) reflexes is **opioid-mediated** (naloxone blocks it); long-term cortical plasticity.

Modalities

Modality	Target and method	Main use
SNM	Tined lead in **S3** foramen + implanted generator	Refractory OAB, NOUR (non-obstructive urinary retention)
PTNS	34-gauge needle above medial malleolus	Refractory OAB

Modality	Target and method	Main use
TTNS (transcutaneous tibial)	Surface patch, home use	OAB (less effective)
Implantable tibial	Subcutaneous device over tibial nerve	OAB, avoids clinic visits
Pudendal	Tined lead in Alcock's canal	IDO (idiopathic detrusor overactivity), NDO (neurogenic DO), SNM failure
DGNS (dorsal genital nerve stimulation)	Dorsal penile/clitoral nerve	Experimental; NDO
IVES (intravesical electrical stimulation)	Intravesical cathode	Underactive bladder in children
SARS (sacral anterior root stimulator)	Anterior roots + dorsal rhizotomy	Complete SCI (spinal cord injury)

A. Sacral neuromodulation

- **Third line** for OAB (AUA (American Urological Association) and EAU (European Association of Urology) agree). **FDA approved:** UUI (urgency urinary incontinence) 1997; urgency-frequency and NOUR 1999. **Not approved:** bladder pain, pelvic pain, neurogenic bladder.

Indications (third line)

FDA (Food and Drug Administration) approved	Not approved
Refractory UUI (urgency urinary incontinence) 1997; urgency-frequency 1999	Bladder pain, pelvic pain
Non-obstructive retention (e.g. Fowler's syndrome) 1999; faecal incontinence 2011	Neurogenic bladder

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography); responds well. Retention needs some detrusor function.

Technique

- **Test phase** (bladder diary): **PNE (percutaneous nerve evaluation)**: office, temporary lead, 5–7 days; or **staged implant**: permanent tined lead under fluoroscopy, test up to 4 weeks, then generator in the buttock (preferred for NOUR).

Test phase	PNE (percutaneous nerve evaluation)	Staged tined lead
Lead	Temporary, office	Permanent, theatre
Duration	5–7 days	Up to 4 weeks
Use	OAB	More reliable; retention

- **Landmark:** Prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**. **S3 response: perineal "bellows" + great-toe flexion** (S2: calf/foot rotation; S4: bellows only).

Root	Pelvic response	Foot response
S2	—	Calf and foot rotation
S3	**Bellows**, perineal sensation	**Great-toe flexion**
S4	Bellows only	None

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral stimulation no better.

Outcomes

- **Evidence: InSite**: success 61% vs 42% with medical therapy (over medical therapy at 6 months). **ROSETTA**: onabotulinumtoxinA 200 U reduced UUI slightly more (−3.9 vs −3.3 episodes/day) but caused more UTI (urinary tract infection) (24% vs 10%) and CIC (clean intermittent catheterisation) (self-catheterisation). NOUR: most stop catheterising.

Complications and contraindications

- **Problems:** pain, lead migration, loss of effect, infection; **revision >30% at 5 years** (over 30%); costly.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

Recent advances

- **Advances:** rechargeable (≥15 years) devices (**Rechargeable** stimulators: FDA (Food and Drug Administration) 2019); non-rechargeable devices lasting ≥10 years (2021); full-body (**Whole-body**) **MRI (magnetic resonance imaging)-conditional** systems at 1.5 T and 3 T.
- **Implantable tibial nerve stimulators** (BlueWind, eCoin); PTNS (percutaneous tibial nerve stimulation) as an alternative.

B. Tibial nerve stimulation

- **PTNS technique:** 34-gauge needle **3–4 cm (three fingerbreadths) above the medial malleolus**, behind the tibia; ground pad on same leg; 20 Hz, 0–10 mA. Confirm by **toe flexion/fanning** or sole tingling. **30 min weekly × 12 weeks**, then maintenance.
- **SUmiT** (n = 220, validated sham): response **55% vs 21%**. Extension: 77% relapse-free at 3 years with tapering maintenance.
- **OrBIT:** PTNS comparable to extended-release tolterodine, with fewer side effects.
- **TTNS:** home self-treatment; skin impedance limits amplitude, so less effective.
- **Implantable devices** remove clinic visits: **BlueWind** (miniature, external control; benefit sustained at 3 years); **eCoin** (coin-sized, battery-powered, office insertion under local anaesthesia).

C. Other targets

- **Pudendal nerve stimulation:** tined lead via posterior approach. Better than SNM in a blinded crossover; **salvage after failed SNM**. Off-label.
- **DGNS:** purely sensory; acute DO suppression in SCI; **conditional (on-demand)** stimulation possible. Electrode fixation is difficult.
- **IVES (Katona):** intravesical cathode in saline activates Aδ afferents → detrusor contraction. Needs some afferent and muscle function; daily sessions; the sham-controlled trial was negative.
- **Brindley SARS:** stimulates anterior S2–S4 roots → voiding, defecation, erection. **Dorsal rhizotomy** abolishes reflex incontinence and DSD (detrusor–sphincter dyssynergia). Sphincter relaxes faster than detrusor between bursts → **post-stimulus voiding**. Needs intact efferents.

Comparison (third-line OAB)

	PTNS	OnabotulinumtoxinA 100 U	SNM
Invasiveness	Lowest	Moderate	Implant
Drawback	12 weekly visits	UTI, retention/CIC	Revision, cost
Retention	No role	Worsens	**Treats**

- **Also know:** intravaginal stimulation (SUI, stress urinary incontinence); sacral dermatome stimulation; bilateral SNM no better; closed-loop SNM.

Diagram to draw: S3 lead with buttock generator (lead in S3 foramen, IPG in the buttock), and PTNS needle above the medial malleolus.

Key points to remember:

- Afferent modulation explains SNM benefit in both OAB and retention.
- S3: bellows + (plus) great-toe flexion; implant if ≥50% improvement (if test improves ≥50%).
- SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
- ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.

Key points to remember (recent advances angle):

- Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
- Full-body MRI (magnetic resonance imaging)-conditional devices at 1.5 T and 3 T.
- Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
- Standardised tined-lead placement improves outcomes and lowers revision.

Section 7 · Voiding Dysfunction — Short answer questions (SAQ)

Video Urodynamics

Type: SAQ (10 marks)

Source: 2016-Summer Paper 4, Q3d

Definition

- **VUDS (videourodynamics)**: multichannel urodynamics (pressure–flow, with or without EMG (electromyography)) recorded at the same time as fluoroscopy, using **contrast as the filling fluid**.

Set-up

- Vesical and rectal catheters (Pdet (detrusor pressure) = Pves − Pabd), flowmeter, EMG; C-arm (most flexible) or a tilting table. Position for the question: **SUI (stress urinary incontinence) standing**; voiding in the usual position (women sitting, men standing).
- **Screen only at key moments** (provocation, pressure rise, voiding) to limit radiation.

What fluoroscopy adds

Situation	VUDS finding
Neurogenic bladder	**Gold standard** (EAU); DESD (detrusor–external sphincter dyssynergia) and DISD (detrusor–internal sphincter dyssynergia); trabeculation, diverticula
VUR (vesicoureteric reflux)	Volume, pressure and grade at onset; reflux "pops off" pressure, so compliance looks falsely safe
Bladder neck	Only way to see it; **PBNO** (primary bladder neck obstruction) in young men and women
Female obstruction	**Nitti criteria**: radiographic narrowing between bladder neck and meatus during voiding
Dysfunctional voiding	"Spinning-top" urethra at the external sphincter
SUI	Leakage with ALPP (abdominal leak point pressure); open bladder neck at rest suggests ISD (intrinsic sphincter deficiency)
Male obstruction	Localises prostatic obstruction, bladder neck contracture or stricture

PBNO types: I high pressure, low flow; II normal pressure, low flow with bladder-neck narrowing; III delayed bladder-neck opening.

Indications

- Known or suspected NLUTD (neurogenic lower urinary tract dysfunction); unexplained retention in women; young men with LUTS (lower urinary tract symptoms); complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant.
- **Children**: PUV (posterior urethral valves), spinal dysraphism; contrast-enhanced ultrasound VUDS avoids radiation.

Urodynamic risk factors seen

- Impaired compliance; DESD; DISD; high-pressure DO (detrusor overactivity); DLPP (detrusor leak point pressure) >40 cm H_2O; poor emptying with high storage pressure.

Limitations

- Radiation, cost, space, availability, contrast allergy. AUA (American Urological Association) guideline: lowest dose that answers the question.

Diagram to draw: VUDS trace (Pves, Pabd, Pdet, flow, EMG) beside a voiding cystogram image.

> **Key points to remember:**
>
> - VUDS = urodynamics + fluoroscopy with contrast.
> - Only test for bladder neck dysfunction and DISD.
> - Reflux pop-off makes compliance look falsely good.
> - DLPP >40 cm H_2O endangers upper tracts.

Botulinum toxin in urology

Type: SAQ (10 marks)

Source: 2017-Winter Paper 4, Q3a

See 2017-Winter Paper 3, Q3b (full answer).

Key points to remember (recent advances angle):

- Paediatric NDO (neurogenic detrusor overactivity): FDA approved onabotulinumtoxinA for children aged ≥5 years (2021); used from about 2 years of age.
- Other formulations: abobotulinumtoxinA and incobotulinumtoxinA under study for detrusor overactivity; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.
- Standard adult intradetrusor doses for comparison: onabotulinumtoxinA 100 U (idiopathic OAB (overactive bladder)), 200 U (NDO).

Detrussor underactivity.

Type: SAQ (10 marks)

Source: 2021-Summer Paper 4, Q3b (also answered in 2019-Summer Paper 2, Q3d)

Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** a contraction of reduced strength and/or duration, causing prolonged or incomplete emptying. **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** the symptom complex (slow stream, hesitancy, straining, incomplete emptying). DUA is common in elderly men and women having urodynamics.

Pathophysiology and causes

- Failure may be **myogenic** (myocyte loss, fibrosis), **efferent** (poor nerve activation) or **afferent** (reduced sensation cannot sustain the voiding reflex).

Type	Examples
Neurogenic	Cauda equina, disc prolapse, spinal injury, diabetic neuropathy, radical pelvic surgery
Myogenic	Long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing
Drugs	Anticholinergics, opioids
Functional	Fowler syndrome, dysfunctional voiding

Evaluation

- Bladder diary, flow rate, **PVR (post-void residual)**, sacral neurological examination, spinal MRI if deficit, drug review.
- **Pressure-flow study is diagnostic:** low detrusor pressure with low flow.
- **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 Qmax:** <100 weak, 100–150 normal, >150 strong. Also PIP (projected isovolumetric pressure), Watts factor, stop test.
- Ambulatory urodynamics if no contraction is seen.

Management

- **Goals:** protect kidneys; prevent UTI (urinary tract infection) and stones; relieve symptoms. No drug reliably improves contraction.

Aim	Options
Drainage	Timed and double voiding; **CISC (clean intermittent self-catheterisation), the mainstay**; suprapubic catheter if CISC impossible
Lower outlet resistance	α-blockers; pelvic-floor biofeedback; outlet surgery (TURP, transurethral resection of the prostate) in selected obstructed men
Improve contraction	**SNM (sacral neuromodulation)** for non-obstructive retention (Fowler); **sacral anterior root stimulator** with rhizotomy in complete spinal injury
Not recommended	Bethanechol, distigmine (weak evidence; bronchospasm, rarely cardiac arrest); Credé or Valsalva voiding (reflux risk)
Experimental	Intravesical prostaglandins, intravesical electrotherapy, TAC-302, latissimus dorsi myoplasty, stem cells

- SNM and parasympathomimetics need some intact detrusor and sensory function; a fibrosed detrusor responds to nothing.

Diagram to draw: pressure-flow plot with BCI lines at 100 and 150.

Key points to remember:

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; parasympathomimetics not recommended.
- SNM for Fowler syndrome.

Rapid Revision Sheet – Section 7 · Voiding Dysfunction

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Neuromodulation for voiding dysfunction, including sacral neuromodulation (LAQ)

- Afferent modulation explains SNM benefit in both OAB and retention.
- S3: bellows + (plus) great-toe flexion; implant if ≥50% improvement (if test improves ≥50%).
- SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
- ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
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Video Urodynamics (SAQ)

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Detrussor underactivity. (SAQ)

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; parasympathomimetics not recommended.
- SNM for Fowler syndrome.

Section 9 · Andrology & Sexual Medicine — Long answer questions (LAQ)

Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival.

Type: LAQ (25 marks)

Source: 2021-Summer Paper 4, Q2 (also answered in 2019-Summer Paper 1, Q3d and 2019-Summer Paper 2, Q3c)

Definition

- **Azoospermia:** no sperm in the **centrifuged pellet** of the ejaculate on **at least two** semen analyses. Found in about 1% of men and 10–15% of infertile men.
- Not **cryptozoospermia** (rare sperm in pellet) or **aspermia** (no ejaculate).

Classification

Type	Causes	Clues
Pre-testicular	Kallmann syndrome, pituitary disease, **exogenous testosterone**	Low FSH (follicle-stimulating hormone), LH (luteinising hormone)
Testicular: **NOA (non-obstructive azoospermia)**	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy, mumps	Small testes, high FSH
Post-testicular: **OA (obstructive azoospermia)**	CBAVD (congenital bilateral absence of the vas deferens), vasectomy, epididymal or ejaculatory duct obstruction	Normal testes and FSH
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Evaluation (brief)

- Absent vasa → CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners. Volume <1 mL → post-ejaculate urine, then TRUS (transrectal ultrasound).
- **FSH ≤7.6 IU/L and testis long axis >4.6 cm → obstruction likely;** FSH >7.6 and axis ≤4.6 cm → spermatogenic failure.
- **Karyotype and Y microdeletion** in NOA. **AZF (azoospermia factor) a or b complete deletion: no sperm, do not operate.**

Testicular biopsy

Indications

- Azoospermia with **normal testes, palpable vasa and normal FSH**: separates OA from NOA (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Since ICSI (intracytoplasmic sperm injection) needs one sperm, biopsy is now **diagnostic and therapeutic**: plan it as TESE (testicular sperm extraction) and **cryopreserve** sperm.
- **Not needed:** CBAVD with normal FSH (spermatogenesis always present).

Types

Type	Method	Use	Limits
Open biopsy	Small incision in the albuginea, pea-sized tissue	**Gold standard** for histology and retrieval	Open procedure
Percutaneous core biopsy	14 G biopsy gun under cord block	Fresh retrieval in OA	Blind; may injure epididymis or testicular artery; avoid in scarred scrotum
FNA (fine-needle aspiration), i.e. TESA (testicular sperm aspiration)	23 G needle, suction, several passes	OA retrieval; backup to epididymal methods	Few tubules, poor architecture; low yield in NOA
Microdissection biopsy (micro-TESE)	Operating microscope	NOA	Long operation

- **Also know:** multi-site FNA "mapping" in NOA.

Open biopsy technique

1. Local anaesthesia with cord block (or general/spinal); choose the larger testis.
2. 2 cm transverse incision; open tunica vaginalis.
3. 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch or wet prep** at once for sperm; negative → contralateral biopsy.
7. Close albuginea and tunica vaginalis separately; no antibiotics.

Histology patterns

Pattern	Meaning
Normal	Suggests obstruction
Hypospermatogenesis	All stages, fewer cells
Maturation arrest	Stops at a germ-cell stage
SCO (Sertoli cell–only)	No germ cells
Tubular sclerosis/hyalinisation	End-stage (e.g. Klinefelter)

- **Complications:** haematoma (commonest); **inadvertent epididymal biopsy** (most serious: causes obstruction); infection rare.

Surgical sperm retrieval

Indications: CBAVD; unreconstructable obstruction; declined or failed vasectomy reversal; backup at reconstruction; NOA; anejaculation.

Technique	Method	Best for	Notes
PESA (percutaneous epididymal sperm aspiration)	Fine needle into caput	OA	Few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened at 16–25×; micropipette by **capillary action**, never suction	**OA (best)**	Many motile sperm, multiple frozen vials
TESA	23 G needle, suction	OA	Least invasive; immotile sperm

Technique	Method	Best for	Notes
Conventional TESE	Open biopsy, one or more sites	OA, NOA	Blind sampling
Micro-TESE (Schlegel 1999)	Testis opened in an avascular plane; **larger, opaque tubules** excised under the microscope	**NOA (best)**	Most sperm, least tissue

Key principles

- **MESA:** start distally and move **proximally** until motile sperm appear; the most motile sperm are often in the efferent ductules.
- **Micro-TESE:** spermatogenesis in NOA is **focal**. Thin white tubules are Sertoli cell–only; plump ones hold sperm. Embryologist examines tissue in theatre; map positive sites; negative → other testis.

Micro-TESE retrieval by diagnosis

Condition	Retrieval
Post-orchidopexy	74%
AZFc deletion	70%
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
SCO	30%
AZFa/AZFb deletion	**0%**

- Overall sperm found in **about half** of NOA.
- **Before retrieval in NOA:** stop testosterone; repair clinical varicocele; off-label clomiphene, anastrozole or hCG (human chorionic gonadotropin).

Use of retrieved sperm

- **ICSI is essential** (poor motility). Frozen sperm equal fresh, except in spinal cord injury.
- NOA: more miscarriage (sperm DNA damage); genetic counselling. No sperm: donor sperm or adoption.
- **Also know:** post-mortem retrieval needs prior consent; microfluidic sperm sorting.

Diagram to draw: micro-TESE: bivalved testis with plump spermatogenic tubules beside thin Sertoli cell–only tubules.

> **Key points to remember:**
>
> - Azoospermia: no sperm in centrifuged pellet on two samples.
> - Open biopsy is the gold standard; fix in Bouin's, never formalin.
> - OA: MESA; NOA: micro-TESE (sperm in about half).
> - AZFa/b deletion: 0% retrieval, do not operate.

Section 9 · Andrology & Sexual Medicine — Short answer questions (SAQ)

Male menopause (andropause, late-onset hypogonadism)

Asked as:

- Male menopause. — *Summer 2017 Paper IV Q3e (SAQ)*
- Andropause — *Winter 2017 Paper IV Q3e (SAQ)*
- Male menopause — *Summer 2019 Paper IV Q3f (SAQ)*

Type: SAQ (10 marks)

Read as: late-onset hypogonadism (testosterone deficiency of ageing men, "andropause"). Same topic as 2017-Summer Paper 4, Q3e and 2017-Winter Paper 4, Q3e; full answer given here.

Definition

- **Symptoms plus consistently low serum testosterone** in an ageing man.
- "Menopause" (or "Andropause") is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

Pathophysiology

- Testosterone falls slowly from middle age.
- Testosterone in the 70s is about one-third lower than in young men. (Another source: mean testosterone in the seventh decade is about **one-third lower** than in young men.)
- **Testis:** fewer Leydig cells, lower sperm output; FSH (follicle-stimulating hormone) rises about threefold.
- **Centre:** blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises with age**, so free testosterone (bioavailable [free + albumin-bound] testosterone) falls more.
- Worse with obesity, diabetes and **opioids**.

Clinical features

- **Sexual (most specific):** low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood.
- **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

Diagnosis

- **Morning (7–11 am) total testosterone, twice (repeated).** AUA (American Urological Association): **<300 ng/dL** (≈10.4 nmol/L). EAU (European Association of Urology): about **12 nmol/L**; calculated free testosterone if borderline. Calculated bioavailable testosterone when SHBG is abnormal.
- **LH (luteinising hormone):** high in primary, low or normal in secondary hypogonadism. Check **prolactin** if LH is low (Low LH → **prolactin**); pituitary MRI (magnetic resonance imaging) if testosterone <150 ng/dL or prolactin high.
- Baseline: PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

Treatment

- Lifestyle first: weight loss, exercise, treat comorbidities.

- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at 3–6 months (another source: at about 3 months).
- **Contraindications:**
  - Prostate or breast cancer.
  - **Wish for fertility** (TRT stops sperm production [suppresses spermatogenesis]; use hCG [human chorionic gonadotrophin] or clomiphene).
  - Haematocrit >50% (raised haematocrit).
  - Untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (stop/hold if >54%), PSA.

Safety evidence

- **TRAVERSE (2023):** 5,246 men aged 45–80 at high cardiovascular risk. TRT was **non-inferior for major cardiac events**; more atrial fibrillation and pulmonary embolism.
- **Testosterone Trials**: modest sexual benefit, **no cognitive benefit**.

Diagram to draw: hypothalamic–pituitary–testis axis with ageing defects.

> **Key points to remember:**
> - Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
> - SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
> - TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

Klinefelter's syndrome.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 4, Q3b

Definition and genetics

- **47,XXY**: commonest sex chromosome aneuploidy and **commonest genetic cause of male infertility**; 1 in 500–1000 male births.
- Non-mosaic in **80–90%**; the rest mosaic **46,XY/47,XXY** (milder, more sperm) or 48,XXXY variants. Extra X from non-disjunction (maternal age).
- Underdiagnosed: most never diagnosed; increasingly detected on NIPT (non-invasive prenatal testing).

Pathophysiology

- Seminiferous tubules **hyalinise** progressively from puberty; germ cells are lost. Leydig cells fail, so testosterone falls. Puberty starts on time, then stalls.

Clinical features

Age	Features
Infant	Usually normal; micropenis, cryptorchidism, hypospadias
Adolescent/adult	**Small firm testes (<3.5 cm, 1–4 mL)**, gynaecomastia, tall stature with long legs (only ~30%), sparse hair, female fat distribution
Neurodevelopment	Language, learning and attention difficulties

Hormones and tests

- **High FSH (follicle-stimulating hormone) and LH (luteinising hormone), low or low-normal testosterone, high oestradiol** (raised oestradiol:testosterone ratio). Inhibin B low.
- Semen: **azoospermia** in most (~8% have ejaculated sperm).
- **Karyotype** confirms (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine, guideline: karyotype all men with non-obstructive azoospermia or severe oligozoospermia).

Management

- **Fertility: micro-TESE** (microdissection testicular sperm extraction) + ICSI (intracytoplasmic sperm injection). Sperm found in **~40–50%** (68% in one series); offspring are usually chromosomally normal.
- Spermatogenesis declines with age, so retrieve early (late adolescence or young adulthood). Some give clomiphene, anastrozole or hCG (human chorionic gonadotropin) beforehand.
- **TRT (testosterone replacement therapy):** for hypogonadism (bone, muscle, mood, libido). It **suppresses spermatogenesis**, so offer sperm retrieval or banking **before** starting.
- Gynaecomastia: reduction mammoplasty if persistent. Speech, educational and psychological support.

Associated risks

- Osteoporosis, metabolic syndrome, diabetes, VTE (venous thromboembolism), autoimmune disease.

- **Mediastinal (extragonadal) germ cell tumours**; male breast cancer (~1% lifetime).

Diagram to draw: hypothalamic–pituitary–testis axis showing high FSH and LH, low testosterone.

> **Key points to remember:**
> - 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
> - Small firm testes, high FSH and LH, azoospermia.
> - Micro-TESE ~50% success; retrieve sperm before testosterone.
> - Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

Semen analysis nomenclature.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 4, Q3e

Collection (WHO, World Health Organization, 2021, 6th edition)

- Masturbation into a clean wide-mouthed non-toxic container (special condom if needed); complete sample.
- Abstinence **2–7 days** (evidence: 1–2 days best). Keep at 20–37 °C; analyse within **1 hour**, after liquefaction (~30 min).
- At least **two samples, 2–3 weeks apart**, because of variability.

Lower reference limits (5th centile, fertile men)

Parameter	Lower limit
Volume	**1.4 mL**
Concentration	**16 million/mL**
Total count	**39 million/ejaculate**
Total motility	**42%**
Progressive motility	**30%**
Normal forms (strict)	**4%**
Vitality	**54%**
Peroxidase-positive leucocytes	<1 million/mL

- These are reference values, not fertility cut-offs. The 6th edition grades motility as progressive, non-progressive or immotile.

Nomenclature

Term	Meaning
Normozoospermia	All parameters at or above reference
Oligozoospermia	Total count (or concentration) below reference
Asthenozoospermia	Progressive motility below reference
Teratozoospermia	Normal forms below reference
OAT (oligoasthenoteratozoospermia)	All three abnormal
Azoospermia	No sperm, **including centrifuged pellet**
Cryptozoospermia	Sperm absent in fresh smear, present only after centrifugation
Aspermia	No ejaculate (dry orgasm)
Hypospermia	Volume below reference
Necrozoospermia	Low vitality (many dead sperm)
Leucocytospermia	Peroxidase-positive white cells above 1 million/mL
Haemospermia	Blood in ejaculate
Globozoospermia	Round-headed sperm without acrosome

Notes

- **Low volume or aspermia:** post-ejaculatory urine for retrograde ejaculation; TRUS (transrectal ultrasound) for EDO (ejaculatory duct obstruction).

- **Vitality:** eosin–nigrosin, or **HOS (hypo-osmotic swelling)** test, which keeps sperm usable for ICSI (intracytoplasmic sperm injection).
- Only azoospermia is definitive; values overlap between fertile and infertile men.
- **Also know:** Kruger strict morphology; DNA fragmentation; antisperm antibodies (MAR, mixed antiglobulin reaction); semen pH no longer recommended.

Diagram to draw: normal sperm (head, midpiece, tail) beside a WHO reference box.

Key points to remember:

- WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
- Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
- Two samples 2–3 weeks apart; abstinence 2–7 days.

Penile implants.

Type: SAQ (10 marks)

Source: 2023-Winter Paper 4, Q3c (also answered in 2018-Summer Paper 2, Q3d and 2020-Summer Paper 2, Q2)

Indications

- ED (erectile dysfunction) failing, contraindicated for, or declining PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injections or vacuum devices.
- Post-prostatectomy, diabetes, pelvic radiotherapy; **Peyronie disease with ED**; corporal fibrosis (**ischaemic priapism >36 h**, implant removal).
- **Contraindications:** active infection (urinary, skin, systemic), uncontrolled diabetes, unrealistic expectations, reversible ED.

Types

Type	Examples	Features
Malleable (semirigid) rods	Tactra (nitinol), Genesis (silver)	Simple, cheap, rarely fail; always firm; erode more in insensate men
2-piece inflatable	Ambicor (discontinued)	No reservoir; poor flaccidity
3-piece inflatable	AMS 700 (CX, LGX), Titan	Cylinders, scrotal pump, reservoir; **best rigidity and flaccidity**

Preparation and consent

- Sterile urine; **blood glucose <200 mg/dL on the day**; chlorhexidine–alcohol skin preparation; vancomycin or cephalosporin plus gentamicin.
- **Consent:** the erect penis may be shorter; sensation and ejaculation may reduce; infection or failure may need reoperation.

Surgical steps (3-piece, penoscrotal)

- Catheter → upper scrotal incision → corporotomy between stay sutures → dilate proximally and distally (instruments dorsolateral) → **goalpost test** for crossover → measure → cylinders equal to or 1 cm shorter.
- Pump in a dependent dartos pouch; reservoir in the **space of Retzius**, or **ectopic submuscular** after pelvic surgery (lockout valve prevents autoinflation).
- Infrapubic approach: reservoir placed under vision; dorsal nerve at risk. Activation at 4–6 weeks.

Complications

- **Infection 1–3%** (biofilm): **antibiotic-coated devices halve it** (InhibiZone rifampin–minocycline; hydrophilic Titan dipped). **No-touch technique** <0.5%. Treat by removing **all** parts; **Mulcahy salvage** succeeds in **84%**.
- **Mechanical failure:** repair about **15% at 5 years, 30% at 10**.
- Crural perforation, crossover, urethral injury; erosion; glans necrosis; reservoir herniation.

Outcomes

- **Satisfaction 80–90%** (patient and partner), highest of any ED treatment.

Diagram to draw: 3-piece inflatable implant: cylinders, scrotal pump, reservoir in the space of Retzius.

Key points to remember:

- 3-piece inflatable: best rigidity and flaccidity.
- Glucose <200 mg/dL; coated device; no-touch.
- Infected implant: remove all; Mulcahy salvage 84%.
- Repair 15% at 5 years; satisfaction 80–90%.

Rapid Revision Sheet – Section 9 · Andrology & Sexual Medicine

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival. (LAQ)

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Open biopsy is the gold standard; fix in Bouin's, never formalin.
- OA: MESA; NOA: micro-TESE (sperm in about half).
- AZFa/b deletion: 0% retrieval, do not operate.

Male menopause (andropause, late-onset hypogonadism) (SAQ)

- Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
- SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

Klinefelter's syndrome. (SAQ)

- 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
- Small firm testes, high FSH and LH, azoospermia.
- Micro-TESE ~50% success; retrieve sperm before testosterone.
- Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

Semen analysis nomenclature. (SAQ)

- WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
- Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
- Two samples 2–3 weeks apart; abstinence 2–7 days.

Penile implants. (SAQ)

- 3-piece inflatable: best rigidity and flaccidity.
- Glucose <200 mg/dL; coated device; no-touch.
- Infected implant: remove all; Mulcahy salvage 84%.
- Repair 15% at 5 years; satisfaction 80–90%.

Section 10 · Testis, Penis & Urethra — Short answer questions (SAQ)

Testis sparing surgeries for CA testis

Type: SAQ (10 marks)

Source: 2016-Summer Paper 4, Q3b

Definition and rationale

- **Partial orchidectomy**: inguinal excision of the tumour, preserving the rest of the testis. The standard remains **radical inguinal orchidectomy**; sparing aims to keep androgen production and avoid lifelong testosterone.

Indications

Suspected germ cell tumour (GCT), all of:

- Synchronous bilateral tumours or tumour in a **solitary testis**.
- Tumour **≤2 cm** (up to 2–3 cm, under 30% of testicular volume), preferably polar.
- Adequate preoperative testosterone.
- Not with a normal contralateral testis.

Suspected benign or indeterminate lesions:

- Small impalpable lesion (<10 mm) with normal markers: most benign (cyst, infarct, Leydig nodule), but 20–50% are small seminomas. Lesions <5 mm are rarely malignant.
- **Prepubertal**: teratoma and epidermoid cyst with normal AFP (alpha-fetoprotein): partial orchidectomy is the standard.

Preoperative work-up

- Ultrasound of both testes; AFP, beta-hCG (human chorionic gonadotropin), LDH (lactate dehydrogenase); testosterone, LH (luteinising hormone); **sperm banking**; consent for radical orchidectomy.

Technique

1. Inguinal incision; early cord control with a soft clamp.
2. Deliver the testis; isolate with towels.
3. **Intraoperative ultrasound** localises the lesion.
4. Cold ischaemia optional if resection <30 minutes.
5. Incise the tunica parallel to the segmental vessels; excise the mass with a rim of normal tubules.
6. **Frozen section** (benign versus malignant) and biopsies of adjacent parenchyma for GCNIS (germ cell neoplasia in situ).
7. Close the tunica; fix the testis in the scrotum.

Management of GCNIS

- **Radiotherapy about 18–20 Gy** to the remnant: prevents invasive GCT, keeps Leydig function, but causes sterility; up to 40% later need testosterone.
- German Testicular Cancer Study Group: no local recurrences with radiotherapy; most men who omitted it recurred.
- Radiotherapy may be deferred for paternity under close follow-up. Local recurrence → completion orchidectomy.

Complications and follow-up

- Haematoma, atrophy, hypogonadism, local recurrence. Follow with self-examination, ultrasound, markers and testosterone.

Diagram to draw: testis with polar tumour excised with a rim of normal tissue.

> **Key points to remember:**
> - Solitary or bilateral tumour, ≤2 cm, normal testosterone.
> - Frozen section plus adjacent biopsies for GCNIS.
> - GCNIS: 18–20 Gy keeps Leydig function but sterilises.

Videoendoscopic inguinal lymphadenectomy

Type: SAQ (10 marks)

Source: 2019-Summer Paper 4, Q3d

Definition

- **VEIL (video-endoscopic inguinal lymphadenectomy):** ILND (inguinal lymph node dissection) through thigh ports, beneath Scarpa's fascia; laparoscopic (Tobias-Machado) or robotic (**RVEIL**).
- **Aim:** open-surgery template without the groin incision that causes flap necrosis.

Indications

- Penile SCC (squamous cell carcinoma), **impalpable nodes**, intermediate/high risk (pT1b+, G3, LVI (lymphovascular invasion)).
- **Mobile, non-bulky palpable nodes.**
- **Not for** bulky, fixed (cN3) or skin-involving nodes → chemotherapy, then open surgery.
- Also vulval cancer, melanoma.

Boundaries (femoral triangle)

- Superior: **inguinal ligament**; lateral: **sartorius**; medial: **adductor longus**; floor: fascia lata; apex: muscles crossing.
- Mark an inverted triangle on the skin **before insufflation**, because gas distorts landmarks.

Technique

```
Position: supine frog-leg / low lithotomy
→ 2 cm camera incision below the femoral triangle apex
→ finger dissection beneath Scarpa's fascia
→ two lateral and medial working ports (+ assistant port)
→ CO2 at 15 mmHg (higher briefly to create space)
→ develop space up to the inguinal ligament
→ remove nodal packet off fascia lata, apex to ligament
→ preserve saphenous vein if possible; clip tributaries
→ frozen section → deep dissection medial to femoral vein if positive
→ bag retrieval, suction drain in dependent part
```

- **Template:** superficial ± deep; Daseler's five zones; Cloquet's node is the highest deep node.
- **Never dissect lateral to the femoral artery** (femoral nerve).

Outcomes vs open

	Open ILND	VEIL/RVEIL
Wound infection, necrosis, dehiscence	Higher	**Lower**
Hospital stay, drain time	Longer	Shorter
Operative time	Shorter	About 2 h (slightly longer per side)
Nodal yield, recurrence	Reference	**Similar**
Evidence	Retrospective	Retrospective, selection-biased

- Commonest complication: **lymphocele**, then lymphoedema and cellulitis.

Robotic VEIL

- 3D vision, wristed instruments, ICG (indocyanine green) for lymphatics; adequate dissection on open re-exploration (phase I). No VEIL vs RVEIL comparison.
- **Also know:** DSNB (dynamic sentinel node biopsy) first for impalpable nodes (EAU-ASCO, European Association of Urology–American Society of Clinical Oncology); modified (Catalona) template; complete nodal management within 3 months.

Diagram to draw: femoral triangle with port sites below the apex.

> **Key points to remember:**
> - Same template as open: ligament, sartorius, adductor longus.
> - Fewer wound complications; equal nodal yield.
> - Not for bulky, fixed or skin-involving nodes.

Surgical management of primary scrotal cancer.

Type: SAQ (10 marks)

Source: 2021-Summer Paper 4, Q3c

Introduction

- Rare. **SCC (squamous cell carcinoma)** is commonest. **Percivall Pott (1775)** linked it to chimney soot: the first occupational cancer.
- **Risks:** soot, tar, mineral oils, arsenic, PUVA (psoralen + ultraviolet A), HPV (human papillomavirus), chronic inflammation.
- **Other types:** EMPD (extramammary Paget disease), BCC (basal cell carcinoma), melanoma, sarcoma.
- Lymph drains to **superficial inguinal nodes**, crossing the midline: both groins at risk.

Evaluation

- Nodule or ulcer on the anterolateral scrotum, often late; **biopsy** confirms.
- Examine both groins; FNAC (fine-needle aspiration cytology) of palpable nodes; CT (computed tomography) for staging.

Staging (Lowe modification of Ray and Whitmore)

Stage	Extent
A1	Confined to scrotum
A2	Invades testis, cord, penis, pubis or perineum
B	Inguinal nodes only
C	Pelvic nodes, no distant spread
D	Distant metastases

Management of the primary

1. **Wide local excision with a 2 cm margin**, full thickness including dartos; frozen sections for margins.
2. Testes and cord preserved unless invaded. Hemi- or total scrotectomy for large tumours.
3. **Reconstruction:** scrotal laxity usually allows primary closure. Defect >50%: local or thigh flaps, split-skin graft, or thigh pouches for the testes.
4. Stage A2: en bloc removal of involved testis, penis, perineum or pubis.

Management of nodes (extrapolated from penile cancer)

- **Palpable, cytology positive:** ILND (inguinal lymph node dissection) ± pelvic dissection. Nodal status is the main prognostic factor.
- **Impalpable:** surveillance; dynamic sentinel node biopsy or modified ILND for high-grade/invasive tumours. Routine prophylactic dissection is not advised.
- **Bulky/fixed nodes:** neoadjuvant cisplatin-based chemotherapy, then surgery; RT (radiotherapy) for palliation.

Other histologies

- **EMPD:** wide excision (up to 3 cm) guided by **mapping biopsies**; screen for associated GU (genitourinary) and other cancers.
- **Also know:** BCC (excision or Mohs); melanoma (excision by Breslow depth ± sentinel node); sarcoma (wide excision).

Diagram to draw: elliptical wide excision with 2 cm margin and inguinal node drainage.

Key points to remember:

- Pott 1775: chimney sweeps, first occupational cancer.
- Wide local excision, 2 cm margin; testes spared unless invaded.
- Nodes: managed as in penile cancer; nodes decide prognosis.

Muscle assisted urethroplasty

Asked as:

- Muscle assisted urethroplasty — *Winter 2023 Paper IV Q3b (SAQ)*
- Muscle assisted urethroplasty — *Practice Set 5 Paper IV Q3a (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3b)

Read as muscle-flap-assisted graft urethroplasty: a vascularised muscle flap used as bed or cover for a graft. "Muscle-sparing" bulbar urethroplasty is noted at the end.

Rationale

- A free graft (e.g. **BMG [buccal mucosal graft]**) has no blood supply. It survives by **imbibition** (about 48 h, nutrients from the bed), then **inosculation** (new microcirculation). So **"a graft lives on its bed"**.
- **Poor beds:** radiation, multiple failed repairs, fistula, pelvic-fracture scar, infection. Here a **muscle flap brings blood supply**, fills dead space and separates suture lines.

Muscle flaps used

Flap	Supply	Use
Gracilis (muscle-only, axial, multiple pedicles)	Medial circumflex femoral branch, proximal thigh	Irradiated posterior urethra, rectourethral fistula, complex perineal repair
Bulbospongiosus	Perineal branches	Covers a ventral graft; support in bulbar repair
Rectus abdominis (VRAM, vertical rectus abdominis myocutaneous)	Inferior epigastric	Large pelvic or perineal defects

Technique (gracilis-assisted BMG)

1. Lithotomy; perineal exposure; excise scar or fistula.
2. Close rectal side; urethra opened.
3. Gracilis harvested through a medial thigh incision; pedicle preserved; tunnelled to perineum.
4. Muscle interposed between rectum and urethra, or used as the graft bed.
5. **BMG quilted onto the muscle** as onlay; urethra closed over a catheter.
6. Catheter drainage; urethrogram before removal.

Results (Chapter 86)

- Gracilis with or without BMG for rectourethral fistula: **non-irradiated, all closed**, no later strictures. **Irradiated:** most closed in one stage, but more strictures and permanent bowel diversion.
- Muscle interposition is integral; BMG is invaluable for urethral closure.

Complications

- Thigh wound problems; graft contraction, restricture, fistula recurrence.

"Muscle-sparing" bulbar urethroplasty (Barbagli)

- The **bulbospongiosus muscle, central perineal tendon and perineal nerves** are left intact; dorsal BMG onlay through a lateral approach.
- Aim: less post-void dribbling and ejaculatory dysfunction.

Diagram to draw: gracilis flap tunnelled to the perineum, lying between rectum and urethra with BMG on it.

Key points to remember:

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

Robotic RPLND

Type: SAQ (10 marks)

Source: 2025-Winter Paper 4, Q3e

See 2025-Winter Paper 3, Q3f (full answer).

Key points to remember (recent advances angle):

- Expanding indication: primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy.
- Robotic nerve sparing preserves antegrade ejaculation in ≥90% after primary surgery.
- Serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND (retroperitoneal lymph node dissection).

Gracilis myocutaneous flap in Uro.

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 4, Q3d (gracilis-assisted urethroplasty also in 2023-Winter Paper 4, Q3b)

Definition

- A pedicled **axial** flap of gracilis muscle, with or without skin, rotated from the medial thigh to bring **vascularised bulk** to the perineum, groin or vagina.

Anatomy

Feature	Detail
Origin–insertion	Inferior pubic ramus and body of pubis → medial proximal tibia (pes anserinus)
Blood supply	**Mathes–Nahai type II**: dominant pedicle from the **medial circumflex femoral artery** (profunda femoris), entering about 8–10 cm below the pubic tubercle; minor distal pedicles from the superficial femoral artery
Nerve	Anterior branch of the **obturator nerve**
Skin paddle	Reliable over the proximal two-thirds only
Function	Expendable (other adductors compensate)

Why it is used

- Brings blood supply into **irradiated or scarred** fields, fills dead space and **separates suture lines**. Near the perineum, easy to tunnel.

Uses in urology

1. **Rectourethral fistula** (perineal): non-irradiated all closed; irradiated mostly closed, but more strictures and permanent colostomy.
2. **Complex or irradiated VVF (vesicovaginal fistula)** with tissue loss.
3. **Urethroplasty** in a poor bed: muscle as bed for BMG (buccal mucosal graft).
4. Extensive perineal or scrotal loss after **Fournier gangrene**.
5. **Groin cover** of femoral vessels after inguinal node dissection with skin excision.
6. Vaginal reconstruction after exenteration.

Technique

1. Lithotomy; line from pubic tubercle to medial femoral condyle; gracilis lies just behind it.
2. Medial thigh incision; skin paddle over the proximal muscle if needed.
3. Divide the distal tendon; ligate minor pedicles; **preserve the dominant pedicle**.
4. Tunnel subcutaneously to the perineum; rotate without twisting.
5. Interpose and fix; drain the donor site.

Complications

- Donor seroma, dehiscence, **distal skin-paddle necrosis**, thigh numbness; flap loss rare.
- **Also know:** alternatives: Martius, omentum, VRAM (vertical rectus abdominis myocutaneous), Singapore flap.

Diagram to draw: gracilis tunnelled on its proximal pedicle between urethra and rectum.

> **Key points to remember:**
>
> - Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
> - Skin paddle unreliable over the distal third.
> - Main uses: rectourethral fistula, complex VVF, Fournier, groin cover.

Rapid Revision Sheet – Section 10 · Testis, Penis & Urethra

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Testis sparing surgeries for CA testis (SAQ)

- Solitary or bilateral tumour, ≤2 cm, normal testosterone.
- Frozen section plus adjacent biopsies for GCNIS.
- GCNIS: 18–20 Gy keeps Leydig function but sterilises.

Videoendoscopic inguinal lymphadenectomy (SAQ)

- Same template as open: ligament, sartorius, adductor longus.
- Fewer wound complications; equal nodal yield.
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- Pott 1775: chimney sweeps, first occupational cancer.
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- Serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND (retroperitoneal lymph node dissection).

Gracilis myocutaneous flap in Uro. (SAQ)

- Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
- Skin paddle unreliable over the distal third.
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Section 11 · Stones, Obstruction & Trauma — Long answer questions (LAQ)

Urolithiasis: etiopathogenesis, medical management of (recurrent) stone disease and recent advances in RIRS instrumentation

Asked as:

- Describe etiopathogenesis of stone formation and medical management of urolithiasis. — *Summer 2020 Paper IV Q2 (LAQ)*
- Describe medical management of recurrent stone disease and discuss recent advances in instrumentations of RIRS. — *Summer 2016 Paper IV Q2 (LAQ)*

Type: LAQ (25 marks)

Introduction

- Urolithiasis is a **chronic metabolic disease** (chronic metabolic disorder) punctuated by stone episodes; recurrence is up to 50% within 5 years (another source: about half recur within 5–10 years). Removal is surgical; recurrence is metabolic. Recurrent formers need a **metabolic evaluation** and targeted therapy.
- **Composition**: calcium oxalate about 60%, mixed with apatite about 20%, uric acid and struvite about 7% each, cystine 1–3%.
- RIRS (retrograde intrarenal surgery) is flexible ureteroscopy inside the kidney.

Aetiology: risk factors

- **Epidemiological**: male sex (gap narrowing), hot climate and occupation (low urine volume), diet (animal protein, salt), family history.
- **Systemic**: **metabolic syndrome**, obesity, diabetes (insulin resistance lowers urine pH), gout, primary hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass, sarcoidosis.
- **Anatomical**: PUJ (pelviureteric junction) obstruction, horseshoe kidney, calyceal diverticulum, medullary sponge kidney (stasis plus metabolic defects).
- **Drugs**: direct (indinavir, triamterene); indirect (acetazolamide, topiramate, vitamin C and D excess).

Physicochemistry: state of saturation

State	Concentration product (CP)	What happens
Undersaturated	< Ksp (solubility product)	No crystals; stones may dissolve
Metastable	Between Ksp and Kf (formation product)	No spontaneous nucleation; growth on existing crystals; inhibitors work. **Zone of therapy**
Unstable	> Kf	Spontaneous nucleation, growth, aggregation

- Crystals form in metastable urine by local high CP, stasis, or **heterogeneous nucleation** on cells, debris or other crystals (needs less energy).

Steps of stone formation

1. **Supersaturation** (low volume, high calcium, oxalate, uric acid; low citrate).

2. **Nucleation**, then **growth** and **aggregation** (aggregation can occlude a tubule within the 5–7 minute nephron transit).
3. **Retention**: fixed-particle theory. **Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) suggest a secondary cause (enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria).

Inhibitors, promoters, matrix

- **Inhibitors**: **citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm-Horsfall protein**, osteopontin, bikunin. There is **no inhibitor of uric acid**.
- **Matrix** (about 2.5% of stone weight; matrix substance A): role uncertain.

Pathogenesis by abnormality

Abnormality	Mechanism
Hypercalciuria (commonest)	Absorptive (gut), renal leak (high PTH (parathyroid hormone)), resorptive (primary hyperparathyroidism)
Hyperoxaluria	Primary (hepatic enzyme defect, PH1 AGXT), **enteric** (fat binds calcium, freeing oxalate), dietary
Hyperuricosuria	Urate promotes calcium oxalate nucleation
Hypocitraturia	Distal RTA, diarrhoea, hypokalaemia, animal protein (intracellular acidosis)
Uric acid	**Low urine pH** (<5.5; pKa 5.5) more than excess uric acid
Cystine	Autosomal recessive defect in renal cystine reabsorption
Struvite	**Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate + carbonate apatite; biofilm

Medical management

1. Acute colic and MET (medical expulsive therapy)

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids only if refractory.
- **α-blocker** (tamsulosin; silodosin faster) for **distal ureteric stones 5–10 mm**: relaxes ureteric muscle. **SUSPEND (2015)** was negative because it included all sizes and sites. Trial of passage 4–6 weeks.

2. Dissolution: uric acid stones (radiolucent) dissolve with oral potassium citrate (EAU (European Association of Urology): pH 7.0–7.2 for dissolution).

3. Prevention: general measures (every former)

- **Fluid** for urine output **≥2.5 L/day**. **Borghi 1996**: urine >2 L cut 5-year recurrence (12.1% vs 27% recurrence at 5 years).
- **Normal calcium**, low sodium (≤2.3 g/day), less animal protein, more fruit and vegetables (DASH (Dietary Approaches to Stop Hypertension)-type diet). **Borghi 2002**: this diet halved recurrence compared with a low-calcium diet. Restricting calcium raises oxalate absorption.
- Citrus juices (lemonade, orange) are adjuncts only.

4. Metabolic evaluation that guides therapy (high-risk and recurrent formers) (see 2015-Summer Paper 2, Q2)

- **Stone analysis** (infrared spectroscopy or X-ray diffraction); serum chemistry: serum calcium, uric acid, creatinine, electrolytes, PTH (parathyroid hormone) if calcium is high-normal.

- One or **two 24-h urines** on usual diet, at least 4 weeks after an episode: volume, calcium, oxalate, citrate, uric acid, pH, sodium, cystine.
- Targets: volume >2.5 L, calcium <250 mg (women) or <300 mg (men), oxalate <40 mg, citrate >320 mg, uric acid <600 mg.
- Repeat within 6 months of therapy, then yearly.

5. Selective drug therapy

Abnormality	Drug and dose (treatment)	Rationale (mechanism and notes)
Hypercalciuria	**Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg	Distal calcium reabsorption plus volume contraction. Add potassium citrate: hypokalaemia causes hypocitraturia (hypokalaemia → hypocitraturia)
Primary hyperparathyroidism	Parathyroidectomy	Cures resorptive hypercalciuria
Hypocitraturia, distal RTA (renal tubular acidosis), chronic diarrhoea	**Potassium citrate 30–60 mEq/day**	Alkali load reduces tubular (proximal) citrate reabsorption; avoid urine pH ≥7
Hyperuricosuria / hyperuricosuric calcium oxalate	Allopurinol 300 mg/day	Watch hypersensitivity; reduce dose in renal impairment
Uric acid stones	Potassium citrate to **pH 6.0–6.5**; allopurinol only for hyperuricosuria or failed alkalinisation	A pH disease: pKa 5.5; dissolution possible
Enteric hyperoxaluria	Low fat and oxalate; **calcium with meals**; liquid potassium citrate; magnesium; cholestyramine	Calcium binds oxalate in the gut (binds gut oxalate)
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran** (RNA (ribonucleic acid) interference against glycolate oxidase)	ILLUMINATE-A: 84% vs 0% reached near-normal urine oxalate
Cystinuria	Urine ≥3 L/day; low sodium and protein; potassium citrate to pH **7.0–7.5**; then **tiopronin** (or D-penicillamine)	Aim cystine <250 mg/L; mixed disulfide: thiols form soluble mixed disulfides
Struvite	Complete clearance; culture-specific antibiotics; AHA (acetohydroxamic acid) 250 mg three times daily if surgery impossible	Urease inhibitor; risk of thromboembolism

- **Also know:** recurrent calcium stones with a normal work-up: thiazide or potassium citrate (AUA (American Urological Association)); citrus juices are adjuncts.
- **Trials**: Ettinger 1988 (chlorthalidone) and Barcelo 1993 (potassium citrate) reduced new stones. **NOSTONE (2023)**: hydrochlorothiazide 12.5–50 mg did not reduce recurrence, which questions the older trials.
- **Follow-up**: repeat 24-h urine within 6 months, then yearly; serum potassium 1–2 weeks after starting a thiazide or citrate; imaging for new stones.

Recent advances in RIRS instrumentation

Advance	Principle	Benefit or limitation
Digital flexible ureteroscopes	CMOS (complementary metal-oxide semiconductor) chip at the tip; LED (light-emitting diode) light	Sharper image that does not degrade; tip larger (~8.5 Fr); no more durable
Single-use scopes (LithoVue, 2016)	7.7 Fr tip, ≥270° deflection, new each case	No repair or reprocessing; cheaper below ~99 cases a year; for lower pole and difficult cases
Thulium fibre laser (about 1940 nm)	Long pulse, very high frequency	Fine dust, minimal retropulsion; faster than Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) at low settings
Pulse-modulated (Moses) Ho:YAG	Pulse shaped through a vapour bubble	Less retropulsion, faster dusting
Access sheaths (12/14 Fr)	Outflow channel	Lower intrarenal pressure and temperature; severe ureteric injury in ~1 in 8
Suction sheaths and FANS (flexible and navigable suction access sheath)	Vacuum at the sheath; flexible tip follows the scope into calyces	Aspirates dust; lower pressure; better stone-free rates in early studies
Intrarenal pressure monitoring	Sensor in the wire or scope (LithoVue Elite)	High pressure causes backflow, SIRS (systemic inflammatory response syndrome) and sepsis; safe limits not yet defined
Robotic flexible ureteroscopy (Avicenna Roboflex)	Seated console controls the scope	Ergonomics, less radiation to surgeon

- **Laser safety**: tissue damage above 43 °C, so keep power ≤10 W in the ureter and ≤25 W in the kidney, with continuous cool irrigation.
- **Also know:** dusting versus fragment-and-basket; nitinol tipless baskets; anti-retropulsion devices; Monarch robotic platform for PCNL (percutaneous nephrolithotomy) access.

Diagram to draw: saturation zones (Ksp, Kf) and Randall plaque on a papilla.

Diagram to draw: 24-h urine → abnormality → drug flowchart.

Key points to remember:

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor; ductal plugs suggest secondary cause.
- Urine ≥2.5 L/day; Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide plus potassium citrate for hypercalciuria; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

Energy sources for breaking stones in urology

Asked as:

- Describe energy sources for breaking stones in Urology — *Practice Set 1 Paper IV Q2 (LAQ)*
- Describe energy sources for braking stones in uro — *Practice Set 5 Paper IV Q1 (LAQ)*

Type: LAQ (25 marks)
(also covered in 2025-Winter Paper 4, Q2 and Q3b; also answered in Practice Set 1 Paper 4, Q2)

Read as "energy sources for breaking stones in urology".

Introduction

- Stones are **brittle**: they crack under tension and shear. Tissue is **elastic** and survives the same stress. Every lithotripter exploits this difference.
- Energy reaches the stone **extracorporeally** (SWL, shock wave lithotripsy) or **intracorporeally** through an endoscope.

Classification

Group	Sources
Extracorporeal	SWL: electrohydraulic, electromagnetic, piezoelectric; BWL (burst wave lithotripsy, investigational)
Intracorporeal mechanical	Ballistic (pneumatic), ultrasonic, combined ultrasonic–pneumatic
Intracorporeal electrical	EHL (electrohydraulic lithotripsy)
Intracorporeal light	Lasers: Ho:YAG (holmium:yttrium-aluminium-garnet), TFL (thulium fibre laser), pulsed Tm:YAG (thulium:YAG); historic pulsed dye and FREDDY

1. Shock wave lithotripsy

- **Shock wave:** peak positive pressure 30–100 MPa, then a tensile tail of about −10 MPa. It is generated at F1, focused, and delivered to the stone at F2.

Generator	Mechanism	Strength	Weakness
Electrohydraulic (original HM3, 1980)	Underwater spark at F1 of an ellipsoid reflector	Very effective	Electrode life <40,000 shocks; focus drifts
Electromagnetic (modern standard)	Coil pushes a metal membrane; lens or reflector focuses	Consistent; high energy at F2; long life	—
Piezoelectric	Many ceramic elements on a spherical dish	Long life; little pain	Small focal zone, low energy

- **Fracture mechanisms:** shear and tear forces; **spallation** at the distal face; quasistatic squeezing; **cavitation** (bubble collapse erodes the surface); dynamic squeezing (most powerful). A focal zone wider than the stone works best.
- **Technique:** 60–90 shocks/min beats 120/min, because bubble clouds shield the stone and injure the kidney; voltage ramping; bubble-free gel coupling.
- Poor results: calcium oxalate monohydrate, brushite, cystine, density >1000 HU (Hounsfield units), skin-to-stone distance >10 cm.

2. Ballistic (pneumatic) lithotripsy

- Compressed air drives a projectile against a metal probe, a **jackhammer** (about 12 Hz).

- Breaks all stones cheaply with little ureteric injury. Drawbacks: **retropulsion**; rigid probes only, as flexible probes block deflection.

3. Ultrasonic lithotripsy

- A piezoceramic transducer vibrates a hollow probe at ultrasonic frequency; contact drills the stone.
- **Suction** through the hollow probe removes fragments at once. Best for soft stones in PCNL (percutaneous nephrolithotomy); weak on hard stones; needs irrigation for cooling.

4. Combined devices

- Ultrasonic plus ballistic in one probe (for example Trilogy, ShockPulse-SE): ballistic cracks hard stones while ultrasound and suction clear fragments. Faster PCNL.

5. Electrohydraulic lithotripsy

- A spark at the electrode tip forms a **vapour bubble**; expansion and collapse send pressure waves in all directions.
- Cheap; 3–5 Fr flexible probes. Poorly focused, so it injures the ureter and the scope; weak on cystine. AUA (American Urological Association) advises against it as first-line ureteric lithotripsy; still used for bladder stones.

6. Lasers

- **Pulse duration decides mechanism.** Short pulses (<10 µs: pulsed dye, FREDDY) are photomechanical, with cavitation shocks. Long pulses (>250 µs: holmium, thulium) are **photothermal**: the stone is heated until it melts, decomposes or flashes its water to steam.

	Ho:YAG	TFL
Wavelength	**2120 nm** (often quoted 2100)	1940 nm (water absorption peak)
Frequency	Up to about 120 Hz	Up to about 2000 Hz
Peak power	High	Low → less retropulsion
Fibre	≥200 µm	50–150 µm (more flow and deflection)
Product	Fragments and dust	Finer dust

- **Holmium:** standard since the 1990s; breaks **every** composition; flexible quartz fibres. Fragmenting 0.6–1.0 J × 6–10 Hz; dusting 0.2–0.3 J × 40–80 Hz; "popcorning" in a calyx.
- **Moses effect:** the start of the pulse boils a vapour channel through which the rest reaches the stone; pulse modulation reduces retropulsion.
- **TFL RCTs (randomised controlled trials):** superior to low-power (30 W) holmium; equal stone-free rate to 120 W pulse-modulated holmium.
- **Retropulsion** rises with pulse energy, not frequency, and falls with longer pulses.

Safety

- Laser heat follows total power: ≤10 W in the ureter, pauses, high flow, access sheath. Eyewear for all.
- Never fire without seeing the fibre tip on the stone.
- SWL: haematoma, arrhythmia, steinstrasse. Contraindications: pregnancy, coagulopathy or anticoagulants, untreated infection, aneurysm, distal obstruction.

Choice by site

Site	Usual energy
Ureter	Laser (semirigid or flexible); pneumatic distally; SWL
Kidney	SWL; laser via flexible URS (ureteroscopy); PCNL with ultrasonic, combined or laser
Bladder	Laser, pneumatic, EHL, mechanical

Recent advances

- **BWL:** handheld ultrasound bursts (5–100 cycles) break stones without anaesthesia. **Ultrasonic propulsion** moves fragments.
- Suction sheaths; AI (artificial intelligence)-adjusted laser settings.

Diagram to draw: SWL electrohydraulic ellipsoid with F1 and F2, stone at F2.

> **Key points to remember:**
> - SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
> - Holmium 2120 nm, photothermal, breaks every composition.
> - TFL 1940 nm: low peak power, finer dust, less retropulsion.
> - EHL is poorly focused, so it is not first-line in the ureter.

Section 11 · Stones, Obstruction & Trauma — Short answer questions (SAQ)

Tubeless PCNL

Asked as:

- Tubeless PCNL — *Summer 2016 Paper IV Q3e (SAQ)*
- Tubeless PCNL — *Practice Set 6 Paper IV Q3e (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2016-Summer Paper 4, Q3e and 2025-Winter Paper 2, Q3e)

Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and (higher) cost. Most tracts stop bleeding without tamponade.

Selection

Suitable	Keep a nephrostomy
Complete clearance, single tract	Pyonephrosis or infected stone
Minimal bleeding	Significant bleeding
Intact collecting system	Significant perforation of the collecting system (significant collecting-system injury)
Supracostal access, bilateral cases (shown safe)	Residual stone needing a second look

Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent through the urethra or a string.
3. Withdraw the sheath under vision; the **safety wire is removed last**.
4. Watch the tract; apply pressure (rib as backstop in thin patients). Fibrin glue in the tract did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

Evidence

- Meta-analyses of randomised trials: shorter stay and operative time.
- Less pain than large tubes (**>16 Fr**), but **no benefit against (over) small 8–9 Fr tubes**.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents are removed (come out) before discharge without cystoscopy.
- Children: insufficient evidence.

Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: re-insert a balloon or Kaye catheter, then **angioembolisation**.

Current position

- Choose tubeless, small tube (≤10 Fr) or stent case by case; both stents and small tubes are acceptable (AUA, American Urological Association). Pairs well with mini-PCNL.

Diagram to draw: kidney with sealed tract and double-J stent.

> **Key points to remember:**
> - Tubeless = stent only; totally tubeless = neither.
> - Benefit is mainly over large (>16 Fr) tubes.
> - Keep a tube for bleeding, perforation, infection or second look.

Miniperc: miniaturised PCNL, its nomenclature and newer PCNL techniques

Asked as:

- Miniperc — *Winter 2017 Paper IV Q3d (SAQ)*
- Newer PCNL techniques and nomenclature of miniaturised PCNL — *Practice Set 4 Paper IV Q3a (SAQ)* (mini-PCNL also answered in 2022-Summer Paper 1, Q1; prone versus supine in 2024-Winter Paper 1, Q3f)
- Miniperc — *Practice Set 6 Paper IV Q3f (earlier version) (SAQ)* (also answered in 2017-Winter Paper 4, Q3d and 2022-Summer Paper 1, Q1)

Type: SAQ (10 marks)

Definition and why miniaturise

- PCNL (percutaneous nephrolithotomy) through a **smaller tract** than standard, aiming to reduce bleeding and renal trauma. First "mini-perc" by **Jackman (1998)**, 13 Fr, in children.
- In PCNL, bleeding rises with tract size. Smaller tracts mean less bleeding and pain, at the cost of longer operating time and poorer vision.

Nomenclature (by tract or outer sheath size)

Name (designation)	Tract (size)	Features
Standard	24–34 Fr	Ultrasonic or pneumatic lithotripsy; graspers
Mini	15–22 Fr	First "mini-perc": Jackman 1998, 13 Fr; irrigation washes fragments out ("vacuum-cleaner" effect)
Super-mini	About 10–14 Fr	Sheath with **active suction** of fragments
Ultra-mini	11–14 Fr	3 Fr inline telescope in 11–13 Fr sheath
Micro-perc	<10 Fr	Single-step "all-seeing needle" with optics; laser only, fragments pass spontaneously

- **Problem:** the nomenclature is contested: names overlap and cut-offs are contested (paediatric chapter: mini <20–24 Fr), so many authors report the **actual sheath size in Fr**.
- The **working channel**, not sheath size, decides which instruments pass (decides what passes): the smallest systems take only a laser fibre.

Instruments and technique

- Mini nephroscopes (e.g. 12 Fr telescope, 15–18 Fr sheath, 6 Fr channel); laser (holmium/thulium) or small pneumatic probes.
- **Fragment removal**: irrigation outflow around the scope washes fragments out (**"vacuum-cleaner" effect**), or **active suction sheaths** (super-mini PCNL).
- Ultrasound or fluoroscopic access; supine or prone; ECIRS (endoscopic combined intrarenal surgery) possible.
- Exit: often **tubeless** or small nephrostomy (≤10 Fr: less pain).

Newer techniques

1. **Supine PCNL and ECIRS (endoscopic combined intrarenal surgery):** Galdakao-modified supine; simultaneous retrograde flexible ureteroscopy watches puncture and dilatation; fewer tracts.
2. **Ultrasound-guided access:** less radiation; shows pleura and viscera.

3. **Single-step or balloon dilatation.**
4. **Suction sheaths:** lower pressure, faster clearance.
5. **Lasers:** high-power and pulse-modulated holmium, TFL (thulium fibre laser) for dusting through small tracts.
6. **Flexible nephroscopy** after bulk removal: higher stone-free rate.
7. **Tubeless or small (≤10 Fr) nephrostomy:** less pain.

- **Also know:** robotic or artificial intelligence-assisted puncture under study.

Advantages over standard PCNL

- Less haemoglobin drop, **fewer transfusions**, shorter stay; less parenchymal damage; useful in children.
- Children: a 24–30 Fr tract in a child equals about a 72 Fr tract in an adult, so miniaturisation matters most here.

Disadvantages

- **Longer operative time**, poorer visualisation (vision), slower fragment clearance.
- **Intrarenal pressure**: high-pressure irrigation through a small (tight) sheath, especially when the scope nearly fills it, can raise pelvic pressure → (causes) pyelovenous backflow, **fever and sepsis**. Poor outflow through small sheaths raises intrarenal pressure. Keep outflow free, use low-pressure irrigation, or suction sheaths.

Evidence

- Mini vs standard (meta-analysis): **non-inferior stone-free rate**, less bleeding (fewer transfusions), shorter stay, longer surgery (operation); same complications (adverse events).
- Stones >2 cm: equal stone-free rate and fever; mini bleeds less than **30 Fr** tracts, but **every advantage disappears versus (against) 24 Fr** (against a **24 Fr** tract these advantages disappear).
- Mini-PCNL versus RIRS (retrograde intrarenal surgery), large matched study: similar stone-free and retreatment rates; RIRS shorter stay.

Indications and contraindications

- Renal stones **1–2 cm**, **lower pole stones** >1 cm (alternative to RIRS (retrograde intrarenal surgery)), failed SWL (shock wave lithotripsy) or RIRS, **paediatric stones**, calyceal diverticular stones; selected larger stones or extra staghorn tracts.
- **Contraindications**: as for PCNL (untreated infection, uncorrected coagulopathy, pregnancy).

Diagram to draw: tract sizes compared (standard, mini, ultra-mini, micro) — scale bar of tract sizes from micro to standard.

> **Key points to remember:**
>
> - Standard 24–34, mini 15–22 Fr; ultra-mini 11–14 Fr; micro <10 Fr; Jackman 1998: first mini-perc (13 Fr).
> - Fewer transfusions and shorter stay, but longer operation (time).
> - Working channel decides instruments; suction lowers pressure.
> - Watch intrarenal pressure; advantage lost against 24 Fr tracts (equal to 24 Fr).

Struvite stones.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 4, Q3a

Definition and epidemiology

- **Infection stones:** magnesium ammonium phosphate hexahydrate ($MgNH_4PO_4 \cdot 6H_2O$), usually mixed with **carbonate apatite**.
- 5–15% of stones (falling); female:male ~2:1; highest in India and Pakistan. Risk: neurogenic bladder, spinal injury, diversion, catheters, obstruction, diabetes, elderly.

Pathogenesis

```
Urease-producing bacteria
  Urea + H2O -> 2NH3 + CO2
  NH3 -> NH4+ (urine pH > 7.2)
  CO2 -> carbonate; phosphate -> PO4(3-)
  + Mg -> struvite;  + Ca -> carbonate apatite
```

- Sterile urine has no urease, so infection is essential. Ammonia damages the GAG (glycosaminoglycan) layer; bacteria live inside the stone in a **biofilm**, so antibiotics cannot sterilise a retained stone.
- **Organisms: Proteus mirabilis** (commonest), Klebsiella, Pseudomonas, Providencia, Staphylococcus, **Ureaplasma** (easily missed). E. coli is rarely urease-positive.

Clinical features

- Recurrent UTI (urinary tract infection), fever, flank pain, haematuria; often few symptoms. Grow fast into **staghorn** calculi. Untreated: pyonephrosis, XGP (xanthogranulomatous pyelonephritis), sepsis, renal loss.

Evaluation

- Urine pH >7; culture (urine and stone cultures often differ). NCCT (non-contrast computed tomography): burden and anatomy; DMSA (dimercaptosuccinic acid) or renogram for split function. Stone analysis; 24-h urine (many have hypercalciuria or hypocitraturia).

Management

- **Complete stone clearance** is the goal: each fragment is an infected nidus.
- Culture-specific antibiotics before surgery; drain an obstructed infected kidney first.
- **PCNL (percutaneous nephrolithotomy): first line** for staghorns (AUA, American Urological Association). Add flexible nephroscopy, ECIRS (endoscopic combined intrarenal surgery) or second look. SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery) for small stones only.
- Anatrophic nephrolithotomy rarely; nephrectomy for a poorly functioning kidney.
- **Prevention:** surveillance cultures, suppressive antibiotics, fluids, correct stasis or metabolic defects.
- **AHA (acetohydroxamic acid)** 250 mg three times daily (urease inhibitor), only when surgery is exhausted. Side effects in up to 30% (haemolytic anaemia, DVT (deep vein thrombosis)); avoid if creatinine >2.5 mg/dL or pregnancy.
- **Also know:** hemiacidrin irrigation (historical); matrix stones (radiolucent, Proteus).

Diagram to draw: urease cascade; staghorn with PCNL tract.

Key points to remember:

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

Role of open and laparoscopic stone surgeries in modern era of endourology.

Type: SAQ (10 marks)

Source: 2021-Summer Paper 4, Q3f

Introduction

- SWL (shock wave lithotripsy), URS (ureteroscopy) and PCNL (percutaneous nephrolithotomy) treat almost all stones. Open surgery is now a small minority in centres with full endourology, but remains in resource-limited settings.
- **AUA (American Urological Association)/Endourological Society 2016:** consider laparoscopic, robotic or open surgery **only when SWL, URS and PCNL fail or are expected to fail**; not first line.

Indications

Category	Examples
Endourology failed or unlikely	Very complex staghorn needing many PCNL sessions; failed multiple procedures
Anatomical problem treated together	**PUJ (pelviureteric junction) obstruction + stone:** pyeloplasty with pyelolithotomy; anterior calyceal diverticulum; infundibular stenosis
Difficult access	Pelvic or ectopic kidney (laparoscopy-assisted PCNL); morbid obesity; skeletal deformity
Poor renal function	**Nephrectomy** for a non-functioning kidney; partial nephrectomy for a non-functioning pole
Large impacted ureteric stone	Laparoscopic ureterolithotomy after failed URS/SWL
Concomitant surgery	Other open or laparoscopic procedure
Bladder	Very large stone; with open prostatectomy, diverticulectomy or augmentation; small bladder; urethral stricture

Procedures

- **Pyelolithotomy:** simple, extended (Gil-Vernet, sinus dissection) or coagulum.
- **Anatrophic nephrolithotomy (Smith and Boyce):** for complex staghorn. Clamp renal artery; cool with ice slush; give mannitol; incise along **Brödel's avascular line**; remove stone; repair infundibula; close.
- Radial nephrotomy, ureterolithotomy, cystolithotomy.

Laparoscopic and robotic surgery

- Transperitoneal or retroperitoneal pyelolithotomy and ureterolithotomy; robotic anatrophic nephrolithotomy reported.
- Best for large pelvic stone in an extrarenal pelvis: high single-stage clearance, less bleeding than PCNL. Drawbacks: longer operation, calyceal stones need a flexible scope.

Feature	Open	Laparoscopic/robotic
Morbidity	Large incision, pain, long stay	Less pain, shorter stay
Kidney	Ischaemic loss of function (anatrophic)	Similar repair, less wound morbidity
Later surgery	Adhesions make redo difficult	Fewer

Complications

- Bleeding, urinary leak, infection, loss of renal function, residual fragments.

Diagram to draw: Brödel's line incision for anatrophic nephrolithotomy.

> **Key points to remember:**
> - Open/laparoscopic only when endourology fails or is unsuitable.
> - Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
> - Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

Diet in stone disease

Asked as:

- Diet in stone disease. — *Winter 2023 Paper IV Q3f (SAQ)*
- Diet in stone disease. — *Practice Set 5 Paper IV Q3f (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3f)

Principle

- Diet lowers urinary **supersaturation** by **dilution**, by reducing promoters (calcium, oxalate, uric acid, sodium) and by raising inhibitors (citrate).
- General advice for all; recurrent formers tailored to **24-hour urine**.

General measures

Measure	Target	Mechanism and evidence
Fluid	Urine **≥2.5 L/day** (about 3 L intake), spread through the day	Most important single factor. **Borghi 1996 RCT (randomised controlled trial):** recurrence 12.1% vs 27% at 5 years
Normal calcium	1000–1200 mg/day, with meals	Binds oxalate in the gut. **Restriction increases stones**. **Borghi 2002:** normal-calcium, low-salt, low-protein diet halved stone events vs low-calcium diet
Low sodium	Salt about 4–5 g/day	Sodium increases urinary calcium, lowers citrate
Limit animal protein	0.8–1.0 g/kg/day	Acid and purine load: hypercalciuria, hypocitraturia, hyperuricosuria
Fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension] diet)	Daily	Alkali and citrate; DASH linked to fewer stones
Weight	BMI (body mass index) 18–25	Obesity: low urine pH, more oxalate and uric acid

- **Citrus:** lemonade (most evidence) and orange juice raise citrate; orange juice adds alkali. **Volume matters more than drink type.**
- Avoid sugar-sweetened drinks and **high-dose vitamin C** (converted to oxalate).

Stone-specific diet

Stone or abnormality	Diet
Hyperoxaluria	Avoid large portions of **spinach, beets, chocolate, nuts, tea**; calcium with meals
Enteric hyperoxaluria	Strict low oxalate and **low fat**; **calcium up to 1 g/day with meals**; fluid; magnesium
Uric acid	Less animal protein (purines give about 30% of urinary uric acid); fruit and vegetables; weight loss
Cystine	Fluid **3–5 L/day** including night; sodium **≤2300 mg/day**; less animal protein (not in children)
Struvite	Infection control; diet has little role

Adherence

- Poor: only half reach 2.5 L at 6 months; apps and smart bottles help.

Diagram to draw: "stone-prevention plate": water, fruit and vegetables, normal dairy, little meat and salt.

Key points to remember:

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

Newer methods of stone analysis and their significance in management of calculus disease

Asked as:

- Newer methods of stone analysis and their significance in management of calculus disease. — *Winter 2024 Paper IV Q3a (SAQ)*
- Newer methods of stone analysis and their significance in management of calculus disease. — *Practice Set 6 Paper IV Q3a (earlier version) (SAQ)*

Type: SAQ (10 marks)

(also answered in 2024-Winter Paper 4, Q3a)

Why analyse stones

- AUA (American Urological Association) and EAU (European Association of Urology): analyse a stone at least once in every former; **repeat if stones recur on treatment** (evolving metabolic condition).
- Composition points to the cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection. Cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine) and xanthine are diagnostic in themselves.

Why wet chemical analysis is obsolete

- Detects ions, not **crystal phases**, so it cannot separate COM (calcium oxalate monohydrate, whewellite) from COD (dihydrate, weddellite), or apatite from brushite.
- High error rates; misses rare stones (2,8-DHA [2,8-dihydroxyadenine], drugs); needs a large sample.

Newer methods

Method	Principle	Comment
FTIR (Fourier-transform infrared) spectroscopy	Each compound absorbs a characteristic infrared spectrum	Commonest reference method; small (tiny) sample; identifies (detects) proteins and drugs
XRD (X-ray diffraction)	Crystal lattice diffracts X-rays in a unique pattern	Gold standard for crystals; misses amorphous material
Polarising microscopy	Crystal optics (birefringence)	Cheap; experienced observer (needs experience)
Morpho-constitutional analysis (Daudon)	Stereomicroscopy of surface and section, plus FTIR	Subtypes point to cause (e.g. type Ia COM low urine volume, Ic primary hyperoxaluria / Ic suggests primary hyperoxaluria)
SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)	Surface structure, elements	Research
Micro-CT (micro-computed tomography)	3-D non-destructive structure	Research
Raman spectroscopy	Scattered light spectrum	Research

In vivo prediction

- **DECT (dual-energy computed tomography):** two energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid stones** with near-perfect accuracy.

- **Hounsfield units:** >1000 HU predicts SWL (shock wave lithotripsy) failure; SWL works best below about 900–1000 HU; cystine and brushite are dense.
- Endoscopic recognition (lost when stones are dusted). Dusting loses the specimen: collect fragments.

Significance in management

- **Uric acid:** oral dissolution by urinary alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture, (culture-directed) antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin; avoid SWL.
- **COM, brushite, cystine:** hard, SWL-resistant; prefer endoscopy.
- **Brushite:** often converts (can convert) from calcium oxalate after SWL or infection; avoid over-alkalinisation.
- **2,8-DHA:** APRT (adenine phosphoribosyltransferase) deficiency, treat with allopurinol or febuxostat. **Drug stones** (indinavir, triamterene): stop the drug.

Diagram to draw: FTIR spectrum outline and COM (dumbbell) versus COD (envelope) crystal shapes.

> **Key points to remember:**
> - FTIR and XRD are reference methods; wet chemistry is obsolete.
> - DECT separates uric acid stones in vivo.
> - Composition guides dissolution, SWL choice and prevention.

ECIRS (endoscopic combined intrarenal surgery)

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 4, Q3b

Definition

- **ECIRS:** PCNL (percutaneous nephrolithotomy) and retrograde flexible ureteroscopy performed **at the same time, in one position**, by two surgeons.
- Term and technique popularised by **Scoffone (2008)** in the **GMSV (Galdakao-modified supine Valdivia) position**. Combined access dates back to the 1980s (Lawson's retrograde needle placement, 1983).

Rationale

- Each route covers the other's blind spots: the retrograde scope guides and watches the puncture, and reaches calyces the rigid nephroscope cannot.
- Result: one tract, better clearance.

Positioning

Position	Features
GMSV (preferred)	Supine; flank on a bolster at the table edge; ipsilateral leg extended, contralateral leg in lithotomy
Bart's modified supine	Variant of supine
Prone split-leg	Easier posterior calyces; needs a split-leg table

- **Supine advantages:** airway access, gravity drainage of fragments, no turning, simultaneous retrograde access.

Technique

1. Cystoscopy; UAS (ureteral access sheath) placed; flexible ureteroscope maps the stones and chooses the target calyx.
2. Ultrasound or fluoroscopic puncture **watched endoscopically** (double bull's-eye technique).
3. Wire grasped and brought out: through-and-through access.
4. Tract dilated under vision (balloon) and sheath placed.
5. Antegrade lithotripsy and suction; the retrograde scope **relocates stones** from inaccessible calyces into the nephroscope's view, or lasers them.
6. Final check with both scopes; stent or tubeless exit.

- Mini-ECIRS uses a mini-PCNL tract.

Indications

- Staghorn and complex multi-calyceal stones, combined renal and ureteric stones, anomalous kidneys, failed previous treatment, urinary diversion, obese or cardiopulmonary patients (supine).

Advantages (meta-analysis)

- Single position, **fewer tracts, less bleeding**, fewer ancillary procedures, **higher stone-free rate**; less fluoroscopy; the UAS keeps intrarenal pressure low.

Disadvantages

- Two surgeons and two sets of equipment; cost; longer set-up; ureteric injury from the sheath; less working space in supine.

Diagram to draw: GMSV position with nephroscope from the flank and flexible ureteroscope from below in the same calyx.

> **Key points to remember:**
>
> - Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
> - Retrograde scope guides the puncture and relocates stones.
> - Fewer tracts, less bleeding, higher stone-free rate.

Rapid Revision Sheet – Section 11 · Stones, Obstruction & Trauma

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Urolithiasis: etiopathogenesis, medical management of (recurrent) stone disease and recent advances in RIRS instrumentation (LAQ)

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor; ductal plugs suggest secondary cause.
- Urine ≥2.5 L/day; Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide plus potassium citrate for hypercalciuria; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

Energy sources for breaking stones in urology (LAQ)

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

Tubeless PCNL (SAQ)

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

Miniperc: miniaturised PCNL, its nomenclature and newer PCNL techniques (SAQ)

- Standard 24–34, mini 15–22 Fr; ultra-mini 11–14 Fr; micro <10 Fr; Jackman 1998: first mini-perc (13 Fr).
- Fewer transfusions and shorter stay, but longer operation (time).
- Working channel decides instruments; suction lowers pressure.
- Watch intrarenal pressure; advantage lost against 24 Fr tracts (equal to 24 Fr).

Struvite stones. (SAQ)

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

Role of open and laparoscopic stone surgeries in modern era of endourology. (SAQ)

- Open/laparoscopic only when endourology fails or is unsuitable.
- Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
- Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

Diet in stone disease (SAQ)

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

Newer methods of stone analysis and their significance in management of calculus disease (SAQ)

- FTIR and XRD are reference methods; wet chemistry is obsolete.
- DECT separates uric acid stones in vivo.
- Composition guides dissolution, SWL choice and prevention.

ECIRS (endoscopic combined intrarenal surgery) (SAQ)

- Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
- Retrograde scope guides the puncture and relocates stones.
- Fewer tracts, less bleeding, higher stone-free rate.

Section 12 · Bladder Cancer & Diversion — Long answer questions (LAQ)

Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion

Asked as:

- Orthotopic versus Non Orthotopic bladder. — *Summer 2016 Paper IV Q1 (LAQ)*
- Describe principles of continent urinary diversions. Current status of orthotopic neobladder. — *Practice Set 3 Paper IV Q2 (LAQ)*
- Neobladder. — *Summer 2020 Paper IV Q3a (SAQ)*
- Studer Neobladder — *Summer 2015 Paper IV Q3e (SAQ)*
- Studer Neobladder. — *Practice Set 4 Paper IV Q3c (earlier version) (SAQ)*
- Neobladder. — *Practice Set 4 Paper IV Q3d (earlier version) (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(neobladder also answered in 2020-Summer Paper 4, Q3a; Studer also answered in 2015-Summer Paper 4, Q3e; selection and general points: see Practice Set 4 Paper 2, Q3e)

See Practice Set 4 Paper 2, Q3e (full answer). Studer technique: see Q3c above.

Read as: orthotopic neobladder versus non-orthotopic (heterotopic) urinary diversion after radical cystectomy.

Introduction, definition and types

- A continent diversion stores urine in a bowel reservoir and empties it **at will**: through the native urethra (**orthotopic neobladder**) or by CIC (clean intermittent catheterisation) of a stoma (**continent cutaneous pouch**).
- **Neobladder definition**: an **orthotopic** reservoir of detubularised bowel (usually ileum) joined to the native urethra after radical cystectomy, allowing **volitional voiding** without a stoma.
- Evolution: ileal conduit (Bricker, 1950) → continent cutaneous (Kock, 1982) → orthotopic (Camey, 1979; Hautmann, 1988; Studer, 1989).
- **Ileal conduit** (Bricker, 1950) remains the commonest diversion worldwide; **ureterosigmoidostomy** is the oldest continent diversion.
- AUA (American Urological Association)/SUO (Society of Urologic Oncology) and EAU (European Association of Urology): counsel every patient on all options. **80–90% of men and ~75% of women** are medically eligible for a neobladder, but far fewer receive one.

Group	Examples	How urine is held and emptied
Orthotopic	Studer, Hautmann W, T pouch, Padua, Mainz III	Native rhabdosphincter; voids by relaxing and straining
Incontinent cutaneous	Ileal conduit; transverse colon, sigmoid or jejunal conduit	Stoma and appliance
Continent cutaneous	Indiana pouch; right colon pouch with appendix (Mitrofanoff) or Yang–Monti channel; Kock (historical)	Self-catheterisation through a valved channel
Rectal	Ureterosigmoidostomy, Mainz II (sigma-rectum pouch)	Anal sphincter; urine mixed with stool

Basic principles of a continent reservoir

1. **Reliable outlet.** Orthotopic: intact rhabdosphincter (pudendal nerve) and a healthy urethra. Cutaneous: a stable catheterisable efferent channel with a dependable continence mechanism.
2. **Low pressure throughout filling.** Complete **detubularisation** and cross-folding abolish (abolishes) coordinated peristaltic contractions. Folding into a **sphere** gives the greatest (largest) volume for its (a given) surface area (Laplace: wall tension = pressure × radius). Result: a low-pressure reservoir.
3. **Adequate volume: ≥500 mL when mature.** Bowel stretches with time (ileum more than colon); 44 cm of double-folded ileum holds <200 mL at first (initially) and 500–600 mL within a year (by 1 year).
4. **Unobstructed ureterointestinal anastomosis** to protect the kidneys.
5. **Enough renal function** to clear the acid load reabsorbed by bowel mucosa.

- Continence is a **pressure balance**: low reservoir pressure against adequate outlet resistance.
- **Studer**: **orthotopic low-pressure ileal reservoir** (**Studer, 1989**): detubularised ileal **orthotopic** reservoir with a long **isoperistaltic afferent limb**, joined to the urethra. **Detubularisation** stops coordinated contractions, so pressure stays low; a **sphere** (Laplace) gives most volume for surface area. The **isoperistaltic afferent limb** limits reflux (damps reflux during Valsalva voiding) and accepts **short ureters**. No staples in the reservoir. Continence depends on the **rhabdosphincter**.

Choice of bowel segment

Segment	Points
Ileum (default)	Most compliant, lowest pressure; mesentery reaches the urethra; mucosa atrophies, so absorption falls
Ileum + caecum (Mainz)	Near-ileal compliance; loses ileocaecal valve
Colon (right, sigmoid)	Less distensible, pressure waves → more night leakage; when ileum is short or diseased
Stomach	Secretes acid: helps in renal impairment; haematuria–dysuria; rarely used

Reflux versus obstruction

- Voiding by Valsalva raises pressure equally throughout, so reflux into a low-pressure pouch is benign, while **every antireflux valve can obstruct** (Le Duc tunnel stricture 7–29%; late Kock nipple stenosis).
- Direct refluxing anastomosis (Wallace, Leadbetter): stricture about 3–6%. **USC (University of Southern California) RCT (randomised controlled trial)**, 484 patients: Studer vs T pouch, no difference in renal function or infection at 3 years; T pouch needed more secondary procedures (a USC randomised trial found the antireflux T pouch no better: gave no renal benefit and more reoperations).
- Studer: reflux on cystogram, yet kidneys preserved in about (~) 95%.

Continent cutaneous mechanisms

Mechanism	Example
Flap valve: appendix tunnelled in taenia	**Mitrofanoff** (Penn, right colon pouch); Yang–Monti ileal tube if no appendix
Imbricated ileocaecal valve + tapered ileum	**Indiana** pouch

Mechanism	Example
Intussuscepted nipple	Kock (historical)

- Used when the urethra is unusable (continence wanted but urethra unusable); umbilical stoma preferred.

Patient selection

	Orthotopic only	All continent diversions
Absolute contraindications	Tumour at the urethral margin (frozen section); significant urethral stricture	Severe renal (eGFR <40 mL/min) or hepatic dysfunction (liver failure); physical or mental inability to self-catheterise
Relative	Poor external sphincter; bladder-neck or anterior vaginal wall tumour (women); prostatic stromal invasion (men); irradiated sphincter (irradiated pelvis)	Frailty, poor cognition, social problems, prior radiation or bowel resection

- **Renal function**: creatinine <1.7–2.2 mg/dL or eGFR (estimated glomerular filtration rate) **>35–40 mL/min** (below about 35–40 mL/min = severe renal impairment; another source: <40 mL/min), because the kidney must excrete acid reabsorbed from stored urine. Formal tests: urine pH ≤5.8 after ammonium chloride, osmolality ≥600 mOsm/kg after water deprivation.
- **Urethral recurrence**: in men the strongest predictor is prostatic stromal invasion (prostatic urethral tumour raises urethral recurrence risk); in women, bladder-neck or anterior vaginal wall tumour. **Intraoperative frozen section** of the urethral margin decides. CIS (carcinoma in situ) or multifocality alone do not exclude a neobladder.
- **Age and obesity are not contraindications** (age and obesity alone are not bars; age alone is not a bar); comorbidity, cognition and support decide. A conduit suits the frail or unsupported patient.
- Continent cutaneous pouch: continence wanted but urethra unusable.
- Mark a conduit stoma site in everyone: up to ~10% change plan intraoperatively.

Comparison

Feature	Orthotopic neobladder	Ileal conduit	Continent cutaneous	Ureterosigmoidostomy
Appliance	None	Bag	None; small stoma	None
Body image	Best	Worst	Good	Good
Complexity	High	Lowest	Highest	Moderate
Continence	Day 80–90%; night incontinence 20–50%	Not applicable	High (Indiana, Mitrofanoff)	Depends on anal sphincter
Catheterisation	10–20% of men, ~30% of women	None	Always	None
Bowel used	Long ileal segment (Studer 54–56 cm)	10–15 cm ileum	Right colon + terminal ileum	None isolated

Feature	Orthotopic neobladder	Ileal conduit	Continent cutaneous	Ureterosigmoidostomy
Typical late problems	Night leakage, retention, acidosis	Stomal stenosis (20–24%), parastomal hernia, renal deterioration	Stomal stenosis, stones, difficult catheterisation	Severe acidosis, pyelonephritis, cancer at the anastomosis

Neobladder types

Design	Segment	Ureters
Studer	40–44 cm ileum + afferent limb	Refluxing, isoperistaltic limb
Hautmann W	About 70 cm ileum	Refluxing; largest capacity
T pouch	44 cm + T limb	Antireflux flap valve
Camey II; Padua	Ileum	Antireflux (Le Duc)
Mainz III; sigmoid	Ileocaecal; colon	Tunnelled

	Studer	Hautmann
Ileum	54–56 cm, U + afferent limb	~70 cm folded in a W
Initial capacity	Smaller (smaller early capacity)	Larger: earlier continence
Late	— (less late retention)	More retention, reabsorption

Studer technique (operative; brief)

- Brief: isolate ileum 15–25 cm from the ileocaecal valve; open the distal part on the antimesenteric border, sew into a U, fold into a sphere; end-to-side ureteric anastomoses to the afferent limb; anastomose the dependent point to the urethra.

1. **Isolate 54–56 cm of terminal ileum**, ending **15–25 cm** above the ICV (ileocaecal valve) (another source: **15–20 cm proximal to the ICV**); restore bowel continuity.
2. **Reservoir: distal 40–44 cm** (two 20–22-cm limbs) opened on the antimesenteric border; sew the two limbs (medial edges) into a **U** (back wall).
3. **Afferent limb:** leave the proximal **10–15 cm** tubular as the afferent limb (another source: **about 15 cm**; original 20 cm; 10–12 cm in later figures); close its end.
4. **Fold the U in the opposite direction** (across) and close the anterior wall into a **sphere**; join its most dependent point to the urethra.
5. **Ureters:** spatulated **end-to-side, refluxing** ureteroileal anastomoses to the afferent limb, interrupted 4-0, over stents (**stented**).
6. **Urethral anastomosis** at its most dependent point with pre-placed sutures; catheter and suprapubic tube.

- Capacity rises from <200 to **500–600 mL** in a year.
- **Intracorporeal robotic Studer** follows the same steps.

Postoperative care, voiding training and follow-up

- Mucus irrigation; oral bicarbonate; cystogram before catheter removal.
- **Voiding training**: void **sitting** (sit), relax the pelvic floor, strain gently; **timed voiding** every 2 hours, extended to 3–4 hours as capacity grows (under 200 mL to 500–600 mL in a year). **Night alarm.**
- Annual metabolic panel, B12, upper-tract imaging; urethral wash cytology if CIS (carcinoma in situ).

Current status of orthotopic neobladder

1. Preferred where suitable: closest to a native bladder. AUA/SUO (American Urological Association/Society of Urologic Oncology) and EAU (European Association of Urology): counsel every cystectomy patient on all options and refer to an experienced surgeon.

2. Under-used. 80–90% of men and about 75% of women are eligible, yet only about 9% of US patients receive any continent diversion, and this is falling; continent diversion in RAZOR 21%, iROC 11%. Causes: comorbid patients, surgeon training, intracorporeal robotics.

3. Designs in use. Studer (40–44 cm U plus isoperistaltic afferent limb) and **Hautmann W** (about 70 cm; refluxing since 1997) dominate; T pouch or serous-lined tunnel when antireflux is wanted.

4. Oncological safety.

- Men: urethral recurrence about 7% at 5 years; main risk is prostatic stromal involvement. The neobladder seems protective (meta-analysis HR (hazard ratio) 0.44).
- Women: risk lies with bladder-neck tumour; no skip lesions, so preserve the urethra. **Frozen section** of the urethral margin decides in both sexes.
- Local recurrence <10%, rarely disabling the pouch; nodal disease is not a bar. Selected **post-radiation** patients do well.

5. Function (outcomes)

Outcome	Result
Day continence	80–90% (most by 3–6 months); another source (Studer): 90–98%; about 90%
Night incontinence	20–50% (night continence lower: up to half leak): no sphincter reflex, mucosal water secretion, nocturnal diuresis
Retention needing CIC	Men 4–10%; **women 20–30%** (pouch angulation): men leak, women retain
Self-catheterisation	**10–20% of men, ~30% of women**

6. Complications. Early morbidity equals conduit. Late: hyperchloraemic metabolic acidosis, vitamin B12 deficiency, mucus, ureteroileal stricture, stones, pouch rupture, pouch–vaginal fistula (5–10%), hernia.

7. Robotic intracorporeal neobladder: comparable to open; long-term data lacking.

8. Quality of life: no diversion proven superior; body image favours neobladder.

- **Also know:** nerve sparing and uterine preservation (nerve and uterine sparing) aid continence; intracorporeal robotic Studer; lifelong follow-up (metabolic panel, B12, upper tracts, urethral wash cytology if CIS (carcinoma in situ)).

Metabolic complications (bowel in the urinary tract)

Segment	Disturbance
Ileum, colon	**Hyperchloraemic metabolic acidosis** (ammonium chloride absorbed); hypokalaemia, worse with colon
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis
Stomach	Hypochloraemic, hypokalaemic alkalosis

- Acidosis is severe in most ureterosigmoidostomies; most colonic (Mainz) pouches need alkali. Treat with sodium bicarbonate or citrate, and replace potassium as well.
- **Also know:** osteomalacia; vitamin B12 deficiency and bile-salt diarrhoea after ileal loss (**B12 and bile-salt loss**; liver B12 stores last 3–5 years); stones (ileal conduit 10–12%, continent caecal reservoirs ~20%); bacteriuria; drug reabsorption (methotrexate, phenytoin); ammoniagenic encephalopathy; growth delay in children.

- **Secondary cancer**: ureterosigmoidostomy 6–29% after 10–20 years (yearly endoscopy from 5 years); ileal conduit and ileal neobladder lowest.

Surgical complications

- **Early (≤90 days)**: ileus, urine leak (long pouch suture lines; drain percutaneously), infection, mucus retention; not more than with a conduit.
- **Late**: ureteroileal stricture ~3–6% (Wallace or direct end-to-side) (Studer/neobladder sources: **<3%**); neobladder retention, urethral stricture, rare pouch rupture; ventral and parastomal hernia; **hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, stones, pyelonephritis (commonest late, Studer), urethral recurrence.

Quality of life

- No randomised trial. Body image favours the neobladder, urinary function the conduit; no diversion is superior on HRQoL (health-related quality of life).

Recent advances

- RARC (robot-assisted radical cystectomy): **RAZOR (2018)** non-inferior 2-year progression-free survival (open diversion in both arms); **iROC**: intracorporeal diversion gave more days alive and out of hospital at 90 days.
- Intracorporeal neobladders and nerve- or organ-sparing cystectomy are increasing.

Diagram to draw: Studer neobladder and ileal conduit side by side; Studer U-pouch with afferent limb, ureters and urethral anastomosis; ileal segment marked (afferent limb plus 40–44-cm U), then the folded spherical pouch with afferent limb.

Key points to remember (incl. SAQ angle):

- Three principles: reliable outlet, low pressure (detubularise and make a sphere), ≥500 mL; 500–600 mL by 1 year.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR (estimated glomerular filtration rate) >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.
- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; refluxing ureteric anastomosis is acceptable (USC (University of Southern California) trial).
- Obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- No diversion is proven best for quality of life; counsel all options.

Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy

Asked as:

- Describe in brief intravesical therapy for transitional cell carcinoma of Urinary bladder. — *Summer 2017 Paper IV Q1 (LAQ)*
- Combination Intravesical therapy — *Winter 2025 Paper IV Q3c (SAQ)*
- Combination Intravesical therapy — *Practice Set 6 Paper IV Q3c (earlier version) (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(combination therapy also answered in 2025-Winter Paper 4, Q3c)

Introduction

- NMIBC (non-muscle-invasive bladder cancer: Ta, T1, CIS [carcinoma in situ]) is 70–80% of bladder cancers. "Urothelial carcinoma" now replaces "transitional cell carcinoma".
- Two problems: **recurrence** (40–60%) and **progression** (about 6% in low-grade Ta to 17% in high-grade T1). Intravesical therapy follows complete TURBT (transurethral resection of bladder tumour).
- **Rule:** chemotherapy reduces recurrence only. BCG (bacille Calmette-Guérin) is the only agent that also reduces progression.

Risk-adapted use (AUA [American Urological Association]/SUO)

Risk	Criteria	Therapy
Low	LG (low-grade) solitary Ta ≤3 cm; PUNLMP	Single post-op instillation only
Intermediate	LG Ta recurrent, >3 cm or multifocal; LG T1; HG (high-grade) Ta ≤3 cm	6-week chemotherapy or BCG; maintenance 1 year
High	HG T1; HG Ta large, multifocal or recurrent; any CIS; LVI (lymphovascular invasion); variant histology	BCG induction + maintenance 3 years

- EAU (European Association of Urology) adds a "very high" group; EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables.

1. Single immediate instillation

- **Rationale:** kills floating cells before they implant. So it must be early: **within 6 hours** (no benefit at 24 hours).
- Agents: MMC (mitomycin C) 40 mg, epirubicin 50 mg, **gemcitabine 2 g** (standard in USA). **SWOG S0337:** 4-year recurrence 35% vs 47% with saline.
- **Withhold** after perforation or extensive resection. **Never give BCG** now (intravasation, sepsis).

2. Adjuvant chemotherapy

- **MMC** (alkylating), 40 mg in 20 mL weekly ×6. Optimise it (Au, 2001): empty bladder, fluid restriction, sodium bicarbonate, concentrated dose. Skin rash.
- **Gemcitabine** (nucleoside analogue): minimal absorption, well tolerated. **Anthracyclines** (epirubicin, doxorubicin) cause chemical cystitis.
- Intermediate risk: 6-week induction, then monthly maintenance for 1 year.
- **Gem/Doce:** gemcitabine 1 g, drain, then docetaxel 37.5 mg; weekly ×6, then monthly for 2 years. Used in BCG-unresponsive disease and during BCG shortage.

3. BCG immunotherapy

- **Mechanism:** BCG binds **fibronectin**. Cytokines follow (IFN-γ [interferon-gamma], IL [interleukin]-2, IL-12, TNF-α [tumour necrosis factor-alpha]). A **Th1 (T-helper 1)** response gives cell-mediated tumour kill; TRAIL release is important.
- **Technique:** 1 vial in 50 mL saline. Start **2–4 weeks** after TURBT (re-epithelialisation). Urinalysis first; retain 1–2 hours.
- **SWOG schedule (Lamm, 2000):** 6 weekly induction doses, then 3 weekly doses at 3 and 6 months, then every 6 months to 3 years. Median recurrence-free survival 76.8 vs 35.7 months.
- **EORTC 30962:** full dose; 1 year for intermediate, 3 years for high risk. **NIMBUS:** reduced frequency was inferior.
- CIS complete response up to 84%; BCG beats MMC only with maintenance.
- **Contraindications:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk intravasation); previous BCG sepsis; total incontinence; immunosuppression (now relative). Relative: active UTI (urinary tract infection), liver disease, previous TB (tuberculosis).

BCG toxicity

Grade	Features	Management
1	Cystitis, fever <38.5 °C, <48 h	Anticholinergics, analgesics, NSAIDs (non-steroidal anti-inflammatory drugs)
2	Severe or >48 h	Culture; isoniazid + rifampicin; resume at ½–⅓ dose
3	Sepsis, organ involvement	Stop BCG; isoniazid + rifampicin + ethambutol; steroids only with cover

- BCG is resistant to **pyrazinamide**. BCG sepsis 0.5%.

BCG failure

- **Adequate BCG:** ≥5 of 6 induction + ≥2 of 3 maintenance doses.
- **BCG-unresponsive:** persistent or recurrent CIS or HG Ta/T1 despite adequate BCG, or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Also know:** refractory, intolerant, exposed, delayed relapse.

BCG-unresponsive options

- **Timely radical cystectomy** is the standard (EAU, AUA).
- If unfit or refusing:
  - **Pembrolizumab** (KEYNOTE-057): IV every 3 weeks; CIS complete response 41%.
  - **Nadofaragene firadenovec:** adenovirus with interferon-α2b gene, every 3 months; CIS complete response 53%.
  - **N-803:** IL-15 superagonist with BCG; CIS complete response 71%.
  - Gem/Doce; valrubicin; **TAR-200** (gemcitabine-releasing device); cretostimogene (trials).

Device-assisted therapy

- **EMDA (electromotive drug administration):** current increases MMC tissue delivery 4–7-fold. Sequential BCG + EMDA-MMC improved outcomes in T1.
- **Chemohyperthermia:** RITE (radiofrequency, 42 °C); HIVEC (heated recirculating MMC).
- **Also know:** UGN-102 MMC gel; interferon with low-dose BCG.

Combination intravesical therapy

Rationale

- Single agents fail: chemotherapy reduces **recurrence** only; BCG (bacille Calmette-Guérin) reduces **progression** but fails in about a third.
- Combinations add **different mechanisms**, overcome resistance, and help (are useful) during **BCG shortage** and in **BCG-unresponsive** NMIBC (non-muscle-invasive bladder cancer).

1. Sequential chemotherapy (doublets)

Regimen	Method	Results
Gem/Doce (gemcitabine–docetaxel)	Gemcitabine 1 g in 50 mL (60–90 min) → drain → docetaxel 37.5 mg (60–120 min); weekly ×6, then monthly for 2 years	High-grade recurrence-free survival about 65% (at 1 year), 52% (at 2 years); CIS (carcinoma in situ) complete response 60% at 1 year; **84% avoid cystectomy**
Gem/MMC (mitomycin C)	Gemcitabine 1 g → MMC 40 mg	Durable recurrence-free about 38%

- **Why it works:** gemcitabine (antimetabolite) and docetaxel (microtubule stabiliser) act on different cell-cycle phases.
- Oral sodium bicarbonate and antiemetics aid tolerance. No level 1 evidence; Phase III BRIDGE (versus BCG in BCG-naive) is testing it (compares it with BCG).

2. Chemotherapy with BCG

- Sequential MMC–BCG: **no benefit** over BCG alone.
- **Exception: BCG alternating with EMDA (electromotive drug administration)-MMC** in T1: lower recurrence and progression than BCG alone. A later trial was non-significant.
- Chemohyperthermia (RITE, radiofrequency-induced thermochemotherapeutic effect) potentiates BCG.

3. BCG with immunomodulators

- **BCG + interferon-α (α2b):** no gain over BCG; allows reduced BCG dose (BCG dose reduction to one-third).
- **N-803 (nogapendekin alfa)** = (an) IL-15 (interleukin-15) superagonist activating CD8 T and NK (natural killer) cells. With BCG in BCG-unresponsive CIS (carcinoma in situ): complete response 71%, 45% at 12 months. FDA (Food and Drug Administration)-approved (in 2024; chapter: "under review").

4. BCG with checkpoint inhibitors (BCG-naive high risk)

- **CREST** (subcutaneous sasanlimab + BCG) and **POTOMAC** (durvalumab + BCG): improved event-free/disease-free (or) survival over BCG alone.
- **ALBAN** (atezolizumab + BCG): negative. KEYNOTE-676 (pembrolizumab) ongoing.

5. Device-assisted and novel combinations

- **Chemohyperthermia** (MMC at 42 °C: RITE, radiofrequency-induced thermochemotherapeutic effect; HIVEC, hyperthermic intravesical chemotherapy) increases uptake.
- **TAR-200:** gemcitabine-releasing intravesical device (± cetrelimab); high (very high early) complete response in BCG-unresponsive CIS (SunRISe-1).
- **Cretostimogene** (oncolytic adenovirus) + pembrolizumab: about two-thirds complete response at 12 months.

Toxicity of combinations

- Additive cystitis; immune-related events with checkpoint inhibitors.

Diagram to draw: BCG mechanism (fibronectin → cytokines → Th1 → kill) with SWOG schedule timeline; Gem/Doce sequence timeline (instil, dwell, drain, instil).

Key points to remember:

- Single instillation within 6 h; never BCG immediately (SWOG S0337 gemcitabine).
- Only maintenance BCG reduces progression; SWOG 3-year schedule.
- BCG-unresponsive: cystectomy; pembrolizumab, nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

Discuss bladder preservation surgery for TCC of urinary bladder.

Type: LAQ (25 marks)

Source: 2019-Summer Paper 4, Q2

Read as: bladder-preserving treatment of muscle-invasive TCC (transitional cell carcinoma, urothelial carcinoma), including radical TURBT (transurethral resection of bladder tumour), partial cystectomy and trimodality therapy.

Introduction

- Standard for MIBC (muscle-invasive bladder cancer): cisplatin NAC (neoadjuvant chemotherapy) + RC (radical cystectomy) with PLND (pelvic lymph node dissection).
- RC is morbid and patients are often elderly. Preservation suits **(1) unfit patients** and **(2) fit, motivated patients with limited disease**.
- **Why single modalities fail:** they leave either the remaining urothelium or occult nodes untreated (cT2 understaged to pT3 in up to about 40%; occult nodes about 25%).

Options

Option	Status
Radical TURBT alone	Not a primary treatment
Partial cystectomy + PLND	Highly selected solitary tumours; urachal cancer
RT (radiotherapy) alone	Inferior; quasi-palliative
Chemotherapy alone	Not recommended
TMT (trimodality therapy)	**Best-studied, guideline-accepted**

A. Radical TURBT

- Rationale: about 10% are pT0 at cystectomy. Only for tumour limited to superficial muscle, with negative re-resection.
- About a third progress to MIBC on long follow-up; selected series report 5-year CSS (cancer-specific survival) about 82%.

B. Partial cystectomy

Ideal candidate:

- Small (**<3 cm**), **solitary** tumour in a mobile part of bladder (dome).
- **No CIS (carcinoma in situ)**, no hydronephrosis.
- 1–2 cm margin achievable away from the ureteric orifice and trigone; adequate residual capacity.
- **Treatment of choice for urachal adenocarcinoma** (en bloc dome, urachus and umbilicus).

Contraindications: CIS, multifocal tumour, trigone/bladder neck tumour, small bladder, prostatic urethral involvement.

Technique: NAC if cisplatin-eligible → PLND → cystotomy away from tumour → full-thickness excision with perivesical fat, **frozen-section margins** → reimplant ureter if needed → watertight closure → irrigation (prevents seeding) → drain; cystogram before catheter removal.

Outcomes: 5-year OS (overall survival) 67–70%, intact bladder 65–74%; salvage RC up to a quarter; late recurrence → lifelong follow-up.

C. Trimodality therapy

Components: (1) **maximal, visibly complete TURBT**; (2) **RT** to bladder ± pelvic nodes; (3) concurrent **radiosensitising chemotherapy**.

Selection criteria:

- cT2 (selected cT3–4a), small solitary tumour.
- **No hydronephrosis, no extensive CIS**, visibly complete TURBT.
- Good bladder function; motivated, compliant patient.
- Only 6–19% of MIBC meet strict criteria.

Radiosensitising regimens:

Regimen	Evidence
Mitomycin C + 5-FU (fluorouracil)	**BC2001**: 2-year locoregional DFS (disease-free survival) **67% vs 48%** vs RT alone, no excess late toxicity
Cisplatin (± 5-FU or paclitaxel)	MGH (Massachusetts General Hospital)/RTOG (Radiation Therapy Oncology Group) protocols
Gemcitabine	Cisplatin-ineligible
Carbogen + nicotinamide (BCON)	Hypoxia modification

Radiation schedules:

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Split course (MGH/RTOG):  TURBT → chemoRT ~40 Gy → cystoscopy + biopsy
                           CR → consolidation to 60–66 Gy
                           Residual tumour → immediate cystectomy
Continuous (current):      TURBT → chemoRT 64 Gy/32 fr or 55 Gy/20 fr
                           → cystoscopy and imaging at 3–4 months
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Response assessment: cystoscopy with biopsy of tumour site, cytology, CT (computed tomography)/MRI (magnetic resonance imaging). **CR (complete response)** → surveillance. Non-response → salvage RC.

Salvage cystectomy:

- Needed in **10–30%** for persistent or recurrent muscle-invasive disease; it is the designed safety net.
- NMIBC (non-muscle-invasive) recurrence → TURBT + intravesical BCG (bacillus Calmette–Guérin).

Outcomes:

- 5-year CSS 50–84%, OS 36–74%.
- MGH/RTOG: CR 69%; 5-year DSS (disease-specific survival) 71%; 80% of survivors keep the bladder.
- Propensity-matched RC vs TMT: equal 5-year metastasis-free survival (about 75%) and CSS (81% vs 84%).
- Low late toxicity; good quality of life.

Comparison with radical cystectomy

	Radical cystectomy	TMT
Staging	Pathological	Clinical only
Nodes	Removed	Irradiated or not treated

	Radical cystectomy	TMT
Function	Diversion; sexual dysfunction	Native bladder; bowel/bladder toxicity
Follow-up	Upper tract, urethra	**Intensive cystoscopy for 10 years**
Evidence	Standard	No completed RCT (SPARE closed for poor accrual)

Follow-up after preservation

- Cystoscopy every 3–4 months for 2 years, 6-monthly years 3–4, then yearly to 10 years; CT urogram and chest CT every 3–6 months for 2 years, then yearly to 5 years; cytology.

Recent advances

- **Immunotherapy + chemoradiation:** PD-1/PD-L1 (programmed death-1 / ligand-1) inhibitors with TMT under trial (e.g. SWOG/NRG S1806 with atezolizumab; KEYNOTE-992 with pembrolizumab); results awaited.
- Hypofractionation 55 Gy/20 non-inferior (BC2001–BCON meta-analysis: same toxicity).
- Bladder sparing after complete response to NAC (± checkpoint inhibitor) with biomarker selection (DNA repair gene mutations, circulating tumour DNA); RETAIN: about half recurred.
- **Also know:** robotic partial cystectomy; molecular subtypes.

Diagram to draw: TMT flowchart from TURBT to chemoradiation, response check, surveillance or salvage cystectomy.

Key points to remember:

- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.
- Partial cystectomy: solitary <3 cm, no CIS, frozen-section margins, always PLND.

Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer?

Type: LAQ (25 marks)

Source: 2025-Winter Paper 4, Q1

Introduction

- After RC (radical cystectomy) alone, nearly half of MIBC (muscle-invasive bladder cancer) patients develop metastases from occult micrometastases.
- Cure improves with **systemic therapy before surgery** and a **meticulous lymph node dissection** (accurate staging, possible therapeutic value).
- UTUC (upper tract urothelial carcinoma) behaves alike, but RNU (radical nephroureterectomy) removes a kidney, changing chemotherapy timing.

A. Neoadjuvant therapy in MIBC

Rationale

- Treats micrometastases early; better tolerated before surgery, so more complete it.
- Downstages; **pT0 at cystectomy** gives about 90% 5-year recurrence-free survival.
- Drawback: delays surgery in non-responders.

Selection: cT2–T4a N0 M0, cisplatin-eligible. **Cisplatin-ineligible** if any one: GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) performance status ≥2, hearing loss grade ≥2, neuropathy grade ≥2, NYHA (New York Heart Association) class III–IV failure.

Trial	Regimen	Result
SWOG 8710 (Southwest Oncology Group)	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×3 then RC vs RC	pT0 38% vs 15%; 5-year OS (overall survival) 57% vs 43%, **P = 0.06**
BA06 30894 (EORTC/MRC, European Organisation for Research and Treatment of Cancer/Medical Research Council)	CMV (cisplatin, methotrexate, vinblastine) ×3	16% relative reduction in mortality; 10-year OS 36% vs 30%
ABC (Advanced Bladder Cancer) meta-analysis	Cisplatin combinations	OS HR (hazard ratio) 0.86; **5% absolute benefit** at 5 years
VESPER	ddMVAC (dose-dense MVAC) ×6 vs GC (gemcitabine–cisplatin) ×4	Organ-confined response 77% vs 63%; 3-year PFS (progression-free survival) 66% vs 56%; more toxicity
NIAGARA	Durvalumab + GC, then adjuvant durvalumab, vs GC	Better event-free survival (HR 0.68) and OS (HR 0.75)

- Single-agent cisplatin, carboplatin and neoadjuvant radiotherapy give **no survival benefit**.
- **Guidelines (EAU, European Association of Urology; AUA, American Urological Association; NCCN, National Comprehensive Cancer Network):** cisplatin-based NAC (neoadjuvant chemotherapy) for all eligible patients; ddMVAC 3–6 cycles or GC 4 cycles. NIAGARA has since added perioperative durvalumab + GC (the chapter predates it and calls immunotherapy unproven).
- **Cisplatin-ineligible:** proceed to RC; neoadjuvant pembrolizumab (PURE-01) and atezolizumab (ABACUS) are investigational.
- **Adjuvant** (pT3–4 or N+): cisplatin chemotherapy if not given before; **nivolumab** (CheckMate 274: disease-free survival 20.8 vs 10.8 months).

B. Lymphadenectomy in MIBC

- **Primary drainage:** obturator, internal and external iliac, presacral. **Secondary:** common iliac, para-aortic, paracaval.
- Node positivity: pT2 8–26%, pT3 33–50%, pT4 36–80%.

Template	Extent
Standard	Bilateral external and internal iliac and obturator nodes; genitofemoral nerve laterally, node of Cloquet caudally, up to the common iliac bifurcation
Extended	Adds common iliac and presacral nodes, up to the aortic bifurcation
Super-extended	Up to the inferior mesenteric artery

- **Evidence:** German phase III (limited vs extended): no survival benefit. **SWOG S1011** (most had NAC): median nodes 24 vs 39, **no difference** in disease-free survival (HR 1.10) or OS; more grade ≥3 events. So **standard bilateral PLND (pelvic lymph node dissection)** is the standard of care.
- **Node count:** fewer than 10 nodes doubled the risk of death (SWOG 8710 reanalysis). **Also know:** lymph node density, extranodal extension (prognostic).
- Unexpected nodes at RC: frozen section, proceed if resectable within the pelvis. Clinically node-positive: chemotherapy first, then RC in responders.

C. Neoadjuvant and adjuvant therapy in UTUC

- **Why neoadjuvant:** CKD (chronic kidney disease) is common and GFR falls after RNU, losing cisplatin eligibility. Give platinum **while both kidneys work**.
- Biopsy under-stages, so select by high-risk features (high grade, invasion on imaging, hydronephrosis).
- **EA8141** (phase II): accelerated MVAC, or gemcitabine–carboplatin if CrCl (creatinine clearance) 30–49; complete response 14%, below ypT2N0 60%.
- **AUA/SUO (Society of Urologic Oncology):** neoadjuvant platinum is a standard option for high-risk UTUC, especially if postoperative CrCl will fall below 50.
- **Adjuvant: POUT** (phase III, pT2–T4 or N+ after RNU): gemcitabine–platinum vs surveillance; disease-free survival HR 0.45, 3-year 71% vs 46%. Weaker with carboplatin, favouring neoadjuvant cisplatin. Adjuvant nivolumab (CheckMate 274) also included UTUC.

D. Lymphadenectomy in UTUC

- Node positivity 12–25%. Lymph drains **right to left and top to bottom**, so the template depends on side and level.

Primary site	Right	Left
Renal pelvis, upper two-thirds ureter	Hilar, paracaval, retrocaval, **interaortocaval**	Hilar, para-aortic
Lower third ureter	Ipsilateral common, external, internal iliac, obturator, presacral	Same

- Value: pN0 better than pNx; ≥8 nodes improved disease-specific survival above pT1. Indicated for high-grade or invasive disease; robotic yield equals open. Risk: chylous leak.

Diagram to draw: standard vs extended pelvic templates, and right vs left UTUC templates.

Key points to remember:

- Cisplatin combinations only: ABC 5% absolute 5-year survival gain; SWOG 8710 P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab improved event-free and overall survival.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.

Section 12 · Bladder Cancer & Diversion — Short answer questions (SAQ)

Optical coherence tomography in carcinoma bladder

Asked as:

- Optical coherence tomography in carcinoma bladder. — *Summer 2017 Paper IV Q3c (SAQ)*
- Optical coherence tomography in carcinoma bladder. — *Practice Set 5 Paper IV Q3c (earlier version) (SAQ)*

Type: SAQ (10 marks)

Introduction

- **WLC (white-light cystoscopy)** is the gold standard but subjective: many biopsies are negative and small lesions are missed. WLC misses flat lesions and cannot show depth.
- **OCT (optical coherence tomography)** is an **"optical biopsy"** technique listed among enhanced cystoscopic methods for NMIBC (non-muscle-invasive bladder cancer): real-time cross-sectional imaging of the bladder wall. It works like ultrasound but uses light.

Principle

- OCT is the **optical analogue of B-mode ultrasound**. **Near-infrared light** (about 1300 nm) is shone on tissue. It measures the echo delay of back-scattered **near-infrared light** instead of sound: using **low-coherence interferometry**: back-scattered light is measured against a reference beam.
- The delay of reflected light gives depth. Scanning across builds a cross-sectional image.
- **Resolution about 10–20 µm** (near histology). **Depth about 2–3 mm** (another source: only **1–2 mm depth**): urothelium, lamina propria and superficial muscle.
- First described in 1991 (ophthalmology).

Technique

- A fibre-optic probe passes through the cystoscope's working channel (channel) and touches the mucosa.
- WLC or blue light finds the lesion; the probe is placed **in contact** with it and read at once.

Appearance (image interpretation)

Normal	Tumour
Thin, dark urothelium	Thickened urothelium
Bright lamina propria; clear boundary	Layers lost
Muscle visible below	Broken boundary = invasion (T1)

Tissue	OCT appearance
Normal	Three layers: thin low-scatter urothelium, bright lamina propria, darker muscularis
Inflammation	Layers preserved but thickened
Papillary tumour (Ta)	Thickened urothelium, papillary pattern; lamina propria boundary intact
Invasive (T1)	**Loss of the urothelium–lamina propria boundary**

Uses (clinical uses)

- Tell tumour from inflammation at suspicious areas / separate benign from malignant flat lesions; fewer unnecessary biopsies.
- **Staging Ta vs T1** (lamina propria invasion); estimate Ta versus T1 during TURBT; helps in diverticula and thin walls.
- Check margins and completeness of TURBT (transurethral resection of bladder tumour).
- **UTUC (upper tract urothelial carcinoma):** through the ureteroscope, OCT correlates well with final grade and stage.

Accuracy (evidence) and limitations

- Small studies show **high sensitivity but lower (moderate) specificity**. Inflammation, scar, recent TURBT or BCG (bacille Calmette-Guérin; post-BCG change) give false positives (mimic tumour). Adding fluorescence cystoscopy (OCT to blue light) improves specificity.
- Small (tiny) field of view: needs WLC or PDD (photodynamic diagnosis) to guide it.
- Cannot judge deep muscle invasion (too shallow to stage muscle invasion); grading unreliable (no cellular detail, so cannot grade); cost; learning curve.
- Still investigational; not in routine guideline use (not recommended routinely by guidelines).

Other enhanced cystoscopy (techniques)

Technique	Principle
PDD (blue light, HAL, hexaminolevulinate)	Tumour porphyrins fluoresce red; finds (detects) about one-third more CIS (carcinoma in situ)
NBI (narrow-band imaging)	415/540 nm light absorbed by haemoglobin; shows vessels; many false positives
CLE (confocal laser endomicroscopy)	Cellular-level (cell-level surface) images after fluorescein

- **Also know:** Raman spectroscopy.

Diagram to draw: OCT image showing normal (three bladder) layers vs tumour with lost layers (loss of layering).

> **Key points to remember:**
> - Near-infrared interferometry ("ultrasound with light"); 10–20 µm resolution; 2–3 mm depth (another source: depth 1–2 mm).
> - Loss of layering suggests tumour.
> - Role: Ta vs T1; cannot assess deep muscle.
> - High sensitivity, moderate specificity, false positives from inflammation; investigational.

Re-look (second-look) TUR bladder tumour

Asked as:

- Re-look TUR bladder tumour. — *Summer 2017 Paper IV Q3d (SAQ)*
- Re-look TUR bladder tumour. — *Practice Set 5 Paper IV Q3d (earlier version) (SAQ)*

Type: SAQ (10 marks)

Read as second-look (repeat or restaging) TURBT.

Definition

- A **planned second (restaging) TURBT (transurethral resection of bladder tumour)** of the primary site, including detrusor muscle, a few weeks after the first. It completes resection and corrects staging. It is a quality indicator of NMIBC (non-muscle-invasive bladder cancer) care.

Rationale

- **TURBT is a staging operation first.** Completeness and muscle in the specimen are the quality markers.
- **Incomplete resection** — residual tumour is common:
  - At the resection site in **≥40%** when re-resected within weeks; tumour at first check cystoscopy in up to 45%.
  - **HG (high-grade) Ta:** residual up to 50%, upstaging up to 15%.
  - **Understaging:** in **HG T1**, re-resection shows worse disease or upstaging to muscle invasion in **25–30%**. This rises to about half if the first specimen had **no muscle**.
  - Understaging is about twice as likely without muscle.

Indications (AUA [American Urological Association] / EAU [European Association of Urology])

Recommended	Consider	Not needed
Incomplete first (initial) resection	HG Ta, large or multifocal (AUA) (residual up to 50%, upstaging up to 15%)	Low-grade Ta
All T1 tumours	Outside resection with no slides for review	Primary CIS (carcinoma in situ)
No detrusor muscle in specimen (except low-grade Ta and primary CIS)		Clear muscle invasion
		Cystectomy already planned

Timing

- **2–6 weeks** after the first TURBT (EAU); **within 6 weeks** (AUA), before BCG (bacille Calmette–Guérin). This allows healing yet avoids delaying intravesical BCG.

Technique

- Resect the **primary-site scar and base deep into detrusor**, plus any residual tumour; send the base / specimens separately. Use blue light or NBI (narrow-band imaging) if available; it helps.

Benefits / evidence

- Corrects stage and **changes treatment in about one-third**.

- **Divrik RCT (randomised controlled trial, 2010)** in T1: re-TUR improved recurrence-free and progression-free survival (lower recurrence and progression).
- Better BCG response with minimal residual tumour / when BCG follows re-TUR.

Result-based plan (decision after re-TUR)

- **Residual T1 at re-resection** means (predicts) progression in up to **80%**: consider early cystectomy.

```
No tumour (T0) / Ta       → BCG induction + maintenance
Residual HG T1            → early cystectomy (or BCG with close review)
Muscle invasion (≥T2)     → treat as muscle-invasive: neoadjuvant chemotherapy + cystectomy
```

Recent advances

- **En bloc resection** gives muscle (more detrusor) more often, less residual tumour, and may reduce re-TUR. It is not yet a proven substitute.
- Expert pathology review changes management in about one-third.
- **Also know:** maximal TURBT before trimodality therapy.

Diagram to draw: flowchart: T1 or no muscle → re-TUR 2–6 weeks → T2: cystectomy pathway / residual T1: consider cystectomy / T0–Ta: BCG; and bladder wall showing first resection scar and deep second resection into detrusor.

Key points to remember:

- Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
- Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
- Residual T1 → progression (progresses) in up to 80%.

Secondary tumours after urinary diversion

Asked as:

- Secondary tumors after urinary diversion. — *Summer 2021 Paper IV Q3e (SAQ)*
- Secondary tumors after urinary diversion. — *Practice Set 4 Paper IV Q3e (earlier version) (SAQ)* (also answered in 2021-Summer Paper 4, Q3e)

Type: SAQ (10 marks)

Definition

- New cancers arising in or next to bowel used for urinary diversion or bladder augmentation, usually after a **long latency** (10–20 years or more). Distinct from recurrent urothelial cancer.

Risk by diversion (German registry, Kälble 2011)

Diversion	Risk	Surveillance
Ureterosigmoidostomy	2.6% (6–29% in long-term series)	Regular colonoscopy **from 5 years**
Augmentation cystoplasty	1.6%	Endoscopy, often from 10 years
Colonic neobladder	1.3%	—
Ileocaecal pouch	0.14%	If symptomatic
Ileal neobladder	0.05%	No routine
Ileal conduit	0.02%	No routine

- **Pattern:** risk is highest where urothelium meets colon and is bathed in faeces; purely ileal diversions are lowest.

Pathogenesis

- **Urothelium–colonic junction** exposed to urine and faeces; tumours arise at the ureterocolic anastomosis.
- **N-nitrosamines** from bacterial action on urinary nitrates (probably enhancers, not sole cause); chronic inflammation; growth factors (EGF, epidermal growth factor); ornithine decarboxylase; abnormal mucin.
- Removing the urothelium prevents tumours experimentally.
- **Defunctioned anastomoses still form cancer:** when converting a ureterosigmoidostomy, excise the old anastomosis.

Pathology

- Mostly **adenocarcinoma**; also adenomatous polyps, urothelial carcinoma, sarcoma.
- **Augmentation:** adenocarcinoma and aggressive urothelial carcinoma, often advanced at diagnosis; earliest at 4 years. **Gastrocystoplasty** carries relatively high risk. Exstrophy and immunosuppression add risk.
- **Nephrogenic adenoma:** benign; commonest tumour after cystoplasty.

Presentation

- Haematuria, rectal bleeding, change in bowel habit, ureteric obstruction (hydronephrosis), recurrent infection, mass.

Management

- Biopsy and staging.

- **Ureterosigmoidostomy:** resection of the segment with the anastomosis; conversion to conduit or continent diversion.
- **Augmentation:** partial or total cystectomy with new diversion.
- **Prevention:** prefer ileal segments; counsel patients; lifelong follow-up.

Also know: after cystectomy, recurrent urothelial cancer may arise in the upper tracts or a retained urethra (urethral recurrence about 7–9%); needs separate surveillance.

Diagram to draw: ureterosigmoid anastomosis with tumour at the urothelial–colonic junction.

Key points to remember:

- Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
- Mostly adenocarcinoma at the anastomosis.
- Excise defunctioned anastomoses.

VI-RADS

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 4, Q3b

Definition

- **VI-RADS (Vesical Imaging–Reporting and Data System):** a 5-point score for the likelihood of **muscle (detrusor) invasion** by bladder cancer on mpMRI (multiparametric magnetic resonance imaging). Panebianco et al., Eur Urol 2018.
- Standardises bladder MRI reporting, as PI-RADS (Prostate Imaging Reporting and Data System) does for the prostate.

Technique

- Multiplanar **T2W (T2-weighted)**, **DWI (diffusion-weighted imaging)** with ADC (apparent diffusion coefficient) map, and **DCE (dynamic contrast-enhanced)** T1.
- Done **before TURBT (transurethral resection of bladder tumour)**: resection oedema mimics invasion. Distend the bladder adequately.

Principle

- On T2W the detrusor is a **low-signal (dark) line**; urine, mucosa and submucosa are bright. The question is whether tumour **interrupts that dark line**.

Scoring

Score	Invasion	T2W	DCE	DWI
1	Highly unlikely	<1 cm, ± stalk; dark line intact	No early muscle enhancement	Muscle continuous
2	Unlikely	>1 cm with stalk and thick inner layer, or sessile; line intact	Inner layer enhances only	As 1
3	Equivocal	No category 2 features; line intact	No muscle interruption	As 1
4	Likely	**Line interrupted**	Enhancement extends into muscle	High signal into muscle
5	Highly likely	**Into perivesical fat**	Through wall into fat	High signal through wall and fat

- **Cut-off ≥3 = suspicious for MIBC (muscle-invasive bladder cancer)**; the "equivocal" category already counts as positive.
- Sensitivity and specificity roughly 80–90%; DWI is the main functional sequence.

Clinical uses

- Local staging before TURBT; earlier referral for neoadjuvant chemotherapy and cystectomy.
- Patients who cannot have iodinated contrast (renal impairment, allergy).
- **MRI-first pathway** (UK BladderPath trial) shortened time to correct treatment for MIBC.
- Response after neoadjuvant chemotherapy or bladder preservation: under study.

Limitations

- Needs an experienced radiologist; post-TURBT or BCG (bacille Calmette–Guérin) inflammation; small, bladder-neck or diverticular tumours; poor for CIS (carcinoma in situ) and nodes; does not replace histology.

Diagram to draw: bladder wall with tumour breaching the dark detrusor line (VI-RADS 4).

Key points to remember:

- Five points; tests the dark muscle line on T2W.
- ≥3 suspicious for MIBC; 4 line broken, 5 fat.
- Scan before TURBT.
- Panebianco 2018; accuracy about 80–90%.

Virtual cystoscopy

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 4, Q3a

Definition and principle

- **Virtual cystoscopy (VC)** is a computer-generated, three-dimensional (3D) endoluminal view of the bladder. It is reconstructed from thin-section **CT (computed tomography)** or **MRI (magnetic resonance imaging)** data of a distended bladder.
- Software "flies through" the lumen using **surface or volume rendering**; multiplanar images also show wall thickness and extravesical spread.

Technique

Step	CT virtual cystoscopy	MR virtual cystoscopy
Distension	Air or CO2 through a catheter (negative contrast), or excretory-phase contrast (positive contrast, no catheter)	Urine-filled bladder; urine bright on T2
Acquisition	Thin slices, supine and prone (moves dependent urine off the posterior wall)	3D T2 or T1 contrast sequences
Drawback	Radiation; catheter for gas	Cost, time, motion artefact

Advantages

- Non-invasive; no anaesthesia.
- Sees where a scope cannot: through a **tight urethral stricture**, inside **diverticula**, beyond a large median lobe.
- With **CTU (CT urography)**, upper tracts and staging are imaged in one study (Chapter 5: high accuracy for bladder tumours if well distended).

Limitations

- Detects polypoid tumours ≥5 mm well; misses **small and flat lesions, including CIS (carcinoma in situ)**.
- No colour or mucosal detail, **no biopsy and no treatment**.
- False positives: trabeculation, clot, debris, folds of an underdistended bladder, prostatic impression.

Indications (selective)

- Unfit for, or refusing, cystoscopy; urethral stricture; diverticulum; bladder evaluation during CTU for haematuria.
- **Guideline position:** EAU (European Association of Urology) and AUA (American Urological Association) do not recommend VC in place of cystoscopy. Flexible cystoscopy remains the standard.

Related advances

- **Also know:** MRI **VI-RADS (Vesical Imaging–Reporting and Data System)** scores muscle invasion (Chapter 6); blue-light and narrow-band cystoscopy improve CIS detection (Chapter 8); virtual ureteroscopy uses the same principle.

Diagram to draw: thin-slice CT of a gas-distended bladder, then a 3D endoluminal fly-through showing a polypoid tumour.

Key points to remember:

- 3D endoluminal reconstruction from thin-section CT or MRI.
- Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
- No biopsy; does not replace cystoscopy.
- Useful beyond strictures and inside diverticula.

Bladder transplantation

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 4, Q3f

Rationale

- After cystectomy, or when the bladder cannot be reconstructed, the usual substitute is **bowel**. Bowel in the urinary tract causes mucus, metabolic acidosis, stones, infection and malignancy risk.
- Chapter 135: two frontiers aim to **remove bowel from urinary reconstruction**: tissue engineering and bladder transplantation.

Tissue engineering (why it has not succeeded)

- **Atala et al. (2006):** first seeded biodegradable construct for neurogenic augmentation. A later multi-institution trial showed **no significant gain in compliance or capacity** (Chapter 48).
- Engineered conduit trial: **half developed stenosis or stricture**, because of stiff material, poor cell coverage and **no blood supply** (Chapter 135).
- Engineered bladder: storage only (Chapter 111). See 2022-Summer Paper 4, Q2.

Bladder transplantation: development (Chapter 135)

- A **VCA (vascularised composite allograft)**: the donor bladder is transplanted with its own blood vessels.
- Stepwise preclinical work at USC (University of Southern California): pigs, perfused cadavers, then brain-dead human donors.

Technique (robotic)

1. **Robotic retrieval** of the bladder on its vesical vascular pedicles from the donor.
2. **Back-table vascular reconstruction**.
3. **Robotic implantation**: vessels to recipient iliac vessels; ureters joined to the graft.
4. **ICG (indocyanine green) fluorescence** confirms perfusion. Vascularity and mucosal integrity held for over 12 hours.

First clinical case

- Reported in **2025** from Los Angeles: a combined kidney and bladder transplant from a deceased donor, in a dialysis patient who had lost most of his bladder.

Problems

- **Lifelong immunosuppression**: infection, malignancy, drug nephrotoxicity. This is the major limitation (Chapter 135), especially after cancer.
- **Denervated graft**: stores urine; voiding may need straining or CIC (clean intermittent catheterisation).
- Rejection monitoring, donor supply, unknown long-term function.

Likely candidates

- Already immunosuppressed patients (kidney transplant recipients) with an absent or non-functioning bladder.

Diagram to draw: donor bladder on its vascular pedicles anastomosed to recipient iliac vessels, ureters entering the graft.

Key points to remember:

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

Artificial urinary bladder

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 4, Q3d

Read as a man-made bladder substitute (prosthetic, engineered or transplanted). Bowel neobladder: see Practice Set 4 Paper 2, Q3e.

Need

- Bowel substitutes absorb urine and secrete mucus, causing acidosis, mucus, stones, infection, perforation and cancer. The aim is a substitute that **is bladder-like, not bowel**.

Requirements of an ideal substitute

- Low-pressure storage of adequate volume; continence.
- Complete, voluntary emptying; no reflux or upper-tract damage.
- Biocompatible: no encrustation, infection, rejection or cancer risk; durable.

Approaches

Approach	Principle	Status
Alloplastic (prosthetic) bladder	Synthetic reservoir (silicone, polymers) joined to ureters and urethra	Failed: encrustation, infection, leak at the tissue–prosthesis junction, migration, fibrosis
Tissue-engineered bladder	Autologous urothelial and muscle cells seeded on a biodegradable scaffold, omental wrap	Atala (2006): 7 patients improved; phase II trial showed no significant gain in capacity or compliance; experimental
Engineered conduit	Seeded scaffold as urinary conduit	Neo-Urinary Conduit trial: half developed stenosis or stricture; stiff, no blood supply
Bladder transplantation	Robotic retrieval and transplant of a vascularised composite bladder allograft	Feasible in models and brain-dead donors; first human case reported in 2025; needs immunosuppression

Why progress is slow

- **Vascularity:** thick constructs die centrally.
- **Innervation and contraction:** an engineered bladder stores urine but cannot yet empty voluntarily.
- **Urine–material interface:** foreign material encrusts and becomes infected.
- Fibrosis and contracture.

Alternatives in clinical use

- Augmentation (bowel, ureterocystoplasty, autoaugmentation); continent or orthotopic diversion; ileal conduit.
- **Also know:** neuroprosthetic emptying (sacral anterior root stimulator), artificial urinary sphincter, intraurethral valve-pump devices for underactive bladders.

Future

- Bioprinted vascularised constructs, stem cells (urine-derived, iPSCs, induced pluripotent stem cells), smart scaffolds.

Diagram to draw: biopsy → cell culture → seeding on bladder-shaped scaffold → omental wrap → anastomosis to bladder base.

Key points to remember:

- Prosthetic bladders failed (encrustation, infection, leak).
- Atala 2006 promising; later trial no gain; still experimental.
- Barriers: vascularity, innervation, urine–material interface.

Rapid Revision Sheet – Section 12 · Bladder Cancer & Diversion

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion (LAQ)

- Three principles: reliable outlet, low pressure (detubularise and make a sphere), ≥500 mL; 500–600 mL by 1 year.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR (estimated glomerular filtration rate) >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.
- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; refluxing ureteric anastomosis is acceptable (USC (University of Southern California) trial).
- Obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- No diversion is proven best for quality of life; counsel all options.

Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy (LAQ)

- Single instillation within 6 h; never BCG immediately (SWOG S0337 gemcitabine).
- Only maintenance BCG reduces progression; SWOG 3-year schedule.
- BCG-unresponsive: cystectomy; pembrolizumab, nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

Discuss bladder preservation surgery for TCC of urinary bladder. (LAQ)

- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.
- Partial cystectomy: solitary <3 cm, no CIS, frozen-section margins, always PLND.

Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? (LAQ)

- Cisplatin combinations only: ABC 5% absolute 5-year survival gain; SWOG 8710 P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab improved event-free and overall survival.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.

Optical coherence tomography in carcinoma bladder (SAQ)

- Near-infrared interferometry ("ultrasound with light"); 10–20 μm resolution; 2–3 mm depth (another source: depth 1–2 mm).
- Loss of layering suggests tumour.
- Role: Ta vs T1; cannot assess deep muscle.
- High sensitivity, moderate specificity, false positives from inflammation; investigational.

Re-look (second-look) TUR bladder tumour (SAQ)

- Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
- Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
- Residual T1 → progression (progresses) in up to 80%.

Secondary tumours after urinary diversion (SAQ)

- Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
- Mostly adenocarcinoma at the anastomosis.
- Excise defunctioned anastomoses.

VI-RADS (SAQ)

- Five points; tests the dark muscle line on T2W.
- ≥3 suspicious for MIBC; 4 line broken, 5 fat.
- Scan before TURBT.
- Panebianco 2018; accuracy about 80–90%.

Virtual cystoscopy (SAQ)

- 3D endoluminal reconstruction from thin-section CT or MRI.
- Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
- No biopsy; does not replace cystoscopy.
- Useful beyond strictures and inside diverticula.

Bladder transplantation (SAQ)

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

Artificial urinary bladder (SAQ)

- Prosthetic bladders failed (encrustation, infection, leak).
- Atala 2006 promising; later trial no gain; still experimental.
- Barriers: vascularity, innervation, urine–material interface.

Section 13 · Kidney & UTUC — Long answer questions (LAQ)

Radiology for solid renal masses.

Type: LAQ (25 marks)

Source: 2018-Summer Paper 4, Q1

Introduction

- Most renal masses are now found incidentally. Imaging is the **strongest predictor of malignancy**; size and **enhancement** are the two key variables.
- About 20% of solid enhancing cT1 masses are benign (oncocytoma, fat-poor AML (angiomyolipoma)).
- **Goals:** solid or cystic? enhancing? fat? subtype? stage and surgical anatomy.

A. CT (computed tomography): the standard test

Renal mass protocol (dedicated, thin slices):

Phase	Timing	Use
Non-contrast	–	Baseline HU (Hounsfield units), fat, calcification, haemorrhage
Corticomedullary	30–70 s	Hypervascular masses; arteries; clear cell vs other subtypes
Nephrographic	90–120 s	**Most sensitive** for detection (tumour enhances less than parenchyma); renal vein and IVC (inferior vena cava) thrombus
Excretory	7–10 min	Collecting system relation

Enhancement thresholds (rise from unenhanced value):

Rise	Meaning
≥20 HU	Enhancing: solid tumour
10–20 HU	Equivocal (MRI, magnetic resonance imaging, or CEUS)
<10 HU	Non-enhancing: cyst

- **Pitfall:** pseudoenhancement of small (<10–15 mm) cysts; dual-energy CT iodine maps help.
- **Macroscopic fat = AML:** fat ≤ −10 to −20 HU on unenhanced CT is diagnostic. Fat **with calcification** suggests RCC (renal cell carcinoma).
- **Fat-poor AML:** hyperdense on unenhanced CT, homogeneous, avid early enhancement and washout.
- **Subtype clues:** clear cell RCC enhances early, avidly and heterogeneously (necrosis); papillary RCC is homogeneous and **hypoenhancing**; chromophobe and oncocytoma intermediate, with a central scar. Overlap is large.

B. MRI

- **Indications:** contrast allergy, CKD (chronic kidney disease), equivocal CT enhancement, young patients, thrombus. Enhancement = signal rise ≥15–20%. Macrocyclic gadolinium carries very low NSF (nephrogenic systemic fibrosis) risk.

Feature	Clear cell	Papillary	Chromophobe / oncocytoma	AML
T2 signal	High, heterogeneous	**Low**, homogeneous (haemosiderin)	Central T2-bright scar	Fat signal
Enhancement	Avid (>200%)	Weak	Intermediate / avid	Avid
Diffusion (ADC, apparent diffusion coefficient)	High	Lowest	Intermediate	–
Chemical shift	**Signal drop** (intracellular lipid, ~59%)	No drop	–	**India-ink** artefact; loss on fat saturation

- **ccLS (clear cell likelihood score, 1–5):** structured MRI algorithm for solid cT1a masses using T2 signal, corticomedullary enhancement and microscopic fat, plus diffusion and **segmental enhancement inversion** (favours oncocytoma). Score 4–5 suggests clear cell RCC; 1–2 a non-clear-cell tumour. It guides biopsy versus surveillance.

C. Ultrasound and CEUS (contrast-enhanced ultrasound)

- Grey-scale: cyst versus solid; poor below 2 cm. AML is hyperechoic, but so are some small RCCs.
- **CEUS** (microbubbles, not nephrotoxic): shows enhancement; used in CKD or allergy, indeterminate lesions, ablation and surveillance.

D. Cystic masses: Bosniak 2019 (CT and MRI)

Class	Key feature	Malignancy / action
I	Simple, thin wall ≤2 mm, no septa	Benign; none
II	1–3 thin septa; homogeneous hyperdense (≥70 HU unenhanced)	Benign; none
IIF	Wall or septa 3 mm, or ≥4 thin septa	<10%; follow 6, 12 months, then yearly for 5 years
III	Wall or septa ≥4 mm, or irregular	~50%; surgery or surveillance
IV	**Enhancing nodule**	~90%; treat

E. Functional and molecular imaging

- **99mTc-sestamibi SPECT/CT** (single-photon emission CT): mitochondria-rich **oncocytoma** and hybrid oncocytic or chromophobe tumours take it up; clear cell RCC does not.
- **Girentuximab PET (positron emission tomography)** targets CA-IX (carbonic anhydrase IX) on clear cell RCC. **ZIRCON** (^{89}Zr-girentuximab, indeterminate cT1 masses): about 85% sensitivity and 87% specificity for clear cell RCC.
- **FDG-PET** (fluorodeoxyglucose PET): not recommended (low sensitivity). PSMA (prostate-specific membrane antigen) PET: experimental.

F. Renal mass biopsy

- **Indications:** result may change management; suspected lymphoma, metastasis or infection; before ablation or surveillance; metastatic disease. Not needed if surgery is planned regardless.
- Cores: PPV (positive predictive value) 98.8%; subtype accuracy ≥90%, grade 60–80%; non-diagnostic ~14%. Seeding almost nil. Oncocytoma cannot be confirmed on cores.

G. Staging and surgical planning

- **Venous thrombus:** CT sensitivity 78% (renal vein), 96% (IVC). MRI or multiplanar CT shows the cephalad tip; enhancing thrombus is tumour, non-enhancing is bland.

Mayo level	Extent	T stage
0	Renal vein	T3a
I	IVC ≤2 cm above renal vein	T3b
II	IVC below hepatic veins	T3b
III	Intrahepatic IVC, below diaphragm	T3b
IV	Above diaphragm or atrium	T3c

- Nodes ≥2 cm usually malignant. Chest CT; bone scan only if symptomatic.
- **Nephrometry:** R.E.N.A.L. (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location), PADUA, C-index: predict complexity and complications; guide partial versus radical nephrectomy.
- **After ablation:** persistent or new enhancement = residual or recurrence.
- **Recent:** radiomics and AI (artificial intelligence).

Diagram to draw: algorithm: incidental mass → renal-protocol CT/MRI → fat (AML) / cyst (Bosniak) / enhancing solid → biopsy or treat.

Key points to remember:

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; ^{89}Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

Section 13 · Kidney & UTUC — Short answer questions (SAQ)

Nephron Saving Surgery (NSS)

Type: SAQ (10 marks)

Source: 2017-Winter Paper 4, Q3f

Definition

- Complete tumour removal with **maximal preservation of functioning parenchyma**, mainly PN (partial nephrectomy).
- **Rationale**: RN (radical nephrectomy) causes CKD (chronic kidney disease), linked to cardiovascular events and death; PN gives equal cancer control with better function.

Indications

Category	Situations
Imperative	Solitary kidney, bilateral tumours, RN would mean dialysis
Relative	Contralateral kidney threatened (stone, stenosis, diabetes, hypertension); hereditary RCC (renal cell carcinoma)
Elective	**cT1a with normal contralateral kidney: standard of care**

- Increasingly used for cT1b–T2 when feasible. AUA (American Urological Association) 2021: RN only if high complexity, no CKD or proteinuria, and post-RN eGFR (estimated glomerular filtration rate) likely >45.

Pre-operative assessment

- Multiphase CT (computed tomography) or MRI; **R.E.N.A.L. nephrometry score** (radius, exophytic, nearness to collecting system, anterior/posterior, location) predicts complexity.

Technique (open, laparoscopic or robotic)

1. Expose kidney; **intraoperative ultrasound** maps the tumour.
2. Hilar control; clamp artery (± vein), selective, or off-clamp.
3. Excise with a thin rim, or **enucleate** along the pseudocapsule (familial, multifocal tumours).
4. Close collecting system and vessels; renorrhaphy; unclamp; drain.

- **Robotic PN**: shorter warm ischaemia and fewer conversions than laparoscopy.

Ischaemia

- Aim warm ischaemia **<25–30 minutes**; **cold ischaemia (ice slush)** if longer is expected, especially in a solitary kidney.
- Function depends mainly on **parenchyma preserved**. Randomised trials: no functional benefit of off-clamp, hypothermia or mannitol with a normal contralateral kidney.
- **Trifecta**: negative margin + no major complication + warm ischaemia ≤25 minutes.

Complications

- **Urine leak** (stent, drain), **bleeding/pseudoaneurysm** (angioembolisation), AV (arteriovenous) fistula, renal insufficiency, hyperfiltration proteinuria (ACE (angiotensin-converting enzyme) inhibitors).

Outcomes

- Local recurrence 1–2% (elective), 3–5% (imperative). Margin width does not matter if negative.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011)**: RN vs PN for ≤5 cm: PN better function, similar oncology; overall survival slightly better with RN, but trial flawed.
- **Also know:** thermal ablation for <3 cm; surveillance in frail patients.

Diagram to draw: tumour excised with rim, clamp on renal artery, two-layer renorrhaphy.

> **Key points to remember:**
>
> - Imperative, relative, elective; elective PN is standard for cT1a.
> - Warm ischaemia ≤25 min; preserved parenchyma matters most.
> - EORTC 30904: PN better function, similar cancer control.

Immuno modulation in Renal, cell carcinoma

Type: SAQ (10 marks)

Source: 2025-Winter Paper 4, Q3a (also answered in 2024-Winter Paper 2, Q3e)

Assumed meaning: immunomodulation (immunotherapy) in renal cell carcinoma.

Rationale

- RCC (renal cell carcinoma) is immunogenic: rare (<1%) spontaneous regression after nephrectomy, tumour-infiltrating T cells.
- **Checkpoints:** PD-1 (programmed death-1) on T cells binds PD-L1 on tumour and switches T cells off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) damps priming. Blockade **releases the brake** on existing immunity.
- VEGF (vascular endothelial growth factor) suppresses immunity, so TKIs (tyrosine kinase inhibitors) synergise.

Cytokines (non-specific stimulation)

	IFN-α (interferon-alpha)	High-dose IL-2 (interleukin-2)
Response	10–15%	15–20%
Durable complete response	<2%	**7–9%**
Toxicity	Flu-like	**Capillary leak**; intensive care
Role	Obsolete	Selected fit patients

- High-dose IL-2: 600,000–720,000 IU/kg intravenously every 8 hours, maximum 15 doses; over 60% of complete responders stay disease-free.

Checkpoint inhibitors (clear cell)

Trial	Regimen vs comparator	Result
CheckMate 025	Nivolumab vs everolimus, second line	OS (overall survival) 25.0 vs 19.6 months
CheckMate 214	Nivolumab + ipilimumab vs sunitinib	Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9%
KEYNOTE-426 / CheckMate 9ER	Pembrolizumab + axitinib / nivolumab + cabozantinib	Better PFS (progression-free survival) and OS
CLEAR	Pembrolizumab + lenvatinib	PFS 23.9 vs 9.2 months; response 71%

- **First line:** IO/TKI (immuno-oncology plus TKI) for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor only.
- After IO, switch to a VEGF-targeted TKI; IO rechallenge is not recommended.

Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC (including resected metastases): disease-free survival HR 0.63; later OS benefit.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis. Grade 2: hold, prednisolone; grade 3–4: stop, high-dose steroids.

Also know: PD-L1 does not select; sarcomatoid responds well; vaccines and allogeneic transplant failed.

Diagram to draw: T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

Key points to remember:

- High-dose IL-2: durable complete response 7–9%.
- CheckMate 214: nivolumab + ipilimumab, OS HR 0.63.
- KEYNOTE-564: adjuvant pembrolizumab.

Zero-ischaemia partial nephrectomy (laparoscopic and robotic)

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 4, Q3e (clamping strategies and laparoscopic partial nephrectomy haemostasis also in 2019-Summer Paper 3, Q1)

Definition and rationale

- **Zero-ischaemia PN (partial nephrectomy):** excision and renorrhaphy **without clamping the main renal artery**, so the kidney stays perfused (Gill and colleagues, 2011).
- **Why:** warm ischaemia injures nephrons (historical limit 30 minutes; trifecta ≤25). Most relevant in a **solitary kidney or CKD (chronic kidney disease)**.

Techniques

Technique	Method
Off-clamp	Hilum looped for rescue but not clamped; cold-scissor excision with strong suction
Anatomical (super-selective)	Microdissect the higher-order arterial branches entering the tumour; occlude only these with small bulldogs
Selective segmental clamping	Only the tumour's segment is ischaemic ("minimal ischaemia")
Parenchymal compression	Parenchymal clamp across a polar tumour
Enucleation	Pseudocapsule plane bleeds less, so suits off-clamp work

Aids

- **3D (three-dimensional) CT (computed tomography) reconstruction** maps segmental arteries; intraoperative Doppler ultrasound.
- **NIRF (near-infrared fluorescence) with ICG (indocyanine green)** on robotic consoles: after selective clamping, perfused kidney glows and the ischaemic tumour zone stays dark.
- **Robot:** wristed instruments and fast suturing ease off-clamp work.
- **Haemostasis:** clip or suture bed vessels as met; sliding-clip or barbed renorrhaphy; clamp if bleeding hides the margin.

Selection

- **Ideal:** small (cT1a), exophytic, peripheral tumour; experienced surgeon.
- **Caution:** endophytic, hilar or central tumours.

Evidence

- Off- versus on-clamp: **better early eGFR (estimated glomerular filtration rate), more blood loss**; **no difference beyond the early period**.
- **Robotic RCTs (randomised controlled trials, e.g. CLOCK):** no difference in eGFR or split function up to 2 years.
- RCT of super-selective clamping with NIRF: no 6-month gain; stopped for futility.

- **Concept:** parenchymal volume preserved, not ischaemia time within accepted limits, governs function.
- **Also know:** early unclamping; hypothermia for long ischaemia; mannitol no benefit.

Diagram to draw: hilum with a micro-bulldog on a tertiary branch; tumour zone dark on NIRF.

Key points to remember:

- Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
- NIRF with ICG confirms the right branch.
- RCTs: no long-term eGFR gain; more blood loss.
- Parenchyma preserved matters more than clamp time.

Rapid Revision Sheet – Section 13 · Kidney & UTUC

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Radiology for solid renal masses. (LAQ)

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; ^{89}Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

Nephron Saving Surgery (NSS) (SAQ)

- Imperative, relative, elective; elective PN is standard for cT1a.
- Warm ischaemia ≤25 min; preserved parenchyma matters most.
- EORTC 30904: PN better function, similar cancer control.

Immuno modulation in Renal, cell carcinoma (SAQ)

- High-dose IL-2: durable complete response 7–9%.
- CheckMate 214: nivolumab + ipilimumab, OS HR 0.63.
- KEYNOTE-564: adjuvant pembrolizumab.

Zero-ischaemia partial nephrectomy (laparoscopic and robotic) (SAQ)

- Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
- NIRF with ICG confirms the right branch.
- RCTs: no long-term eGFR gain; more blood loss.
- Parenchyma preserved matters more than clamp time.

Section 14 · Adrenal & Retroperitoneum — Long answer questions (LAQ)

Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management

Asked as:

- Diagnosis and management for Adrenal Carcinoma. — *Summer 2018 Paper IV Q2 (LAQ)*
- Approach to evaluation of adrenal mass/s. — *Summer 2022 Paper IV Q3f (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction and epidemiology

- **Adrenal incidentaloma:** mass ≥1 cm found on imaging done for another reason.
- Two questions: **Is it malignant? (imaging)** and **Is it functioning? (biochemistry).**
- ACC (adrenocortical carcinoma): rare (0.7–2 per million per year), aggressive cortical tumour. **Bimodal** (children <10 years; adults 40–50 years); slight female excess.
- **Syndromes:** Li–Fraumeni (TP53), Beckwith–Wiedemann (IGF-2, insulin-like growth factor 2), Lynch, MEN-1 (multiple endocrine neoplasia 1), Carney complex, FAP (familial adenomatous polyposis). Surgery is the only cure.

Differential diagnosis of an adrenal mass

- Non-functioning adenoma (commonest); cortisol-secreting adenoma (5.3%); phaeochromocytoma (5.1%); aldosteronoma (1%); ACC (adrenocortical carcinoma, 4.7%); metastasis (2.5%); myelolipoma, cyst, haemorrhage.
- **Bilateral:** metastases, lymphoma, **tuberculosis** and histoplasmosis, haemorrhage, CAH (congenital adrenal hyperplasia), macronodular hyperplasia, bilateral phaeochromocytoma (MEN-2 [multiple endocrine neoplasia type 2], VHL [von Hippel–Lindau]).

Clinical features of ACC

- **Functional in 50–79% of adults** (90% of children):
  - Cortisol (commonest): rapidly progressive Cushing syndrome, often with virilisation.
  - Androgens: virilisation in women. **Oestrogen in a man (feminisation) suggests carcinoma.**
  - Aldosterone: rare.
- Non-functional: pain, fullness, weight loss, or incidental.

Hormonal work-up (all lesions; ESE–ENSAT, European Society of Endocrinology–European Network for the Study of Adrenal Tumours)

Axis / hormone	Tests
Glucocorticoid (cortisol)	**1 mg (1-mg) overnight dexamethasone suppression test (DST)** (or 24-h urine or late-night salivary cortisol): ≤1.8 µg/dL (50 nmol/L) excludes; >1.8 = **MACS** (mild autonomous cortisol secretion); >5 µg/dL autonomous; ACTH (adrenocorticotropic hormone)
Sex steroids and precursors	DHEA-S (dehydroepiandrosterone sulphate), androstenedione, testosterone, 17-OH-progesterone, 11-deoxycortisol; oestradiol in men and postmenopausal women (DHEA-S etc. only if ACC suspected)
Mineralocorticoid (aldosterone)	**ARR** (aldosterone-to-renin ratio; aldosterone:renin ratio) if hypertension or hypokalaemia

Axis / hormone	Tests
Catecholamine (catecholamines)	**Plasma free or 24-h urinary fractionated metanephrines** (plasma or urine **metanephrines**): must exclude phaeochromocytoma
Bilateral	17-hydroxyprogesterone (CAH); exclude adrenal insufficiency

- Why: confirms cortical origin, plans steroid cover, gives **tumour markers** for follow-up.
- **Exclude phaeochromocytoma before biopsy or surgery.** Biopsy only for suspected metastasis with known cancer.

Imaging

- CT (computed tomography), HU (Hounsfield units), MRI (chemical-shift magnetic resonance imaging), MDT (multidisciplinary team), FDG-PET (fluorodeoxyglucose positron emission tomography).

```
Unenhanced CT -> <=10 HU or MRI signal drop-out -> lipid-rich adenoma
              -> >10 HU -> washout CT (15 min): APW >60% or RPW >40% -> adenoma
                         -> indeterminate -> MDT: repeat imaging 6-12 months, FDG-PET or
surgery
```

- **APW** (absolute percentage washout) = (enhanced − delayed) ÷ (enhanced − unenhanced) × 100; **RPW** (relative) = (enhanced − delayed) ÷ enhanced × 100.
- Malignant features: heterogeneous, irregular, invasion, >4 cm. Macroscopic fat = myelolipoma.
- **Size and malignancy:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm 19.5% (another source: carcinoma in ~2% <4 cm, ~6% at 4–6 cm, ~25% >6 cm).
- **ACC size:** mean 10–12 cm; >90% exceed 5 cm.
- **CT in ACC:** irregular, heterogeneous, necrosis, calcification; **unenhanced >10 HU (Hounsfield units)** (mean ~39 HU vs ~8 HU in adenoma); **slow washout** (absolute <60%, relative <40% at 15 min). Growth ~1.9 cm/year.
- **MRI (magnetic resonance imaging):** no chemical-shift signal loss; best for IVC (inferior vena cava) thrombus and invasion.
- **FDG-PET/CT** (fluorodeoxyglucose positron emission tomography): avid, but cannot separate ACC from metastasis.
- **Staging:** CT chest, abdomen, pelvis; bone or brain only if symptomatic.
- **No biopsy** if ACC is suspected (seeding, poor accuracy); only for unresectable or metastatic disease, after excluding phaeochromocytoma.

ESE–ENSAT 2023 (European Society of Endocrinology–European Network for the Study of Adrenal Tumors) for incidentaloma

- Homogeneous **≤10 HU: benign, no further imaging whatever the size**; metanephrines may be omitted.
- MACS: screen for hypertension, diabetes and osteoporosis; surgery individualised.
- Management: surgery for functioning tumour, suspicious imaging, >4 cm in fit patients, growth ≥5 mm and >20%.

Pathology: Weiss score (≥3 of 9 = malignant)

Domain	Criteria
Nuclear / cytology	High nuclear grade; >5 mitoses per 50 high-power fields; atypical mitoses
Architecture	≤25% clear cells; diffuse architecture; necrosis
Invasion	Venous; sinusoidal; capsular

- SF-1 (steroidogenic factor 1) confirms cortical origin. **Ki-67** and mitotic rate grade the tumour (low vs high grade).

ENSAT staging

Stage	Definition	5-year disease-specific survival (ENSAT 2009)
I	T1 (≤5 cm), N0 M0	~80%
II	T2 (>5 cm), N0 M0	~60%
III	T3 (periadrenal fat) or T4 (adjacent organ, venous thrombus) or N1	~50%
IV	Distant metastasis only	~13%

- ENSAT differs from older UICC (Union for International Cancer Control) staging by moving T4 and N1 without metastasis into stage III.

Management of ACC

1. Surgery (stages I–III)

- **Open transperitoneal adrenalectomy** for suspected ACC, especially >6 cm or invasive. Laparoscopy risks capsule rupture and peritoneal carcinomatosis; only for small organ-confined tumours by experts.
- **R0, intact capsule:** rupture gives dismal results. **En bloc** removal of involved kidney, liver, spleen, pancreas or IVC; IVC thrombectomy (bypass if needed).
- **Locoregional lymphadenectomy** (periadrenal fat, renal hilum, para-aortic or paracaval) improves staging and may reduce recurrence.
- High-volume surgeon (ESE–ENSAT: >6, preferably ≥20 adrenalectomies a year). Hydrocortisone cover if cortisol-secreting.

2. Adjuvant therapy

- **Mitotane** (adrenolytic DDT (dichlorodiphenyltrichloroethane) derivative) for **high risk**: stage III, R1/Rx, or Ki-67 >10% (ESE–ENSAT 2018), for at least 2 years.
  - Target blood level **14–20 mg/L**; needs **glucocorticoid replacement** (it destroys the cortex and raises cortisol-binding globulin).
  - Toxicity: gastrointestinal, neurological, hepatic, hypothyroidism, gynaecomastia.
- **ADIUVO** (RCT, randomised controlled trial; R0, stage I–III, Ki-67 ≤10%): no recurrence-free survival benefit. So observe low-risk patients.
- **Adjuvant radiotherapy** (tumour bed): R1 or Rx, or stage III. It cuts local recurrence but not overall survival.

3. Advanced or metastatic (stage IV)

- **EDP-M** (etoposide, doxorubicin, cisplatin + mitotane) is first line. **FIRM-ACT** (2012): versus streptozotocin–mitotane, response 23% vs 9% and PFS (progression-free survival) 5.0 vs 2.1 months; overall survival not significantly better.
- Mitotane alone for low-volume, slow disease.
- **Surgery:** resect oligometastases or late recurrence; debulk only if >90% removable.
- **Hormone control:** metyrapone, ketoconazole, mifepristone, etomidate. Radiotherapy for bone and brain metastases.

Follow-up

- CT chest, abdomen, pelvis and baseline-raised hormones **every 3 months for 2 years**, then 6-monthly to 5 years.

Prognosis

- Overall 5-year survival **20–47%**. Bad factors: stage, R1, high Ki-67 or mitoses, cortisol excess, age, size >12 cm.

Recent advances

- Molecular classes (CIMP, CpG island methylator phenotype) predict prognosis. Checkpoint inhibitors give modest responses; IGF-1R (IGF-1 receptor) inhibitor linsitinib failed (GALACCTIC). ADIUVO-2 tests mitotane ± platinum–etoposide in high risk.

Diagram to draw: imaging and biochemistry algorithm above; algorithm: adrenal mass → hormones + CT (HU, washout) → suspected ACC → open R0 en bloc resection + LND (lymph node dissection) → risk-based adjuvant mitotane or radiotherapy.

Key points to remember:

- ≤10 HU = lipid-rich adenoma. APW >60%, RPW >40%.
- 1-mg DST: >1.8 µg/dL = MACS.
- Rule out phaeochromocytoma before biopsy.
- Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
- Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
- Open, R0, intact capsule, en bloc + lymphadenectomy.
- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

Rapid Revision Sheet – Section 14 · Adrenal & Retroperitoneum

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Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management (LAQ)

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- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

Section 15 · Renal, Nephrology & Transplant — Long answer questions (LAQ)

Allograft rejection, immunobiology of renal transplant, marginal kidney donor and newer immunosuppressant drugs

Asked as:

- Discuss pathophysiology of allograft rejection. Discuss newer immunosuppressive drugs in renal transplant. — *Winter 2017 Paper IV Q2 (LAQ)*
- Describe immunobiology of renal transplant and newer immunosuppressant drugs. — *Summer 2020 Paper IV Q1 (LAQ)*
- Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs — *Practice Set 3 Paper IV Q1 (LAQ)*
- Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation — *Summer 2015 Paper IV Q2 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2020-Summer Paper 4, Q1)

Introduction

- Rejection is the recipient's immune response against donor (allo) antigens. The recipient's immune system recognises donor antigens as foreign. Immunosuppression has cut acute rejection: it has fallen from at least 60% (pre-cyclosporine era; "over 60%") to **below 10%**. Slow, subclinical **antibody-mediated injury** (**antibody injury**) is now the main problem and the main cause of late graft loss.
- Organ shortage drives use of **marginal donors**. Their kidneys carry more DGF (delayed graft function) and shorter graft survival, but transplantation still beats dialysis.

Part A: Immunobiology and pathophysiology of rejection

Transplant (target) antigens

Antigen	Features
HLA (human leucocyte antigen) = MHC (major histocompatibility complex)	Class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells. Class II mismatch is most immunogenic
ABO blood group	On endothelium; incompatibility causes hyperacute rejection unless desensitised
Minor and non-HLA antigens	Rejection despite HLA identity

- **Sensitisation**: transfusion, pregnancy or previous transplant create (creates) preformed anti-HLA antibodies.
- **Testing**: CDC (complement-dependent cytotoxicity) and flow crossmatch (donor lymphocytes with recipient serum); **Luminex single-antigen beads** define each recipient's antibodies and allow a **virtual crossmatch**; cPRA (calculated panel-reactive antibody) expresses sensitisation.

Innate immunity: the first trigger

- **Ischaemia–reperfusion injury** activates innate immunity and up-regulates MHC: it releases danger signals (DAMPs, damage-associated molecular patterns). These activate

Toll-like receptors, complement and macrophages, and up-regulate MHC on the graft. Hence DGF (delayed graft function) raises rejection risk.

- **Dendritic cells** bridge innate and adaptive immunity: activated, they up-regulate MHC and costimulatory B7 and migrate to lymph nodes.

Allorecognition (how T cells see the graft)

Pathway	Mechanism	Main role
Direct	Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells	Very strong (high T-cell frequency); strong early acute TCMR (T-cell-mediated rejection) / early acute T-cell rejection
Indirect	Recipient APCs (antigen-presenting cells) present **processed donor peptides on self-MHC**	Lifelong; helps B cells make DSA (donor-specific antibody), so alloantibody and chronic rejection
Semi-direct	Recipient APCs acquire **intact donor MHC** (membrane transfer) and present it	Links the two pathways (both pathways)

Three-signal model of T-cell activation

1. **Signal 1** (antigen): TCR (T-cell receptor) binds MHC–peptide → **calcineurin** dephosphorylates NFAT (nuclear factor of activated T cells) → IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation)**: **CD28 binds CD80/86 (B7)**. Signal 1 without signal 2 causes **anergy**. CTLA-4 (cytotoxic T-lymphocyte antigen-4) is the natural brake.
3. **Signal 3**: IL-2 binds **CD25** (IL-2 receptor) → **mTOR (mammalian target of rapamycin)** → cell cycle G1 to S → clonal proliferation, which needs de novo purine synthesis.

Effector mechanisms

- **Cellular**: CD4 Th1 cells (IL-2, IFN-γ (interferon-γ)) recruit macrophages; CD8 cytotoxic T cells kill by perforin and granzyme → **tubulitis, endarteritis**.
- **Humoral**: B cells with T-cell help become plasma cells → **DSA (donor-specific antibody)** binds endothelium → complement, leaving (leaves) **C4d** in (on) peritubular capillaries → **microvascular inflammation** (glomerulitis, capillaritis). NK (natural killer) cell injury is complement-independent, so AMR (antibody-mediated rejection) may be C4d-negative.
- **Regulation**: **Tregs** (regulatory T cells, FoxP3+) suppress responses; the basis of tolerance research.

Types of rejection (brief; detail in 2020-Summer Paper 1, Q3d; context for drug choice)

Type	Timing	Mechanism and features
Hyperacute	Minutes–hours	Preformed ABO/HLA antibody → complement, thrombosis; graft turns blue; **prevented by crossmatch**
Accelerated acute	2–5 days	Memory (anamnestic) response in sensitised recipients
Acute TCMR	Days–months (mostly first 6 months)	Interstitial infiltrate, tubulitis, arteritis
Acute AMR	Any time	DSA, C4d, microvascular inflammation
Chronic active AMR	Months–years	**Transplant glomerulopathy** (GBM (glomerular basement membrane) double contours), arterial intimal fibrosis; commonest cause of late graft loss

- **Also know: subclinical** rejection (histology, stable function). **Risk factors**: HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF, **non-adherence**,

retransplant.

Banff classification (biopsy is the gold standard; Banff grades biopsy findings and classifies TCMR and AMR)

Category	Key criteria
Borderline	Foci of tubulitis with mild interstitial inflammation
TCMR IA / IB	Interstitial inflammation >25% with moderate (IA) or severe (IB) tubulitis
TCMR IIA / IIB	Mild-moderate (IIA) or severe (IIB) intimal arteritis
TCMR III	Transmural arteritis or fibrinoid necrosis
Active AMR	All three: (1) tissue injury (microvascular inflammation, arteritis, TMA (thrombotic microangiopathy)); (2) antibody–endothelium interaction (C4d, microvascular inflammation or gene transcripts); (3) **circulating DSA**

Diagnosis and monitoring

- Rising creatinine; **de novo DSA** (donor-specific antibody) predicts graft loss; biopsy; **dd-cfDNA (donor-derived cell-free DNA)** (negative predictive value >90%) avoids unnecessary biopsy; gene-expression panels.
- **Exclude**: obstruction, leak, renal artery stenosis, CNI (calcineurin inhibitor) toxicity, BK virus nephropathy.

Treatment

- **TCMR**: IV (intravenous) pulse steroids; **rATG (rabbit antithymocyte globulin)** / ATG (antithymocyte globulin) for Banff II–III.
- **AMR** (not standardised): steroids, **IVIG (intravenous immunoglobulin)**, **plasmapheresis**, rituximab; eculizumab in selected patients.

Part B: Marginal kidney donor

Categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002: age **≥60**, or **50–59** with **≥2** of **hypertension**, **creatinine >1.5 mg/dL**, **death from CVA (cerebrovascular accident)**. About 70% higher risk of graft loss.

2. KDPI (Kidney Donor Profile Index): percentile of a 10-factor risk index. **KDPI >85%** replaced ECD in the US (December 2014). Low-KDPI kidneys go to recipients with the longest EPTS (Estimated Post-Transplant Survival): **longevity matching**.

3. DCD (donation after circulatory death): Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of therapy	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled

- Warm ischaemia needs rapid cannulation and cooling, and DCD kidneys usually go on machine perfusion. DGF is **higher (roughly 40–50%)**, but long-term survival is similar to DBD (donation after brain death).

4. Others: **AKI (acute kidney injury)** donors; **HCV (hepatitis C virus)-viraemic** donors into negative recipients with DAAs (direct-acting antivirals); **paediatric donors <15 kg** (en

bloc); older or hypertensive living donors (KDIGO (Kidney Disease: Improving Global Outcomes) 2017 risk-based approach).

- **Absolute exclusions**: metastatic or haematological cancer, melanoma, untreatable viral infection. Treated bacterial infection is acceptable; screen by serology and nucleic acid testing.

Assessment and optimisation

- **Remuzzi biopsy score** (glomeruli, tubules, interstitium, arteries, each 0–3): **0–3 single**, **4–6 dual kidney transplant**, **≥7 discard**.
- **Cold storage** at 4 °C in UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution slows metabolism.
- **HMP (hypothermic machine perfusion)** reduces DGF (**Moers, NEJM 2009**: about 21% vs 27%; greatest benefit in ECD kidneys); Campbell notes no long-term survival difference. Normothermic perfusion allows viability testing (limited experience).
- **ESP (Eurotransplant Senior Program)**: donors **≥65** to recipients **≥65**, prioritising short cold ischaemia over HLA (human leucocyte antigen) matching. Informed consent is essential.
- **Ojo (JASN 2001)**: ECD transplantation improves survival over dialysis, although graft survival is lower than with standard kidneys.
- With DGF, give antibody induction and delay nephrotoxic CNI (calcineurin inhibitor) therapy.

Part C: Newer immunosuppressive drugs (newer immunosuppressants)

Standard immunosuppression (current practice)

- **Three-signal model**: signal 1 is antigen (TCR (T-cell receptor) meeting MHC (major histocompatibility complex)); signal 2 is costimulation (CD28–CD80/86); signal 3 is IL-2 receptor signalling through mTOR, driving proliferation.
- **Induction** (>90%; over 90% receive induction) plus **tacrolimus + mycophenolate** ± steroids (mycophenolate: GI (gastrointestinal) upset, teratogenic). Acute rejection is below 10%; chronic **AMR (antibody-mediated rejection)** is now the main problem. Newer drugs aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

Drug	Mechanism (site)	Key points
Belatacept	**CTLA4-Ig**: binds CD80/86, blocks **CD28 costimulation (signal 2)**	FDA (Food and Drug Administration) 2011; IV monthly: IV 10 mg/kg induction, then **5 mg/kg 4-weekly**; CNI (calcineurin inhibitor)-free. **BENEFIT**: better GFR (glomerular filtration rate), more early acute rejection; 7-year better patient and graft survival. **Contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD (post-transplant lymphoproliferative disorder))
Extended-release tacrolimus	CNI: binds FKBP12 → inhibits calcineurin (**signal 1**), so less IL-2 transcription	Once daily, better adherence, steadier levels; nephrotoxicity, NODAT (new-onset diabetes after transplant), tremor
Mycophenolate	Inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis	Standard partner of tacrolimus

Drug	Mechanism (site)	Key points
mTOR inhibitors (sirolimus, everolimus)	Bind FKBP12 → block mTOR (**signal 3**), arresting G1 to S	Not nephrotoxic; fewer CMV (cytomegalovirus)/BK infections; antitumour effect. **TRANSFORM (2018)**: everolimus + reduced (low-dose, low) CNI non-inferior. Poor wound healing, **lymphocele**, proteinuria, dyslipidaemia
Basiliximab	Anti-CD25 (IL-2 (interleukin-2) receptor), non-depleting (**signal 3**)	Low-risk induction, **20 mg IV (intravenous) days 0 and 4**
rATG (rabbit antithymocyte globulin); **alemtuzumab**	T-cell depletion; anti-CD52	High-risk induction, DGF; infection (CMV), PTLD
Rituximab; bortezomib	Anti-CD20 (B cells); proteasome inhibitor (plasma cells)	AMR, desensitisation
Eculizumab	Anti-C5 (complement)	Selected AMR; aHUS (atypical haemolytic uraemic syndrome) recurrence
Imlifidase	Enzyme that **cleaves IgG**	**EMA (European Medicines Agency) 2020**: crossmatch-positive, highly sensitised patients
Anti-IL-6 therapy (tocilizumab, clazakizumab)	Blocks IL-6 pathway	Chronic active AMR

- **Also know:** mycophenolate inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis; anti-CD40 costimulation blockade under trial.
- **Voclosporin** is approved for lupus nephritis, **not transplantation**.

Recent advances

- **Tolerance induction** by mixed chimerism (combined kidney and donor stem-cell transplant) and Treg therapy; normothermic repair; **pig-kidney xenotransplant** into a living human (March 2024).

Diagram to draw: three-signal T-cell activation (APC–T-cell synapse showing signals 1–3) with drug sites (CNI at signal 1 / after signal 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis).

Key points to remember:

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, avoid (contraindicated) if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

Section 15 · Renal, Nephrology & Transplant — Short answer questions (SAQ)

Brain-stem death: guidelines for declaration and the Board of Medical Experts (brain dead committee)

Asked as:

- Brain dead committee. — *Summer 2018 Paper IV Q3c (SAQ)*
- Guidelines for declaration of brain stem death. — *Summer 2022 Paper IV Q3b (SAQ)*

Type: SAQ (10 marks)
(also answered in 2018-Summer Paper 4, Q3c)

Assumed meaning: the Board of Medical Experts that certifies brain-stem death under Indian law.

Definition and legal basis

- **Brain-stem death:** irreversible loss of consciousness and of the capacity to breathe; equivalent to death of the person.
- **India: THOA (Transplantation of Human Organs Act) 1994** legalised brain-stem death in India; **amended 2011** (Transplantation of Human Organs and Tissues Act); **Rules 2014**.
- Internationally: Harvard criteria (1968), AAN (American Academy of Neurology) criteria (updated 2023).

Composition of the Board of Medical Experts (India; Rules 2014)

Member	Role
1. Registered medical practitioner **in charge of the hospital**	Hospital authority
2. **Independent specialist**, nominated by the in-charge from a panel approved by the Appropriate Authority	Independent check
3. **Neurologist or neurosurgeon** from the approved panel (if unavailable: surgeon, physician, anaesthetist or intensivist)	Expert
4. Medical practitioner **treating the patient**	Clinical history

- **No member may belong to the transplant team.**

Certification procedure

- **Preconditions:** known irreversible cause, apnoeic coma on (a) ventilator; exclude **sedatives, relaxants**, hypothermia, metabolic or endocrine causes, shock.

Tests: all reflexes absent

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Fixed pupils, no light reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic	Turn head: eyes move with head	VIII, III, VI
Oculovestibular	Ice-cold water in each ear: no eye movement	VIII, III, VI
Motor in cranial nerve area	Supraorbital pressure: no grimace	V, VII
Gag and cough	Pharynx (touch) and tracheal suction: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial CO_2; arterial carbon dioxide tension) ≥60 mmHg** (or ≥20 mmHg above baseline). Abort if (for) hypotension, desaturation or arrhythmia.
- **Two sets of tests, at least 6 hours apart**, recorded on **Form 10**.
- **Ancillary tests** if examination is impossible (facial trauma): angiography, perfusion scan, EEG (electroencephalogram).

Linked duties (after declaration)

- **Mandatory request:** ICU (intensive care unit) staff must inform relatives of the option to donate (mandatory request for donation by ICU staff); a **transplant coordinator** counsels; next of kin give consent.
- **NOTTO** (National Organ and Tissue Transplant Organisation), with regional and state bodies (ROTTO, SOTTO: regional and state organ and tissue transplant organisations), allocates organs and keeps the registry.

Diagram to draw: flowchart: coma → preconditions → test 1 → 6 h → test 2 → Form 10 → consent → NOTTO.

> **Key points to remember:**
> - 4-member (Four-member) Board; no transplant-team member.
> - Two tests ≥6 h apart; Form 10.
> - Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.

Vascular access for dialysis.

Type: SAQ (10 marks)

Source: 2020-Summer Paper 4, Q3b

Introduction

- Haemodialysis needs blood flow of **250–350 mL/min**; access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Types

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino, 1966)	Radial artery to cephalic vein at wrist (vein end to artery side)	6–8 weeks	**First choice**: best patency, fewest infections; most fail to mature
Brachiocephalic AVF	Elbow	6–8 weeks	Higher flow, more steal
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene) loop or straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular	At once	Bridge or last resort; infection, central stenosis

- Non-tunnelled 10–12 Fr jugular or femoral catheters: acute dialysis only.

Planning

- **Hierarchy**: AVF → AVG → CVC; **distal before proximal; non-dominant arm first**. KDOQI (Kidney Disease Outcomes Quality Initiative) 2019: individual "ESKD Life-Plan".
- **Timing**: refer at eGFR (estimated glomerular filtration rate) <30; create AVF about **6 months** before dialysis.
- **Vein preservation**: no venepuncture, cannula or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment**: pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s"**: flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications

Complication	Treatment
Failure to mature (juxta-anastomotic stenosis)	Angioplasty, ligate side branches
Stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then fix stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft

- **Also know:** brachiobasilic transposition AVF; venous hypertension; high-output cardiac failure.

Monitoring

- Thrill and bruit, arm-elevation test, prolonged bleeding after needling, rising venous pressure; duplex when abnormal.

Recent advances

- **Endovascular AVF** (WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels.

Diagram to draw: radiocephalic end-to-side fistula at the wrist.

> **Key points to remember:**
> - AVF first, distal, non-dominant, 6 months ahead.
> - Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
> - Thrombosis sits on stenosis; steal → banding or DRIL.

Female kidney donors: special precautions

Asked as:

- Female kidney donors: enumerate special precautions — *Winter 2023 Paper IV Q3d (SAQ)*
- Female kidney donors: enumerate special precautions — *Practice Set 4 Paper IV Q3a (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3d)

Introduction

- Most living donors, especially in India, are **women** (wives, mothers). Guidance: **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** living donor guideline.

Before donation

1. **Free, uncoerced consent:** interview the donor alone; independent donor advocate; family pressure and gender inequity screened. In India the THOA (Transplantation of Human Organs Act) authorisation committee verifies this.
2. **Pregnancy:** pregnancy test before contrast CT (computed tomography) and surgery; **reliable contraception** during work-up. Pregnant women do not donate.
3. **Obstetric history:** previous **pre-eclampsia, gestational hypertension or gestational diabetes** signals higher future hypertension, diabetes and kidney risk; evaluate and counsel.
4. **Future pregnancy counselling (KDIGO):** donation **increases gestational hypertension and pre-eclampsia** (Garg et al., NEJM 2015: about 11% vs 5% in non-donors). Maternal and fetal outcomes are otherwise similar (Chapter 91).
5. **Oestrogen therapy:** combined OCPs (oral contraceptive pills) and HRT (hormone replacement therapy) raise **VTE (venous thromboembolism)** risk; stop them some weeks before surgery and give barrier or progestogen-only contraception; VTE prophylaxis.
6. **Gynaecological and cancer screening:** cervical cytology, breast examination or mammography (age-appropriate); pelvic ultrasound if indicated.
7. **Urine tests:** repeat urinalysis **outside menstruation** to avoid false haematuria; exclude recurrent UTI (urinary tract infection).
8. **GFR (glomerular filtration rate):** low muscle mass makes creatinine-based estimates unreliable; **measured GFR** preferred. Note nephron mismatch with a large recipient.
9. Correct **iron-deficiency anaemia**.

Operative

- Laparoscopic donor nephrectomy; **Pfannenstiel** extraction for cosmesis; transvaginal extraction possible.

After donation

- **Defer pregnancy** until recovery and stable function; many programmes advise at least 6–12 months (another source: often 6–12 months).
- In pregnancy: inform obstetrician; **high-risk antenatal care**; monitor BP (blood pressure), proteinuria and creatinine; low-dose aspirin may be considered.
- **Lifelong annual BP, creatinine and proteinuria**; avoid NSAIDs (non-steroidal anti-inflammatory drugs).

Diagram to draw: flowchart: consent → pregnancy test → hormone review → GFR → surgery → pregnancy follow-up.

Key points to remember:

- Ensure free consent (Indian gender gap).
- Pregnancy test; contraception; stop oestrogens.
- Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
- Defer pregnancy; high-risk antenatal follow-up.

Hospital authorization committee for live related kidney donation.

Type: SAQ (10 marks)

Source: 2023-Winter Paper 4, Q3e

Indian law; not covered in the chapter notes.

Legal framework

- **THOA (Transplantation of Human Organs Act) 1994**; amended 2011 as THOTA (Transplantation of Human Organs and Tissues Act); **Rules 2014**.
- Purpose: **prevent organ trade** while allowing genuine donation.

Donor categories

Donor	Approval
Near relative: spouse, son, daughter, father, mother, brother, sister, grandparents, grandchildren	Verified by the hospital's competent authority
Other than near relative (affection or attachment)	**Authorisation Committee approval mandatory**
Swap (paired) donation between near-relative pairs (2011)	Authorisation Committee
Foreign donor or recipient	Authorisation Committee; embassy certifies relationship

Constitution

- **Hospital-based** if a centre does more than **25 transplants a year**; otherwise **district or state committee**.
- **Members:** medical director or superintendent (chair); two senior doctors of the hospital **not in the transplant team**; two persons of high integrity and social standing (preferably one woman), nominated by the State; state committees add health-department nominees.

Functions and procedure

1. Receive the **joint donor–recipient application** with consent forms, fitness certificates and identity and relationship documents.
2. **Personal interview** of donor, recipient and families; proceedings **video-recorded**.
3. Verify **relationship** (documents; **HLA [human leukocyte antigen] or DNA testing** if doubtful).
4. Exclude **commercial dealing**: financial status, reasons, long association, middlemen.
5. Confirm donor understands risks and consents freely, without coercion.
6. Check **domicile** (other-state donors).
7. Decide promptly; decision recorded and displayed; reasons given if refused. **Appeal** to the State Government within **30 days**.

Restrictions

- **Minors and mentally ill people** not accepted as living donors except in exceptional, approved cases.
- **Foreign recipient with unrelated Indian donor** not approved.
- **Penalty** for commercial dealing: imprisonment up to 10 years and fine up to ₹1 crore; doctors' registration removed.

Diagram to draw: flowchart: application → near relative (competent authority) or other (committee) → interview, verify → decision → appeal.

Key points to remember:

- THOA 1994; amended 2011; Rules 2014.
- Hospital-based committee if >25 transplants a year; no transplant team member.
- Non-relatives, swap and foreigners need committee approval.
- Interview video-recorded; appeal within 30 days.

Marginal donor and Remuzzi score for pre-implantation biopsy

Asked as:

- Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys. — *Winter 2024 Paper IV Q3e (SAQ)*
- Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys. — *Practice Set 4 Paper IV Q3b (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2024-Winter Paper 4, Q3e; 2016-Summer Paper 3, Q3f)

"Preemptive" is read as pre-implantation (procurement) biopsy.

Definition

- A **marginal (extended) donor** is one whose kidney carries a **higher risk of DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- Still better survival than dialysis (Ojo, 2001).

Categories

- **ECD (expanded criteria donor), UNOS (United Network for Organ Sharing):**
  - Age **60 or over**, or
  - Age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**.
  - Graft-loss risk about 1.7 times that of a standard donor.
- **KDPI (Kidney Donor Profile Index) >85%** replaced ECD in the US (2014); 10 donor factors (age, height, weight, ethnicity, hypertension, diabetes, cause of death, creatinine, HCV [hepatitis C virus], DCD).
- **Others:** DCD (donation after circulatory death); donor AKI (acute kidney injury); HCV-positive donors; older or hypertensive living donors.

Remuzzi score (Karpinski/Remuzzi)

- Pre-implantation biopsy (ideally at least 25 glomeruli), read on rapid sections.
- **Four components, each 0–3: total 0–12.**

Component	0	1	2	3
Glomerular global sclerosis	None	<20%	20–50%	>50%
Tubular atrophy	None	<20%	20–50%	>50%
Interstitial fibrosis	None	<20%	20–50%	>50%
Arterial and arteriolar narrowing	None	Wall thickness < lumen diameter	Wall = or slightly > lumen	Wall far > lumen, severe narrowing or occlusion

Allocation by score

Total	Grade	Use
0–3	Mild	**Single** kidney transplant
4–6	Moderate	**Dual** kidney transplant
7–12	Severe	**Discard**

- Remuzzi (NEJM 2006): biopsy-guided older-donor kidneys, single or dual, had graft survival similar to young-donor kidneys.
- **Limitations:** sampling error; wedge biopsy overestimates sclerosis; interobserver variation.

Optimising marginal kidneys

- Short cold ischaemia; **HMP (hypothermic machine perfusion)** reduces DGF; old-for-old matching; informed consent.

Diagram to draw: Remuzzi grid with four compartments scored 0–3.

> **Key points to remember:**
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
> - Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
> - 0–3 single, 4–6 dual, ≥7 discard.

Laparoscopic donor nephrectomy (including pros and cons)

Asked as:

- Laparoscopic donor nephrectomy — *Practice Set 2 Paper IV Q3a (SAQ)*
- Pros and cons of lap donor nephrectomy — *Practice Set 2 Paper IV Q3d (SAQ)*

Type: SAQ (10 marks)
(living donation also in 2023-Winter Paper 1, Q3c; the pros and cons are asked again in Practice Set 2 Paper 4, Q3d)

See Practice Set 2 Paper 4, Q3a (full answer).

Introduction

- **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first.**
- Work-up includes measured GFR (glomerular filtration rate) and CT (computed tomography) angiography.

Side selection

- **Left preferred:** longer renal vein, simpler dissection.
- **Right** if the left has multiple arteries or complex veins, or the right is smaller or abnormal: the donor keeps the better kidney. The right vein is short.

Techniques

Technique	Features
Pure transperitoneal laparoscopic	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds of US cases)	Hand port is also the extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	Balloon-made space; no bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed dissection; single umbilical incision; cost

Key steps

1. Reflect colon; divide gonadal, adrenal and lumbar veins (left).
2. Keep periureteric tissue and the gonadal–ureter triangle (ureteric blood supply).
3. Hydration and diuretic before vessel division.
4. Divide ureter, artery, then vein (stapler; see Q3f on Hem-o-lok).
5. Extract quickly; cold flush.

Warm ischaemia

- Arterial occlusion to cold flush: a few minutes, slightly longer than open, without effect on long-term graft function.

Pros and cons (versus open)

Pros	Cons
Less pain, shorter stay, faster (earlier) return to work	Longer operation; learning curve
Better cosmesis; **more people volunteer to donate** (more willing donors)	Slightly longer warm ischaemia

Pros	Cons
Magnified view of hilum	Shorter vessels, especially right vein (on the right)
Equal graft function and survival (graft function equals open)	Pneumoperitoneum lowers renal blood flow; cost

- **Risks (safety):** complications about 16% (mostly gastrointestinal); mortality **3 per 10,000**; conversion 0.26–3.34%. Lifetime ESRD (end-stage renal disease) 90 per 10,000 versus 14 in matched non-donors.

Diagram to draw: left hilum with gonadal, adrenal and lumbar veins and the stapled artery and vein.

Key points to remember (including pros and cons angle):

- Ratner 1995; left kidney preferred (longer vein).
- Hand-assisted in about two-thirds of US cases.
- Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
- Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
- Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
- Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

Steroid-free immunosuppression in renal transplantation

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 4, Q3f (immunosuppression also in Practice Set 3 Paper 4, Q1 and 2020-Summer Paper 4, Q1)

Why avoid steroids

- Steroids block cytokine transcription (IL-1, IL-2 [interleukins]) and redistribute lymphocytes, but long-term use harms:
  - **Metabolic:** PTDM (post-transplant diabetes mellitus), hypertension, dyslipidaemia, weight gain, hence **cardiovascular death** (leading cause of death with a functioning graft).
  - **Bone:** osteoporosis, avascular necrosis; **growth failure** in children.
  - Cataract, infection, poor healing.

Strategies

Strategy	Timing
Steroid avoidance	No steroids, or perioperative doses only
Early (rapid) withdrawal	Stopped within the first week; most studied
Late withdrawal	Months later; more rejection, less favoured

Regimen

- **Induction is essential:** depleting rATG (rabbit antithymocyte globulin) or alemtuzumab; basiliximab (IL-2 receptor blocker) in low risk.
- **Maintenance:** tacrolimus at target levels plus MMF (mycophenolate mofetil).

Selection

- **Suitable:** low immunological risk: first transplant, non-sensitised, no DSA (donor-specific antibody), prompt graft function, good adherence.
- **Unsuitable:** highly sensitised, DSA, retransplant after immunological loss, DGF (delayed graft function), early rejection, **recurring steroid-responsive disease** (IgA nephropathy, lupus), steroids needed for other illness.

Evidence

- **Woodle et al. (2008):** withdrawal after day 7 versus low-dose maintenance: similar 5-year outcomes; **more mild, steroid-sensitive rejection**.
- **HARMONY (Lancet 2016):** steroids stopped on day 8: similar rejection and graft survival, **less PTDM**.
- **Cochrane review (2016):** more acute rejection, but no difference in patient or graft survival up to 5 years.
- **KDIGO (Kidney Disease: Improving Global Outcomes, 2009):** low-risk recipients with induction may stop steroids within the first week; if continued beyond it, continue rather than withdraw.
- **Practice:** over two-thirds still take maintenance steroids (Chapter 91); steroid-free maintenance increasing in low-risk children.

Risks and safeguards

- Early rejection; possibly more chronic injury and recurrence. Safeguards: tacrolimus levels, adherence, DSA monitoring, biopsy for rising creatinine.

Diagram to draw: timeline: induction, steroids stopped by day 7, tacrolimus–MMF maintenance.

Key points to remember:

- Avoidance, early (first week) or late withdrawal.
- Needs induction plus tacrolimus–MMF; low-risk recipients only.
- More mild rejection, same survival; less PTDM (HARMONY).
- KDIGO 2009: stop within week 1, or continue.

Artificial kidney

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 4, Q3e

Read as devices that replace kidney function: the dialyser (classic "artificial kidney") and newer wearable, implantable and bioartificial kidneys.

Definition

- An **artificial kidney** removes solutes and water from blood across a semipermeable membrane. The haemodialysis dialyser is the standard example.
- Pioneered by **Kolff (1943)**; long-term use followed the Scribner shunt (1960) and Brescia–Cimino AV (arteriovenous) fistula (1966).

Principles

Process	Mechanism	Removes
Diffusion	Concentration gradient across membrane	Small solutes (urea, potassium)
Convection (haemofiltration)	Solute dragged with water	Middle molecules
Ultrafiltration	Hydrostatic pressure gradient	Water
Adsorption	Binding to membrane or sorbent	Some proteins, toxins

- **Hollow-fibre dialyser**; blood and dialysate flow **countercurrent** to keep the gradient.
- **Schedule:** blood flow 250–350 mL/min, 3–5 hours, 3–5 times a week; adequacy by Kt/V (urea clearance × time ÷ volume), at least 1.2 per session.
- **Peritoneal dialysis** uses the peritoneum as membrane (glucose or icodextrin draws water).

Limitations of current dialysis

- Intermittent: fluid and solute swings cause cramps.
- No endocrine (erythropoietin, vitamin D activation) or tubular metabolic function.
- Access thrombosis, infection, steal; cardiovascular disease, vascular calcification.
- **Transplantation remains the best renal replacement therapy.**

Newer artificial kidneys

Device	Principle	Status
WAK (wearable artificial kidney)	Belt-worn; sorbent regenerates a small volume of dialysate	Early human trials
Wearable or portable peritoneal dialysis	Continuous sorbent-regenerated peritoneal fluid	Development
Implantable bioartificial kidney	Silicon nanopore haemofilter plus bioreactor of renal tubule cells; driven by blood pressure, no pump	Preclinical
RAD (renal assist device)	Human proximal tubule cells in a hollow-fibre cartridge with continuous dialysis	Phase II in acute kidney injury; not adopted
Decellularised or organoid kidneys; pig kidney xenotransplant	Regenerative and xenograft routes	Experimental (see Practice Set 6 Paper 4, Q2)

Urologist's role

- AV fistula and graft work-up, laparoscopic PD (peritoneal dialysis) catheter placement with omental fixation, transplant.

Diagram to draw: hollow-fibre dialyser with countercurrent blood and dialysate flow.

> **Key points to remember:**
> - Diffusion, convection, ultrafiltration across a semipermeable membrane.
> - Dialysis lacks endocrine and tubular function; transplant is best.
> - Wearable, implantable bioartificial and RAD devices are investigational.

Robotic kidney transplantation

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 4, Q3f

Introduction

- **RAKT (robot-assisted kidney transplantation)**: the graft is implanted intracorporeally on the iliac vessels using the robot.
- Milestones: Hoznek (2002, robot-assisted anastomosis); Giulianotti (2010, obese recipient); **Menon and Ahlawat (Vattikuti–Medanta, 2014)** standardised the transperitoneal technique with **regional hypothermia**, assessed by the IDEAL (idea, development, exploration, assessment, long-term study) framework.

Indications and selection

- Living-donor grafts mainly; increasingly deceased donors.
- **Obese recipients** gain most, as open transplant carries more wound complications.
- **Contraindications (relative):** heavy iliac calcification on non-contrast CT (computed tomography), multiple graft vessels early in experience, previous transplant on that side, dense adhesions.

Technique (Vattikuti–Medanta)

1. Lithotomy, steep Trendelenburg; transperitoneal ports.
2. Periumbilical **GelPOINT** incision for graft and ice insertion.
3. Peritoneal flap; expose the **external iliac vessels**.
4. Graft wrapped in gauze with ice slush; **ice slush in the iliac fossa** for regional hypothermia.
5. **Vein first:** renal vein end-to-side to external iliac vein, to shorten arterial ischaemia.
6. Renal artery end-to-side to external iliac artery; unclamp and reperfuse.
7. Extravesical **Lich–Gregoir** ureteroneocystostomy over a double-J stent.
8. Graft retroperitonealised with the peritoneal flap to prevent torsion.

Advantages

- Small incision: **fewer wound infections, less lymphocele**, less pain, faster recovery, better cosmesis.
- Magnified 3D (three-dimensional) view and precise suturing in deep pelvis or obese patients.

Disadvantages

- **Longer rewarming (anastomosis) time**; cost; learning curve; pneumoperitoneum and Trendelenburg effects.
- No tactile feedback for calcified vessels.

Outcomes

- Multicentre and ERUS (European Association of Urology Robotic Urology Section) series: graft function and survival comparable with open transplant, with lower surgical-site complications.

Complications

- Graft thrombosis, haemorrhage, urine leak, ureteric stricture, lymphocele (less), port-site hernia; DGF (delayed graft function).

Diagram to draw: pelvis with graft on external iliac vessels and ureter to bladder dome; ice slush via GelPOINT.

Key points to remember:

- Vattikuti–Medanta technique with regional hypothermia (2014).
- Vein first, then artery, then Lich–Gregoir.
- Best for obese recipients; graft function equals open.

Rapid Revision Sheet – Section 15 · Renal, Nephrology & Transplant

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Allograft rejection, immunobiology of renal transplant, marginal kidney donor and newer immunosuppressant drugs (LAQ)

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, avoid (contraindicated) if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

Brain-stem death: guidelines for declaration and the Board of Medical Experts (brain dead committee) (SAQ)

- 4-member (Four-member) Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.

Vascular access for dialysis. (SAQ)

- AVF first, distal, non-dominant, 6 months ahead.
- Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis sits on stenosis; steal → banding or DRIL.

Female kidney donors: special precautions (SAQ)

- Ensure free consent (Indian gender gap).
- Pregnancy test; contraception; stop oestrogens.
- Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
- Defer pregnancy; high-risk antenatal follow-up.

Hospital authorization committee for live related kidney donation. (SAQ)

- THOA 1994; amended 2011; Rules 2014.
- Hospital-based committee if >25 transplants a year; no transplant team member.
- Non-relatives, swap and foreigners need committee approval.
- Interview video-recorded; appeal within 30 days.

Marginal donor and Remuzzi score for pre-implantation biopsy (SAQ)

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
- 0–3 single, 4–6 dual, ≥7 discard.

Laparoscopic donor nephrectomy (including pros and cons) (SAQ)

- Ratner 1995; left kidney preferred (longer vein).
- Hand-assisted in about two-thirds of US cases.
- Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.

- Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
- Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
- Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

Steroid-free immunosuppression in renal transplantation (SAQ)

- Avoidance, early (first week) or late withdrawal.
- Needs induction plus tacrolimus–MMF; low-risk recipients only.
- More mild rejection, same survival; less PTDM (HARMONY).
- KDIGO 2009: stop within week 1, or continue.

Artificial kidney (SAQ)

- Diffusion, convection, ultrafiltration across a semipermeable membrane.
- Dialysis lacks endocrine and tubular function; transplant is best.
- Wearable, implantable bioartificial and RAD devices are investigational.

Robotic kidney transplantation (SAQ)

- Vattikuti–Medanta technique with regional hypothermia (2014).
- Vein first, then artery, then Lich–Gregoir.
- Best for obese recipients; graft function equals open.

Section 16 · BPH — Long answer questions (LAQ)

HoLEP (holmium laser enucleation of the prostate)

Asked as:

- Holep — *Winter 2023 Paper IV Q2 (LAQ)*
- HOLEP. — *Summer 2020 Paper IV Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2019-Summer Paper 1, Q3c and 2020-Summer Paper 4, Q3d)

Read as HoLEP (holmium laser enucleation of the prostate) for BPH (benign prostatic hyperplasia).

Introduction and principle

- **HoLEP** removes the whole transition-zone adenoma endoscopically **along the surgical capsule**. It is the endoscopic equivalent of open simple prostatectomy.
- **Evolution:** HoLAP (holmium laser ablation, 1994, slow) → HoLRP (resection, strips removed with a grasper) → **HoLEP (1998)**, made practical by the **tissue morcellator**.
- "Enucleation is enucleation": any energy giving a true **AEEP (anatomic endoscopic enucleation of the prostate)** gives durable results.

Laser physics

- **LASER:** light amplification by stimulated emission of radiation. Ho:YAG (holmium:yttrium–aluminium–garnet) = holmium atoms in a YAG (yttrium–aluminium–garnet) crystal.
- **Wavelength about 2120 nm**; chromophore **water** (absorbed by water); **pulsed**; optical penetration **0.4 mm**. It cuts by vaporisation with a thin coagulation zone; heat spreads 2–3 mm and seals (coagulates) small and medium vessels.
- Flexible quartz fibres; **normal saline** irrigation, so no TUR (transurethral resection) syndrome.

Laser	Wavelength	Chromophore	Penetration
Holmium:YAG	2120 nm	Water	0.4 mm
Thulium fibre	1940 nm	Water	0.2 mm
KTP (potassium-titanyl-phosphate, green light)	532 nm	Haemoglobin	0.8 mm
Nd:YAG	1064 nm	–	About 1 cm (sloughing)

MOSES (pulse modulation)

- **Moses effect:** the first part of each pulse boils a **vapour channel** in the water. Vapour absorbs the light far less, so the rest of the pulse passes through to the tissue ("parting the water").
- **Moses technology** splits each pulse to exploit this: more energy reaches tissue, and it **may shorten operating time**.

Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without a median lobe (AUA [American Urological Association] Statement 38). Only enucleation spans every size band.
- **High bleeding risk or anticoagulated men** (AUA recommendation, Statement 43).
- Large glands otherwise needing open prostatectomy (alternative to open prostatectomy for large glands); retention; failed medical therapy.

Technique (steps)

- **Equipment:** 60–150 W Ho:YAG laser; **550 µm end-firing fibre**; 24–28 Fr continuous-flow laser resectoscope with 30° lens; saline.
- **Settings:** 2.0 J × 40–50 Hz to enucleate; 1.0–2.0 J × 20 Hz to coagulate.
- **Three-lobe (bottom-up, early apical release):**
  1. Median lobe: **bladder-neck incisions at 5 and 7 o'clock** to the capsule, joined proximal to the verumontanum; median lobe enucleated into the bladder. No median lobe: one 6 o'clock incision.
  2. **Perimontanal inverted-U incision**; scope pushes laterally to open the lobe–capsule plane at the apex; **early apical release** to protect the sphincter.
  3. Plane carried (develop the capsular plane) anteriorly to 11–1 o'clock (to 12 o'clock); **apical mucosal strip incised early** to avoid sphincter traction.
  4. Anterior commissure divided; lateral lobes enucleated into the bladder (**en bloc, 2-lobe or 3-lobe**).
  5. **Haemostasis** with a defocused beam.
- **Two-lobe:** 6 and 12 o'clock grooves; each half enucleated with its share of median lobe.
- **En bloc:** whole adenoma in one piece after circumferential apical release; usually paired with early apical release.
- **Morcellation:** nephroscope, two open inflow lines, **full bladder** and good haemostasis. Morcellators: Piranha (rotating), Versacut (oscillating). Fragments sent for histology.
- **Catheter:** 22 Fr three-way; same-day or next-day removal.

Outcomes and trials

- **Versus TURP (transurethral resection of the prostate):** RCT (randomised controlled trial) in glands <100 mL: better symptom score and PVR (post-void residual), less haemoglobin drop (less bleeding), shorter catheter time and stay, longer operation.
- **Meta-analyses:** better **IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, durable to (at) **3–8 years**. **HoLEP is the only endoscopic procedure with efficacy superior to TURP.**
- **Versus open prostatectomy** (Naspro 2006; Kuntz 2008, 5 years): equal function, less morbidity, less blood loss, shorter catheter time and stay.
- **Retreatment rare**: whole transition zone removed.
- **Anticoagulated men:** feasible; slightly longer stay; warfarin worse than DOACs (direct oral anticoagulants).
- **Guidelines:** EAU (European Association of Urology) and AUA list HoLEP as a first-line alternative to TURP and to open surgery for large glands.

Complications

- **Intraoperative:** capsular perforation; ureteric orifice injury (glands >80 mL); **morcellator bladder injury** (stop and drain; intraperitoneal → repair); retained adenoma.
- **Early:** transient storage symptoms and dysuria; **transient (early) SUI (stress urinary incontinence)** from adenoma volume removed, falling over 6 months; persistent SUI rare.
- **Late: retrograde ejaculation about 75%**; BNC (bladder neck contracture) **0–3.2%** (small glands <40 mL, consider bladder-neck incision); meatal stricture; urethral stricture.
- Erectile function unchanged. Transfusion rare; conversion to TURP almost never.

Learning curve (limitation and comparison)

- The main barrier (**steep learning curve**): from 20 supervised cases; **efficiency plateaus after about 50 cases**. Mentoring and simulators help.

Recent advances

- High-power pulse modulation (**Moses pulse modulation**); **ThuLEP** (thulium laser enucleation) and bipolar enucleation.
- **Ejaculation-preserving and median-lobe-only HoLEP** (antegrade ejaculation about 90%).
- **Same-day discharge** and catheter removal.

Diagram to draw: sagittal prostate showing the enucleation plane along the surgical capsule, with 5 and 7 o'clock incisions.

> **Key points to remember:**
>
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
> - Size-independent; preferred (best) in anticoagulated men.
> - Only endoscopic method superior to TURP; retreatment rare.
> - Retrograde ejaculation about 75%; learning curve about 50 cases.

Prostate anatomy, newer surgical concepts in CA prostate and neurovascular bundle preservation in radical prostatectomy

Asked as:

- Discuss in detail anatomy of prostate with special reference to newer concepts in surgical management of CA prostate — *Practice Set 2 Paper IV Q1 (LAQ)*
- Neurovascular bundle preservation in Radical prostatectomy: recent techniques — *Winter 2023 Paper IV Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(anatomy also answered in Practice Set 1 Paper 1, Q2; robotic prostatectomy in 2023-Winter Paper 1, Q3b)

Introduction

- A 20-g fibromuscular gland around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**).
- Modern RP (radical prostatectomy) is anatomical: each new technique gives a clear margin while saving a named structure (nerves, sphincter or anterior support).

Zonal anatomy (McNeal, 1981)

Zone	Share of glands	Surgical relevance
Peripheral (PZ)	About 70%	**70% of cancers**; posterolateral, so ECE (extracapsular extension) runs along the nerves
Central (CZ)	About 25%	Cone around the ejaculatory ducts; few cancers
Transition (TZ)	About 5%	BPH (benign prostatic hyperplasia); anterior cancers missed by DRE (digital rectal examination)
AFMS (anterior fibromuscular stroma)	No glands	Anterior tumours reach it; margin risk in posterior approaches

- There is **no true capsule**, so cancer breaching the gland is "extraprostatic extension".

Fascial planes and nerve sparing (dissection planes)

Plane	Lies (where)	Trade-off (use)
Intrafascial	Between capsule and prostatic fascia	Most nerves (maximal nerves); more positive margins
Interfascial	Between prostatic and levator fascia	**Standard** for organ-confined disease; **standard nerve-sparing plane**
Extrafascial	Outside levator and Denonvilliers fascia	Widest margin; ED (erectile dysfunction) likely

- **Denonvilliers fascia** stays on the rectum when ECE risk is low; it is excised when posterolateral ECE is suspected.
- Chapter 160 (Walsh) calls intrafascial "never used"; Chapter 161 accepts it in selected men.

Neurovascular bundle (NVB)

- **NVB (neurovascular bundle) of Walsh** (Walsh and Donker, 1982): capsular vessels plus **cavernous nerves** (parasympathetic, from the pelvic plexus). The pelvic plexus (S2–S4 parasympathetic plus hypogastric sympathetic) gives the **cavernous nerves**, running

posterolaterally **3–5 mm from the capsule**, marked by capsular vessels. It lies posterolateral, in the lateral pelvic fascia between **prostatic and levator fascia**.

- Nerves are invisible; the **vessels mark them**.
- At the apex they lie at **5 and 7 o'clock**, swinging to 3 and 9 o'clock at the mid-urethra; most nerve injuries occur here.
- Nerve distribution is diffuse (varies): a main posterolateral bundle, a posterior "hammock" and anterolateral fibres (basis of high anterior release).
- **Injury mechanisms:** incision, entrapment (clip or suture), **heat** (**thermal**), **traction**; so dissect athermally without traction.

Vessels and lymphatics

- Inferior vesical artery (urethral and capsular branches). Preserve **accessory pudendal arteries**; they may be the main penile supply.
- DVC (dorsal venous complex): CO_2 pneumoperitoneum tamponades it, so robotic blood loss is low.
- Lymph: obturator, external and internal iliac, common iliac, presacral.

Continence anatomy

Passive	Active
Smooth muscle of bladder neck and urethra; anterior suspensory system (endopelvic fascia, detrusor apron, puboprostatic ligaments, DVC)	Horseshoe striated sphincter at the apex; pudendal nerve branches enter it at the apex

Newer concepts in surgical management

1. **RARP (robot-assisted RP):** 3D, 10× view and wristed instruments removed the suturing barrier. **Yaxley 2016** RCT (randomised controlled trial): equal function and early oncology versus open; less blood loss (300 versus 1400 mL), shorter stay. Surgeon matters more than platform.
2. **Risk-graded nerve sparing (graded, risk-stratified):** each side is graded from full (intrafascial) through partial to none (for example the Tewari four-grade scale). The choice uses grade, positive cores and mpMRI (multiparametric magnetic resonance imaging) capsular contact; full, partial or none per side, from MRI (magnetic resonance imaging) and nomograms.
3. **NeuroSAFE:** intraoperative frozen section of each posterolateral surface (of the whole posterolateral margin); a positive section means resecting that NVB (if positive, that NVB is resected). It aims to allow more nerve sparing without more positive margins; the UK NeuroSAFE PROOF RCT (randomised controlled trial) tested its effect on erectile function.
4. **Anterior-support preservation:**
    - **Retzius-sparing** (Bocciardi, 2010): fully posterior via the pouch of Douglas. Earlier continence at 3 months, equal at 12; more positive margins, so avoid for anterior tumours.
    - **Hood technique** (Tewari 2021): anterior approach keeping detrusor apron, puboprostatic ligaments and DVC; early data.
    - Retzius-sparing and Hood are aimed at continence; potency equal at 12 months.
5. **Apical dissection:** the apex is the commonest positive-margin site. Keep maximal urethral length, watch for an apical lip, divide sharply without cautery; control the DVC distally or late. Posterior (Rocco) reconstruction aids early continence.
6. **PLND (pelvic lymph node dissection) templates:**

Template	Nodes	Drainage captured
Limited	Obturator	—
Standard	+ External iliac	About 38%
Extended	+ Internal iliac	About 63%
+ Common iliac	+ Common iliac	About 75%

- Extended template when nodal risk exceeds 5% (EAU [European Association of Urology]); it stages but does not improve survival. Lymphocele is commonest.

7. **Single-port robot (da Vinci SP, 2018):** one 2.5-cm cannula; extraperitoneal or transvesical; same-day discharge.
8. **Focal therapy anatomy:** ablate the **index lesion**. Transrectal HIFU (high-intensity focused ultrasound) suits posterior PZ lesions; anterior lesions need transperineal cryotherapy or TULSA (transurethral ultrasound ablation). Spare sphincter, NVB and rectum.

- **Also know:** PSMA PET (prostate-specific membrane antigen positron emission tomography) staging; salvage robotic RP; new platforms; ERAS (enhanced recovery after surgery).

Nerve-sparing techniques (NVB preservation)

- **Walsh anatomic (open):** release apex to base from the rectal surface; small clips; **no thermal energy**.
- **High anterior release** (open) and **Veil of Aphrodite** (robotic): lateral prostatic fascia lifted high anteriorly as a "veil". Select by **Tsuzuki criteria**: ≥2 of PSA (prostate-specific antigen) >10 ng/mL, Gleason >6, abnormal DRE (digital rectal examination), perineural invasion, high core involvement → >10% risk, so avoid.
- **Antegrade release (robotic):** incise levator fascia, sweep NVB posterolaterally, clip pedicle at the "groove".
- **Athermal, traction-free:** clips or bulldogs, cold scissors.
- Graded nerve sparing, NeuroSAFE, Retzius-sparing and Hood: see above.

Adjuncts

- **Penile rehabilitation:** PDE5i (phosphodiesterase type 5 inhibitor), then injections.
- **Also know:** sural nerve grafts and nerve stimulation tests (abandoned); amniotic membrane wraps, fluorescence nerve imaging (experimental).

Diagram to draw: axial prostate showing the three fasciae (fascial planes), the NVB (NVBs) at 5 and 7 o'clock and the intra-, inter- and extrafascial planes.

> **Key points to remember:**
> - PZ gives 70% of cancers; no true capsule.
> - Interfascial plane is the standard nerve-sparing plane; apex is the commonest margin and nerve-injury site.
> - Avoid heat and traction; clips, cold scissors.
> - Veil and high release for low-risk sides (Tsuzuki).
> - NeuroSAFE frozen section: more sparing, safe margins.
> - Retzius-sparing: earlier continence, more positive margins.
> - Extended PLND captures about 63% of drainage.

Section 16 · BPH — Short answer questions (SAQ)

Silodosin

Asked as:

- Silodosin — *Summer 2015 Paper IV Q3b (SAQ)*
- Silodosin. — *Practice Set 5 Paper IV Q3b (earlier version) (SAQ)*

Type: SAQ (10 marks)

Introduction: definition and pharmacology

- **Uroselective α1-blocker**, the **most α1A-selective** blocker/agent (well over 100-fold vs α1B — another source: several hundredfold over α1B; about 50-fold vs α1D), for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).
- Sparing vascular α1B gives **minimal hypotension** (little α1B (vascular) blockade means few blood-pressure effects). Blocking α1A in the vas and seminal vesicles causes **failure of emission**.
- Onset: flow improves in (within) 2–6 hours (h), symptoms in (within) 3–4 days. CYP3A4 metabolism (metabolised by CYP3A4).

Mechanism of action

- **α1A receptors** (about 70% of prostatic α1 receptors) dominate in the prostate, bladder neck and prostatic urethra. Noradrenaline on them causes the **dynamic component** of BOO (bladder outlet obstruction).
- Blockade relaxes this smooth muscle.

Dose and contraindications

Item	Detail
Dose	**8 mg once daily with a meal**; no titration
Renal: CrCl (creatinine clearance) 30–50 mL/min	**4 mg daily**
Contraindicated	CrCl <30; severe liver (hepatic) impairment; strong (potent) CYP3A4 inhibitors (e.g. ketoconazole)
Caution	Other α-blockers (avoid), PDE5 (phosphodiesterase 5) inhibitors, planned cataract surgery

Indications, efficacy and other uses

Indication	Evidence
LUTS (lower urinary tract symptoms) due to BPE (benign prostatic enlargement)	Japanese, American and European 12-week RCTs (randomised controlled trials): IPSS (International Prostate Symptom Score) and flow better than placebo; **non-inferior to tamsulosin 0.4 mg** (**Chapple 2011**), not better; effective (works) in severe LUTS
MET (medical expulsive therapy)	Distal ureteric stones 5–10 mm; meta-analyses show higher, faster expulsion than tamsulosin; **SUSPEND (2015)** found no overall tamsulosin benefit
CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)	α-blocker-naive men with voiding symptoms; of recent onset

- **Greatest fall in BOOI (bladder outlet obstruction index; BOO index)** of all α1-blockers (subgroup data).
- Does not shrink the prostate; add a 5-ARI (5α-reductase inhibitor) for large glands.
- **Also know:** before ureteroscopy (easier access); with propiverine for storage symptoms; nocturia (modest).

Adverse effects

- **Ejaculatory dysfunction (signature) / anejaculation**: over a quarter of men in (registration) trials (about 14% vs 2% vs (with) tamsulosin head-to-head), yet few stop the drug; responders do better. Mechanism: **emission failure** (vas and seminal vesicle α1A blockade) and bladder-neck relaxation. Reversible. Counsel young men.
- **Cardiovascular:** dizziness; orthostatic hypotension rare; no QT change; no excess hypotension in the elderly or with antihypertensives (safe with antihypertensives).
- Falls in older men: falls and fractures rise in men ≥66 on any α-blocker.
- **IFIS (intraoperative floppy iris syndrome)**: reported; class effect; warn (tell the ophthalmologist) before cataract surgery (stopping the drug does not remove risk).

Comparison

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction (disorder)	Efficacy in LUTS
Silodosin	α1A (highest); α1A ≫ α1D, α1B	8 mg OD	Lowest	**Highest** (more)	Equal
Tamsulosin	α1A, α1D > α1B (α1A and α1D)	0.4 mg OD	Low	Moderate (less)	Equal
Alfuzosin	Clinically uroselective	10 mg OD	Low	Lowest	
Naftopidil	α1D	25–75 mg	Some	Less	
Doxazosin, terazosin	Non-selective; titrate	1–8 mg, 1–10 mg	Highest	Low	

Diagram to draw: α1A prostate and vas, α1B vessels, α1D bladder; prostate and bladder neck showing α1A receptor sites blocked by silodosin.

Key points to remember:

- Most α1A-selective α-blocker; 8 mg with a meal (food); no titration.
- 4 mg if CrCl 30–50 (halve to 4 mg at CrCl 30–50); avoid if (below) <30.
- Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
- Anejaculation is the signature effect (more anejaculation); least hypotension.

UroLift (prostatic urethral lift)

Asked as:

- Urolift. — *Winter 2024 Paper IV Q3d (SAQ)*
- Urolift. — *Practice Set 6 Paper IV Q3b (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2024-Winter Paper 4, Q3d)

Read as PUL (prostatic urethral lift, UroLift) for BPH (benign prostatic hyperplasia).

Introduction

- A **non-ablative MIST (minimally invasive surgical therapy)**. Permanent implants pull (retract) the lateral lobes apart and open the prostatic urethra.
- **FDA (Food and Drug Administration) approval 2013**; extended to ≤100 mL in 2020.
- **Main advantage:** preserves erectile and **ejaculatory** function.

Implant and mechanism

- Spring-loaded implant with T-shaped ends:
  - **Capsular tab:** nitinol.
  - **Urethral end:** stainless steel (resists encrustation).
  - **Connector:** polyethylene terephthalate suture under tension.
- Delivered through a **19-gauge needle** from a cystoscopic device.
- The gland is more compliant than the capsule, so tension compresses the lobe outward; later local atrophy and fibrosis. Implants are MRI (magnetic resonance imaging)-conditional (≤3 T).

Indications

- **AUA (American Urological Association): prostate 30–80 mL, no obstructing median lobe** (confirm by cystoscopy or ultrasound). EAU (European Association of Urology): <70 mL, no middle lobe, men wanting ejaculation preserved.
- Men wishing to stop medication.
- **Avoid:** obstructing median lobe (MedLift data non-randomised), large gland, active infection.

Technique

1. Local anaesthesia; rigid cystoscope.
2. Compress lateral lobe, fire needle, deploy implant; repeat on the other side.
3. **Usually 4–6 implants**, anterolateral, from bladder neck to apex, avoiding the neurovascular bundles.
4. **Keep implants >1.5 cm distal to the bladder neck** (encrustation).
5. Most go home without a catheter.

Evidence

	L.I.F.T. (2013)	BPH6 (2015)
Design	PUL vs sham, 2:1, 19 centres	PUL vs TURP (transurethral resection of prostate), 10 European centres
Result	IPSS (International Prostate Symptom Score) about halved, twice the sham effect; Qmax (maximum flow) +4 mL/s; no new sexual dysfunction	Composite end point met 52.3% vs 20.0%; antegrade ejaculation 100% vs 60.6%

	L.I.F.T. (2013)	BPH6 (2015)
Durability / retreatment	13.6% surgical retreatment at 5 years	13.6% vs 5.7% at 2 years; TURP better IPSS, Qmax, PVR (post-void residual)

- TURP wins on IPSS, Qmax and PVR (post-void residual).

Complications

- Transient dysuria, haematuria, urgency (<2 weeks); encrustation; rare pelvic haematoma; device malfunction (misfire).
- Lower major adverse events but **higher retreatment than TURP**.

Diagram to draw: axial prostate with implants pulling lateral lobes to the capsule.

> **Key points to remember:**
> - 30–80 mL, no median lobe; local anaesthesia.
> - Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
> - TURP wins (better) on IPSS, flow and durability.

Green light lasers

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 4, Q3b (also answered as KTP laser in Practice Set 1 Paper 4, Q3e; lasers in BPH in 2017-Winter Paper 2, Q3a)

Physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam is passed through a **KTP (potassium titanyl phosphate) or LBO (lithium triborate) crystal**. This doubles its frequency and halves the wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water, so vascular prostate absorbs it avidly.
- Optical penetration **0.8 mm**, tissue effect 1–3 mm, thin (0.2 mm) coagulation rim. The result is bloodless **photoselective vaporisation**.

Platforms

System	Power	Note
KTP	80 W (later 100 W)	Popularised PVP (photoselective vaporisation of the prostate)
LBO HPS (High Performance System)	120 W	Faster vaporisation
LBO XPS	**180 W**	750 µm MoXy fibre: metal tip, fluid cooling

Technique

- 23–24 Fr cystoscope; side-firing fibre emitting at **70°**; saline irrigation.
- Open the bladder neck first, away from the orifices; sweep 15–30° in 1–2 seconds at ≤2 mm.
- Large bubbles mean efficient vaporisation. Pulsatile bleeding: coagulation mode (about 30 W).
- More than 5 kJ/cc predicts complete vaporisation.

Indications (AUA [American Urological Association])

- Prostates <30 cc and 30–80 cc; men at **higher bleeding risk**, who can continue anticoagulants. Not >80 cc (high retreatment).

Evidence: GOLIATH trial

- RCT (randomised controlled trial), 29 European centres, about 270 men, prostate ≤100 cc: 180 W PVP versus TURP (transurethral resection of the prostate).
- PVP **non-inferior** for IPSS (International Prostate Symptom Score), Qmax and complications to 2 years. Shorter catheter time and stay; TURP faster. Retrograde ejaculation (about two-thirds) and reoperation equal.

Complications

- Dysuria (highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum; capsular perforation 0.2–1%.
- No TUR (transurethral resection) syndrome; transfusion very rare; **no tissue for histology**.

Recent advances

- GreenLight enucleation; office and same-day catheter-free pathways.

Diagram to draw: side-firing fibre at 70° vaporising a lateral lobe.

Key points to remember:

- 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration.
- 180 W XPS with MoXy fibre.
- GOLIATH: PVP non-inferior to TURP.
- Best for anticoagulated men; not >80 cc.

Rapid Revision Sheet – Section 16 · BPH

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

HoLEP (holmium laser enucleation of the prostate) (LAQ)

- Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
- Size-independent; preferred (best) in anticoagulated men.
- Only endoscopic method superior to TURP; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Prostate anatomy, newer surgical concepts in CA prostate and neurovascular bundle preservation in radical prostatectomy (LAQ)

- PZ gives 70% of cancers; no true capsule.
- Interfascial plane is the standard nerve-sparing plane; apex is the commonest margin and nerve-injury site.
- Avoid heat and traction; clips, cold scissors.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

Silodosin (SAQ)

- Most α1A-selective α-blocker; 8 mg with a meal (food); no titration.
- 4 mg if CrCl 30–50 (halve to 4 mg at CrCl 30–50); avoid if (below) <30.
- Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
- Anejaculation is the signature effect (more anejaculation); least hypotension.

UroLift (prostatic urethral lift) (SAQ)

- 30–80 mL, no median lobe; local anaesthesia.
- Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
- TURP wins (better) on IPSS, flow and durability.

Green light lasers (SAQ)

- 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration.
- 180 W XPS with MoXy fibre.
- GOLIATH: PVP non-inferior to TURP.
- Best for anticoagulated men; not >80 cc.

Section 17 · Prostate Cancer — Long answer questions (LAQ)

Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate?

Type: LAQ (25 marks)

Source: 2021-Summer Paper 4, Q1 (also answered in 2017-Summer Paper 2, Q3f)

"Focal" is taken to mean localised (organ-confined and locally advanced) disease.

Radiobiology

- RT (radiotherapy) causes DNA double-strand breaks. Prostate cancer has a **low α/β ratio (~1.5 Gy)**, lower than the rectum (~3 Gy). So larger fractions kill more tumour for the same late toxicity: the basis of **hypofractionation**.

EBRT (external beam radiotherapy) techniques

Step	Advance
3D-CRT (three-dimensional conformal RT)	CT-planned beams shaped to the prostate
IMRT (intensity-modulated RT)/VMAT (volumetric modulated arc therapy)	Inverse planning, steep dose gradients; less rectal toxicity; now standard
IGRT (image-guided RT)	Daily fiducials or cone-beam CT (computed tomography); margins 3–7 mm
MRI fusion, PSMA (prostate-specific membrane antigen) PET (positron emission tomography)	Better targeting; PSMA PET changes plans in up to half of high-risk men

Schedule	Dose per fraction	Example	Evidence
Conventional	1.8–2 Gy	78 Gy in 39	Dose escalation (RTOG 0126): better PSA control, no survival gain
Moderate hypofractionation	2.25–4 Gy	**60 Gy in 20**	**CHHiP, PROFIT**: non-inferior
SBRT (stereotactic body RT)	≥5 Gy	36.25 Gy in 5	**HYPO-RT-PC** non-inferior; **PACE-B** equal toxicity

Brachytherapy

Feature	LDR (low dose rate)	HDR (high dose rate)
Source	Permanent ^{125}I (145 Gy) or ^{103}Pd seeds	Temporary ^{192}Ir via afterloader
Monotherapy	Low and favourable intermediate risk	27 Gy in 2 or 34–38 Gy in 4 (single 19 Gy fraction fails)
Quality	Post-implant CT: **D90** (dose covering 90% of gland) **>90%** of prescription	Real-time planning

- **Technique:** TRUS (transrectal ultrasound)-guided perineal template; peripheral loading spares the urethra.
- **Boost** (EBRT 45 Gy + implant) for unfavourable intermediate/high risk: **ASCENDE-RT** halved biochemical failure, with more grade 3 urinary toxicity (18% vs 5%). **RTOG 0232:** adding EBRT to implant gave no benefit in intermediate risk.

- **Contraindications:** IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s, recent TURP (transurethral resection of the prostate), pubic arch interference, prior pelvic RT, inflammatory bowel disease.

ADT (androgen deprivation therapy) with RT

- **Why:** ADT radiosensitises (less DNA repair) and causes apoptosis; RT up-regulates androgen-receptor signalling, which ADT blocks.

Risk group	ADT with EBRT
Low, favourable intermediate	None
Unfavourable intermediate	**4–6 months**
High/locally advanced	**18–36 months** (EORTC 22863, PCS IV); add **abiraterone** if very high risk (STAMPEDE)

- **RT added to ADT** improves survival in locally advanced disease: **SPCG-7, NCIC PR.3/MRC PR07** (about 8–10% absolute).

After radical prostatectomy

- **Adjuvant RT** (pT3/positive margin): EORTC 22911, ARO 96-02 (biochemical), **SWOG 8794** (metastasis-free and overall survival).
- **RADICALS-RT, RAVES, GETUG-AFU 17** (ARTISTIC meta-analysis): **early salvage at PSA (prostate-specific antigen) 0.1–0.2 ng/mL** equals adjuvant RT and spares over 60% of men RT.
- Salvage dose ≥64–66 Gy. **RTOG 9601:** 2 years bicalutamide improved overall survival. **SPPORT:** pelvic nodes + short ADT improve freedom from progression.

Advanced and metastatic disease

Setting	RT role	Evidence
Low-volume de novo mHSPC (metastatic hormone-sensitive)	**RT to prostate** + ADT	**STAMPEDE arm H**: overall survival HR (hazard ratio) 0.68 (3-year 73% → 81%); none in high volume
Oligometastases (≤3–5)	SBRT to all lesions (MDT, metastasis-directed therapy)	STOMP, **ORIOLE** (less progression), SABR-COMET
Painful bone metastases	Palliative EBRT, e.g. **8 Gy single fraction**; cord compression	Equal pain relief to multiple fractions
mCRPC (metastatic castration-resistant), bone-only symptomatic	**Radium-223** (α-emitter, calcium mimic, range <100 µm) 50 kBq/kg 4-weekly × 6	**ALSYMPCA**: overall survival 14.9 vs 11.3 months; fewer skeletal events
mCRPC, PSMA-positive, after taxane and ARPI (androgen-receptor pathway inhibitor)	**^{177}Lu-PSMA-617** (β-emitter radioligand)	**VISION**: survival 15.3 vs 11.3 months (HR 0.62); **TheraP**: better PSA response than cabazitaxel

- **Never combine radium-223 with abiraterone** (more deaths and fractures).
- **Also know:** protons (no proven gain over IMRT); strontium-89, samarium-153 (pain palliation); elective pelvic nodal RT (RTOG 94-13, GETUG-01: no survival gain).

Response assessment

- **Phoenix failure:** PSA nadir + 2 ng/mL. **Benign bounce** after brachytherapy in about a third: watch, do not salvage.

- Biopsy-proven local failure: salvage prostatectomy, brachytherapy, HIFU (high-intensity focused ultrasound) or cryotherapy.

Toxicity

Type	Features and management
Acute (weeks)	Frequency, dysuria, diarrhoea; α-blockers
Late GI (gastrointestinal)	**Radiation proctitis** at 6–18 months: sucralfate/steroid enemas, argon plasma coagulation, hyperbaric oxygen
Late GU (genitourinary)	Cystitis, haematuria, stricture
Sexual	ED (erectile dysfunction), vascular; worse with ADT; PDE5 (phosphodiesterase type 5) inhibitors work in most
Rare	Second cancers, fistula

Recent advances

- **Hydrogel rectal spacer** (late rectal toxicity 2% vs 7%); MR-linac adaptive SBRT; **FLAME** focal boost to the MRI lesion; PSMA radioligands moving earlier (PSMAfore).

Diagram to draw: TRUS-guided perineal template brachytherapy.

Key points to remember:

- α/β ~1.5 Gy: basis of 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.

Discuss advances in management of CA prostate

Type: LAQ (25 marks)

Source: Practice Set 5 Paper 4, Q2 (related: 2025-Winter Paper 3, Q3b; 2024-Winter Paper 4, Q3f)

Introduction

- Three aims drive recent change: **find significant cancer and avoid diagnosing insignificant cancer**; **match local treatment to risk** with less morbidity; give **earlier, intensified and targeted** systemic therapy.

1. Screening and diagnosis

- **ERSPC (European Randomized Study of Screening for Prostate Cancer)** and Göteborg-1: PSA (prostate-specific antigen) screening lowers cancer-specific mortality, but PSA alone causes negative biopsies and overdiagnosis of GG (Grade Group) 1. Hence risk-stratified pathways.
- **Biomarkers** (reflex tests before biopsy):

Test	Sample	Content
PHI (Prostate Health Index)	Blood	(p2PSA ÷ free PSA) × √total PSA
4Kscore	Blood	Four kallikreins + clinical data; % risk of GG ≥2
STHLM3	Blood	4Kscore kallikreins + 232 risk SNPs (single-nucleotide polymorphisms) + clinical variables
PCA3 (prostate cancer antigen 3), SelectMDx	Urine after DRE (digital rectal examination)	RNA ratio (PCA3); DLX1 and HOXC6 (SelectMDx)
ExoDx IntelliScore	Urine, no DRE	Exosomal RNA of PCA3, ERG, SPDEF

- Tissue genomic classifiers (Decipher, Prolaris, Oncotype DX) refine risk for surveillance and adjuvant decisions.
- **MRI-first pathway:** mpMRI (multiparametric magnetic resonance imaging), scored by **PI-RADS (Prostate Imaging–Reporting and Data System)** 1–5, before biopsy.
  - **PROMIS (2017):** mpMRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; NPV (negative predictive value) 89%; 27% could avoid biopsy.
  - **PRECISION (2018):** MRI with targeted biopsy of PI-RADS ≥3 found more significant cancer (38% vs 26%); 28% avoided biopsy.
  - So PI-RADS ≥3 lesions are biopsied, while PI-RADS 1–2 with low PSA density may safely avoid biopsy.
- **Transperineal biopsy** under local anaesthesia: near-zero sepsis; MRI–ultrasound fusion targeting. **Micro-ultrasound** (PRI-MUS scoring) matches mpMRI in registry data.
- **PSMA (prostate-specific membrane antigen) PET (positron emission tomography): proPSMA:** accuracy 92% vs 65% for CT (computed tomography) plus bone scan (sensitivity 85%, specificity 98%). Used for high-risk staging, biochemical recurrence and selecting radioligand therapy.

2. Localised disease

- **AS (active surveillance):** standard for GG1; extended to low-volume favourable intermediate risk (GG2), with MRI and PSA density. **ProtecT (15 years):** prostate cancer mortality about 2–3% in monitoring, surgery and radiotherapy arms; more metastasis on monitoring, no survival difference.

- **Robotic RP (radical prostatectomy): Retzius-sparing** (Bocciardi) gives earlier continence (1 week to 3 months; equal by 12 months); single-port robot; frozen-section-guided nerve sparing (NeuroSAFE).
- **Radiotherapy:**
  - Moderate hypofractionation **60 Gy in 20** (CHHiP) is standard.
  - **SBRT (stereotactic body radiotherapy) 36.25 Gy in 5:** PACE-B showed similar late toxicity at 2 years (chapter); its 5-year report (2024) showed non-inferior biochemical or clinical control. HYPO-RT-PC (42.7 Gy in 7) was also non-inferior.
  - **FLAME:** MRI-lesion boost improved biochemical PFS (progression-free survival) (HR [hazard ratio] 0.45). **Hydrogel rectal spacer** reduces rectal toxicity.
  - High-risk M0: **abiraterone added to ADT (androgen deprivation therapy) and RT (STAMPEDE):** MFS (metastasis-free survival) HR 0.53.
- **Focal therapy:** based on the **index lesion hypothesis** (the largest or highest-grade focus drives progression). Energies: HIFU (high-intensity focused ultrasound), cryotherapy, IRE (irreversible electroporation), focal laser, VTP (vascular-targeted photodynamic therapy, padeliporfin). **HEAT registry (1379 men):** failure-free survival 68% (intermediate risk); continence 90–95%. Still within trials or registries.

3. Recurrent and advanced disease

Setting	Advance	Evidence
High-risk BCR (biochemical recurrence)	PSMA-guided salvage; enzalutamide + leuprolide	**EMBARK:** MFS HR 0.42
mHSPC (metastatic hormone-sensitive)	ADT + ARPI (androgen receptor pathway inhibitor)	LATITUDE, TITAN, ENZAMET, ARCHES
	Triplet (ADT + docetaxel + ARPI)	**PEACE-1, ARASENS**
	Prostate RT, low volume	STAMPEDE arm H
	MDT (metastasis-directed therapy)	STOMP, ORIOLE
nmCRPC (non-metastatic castration-resistant)	Apalutamide, enzalutamide, darolutamide	SPARTAN, PROSPER, ARAMIS
HRR (homologous recombination repair)-mutated mCRPC	**PARP (poly-ADP-ribose polymerase) inhibitors**	**PROfound:** olaparib imaging PFS 7.4 vs 3.6 months; PROpel, MAGNITUDE, TALAPRO-2
PSMA-positive mCRPC	**^{177}Lu-PSMA-617**	**VISION:** OS (overall survival) 15.3 vs 11.3 months (HR 0.62); TheraP: PSA response 65% vs 37% vs cabazitaxel

- **ADT alone is no longer standard** for mHSPC; every fit man gets intensification.
- **Germline and somatic testing** for all metastatic disease; **pembrolizumab** for MSI-high (microsatellite instability-high) tumours.
- **Relugolix:** oral GnRH (gonadotropin-releasing hormone) antagonist; HERO showed better sustained castration than leuprolide.
- **Also know:** radium-223 (ALSYMPCA, OS HR 0.70); PSMAfore (lutetium before chemotherapy); AI in MRI reading and pathology.

Diagram to draw: MRI-first pathway: raised PSA → mpMRI → PI-RADS ≥3 targeted transperineal biopsy; PI-RADS 1–2 with low PSA density → surveillance.

Key points to remember:

- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
- proPSMA: PSMA PET accuracy 92% vs 65%.
- ProtecT: no mortality difference at 15 years; AS is standard for GG1.
- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

Section 17 · Prostate Cancer — Short answer questions (SAQ)

Cryotherapy for carcinoma prostate

Type: SAQ (10 marks)

Source: 2015-Summer Paper 4, Q3f

Principle

- **Transperineal, image-guided** freezing; suits **anterior** tumours. HIFU (high-intensity focused ultrasound) is the main alternative.
- **Joule–Thomson effect**: **argon** freezes, **helium** thaws. Thermocouples guard sphincter and neurovascular bundles; a **urethral warmer** prevents sloughing.
- Generations: liquid nitrogen without imaging; TRUS (transrectal ultrasound) plus warmer; argon/helium 17-gauge probes; focal with MRI (magnetic resonance imaging) fusion.

Cell death

- **Intra- and extracellular ice**, recrystallisation, **vascular stasis and ischaemia**, apoptosis, immune effects.
- **−40 °C**, fast freeze, slow thaw, **two cycles**; the ice-ball edge is sublethal.

Indications

1. **Salvage after radiotherapy** (main role): biopsy-proven local recurrence, no metastases, **PSA (prostate-specific antigen) <10 (ideally <4)**, doubling time ≥16 months.
2. **Primary whole-gland**: men unfit for surgery or radiotherapy.
3. **Focal or hemi-gland**: MRI-visible grade group 2–3 lesion; rests on the index-lesion hypothesis.

Technique

- TRUS-guided transperineal template: place cryoprobes, thermocouples and urethral warmer; freeze anterior to posterior until −40 °C is reached; helium thaw; repeat the cycle.

Outcomes

- Salvage: biochemical control **55–71% at 3–5 years** (Phoenix: nadir + 2).
- **Donnelly, Cancer 2010** (RCT (randomised controlled trial)): **non-inferior** to external beam radiotherapy; worse sexual function.

Complications

	Primary / focal	Salvage
ED (erectile dysfunction)	**Almost always** (whole gland)	Very common
Incontinence	Rare	Single digit
Rectourethral fistula	≤0.8% (focal)	1–3.4%
Retention	Commonest after focal	~4%

- **Also know:** urethral sloughing, stricture; focal ablation preserves erections better (COLD (Cryo On-Line Data) registry).

	Cryotherapy	HIFU	IRE (irreversible electroporation)
Energy	Freezing	Focused ultrasound heat (≥55 °C)	Electric pulses, non-thermal

	Cryotherapy	HIFU	IRE (irreversible electroporation)
Route	Transperineal	Transrectal	Transperineal
Best for	Anterior lesions	Posterior lesions	Lesions near nerves, vessels

Guidelines and advances

- **EAU (European Association of Urology)**: primary or focal only in trials or registries. PSMA (prostate-specific membrane antigen) PET selects salvage patients.

Diagram to draw: prostate with ice ball, −40 °C isotherm inside 0 °C edge.

Key points to remember:

- Argon freezes, helium thaws; −40 °C, two cycles, urethral warmer.
- Main role is salvage: PSA <10, doubling time ≥16 months.
- Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

Saturation biopsies of prostate

Type: SAQ (10 marks)

Source: 2016-Summer Paper 4, Q3a

Definition

- Systematic prostate biopsy with **more cores than the extended 12-core scheme**, usually **18–21 or more** (often about 20–30). It samples the peripheral zone, transition zone, anterior gland and apex.
- Route: transrectal, or **transperineal template mapping biopsy** through a brachytherapy grid with a core every **5 mm**.

Rationale

- Sextant biopsy (Hodge, 1989) under-sampled the lateral peripheral zone. The 12-core scheme (adds apical and far-lateral cores) is today's standard.
- Saturation aims to cut sampling error, mainly for **anterior and apical cancers** (anterior-only cancer 6–20%).

Indications

- **Repeat biopsy** after a negative biopsy with persistent suspicion: rising PSA (prostate-specific antigen), ASAP (atypical small acinar proliferation), negative or unavailable MRI (magnetic resonance imaging).
- **Active surveillance** confirmation and **focal therapy** mapping (shows multifocality).

Templates and technique

Template	Division	Cores
Original Barzell	Octants (anterior/posterior × apex/base, each side)	4–8 per octant
Modified Barzell	Apex and base of each side × 5 sectors	20
Ginsburg	Anterior, mid, posterior sectors per side	4 per sector

- Lithotomy, stepper and grid; general or spinal anaesthesia, or periprostatic block (up to 22 mL of 1% lidocaine transrectally).

Results

- **Initial biopsy**: 12 to 18–21 cores did not raise detection.
- **Repeat setting**: early series gained; a large prospective study did not. Extra cancers are often insignificant (overdiagnosis).
- **PROMIS (2017)** used 5-mm template mapping as the reference: mpMRI (multiparametric MRI) sensitivity 93% vs TRUS (transrectal ultrasound) biopsy 48%.

Complications

- More haematuria, haematospermia, pain and **urinary retention** than standard biopsy; anaesthesia and theatre time. Transperineal route has less infection.

Current position

- Largely replaced by **mpMRI with targeted biopsy**: **PRECISION (2018)** found more significant cancers (38% vs 26%) and 28% avoided biopsy.
- **EAU (European Association of Urology) 2023** prefers the transperineal route. Saturation remains for MRI-negative high suspicion and focal therapy mapping.

Diagram to draw: transperineal grid over the prostate with sampling sectors.

Key points to remember:

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Prostate cancer prevention trials

Type: SAQ (10 marks)

Source: 2017-Summer Paper 4, Q3a

Rationale

- Prostate cancer is common, slow and androgen-dependent. Men with 5α-reductase deficiency have rudimentary prostates. This supported **5ARI (5α-reductase inhibitor)** trials.

Landmark trials

	PCPT (2003)	REDUCE (2010)	SELECT (2009)
Agent	Finasteride 5 mg	Dutasteride 0.5 mg	Selenium 200 µg ± vitamin E 400 IU
Men	18,882; PSA (prostate-specific antigen) ≤3	PSA 2.5–10, prior negative biopsy	35,533
Design	7 years; end-of-study biopsy	Biopsy at 2 and 4 years	Stopped early for futility
All cancers	↓24.8%	↓22.8%	No benefit
High grade	Gleason 7–10 ↑ (6.4% vs 5.1%)	Gleason 8–10 ↑ (0.9% vs 0.6%)	Vitamin E alone ↑ risk

The high-grade controversy

- Proposed artefacts: better PSA performance and a smaller gland make cancer easier to find.
- The FDA (Food and Drug Administration) did **not accept** these. In 2011 it refused a prevention licence and added a **high-grade warning** to 5ARI labels.
- PCPT at 18.4 years: non-significant 25% fewer prostate cancer deaths. Real-world data show no mortality effect.
- **Conclusion:** 5ARIs reduce detection of low-grade cancer, not death.

Vitamins and other agents

- **ATBC** trial: low-dose vitamin E cut prostate cancer 32% in smokers (secondary end point). SELECT then failed.
- **VITAL** (vitamin D): no prostate effect. High-dose zinc, calcium and antioxidants may harm.
- **Aspirin/NSAIDs** (non-steroidal anti-inflammatory drugs): small effect. **Statins:** observational benefit only. **Metformin:** none.
- Diet: whole foods; avoid obesity.

Current position

- No agent is approved for prevention. 5ARIs may be discussed with men taking them for BPH (benign prostatic hyperplasia).

Diagram to draw: bar chart: all cancers ↓ vs high-grade ↑ in PCPT and REDUCE.

Key points to remember:

- PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
- FDA warning; no approved preventive agent.
- SELECT negative; vitamin E alone harmful.

GnRH (LHRH) analogues: agonists and antagonists

Asked as:

- GNRH Analogue — *Winter 2017 Paper IV Q3c (SAQ)*
- LHRH agonist — *Summer 2019 Paper IV Q3b (SAQ)*

Type: SAQ (10 marks)

Read as: GnRH (gonadotrophin-releasing hormone, also LHRH, luteinising hormone-releasing hormone) agonists and antagonists, chiefly in prostate cancer.

Physiology

- **LHRH (luteinising hormone-releasing hormone, GnRH (gonadotrophin-releasing hormone))** from the hypothalamus is released in **pulses**. It (**pulsatile** GnRH) drives (stimulates) pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone). LH → Leydig cells → (Leydig-cell) testosterone (about 95% testicular; about 95% of total).
- **Continuous** exposure (stimulation) **desensitises and down-regulates** the pituitary receptor. This is the basis of agonist therapy.

GnRH (LHRH) agonists

Drugs

Drug	Formulation
Leuprolide	Depot 1, 3, 6 monthly
Goserelin	SC (subcutaneous) implant 1 or 3 monthly
Triptorelin	IM (intramuscular) depot
Histrelin	Implant, 1 year
Buserelin	Nasal or SC

- Substitution at **position 6** of the decapeptide (position-6 substitution) raises potency and half-life; depots last 1 month to 1 year.

Mechanism

```
Agonist → initial LH surge (flare, testosterone up)
        → receptor down-regulation (1–2 weeks)
        → LH falls → castrate testosterone in 2–4 weeks
```

- Initial LH surge, then receptor **down-regulation** → low LH → castration in 2–4 weeks; survival **equals orchidectomy**.

Flare phenomenon

- LH rises up to **10-fold for 10–20 days** → bone pain, **spinal cord compression**, urinary retention (pain).
- Dangerous with spinal or weight-bearing metastases.
- **Prevent with an antiandrogen** (e.g. bicalutamide, flutamide) started with or before the agonist, for **21–28 days**.
- Impending cord compression → **antagonist** (degarelix) or orchidectomy instead.

GnRH antagonists

- Immediate competitive receptor block: LH falls 84% in 24 hours, **no flare**, FSH falls.
- **Degarelix** (SC (subcutaneous) monthly): castrate by day 3 in >95%; injection-site reactions.

- **Relugolix** (oral daily, FDA 2020). **HERO** vs leuprolide: sustained castration 97% vs 89%; faster recovery after stopping; **fewer MACE (major adverse cardiovascular events)** (about 3% vs 6%).
- Preferred for **impending cord compression** or severe bone pain when orchidectomy is unsuitable.
- **Also know:** abarelix restricted (severe allergy); PRONOUNCE (degarelix vs leuprolide, stopped early) showed no MACE difference.

Agonist vs antagonist

Agonist	Antagonist
Flare; needs antiandrogen	No flare; castrate by day 3 (degarelix)
FSH rises	FSH falls
Depot to yearly	Monthly SC or oral (relugolix; HERO)

Castrate level

- **<50 ng/dL** (traditional); **<20 ng/dL** is achievable (orchidectomy median about 15) and may be better (aim **<20 ng/dL**). Check testosterone when PSA (prostate-specific antigen) rises.

Indications in urology

- Metastatic hormone-sensitive prostate cancer (with ARPI (androgen receptor pathway inhibitor) or docetaxel).
- With radiotherapy for intermediate/high-risk and locally advanced disease.
- Recurrent prostate cancer: BCR (biochemical recurrence) with short doubling time; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** BPH (benign prostatic hyperplasia) volume reduction (not routine); central precocious puberty; cryptorchidism (GnRH or hCG (human chorionic gonadotrophin): low success, not first-line); endometriosis; intermittent ADT (androgen deprivation therapy) acceptable (PR.7).

Adverse (side) effects

- Hot flushes (50–80%), loss of libido, ED (erectile dysfunction).
- **Osteoporosis** (DEXA (dual-energy X-ray absorptiometry); calcium, vitamin D, denosumab or zoledronic acid).
- Diabetes, cardiovascular events, sarcopenia, anaemia, gynaecomastia (≤5%), cognitive change.
- Slow testosterone recovery after stopping.

Diagram to draw: hypothalamic–pituitary–testis axis showing agonist and antagonist sites, and the agonist site and flare.

Key points to remember:

- Agonist: surge, then down-regulation; continuous stimulation → down-regulation → castration in 2–4 weeks.
- Flare 10–20 days; cover flare with antiandrogen 21–28 days.
- Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE).
- Castrate <50 ng/dL; aim <20 ng/dL.

Combined Androgen Blockade.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 4, Q3f

Definition and rationale

- **CAB (combined androgen blockade)**, also called maximal or total androgen blockade: castration (orchidectomy or LHRH (luteinising hormone-releasing hormone) agonist) **plus a non-steroidal antiandrogen**.
- **Rationale (Labrie):** castration removes ~90–95% of testosterone, but **adrenal androgens** (DHEA, dehydroepiandrosterone; androstenedione) are converted to DHT (dihydrotestosterone) inside the prostate. An antiandrogen blocks these at the AR (androgen receptor).

Antiandrogens

Drug	Dose	Signature toxicity
Flutamide	250 mg three times daily	Diarrhoea, hepatotoxicity
Nilutamide	300 mg then 150 mg daily	Delayed dark adaptation, pneumonitis
Bicalutamide	**50 mg daily** (with LHRH agent)	Best tolerated; gynaecomastia
Cyproterone (steroidal)	100 mg two to three times daily	Cardiovascular, hepatic; not used for CAB now

- Class effects: gynaecomastia, hepatotoxicity, pneumonitis.

Evidence

- **PCTCG (Prostate Cancer Trialists' Collaborative Group) meta-analysis, 2000** (27 trials, ~8,000 men): with non-steroidal antiandrogens, 5-year survival improved by only **~2–3%** (borderline significance). Cyproterone gave no benefit (slightly worse).
- So the benefit is small, with more cost and toxicity. **AUA (American Urological Association): long-term CAB not recommended**.

Current uses

1. **Flare prevention:** an LHRH agonist raises LH (luteinising hormone) and testosterone for 10–20 days, risking cord compression, pain or obstruction. Start antiandrogen **with or before** the agonist for **21–28 days**. (Antagonists such as degarelix or relugolix cause no flare.)
2. **Antiandrogen withdrawal:** on progression, stopping bicalutamide gives PSA (prostate-specific antigen) fall >50% in 15–30% (mutant AR made the drug an agonist).

Modern intensification (replaces CAB in mHSPC, metastatic hormone-sensitive prostate cancer)

Trial	Addition to ADT (androgen deprivation therapy)	Result
LATITUDE, STAMPEDE (arm G)	Abiraterone + prednisone	OS (overall survival) HR (hazard ratio) 0.62 / 0.63
ENZAMET	Enzalutamide (vs ADT + older antiandrogen)	OS HR 0.67
ARCHES	Enzalutamide	Radiographic PFS (progression-free survival) HR 0.39
TITAN	Apalutamide	OS HR 0.67

Trial	Addition to ADT (androgen deprivation therapy)	Result
PEACE-1, ARASENS	**Triplet**: + docetaxel + abiraterone or darolutamide	OS benefit (ARASENS HR 0.68)

Diagram to draw: testis and adrenal androgen sources, with LHRH agonist and antiandrogen sites of action.

Key points to remember:

- CAB = castration + non-steroidal antiandrogen (adrenal androgens).
- PCTCG: ~2–3% 5-year survival gain only.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC.

PSMA PET Ct scan

Type: SAQ (10 marks)

Source: 2019-Summer Paper 4, Q3a

Principle

- **PSMA (prostate-specific membrane antigen):** transmembrane glycoprotein over-expressed by prostate cancer, rising with grade.
- **PET/CT (positron emission tomography/computed tomography):** positron-labelled ligand binds PSMA; uptake as SUV (standardised uptake value).
- **Tracers:** ^{68}Ga-PSMA-11 (half-life 68 min); ^{18}F-DCFPyL, ^{18}F-rhPSMA-7.3, ^{18}F-PSMA-1007 (110 min, better images; PSMA-1007 little urinary excretion).
- **Specificity >90%**; beats choline and fluciclovine.

Indications

1. Primary staging (high risk):

- **proPSMA (2020):** accuracy **92% vs 65%** vs CT + bone scan.
- Nodal sensitivity only 30–40% against PLND (pelvic lymph node dissection): a negative scan does not exclude micrometastases.
- NCCN (National Comprehensive Cancer Network): may **replace** conventional imaging; AUA (American Urological Association): add-on.

2. BCR (biochemical recurrence):

- Detection rises with PSA (prostate-specific antigen): about **half positive below 0.5 ng/mL**, nearly all above 2.
- **Calais trial:** PSMA 56% vs fluciclovine 26%. **CONDOR:** management changed in 63.9%.
- Guides salvage radiotherapy (disease outside field in about 1 in 5).

3. Intraprostatic detection: PRIMARY trial: adding PSMA PET to MRI (magnetic resonance imaging) improved negative predictive value; 5-point PRIMARY score.

4. Theranostics: selects patients for **^{177}Lu-PSMA-617**. **VISION (2021):** mCRPC (metastatic castration-resistant prostate cancer) after ARPI (androgen receptor pathway inhibitor) and taxane: OS 15.3 vs 11.3 months.

Reporting

- **PROMISE miTNM** (molecular imaging TNM): miT, miN, miM staging plus expression score (vs blood pool, liver, parotid). PSMA-RADS also used.

Pitfalls

False positive	False negative
Coeliac/stellate ganglia, fractures, Paget's disease, fibrous dysplasia	Small nodes (<5 mm)
Other tumours (RCC (renal cell carcinoma), HCC (hepatocellular carcinoma))	PSMA-negative tumours (about 5–10%)
^{18}F-PSMA-1007 benign bone uptake	Neuroendocrine transformation; masking by bladder activity

- **Also know:** no proof yet that PSMA-guided treatment improves survival (stage migration).

Diagram to draw: PSMA ligand binding the cell-surface antigen with fused PET/CT image of a nodal lesion.

Key points to remember:

- proPSMA: accuracy 92% vs 65%.
- BCR: about half positive below PSA 0.5 ng/mL.
- Nodal sensitivity 30–40%: negative scan does not exclude nodes.
- VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

Methods to predict prognosis of patient of Cancer prostate

Type: SAQ (10 marks)

Source: 2019-Summer Paper 4, Q3e

1. Clinical and pathological factors

- **PSA (prostate-specific antigen):** organ-confined about 80% at <4 but <50% at >10 ng/mL; density and velocity add information.
- **Clinical stage (DRE (digital rectal examination), TNM).**
- **Grade group (ISUP (International Society of Urological Pathology)) – strongest single factor:**

Grade group	Gleason	5-year BCR (biochemical recurrence)-free
1	≤6	96%
2	3+4	88%
3	4+3	63%
4	8	48%
5	9–10	26%

- Biopsy: % positive cores, **cribriform and intraductal** patterns, perineural invasion.
- Prostatectomy: margin (extent, grade), EPE (extraprostatic extension), seminal vesicle and node status.

2. Risk groups

D'Amico	Low	Intermediate	High
PSA (ng/mL)	≤10	10–20	>20
Gleason	≤6	7	8–10
Stage	T1–T2a	T2b	T2c

- **NCCN (National Comprehensive Cancer Network):** very low, low, favourable/unfavourable intermediate, high, very high.
- **CAPRA (Cancer of the Prostate Risk Assessment)** score 0–10 (age, PSA, Gleason, stage, % cores); **CAPRA-S** after surgery.

3. Nomograms

- **Partin tables:** organ-confined disease, EPE, seminal vesicle, node involvement.
- **MSKCC (Memorial Sloan Kettering) nomograms:** pre- and post-treatment BCR and 15-year cancer-specific survival.
- **Briganti nomogram:** risk of nodal invasion → decides PLND (pelvic lymph node dissection).

4. Imaging

- mpMRI (multiparametric MRI): PI-RADS (Prostate Imaging Reporting and Data System) score, tumour contact length, EPE.
- PSMA PET (prostate-specific membrane antigen PET) for metastatic burden.

5. Tissue biomarkers

Test	Basis	Use
Decipher	22-RNA classifier	Metastasis risk; post-op radiotherapy

Test	Basis	Use
Oncotype DX GPS	17 genes	Adverse pathology; active surveillance
Prolaris	Cell-cycle progression genes	Progression, cancer death

- Germline BRCA2 (BReast CAncer gene 2): aggressive disease.

6. After treatment

- **PSA doubling time (PSADT):** <3 months highest risk; Pound: time to BCR <2 years, PSADT <10 months, Gleason 8–10 predict metastasis.
- **EAU (European Association of Urology) BCR risk groups:** low risk = PSADT >1 year and grade group <4.
- PSA nadir on ADT (androgen deprivation therapy); **Phoenix** failure (nadir + 2) after radiotherapy.

Diagram to draw: flowchart from risk group to nomogram, biomarker and treatment decision.

> **Key points to remember:**
> - Grade group is the strongest predictor.
> - D'Amico, NCCN and CAPRA combine PSA, grade and stage.
> - Decipher, Oncotype, Prolaris add genomic risk.

PARP inhibitors in Prostate cancer.

Type: SAQ (10 marks)

Source: 2024-Winter Paper 4, Q3f

Introduction

- **PARP (poly-ADP-ribose polymerase)** repairs single-strand DNA breaks. **PARPIs (PARP inhibitors):** olaparib, rucaparib, niraparib, talazoparib.

Mechanism: synthetic lethality

- **Catalytic inhibition** and **PARP trapping** on DNA leave single-strand breaks unrepaired.
- At replication these become **double-strand breaks**, normally repaired by **HRR (homologous recombination repair)**.
- Tumours lacking HRR (e.g. **BRCA2** loss) die; normal cells survive.
- **AR (androgen receptor)–PARP synergy:** AR maintains HRR genes, so AR blockade adds repair deficiency: basis for PARPI + ARPI (androgen receptor pathway inhibitor).

Who benefits: testing

- **HRR alterations in mCRPC (metastatic castration-resistant prostate cancer): up to 20–25%.** Genes: **BRCA2 (largest benefit)**, BRCA1, ATM (weak), PALB2, CHEK2, CDK12.
- **Germline testing** in all metastatic disease, strong family history or Ashkenazi ancestry; **somatic testing** on tissue or ctDNA (circulating tumour DNA), where clonal haematopoiesis can mislead.

Key trials

Trial	Setting	Result
PROfound	Olaparib vs enzalutamide/abiraterone after ARPI	BRCA/ATM cohort: imaging PFS (progression-free survival) **7.4 vs 3.6 months (HR [hazard ratio] 0.34)**; OS (overall survival) 19.1 vs 14.7 months in the final analysis (chapter quotes the interim 18.5 vs 15.1)
TRITON3	Rucaparib vs physician's choice	BRCA: PFS 11.2 vs 6.4 months (HR 0.50); no ATM benefit
PROpel	Olaparib + abiraterone, first-line mCRPC	OS 42.1 vs 34.7 months (P = 0.054, not significant); greatest in BRCA
MAGNITUDE	Niraparib + abiraterone, HRR-altered	BRCA rPFS (radiographic PFS) 19.5 vs 10.9 months (HR 0.55); no benefit without HRR alteration
TALAPRO-2	Talazoparib + enzalutamide, first line	All-comers rPFS not reached vs 21.9 months (HR 0.63); largest in BRCA

- **Also know:** TOPARP-A/B, TRITON2, GALAHAD (phase II).

Approvals (FDA [Food and Drug Administration])

- **Olaparib monotherapy:** HRR-mutated mCRPC after ARPI (2020).
- **Rucaparib:** BRCA-mutated mCRPC.
- **Olaparib or niraparib + abiraterone:** BRCA-mutated; **talazoparib + enzalutamide:** HRR-mutated (both 2023).

Toxicity

- **Anaemia** (commonest grade 3–4; monitor counts), fatigue, nausea; rare **MDS/AML (myelodysplastic syndrome/acute myeloid leukaemia)**, pulmonary embolism.

Recent advances

- **mCSPC (metastatic castration-sensitive prostate cancer)** trials: AMPLITUDE, TALAPRO-3; ATR inhibitors for ATM loss.

Diagram to draw: synthetic lethality: PARP block + BRCA loss → cell death.

> **Key points to remember:**
> - Synthetic lethality; BRCA2 gains most, ATM least.
> - PROfound: olaparib, PFS 7.4 vs 3.6 months.
> - Combinations: PROpel, MAGNITUDE, TALAPRO-2.

PI-RADS and MRI–ultrasound fusion biopsy

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 4, Q3c (mpMRI and PI-RADS also answered in 2024-Winter Paper 1, Q3d; biopsy in Practice Set 4 Paper 2, Q3d)

PI-RADS

- **PI-RADS (Prostate Imaging Reporting and Data System)** v2.1 (2019) scores the likelihood of **clinically significant cancer** (Gleason ≥7, volume >0.5 cm^3 or EPE [extraprostatic extension]) on mpMRI (multiparametric magnetic resonance imaging): T2W (T2-weighted: anatomy), DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient: cellularity), DCE (dynamic contrast-enhanced: vascularity).

Score	Significant cancer
1	Highly unlikely
2	Unlikely
3	Equivocal
4	Likely; <1.5 cm
5	Highly likely; ≥1.5 cm or definite EPE

Zone	Dominant sequence	Tie-breaker at score 3
Peripheral	**DWI**	DCE positive → 4
Transition	**T2W**	DWI 5 → 4 (v2.1: T2W 2 with DWI ≥4 → 3)

- Cancer in about one-third of PI-RADS 3, over half of 4, 90% of 5.
- **Also know:** PI-QUAL (image quality); PRECISE (active surveillance).

Fusion biopsy: principle

- Lesion contoured on MRI → software **co-registers it with live TRUS (transrectal ultrasound)** in three dimensions; **elastic deformation** corrects probe distortion → needle aimed at the displayed target.
- Freehand electromagnetic or mechanical-arm tracking; transrectal or **transperineal** (fewer infections).

	Cognitive	Software fusion	In-bore MRI
Method	Aims from memory of MRI	Software overlay	Needle under MRI
Drawback	Operator-dependent; small lesion in large gland	Cost; registration error	Time, cost

- Meta-analysis (2022): no significant difference among the three; one RCT (randomised controlled trial) favoured fusion over cognitive.

Indications

- PI-RADS ≥3 lesion with raised PSA (prostate-specific antigen), first or repeat biopsy; active surveillance; focal therapy.
- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023:** 2–3 cores per target ± systematic cores.

Evidence

- **PROMIS (2017):** MRI sensitivity 93% versus 48% for TRUS biopsy; 27% avoid biopsy.
- **PRECISION (2018):** MRI pathway found more significant cancer (38% versus 26%); 28% avoided biopsy.

- Targeting reduces overdiagnosis; MRI-invisible cancers keep systematic cores relevant.

Diagram to draw: MRI target on live TRUS with needle path.

Key points to remember:

- Peripheral zone by DWI, transition zone by T2W; DCE only lifts peripheral 3 to 4.
- Fusion: co-registration with elastic deformation.
- Cognitive, fusion, in-bore: similar yield.
- PROMIS and PRECISION: MRI before biopsy.

Indications of genomic testing in carcinoma prostate

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 4, Q3e (PARP [poly-ADP-ribose polymerase] inhibitors also answered in 2024-Winter Paper 4, Q3f)

Types of test

Type	Sample	What it finds	Use
Germline	Blood or saliva	Inherited variants: BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13, MMR (mismatch repair: MLH1, MSH2, MSH6, PMS2)	Therapy, family testing, screening
Somatic	Tumour tissue (usually primary) or ctDNA (circulating tumour DNA)	HRR (homologous recombination repair) genes, CDK12, MSI-H (microsatellite instability-high)/dMMR (MMR deficiency)	Targeted therapy
Gene-expression classifiers	Biopsy or prostatectomy tissue	Decipher (22 RNA markers); Oncotype DX GPS (Genomic Prostate Score, 17 genes); Prolaris (cell-cycle genes)	Prognosis

Indications: germline testing

- **All metastatic** disease (AUA [American Urological Association], NCCN [National Comprehensive Cancer Network]): over 10% carry actionable DNA-repair mutations.
- **High- or very-high-risk** localised disease.
- **Intermediate risk with intraductal or cribriform histology** (consider).
- **Family history:** early-onset or metastatic prostate cancer (relative <60 years); BRCA-related (breast, ovarian, pancreatic) or Lynch-related (colorectal, endometrial, upper tract) cancers; known familial mutation.
- **Ashkenazi Jewish ancestry.**
- **Why:** PARP inhibitor eligibility; caution with active surveillance in BRCA2; carrier screening; cascade testing of relatives with counselling.

Indications: somatic testing

- **Metastatic disease, especially mCRPC (metastatic castration-resistant prostate cancer):** HRR alterations in up to 20–25%.
  - **HRR (BRCA2 most)** → PARP inhibitors (olaparib, rucaparib, niraparib, talazoparib).
  - **MSI-H/dMMR** (2–5%) → pembrolizumab (tumour-agnostic FDA [Food and Drug Administration] approval, 2017); then germline Lynch testing.
- Tissue preferred; ctDNA alternative, but **CHIP (clonal haematopoiesis of indeterminate potential)** gives false calls.

Indications: tissue classifiers

- **Low and favourable-intermediate risk:** active surveillance versus treatment; surveillance intensity. Never the sole basis.
- **Intermediate and high risk:** metastasis risk guides treatment intensity.
- **After prostatectomy:** high Decipher score may favour adjuvant or early salvage radiotherapy (retrospective evidence).

- **Limits:** tumour heterogeneity, retrospective data, cost.
- **Also know:** ConfirmMDx methylation before repeat biopsy; polygenic risk scores.

Diagram to draw: flowchart: metastatic → germline + somatic; localised high-risk or family history → germline; localised low or intermediate → tissue classifier.

Key points to remember:

- Germline: all metastatic, high-risk, family history, Ashkenazi.
- Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
- BRCA2: aggressive; avoid casual active surveillance.
- Decipher, Oncotype GPS, Prolaris: prognosis only.

Rapid Revision Sheet – Section 17 · Prostate Cancer

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate? (LAQ)

- α/β ~1.5 Gy: basis of 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.

Discuss advances in management of CA prostate (LAQ)

- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
- proPSMA: PSMA PET accuracy 92% vs 65%.
- ProtecT: no mortality difference at 15 years; AS is standard for GG1.
- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

Cryotherapy for carcinoma prostate (SAQ)

- Argon freezes, helium thaws; −40 °C, two cycles, urethral warmer.
- Main role is salvage: PSA <10, doubling time ≥16 months.
- Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

Saturation biopsies of prostate (SAQ)

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Prostate cancer prevention trials (SAQ)

- PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
- FDA warning; no approved preventive agent.
- SELECT negative; vitamin E alone harmful.

GnRH (LHRH) analogues: agonists and antagonists (SAQ)

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PSMA PET Ct scan (SAQ)

- proPSMA: accuracy 92% vs 65%.
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Appendix · Original papers: headings, notes and contents

Each source file's own title, note and contents list, exactly as printed, so every word of the originals is kept.

2015-Summer Paper 4 – Recent Advances

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1. Q1. Role of robotics in urology (LAQ, 25)
2. Q2. Marginal kidney donor and newer immunosuppressant drugs in renal transplantation (LAQ, 25)
3. Q3a. Hypospadias fistula: aetiology and management (SAQ, 10)
4. Q3b. Silodosin (SAQ, 10)
5. Q3c. Sacral neuromodulation (SAQ, 10): cross-reference to 2015-Summer Paper 3
6. Q3d. Urethral stents (SAQ, 10)
7. Q3e. Studer neobladder (SAQ, 10)
8. Q3f. Cryotherapy for carcinoma prostate (SAQ, 10)

2016-Summer Paper 4 – Recent Advances

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4. Q3b. Testis-sparing surgery for carcinoma testis (SAQ, 10)
5. Q3c. Sacral neuromodulation (SAQ, 10)
6. Q3d. Video urodynamics (SAQ, 10)
7. Q3e. Tubeless PCNL (SAQ, 10)
8. Q3f. Vesicostomy (SAQ, 10)

2017-Summer Paper 4 – Recent Advances

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1. Q1. Intravesical therapy for transitional cell carcinoma of urinary bladder (LAQ, 25)
2. Q2. Applications of various LASERs in urology (LAQ, 25)
3. Q3a. Prostate cancer prevention trials (SAQ, 10)
4. Q3b. Prepucioplasty (preputioplasty) (SAQ, 10)
5. Q3c. Optical coherence tomography in carcinoma bladder (SAQ, 10)
6. Q3d. Re-look TUR bladder tumour (SAQ, 10)
7. Q3e. Male menopause (SAQ, 10)
8. Q3f. Acute urinary retention in females (SAQ, 10)

2017-Winter Paper 4 – Recent Advances

Contents

1. Q1. Robot and its role in urology (LAQ, 25)
2. Q2. Pathophysiology of allograft rejection; newer immunosuppressive drugs in renal transplant (LAQ, 25)
3. Q3a. Botulinum toxin in urology (SAQ, 10): cross-reference to 2017-Winter Paper 3
4. Q3b. Sacral neuromodulation (SAQ, 10): cross-reference to 2017-Winter Paper 3
5. Q3c. GnRH analogue (SAQ, 10)

6. Q3d. Miniperc (SAQ, 10)
7. Q3e. Andropause (SAQ, 10)
8. Q3f. Nephron saving surgery (SAQ, 10)

2018-Summer Paper 4 – Recent Advances

Contents

1. Q1. Radiology for solid renal masses (LAQ)
2. Q2. Diagnosis and management of adrenal carcinoma (LAQ)
3. Q3a. Struvite stones (SAQ)
4. Q3b. Klinefelter's syndrome (SAQ)
5. Q3c. Brain dead committee (SAQ)
6. Q3d. Chronic non-bacterial prostatitis (SAQ)
7. Q3e. Semen analysis nomenclature (SAQ)
8. Q3f. Combined androgen blockade (SAQ)

2019-Summer Paper 4 – Recent Advances

Contents

1. Q1. Neuromodulation for voiding dysfunction (LAQ)
2. Q2. Bladder preservation surgery for TCC of urinary bladder (LAQ)
3. Q3a. PSMA PET CT scan (SAQ)
4. Q3b. LHRH agonist (SAQ)
5. Q3c. Digital flexible ureteroscopes (SAQ)
6. Q3d. Videoendoscopic inguinal lymphadenectomy (SAQ)
7. Q3e. Methods to predict prognosis of cancer prostate (SAQ)
8. Q3f. Male menopause (SAQ)

2020-Summer Paper 4 – Recent Advances

Contents

1. Q1. Describe immunobiology of renal transplant and newer immunosuppressant drugs. (LAQ)
2. Q2. Describe etiopathogenesis of stone formation and medical management of urolithiasis. (LAQ)
3. Q3a. Neobladder. (SAQ)
4. Q3b. Vascular access for dialysis. (SAQ)
5. Q3c. STING technique for VUR. (SAQ)
6. Q3d. HOLEP. (SAQ)
7. Q3e. Nitrofurantoin. (SAQ)
8. Q3f. Describe evidence based medicine. (SAQ)

2021-Summer Paper 4 – Recent Advances

Contents

1. Q1. Role and impact of radiotherapy methods for management of focal and advanced carcinoma prostate (LAQ)
2. Q2. Define azoospermia; types of testicular biopsies and surgical sperm retrieval (LAQ)
3. Q3a. Operating room safety: the role of surgeon (SAQ)
4. Q3b. Detrusor underactivity (SAQ)
5. Q3c. Surgical management of primary scrotal cancer (SAQ)
6. Q3d. Painful bladder syndrome management (SAQ)
7. Q3e. Secondary tumours after urinary diversion (SAQ)
8. Q3f. Role of open and laparoscopic stone surgeries in the modern era of endourology (SAQ)

2022-Summer Paper 4 – Recent Advances

Contents

1. Q1. Natural orifice endoscopic surgery (LAQ)
2. Q2. Tissue engineering in urology (LAQ)
3. Q3a. Urological applications in nanotechnology (SAQ)
4. Q3b. Guidelines for declaration of brain stem death (SAQ)
5. Q3c. Different types of trocars in laparoscopy (SAQ)
6. Q3d. Current ureter scopes (SAQ)
7. Q3e. General complications related to laparoscopy (SAQ)
8. Q3f. Approach to evaluation of adrenal mass/s (SAQ)

2023-Winter Paper 4 – Recent Advances

Contents

1. Q1. Laproendoscopic single site surgery: LESS (LAQ)
2. Q2. Holep (LAQ)
3. Q3a. Neurovascular bundle preservation in Radical prostatectomy: recent techniques (SAQ)
4. Q3b. Muscle assisted urethroplasty (SAQ)
5. Q3c. Penile implants (SAQ)
6. Q3d. Female kidney donors: enumerate special precautions (SAQ)
7. Q3e. Hospital authorization committee for live related kidney donation (SAQ)
8. Q3f. Diet in stone disease (SAQ)

2024-Winter Paper 4 – Recent Advances

Contents

1. Q1. Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis. (LAQ)
2. Q2. Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T. (LAQ)
3. Q3a. Newer methods of stone analysis and their significance in management of calculus disease. (SAQ)
4. Q3b. Enlist the steps in female to male gender reassignment surgery. (SAQ)
5. Q3c. Newer generation laser technologies. (SAQ)
6. Q3d. Urolift. (SAQ)
7. Q3e. Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys. (SAQ)
8. Q3f. PARP inhibitors in Prostate cancer. (SAQ)

2025-Winter Paper 4 – Recent Advances

Contents

1. Q1. Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? (LAQ)
2. Q2. Describe Basic Energy Modalities in Urological surgery? (LAQ)
3. Q3a. Immuno modulation in Renal, cell carcinoma (SAQ)
4. Q3b. New LASER for stone Treatment (SAQ)
5. Q3c. Combination Intravesical therapy (SAQ)
6. Q3d. Penile Inversion Vaginoplasty Technique (SAQ)
7. Q3e. Robotic RPLND (SAQ)
8. Q3f. Management of Defecation Disorders resulting into LUTS (SAQ)

Practice Set 1 Paper 4 – Recent Advances

Contents

1. Q1. Briefly outline flexible endourological instrument and its role in management of urological disease (LAQ)
2. Q2. Describe energy sources for breaking stones in Urology (LAQ)
3. Q3a. Antioxidants in urological malignancy (SAQ)
4. Q3b. Bipolar energy (SAQ)
5. Q3c. Thalium laser in urology (SAQ)
6. Q3d. Morcellation (SAQ)
7. Q3e. KTP*** laser in urology (SAQ)
8. Q3f. Stem cell therapy (SAQ)

Practice paper compiled from the question bank.

Practice Set 2 Paper 4 – Recent Advances

Contents

1. Q1. Discuss in detail anatomy of prostate with special reference to newer concepts in surgical management of CA prostate (LAQ, 25)
2. Q2. Describe etiopathogenesis of VUR and recent trends in management (LAQ, 25)
3. Q3a. Laparoscopic donor nephrectomy (SAQ, 10)
4. Q3b. Green light lasers (SAQ, 10)
5. Q3c. Single port lap. Surgeries in urology (SAQ, 10)
6. Q3d. Pros and cons of lap donor nephrectomy (SAQ, 10): cross-reference to Q3a
7. Q3e. Holmium laser (SAQ, 10)
8. Q3f. Hemostasis in lap surgery (SAQ, 10)

Practice paper compiled from the question bank.

Practice Set 3 Paper 4 – Recent Advances

Contents

1. Q1. Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs (LAQ, 25)
2. Q2. Describe principles of continent urinary diversions. Current status of orthotopic neobladder. (LAQ, 25)
3. Q3a. Advances in lap surgeries (SAQ, 10)
4. Q3b. Robotic vs std lap surgeries (SAQ, 10)
5. Q3c. Metallic stents in urology (SAQ, 10)
6. Q3d. Gracilis myocutaneous flap in Uro. (SAQ, 10)
7. Q3e. Zero-ischaemia partial nephrectomy (laparoscopic and robotic) (SAQ, 10)
8. Q3f. Steroid-free immunosuppression in renal transplantation (SAQ, 10)

Practice paper compiled from the question bank.

Practice Set 4 Paper 4 – Recent Advances

Contents

1. Q1. Discuss in brief, present concepts in various GU cancers (LAQ, 25)
2. Q2. Robots in urology (LAQ, 25)
3. Q3a. Newer PCNL techniques and nomenclature of miniaturised PCNL (SAQ, 10)
4. Q3b. VI-RADS (SAQ, 10)
5. Q3c. PI-RADS and MRI–ultrasound fusion biopsy (SAQ, 10)
6. Q3d. Artificial intelligence in urology (SAQ, 10)
7. Q3e. Indications of genomic testing in carcinoma prostate (SAQ, 10)

8. Q3f. Gene therapy in urology (SAQ, 10)

Practice paper compiled from the question bank.

Practice Set 5 Paper 4 – Recent Advances

Contents

1. Q1. Describe energy sources for braking stones in uro (LAQ)
2. Q2. Discuss advances in management of CA prostate (LAQ)
3. Q3a. Virtual cystoscopy (SAQ)
4. Q3b. 3D laparoscopy (SAQ)
5. Q3c. Telemedicine in urology (SAQ)
6. Q3d. Simulators in urology (SAQ)
7. Q3e. Virtual and augmented reality in urology (SAQ)
8. Q3f. Bladder transplantation (SAQ)

Practice paper compiled from the question bank.

Practice Set 6 Paper 4 – Recent Advances

Contents

1. Q1. Present concept in management of hypospadias (LAQ, 25)
2. Q2. Discuss recent advances in tissue engineering with respect to urology (LAQ, 25)
3. Q3a. 3D printing in urology (SAQ, 10)
4. Q3b. ECIRS (endoscopic combined intrarenal surgery) (SAQ, 10)
5. Q3c. Molecular targeted therapy in urological cancers (SAQ, 10)
6. Q3d. Artificial urinary bladder (SAQ, 10)
7. Q3e. Artificial kidney (SAQ, 10)
8. Q3f. Robotic kidney transplantation (SAQ, 10)

Practice paper compiled from the question bank.

(Earlier version) Practice Set 3 Paper 4 – Recent Advances

Contents

1. Q1. Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs (LAQ, 25)
2. Q2. Describe principles of continent urinary diversions. Current status of orthotopic neobladder. (LAQ, 25)
3. Q3a. Advances in lap surgeries (SAQ, 10)
4. Q3b. Robotic vs std lap surgeries (SAQ, 10)
5. Q3c. Metallic stents in urology (SAQ, 10)
6. Q3d. Gracilis myocutaneous flap in Uro. (SAQ, 10)
7. Q3e. Newer generation laser technologies. (SAQ, 10)
8. Q3f. New LASER for stone Treatment (SAQ, 10): cross-reference to Q3e

Practice paper compiled from the question bank.

(Earlier version) Practice Set 4 Paper 4 – Recent Advances

Contents

1. Q1. Discuss in brief, present concepts in various GU cancers (LAQ, 25)
2. Q2. Robots in urology (LAQ, 25)
3. Q3a. Female kidney donors: enumerate special precautions (SAQ, 10)
4. Q3b. Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys. (SAQ, 10)
5. Q3c. Studer Neobladder. (SAQ, 10)
6. Q3d. Neobladder. (SAQ, 10): cross-reference to Practice Set 4 Paper 2, Q3e

7. Q3e. Secondary tumors after urinary diversion. (SAQ, 10)
8. Q3f. Urethral stents. (SAQ, 10)

Practice paper compiled from the question bank.

(Earlier version) Practice Set 5 Paper 4 – Recent Advances

Contents

1. Q1. Describe energy sources for braking stones in uro (LAQ)
2. Q2. Discuss advances in management of CA prostate (LAQ)
3. Q3a. Muscle assisted urethroplasty (SAQ)
4. Q3b. Silodosin. (SAQ)
5. Q3c. Optical coherence tomography in carcinoma bladder. (SAQ)
6. Q3d. Re-look TUR bladder tumour. (SAQ)
7. Q3e. STING technique for VUR. (SAQ)
8. Q3f. Diet in stone disease. (SAQ)

Practice paper compiled from the question bank.

(Earlier version) Practice Set 6 Paper 4 – Recent Advances

Contents

1. Q1. Present concept in management of hypospadias (LAQ, 25)
2. Q2. Discuss recent advances in tissue engineering with respect to urology (LAQ, 25)
3. Q3a. Newer methods of stone analysis and their significance in management of calculus disease. (SAQ, 10)
4. Q3b. Urolift. (SAQ, 10)
5. Q3c. Combination Intravesical therapy (SAQ, 10)
6. Q3d. Etiology and management of hypospadias fistula. (SAQ, 10)
7. Q3e. Tubeless PCNL (SAQ, 10)
8. Q3f. Miniperc (SAQ, 10)

Practice paper compiled from the question bank.